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PAPER-III

INTRODUCTION TO NEURO DEVELOPMENTAL DISABILITIES (SECD 03)

B.Ed. Spl. Ed

Madhya Pradesh Bhoj (Open) University

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INTRODUCTION TO NEURO DEVELOPMENTAL DISABILITIES (SECD 03)

Block 1: Learning Disability: Nature, Needs and Intervention

Block 2: Intellectual Disability: Nature, Needs and Intervention

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Block 1: Learning Disability: Nature, Needs and Intervention

Unit 1: Definition, Types and Characteristics

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Unit 1

Definition, Types and Characteristics

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1.1 Introduction

Although the term “*learning disabilities*” has been in use since 1962, there is no single universally accepted definition of the condition. Current descriptions and definitions of learning disabilities are found in the World Health Organization’s disabilities document, in legislation and policy pertaining to education, disability issues, psychology, medicine and human rights. In addition, groups such as the Learning Disabilities Associations and the National Joint Committee on Learning Disabilities have put forward their own definitions. While these definitions contain some common features, they are not consistent nor are they written in language readily understood and applied by those who have learning disabilities, their families and those who work in the relevant helping professions. This lack of a consistent definition represents a major barrier for people who have learning disabilities. To address this and other related issues, LDAO’s **Promoting Early Intervention for Learning Disabilities (PEI)** Project was established in 1999. The first deliverable for the project was a new definition of learning disabilities, meeting the following minimal criteria:

the definition must be scientifically sound, based on and supported by current research findings, easily understood and utilized by all those who need to understand and use it, practical, and inclusive of the varying types of learning disabilities.

The new definition was developed by the Definition Working Group, a group of individuals representing all the relevant fields. The definition has been approved by the Steering Committee of the Promoting Early Intervention Project and subsequently endorsed by the LDAO Board of Directors. This supporting document has been developed to assist all those who utilize the definition of learning disabilities to understand the definition and its application and the underlying principles.

What are learning disabilities?

There are many different neuropsychological or neurobiological impairments or difficulties, which are collectively described as “learning disabilities”. Learning disabilities are not a single or uniform condition and they do not affect solely one particular type of skill or area of learning. Other definitions have often used the term “*heterogeneous*” to describe this range. This definition uses “*variety of disorders*” instead, a more accessible term. Furthermore, it is important to note the diversity of the impact of learning disabilities. They range from mild to moderate to severe and an individual can have a number of different types of difficulties in different areas and at different levels of severity or complexity.

1.2 Establishing a context

The use of the term *disorder* indicates that the manifestations of learning disabilities are outside the normal range or continuum of human function, in terms of an individual’s ability to process information and communicate. The word disorder is usually used to describe a “lack or loss of normal function”. It is a term that appears in many psychological, psychiatric and other medical documents and manuals, including the DSM-IV. The level of observed diversity is often described as *clinically significant*. The World Health Organization has, in the past, used a three-tier hierarchy of terms to describe the variations in human function, as related to disability issues. These were impairment, disability and handicap.

Impairment was defined as “loss or abnormality of psychological, physiological or anatomical structure or function”. This made impairment “intrinsic to the individual” (a phrase that appears

in a number of other definitions of learning disabilities). Impairment is either functional or medical in nature and a learning disability is usually considered to be a neurological impairment. Impairment became a *disability*, when the individual with impairment was expected to carry out regular or routine tasks that relied on the use of skills or knowledge in the area(s), affected by the impairment. The third level in this hierarchy was the term “*handicap*”, which described the impact of the disability, when there were no accommodations or supports offered to the individual concerned. Most jurisdictions have eliminated the term “handicap” from legislation and regular use. However, this term still appears in the Ontario Human Rights Code and therefore it cannot be eliminated from usage until such time that the Human Rights Code is amended.

In December 2000, the World Health Organization released its new disability-related document, the *International Classification of Functioning, Disability and Health*. In addition to eliminating all references to handicap, this document also stressed the importance of matching the individual’s strengths and needs to the demands of his or her environment. The LDAO definition also includes this important concept. (See section 3 of the definition) People with learning disabilities usually have difficulties acquiring, retaining, processing, understanding, organizing and using information in all of its diverse forms. Within the educational system in the Province of Ontario, learning disabilities are included among “communication exceptionalities”. In lay terms, they are often described as problems with information processing, especially verbal and non-verbal, symbolic and concrete information.

1.3 The ongoing debate about a discrepancy between intelligence and achievement

When the term learning disabilities came into use in 1962, Dr. Samuel Kirk introduced the concept of psychological processing disorders, which he said interfered with academic achievement. This was accompanied by an exclusion concept that learning disabilities could not primarily be due to some other condition, including developmental disabilities (mental retardation). In 1965, Dr. B. Bateman expanded these concepts by emphasizing the importance of underachievement and a discrepancy between estimated potential and actual performance as key requirements for the identification of a learning disability.

Since the nineteen sixties most definitions of learning disabilities have included references to these key concepts of average intelligence and a measurable ability-achievement discrepancy. This new definition does not restrict itself to the concept of a discrepancy between a global I.Q. calculation and achievement. However, the new definition does propose that learning disabilities are demonstrated by:

- impairments in one or more psychological processes related to learning, in combination with **otherwise average abilities essential for thinking and reasoning**, as well as
- **Unexpectedly low academic achievement**, or
- **Average or above-average achievement**, attained only at the expense of **unrealistically high levels of effort** and/or educational **support**.

Discrepancy factors have enabled diagnosticians to document the learning problems of people whose performances on measures of academic achievement or measures of specific processing skills are out of line with their level of measured intelligence. While a discrepancy may be observed at all levels of intellectual functioning, learning disabilities have been generally linked to average to above average intelligence. Applying a discrepancy formula enabled psychologists to deal with the inherent unreliability of many measures of ability by applying statistical procedures that take these sources of unreliability into account. As a result, psychologists are able to arrive at a more accurate differential diagnosis. For people with learning disabilities and their families it has been particularly important to differentiate between intellectual or developmental disabilities and learning disabilities.

Furthermore, this distinction has been crucial in ensuring that educational programs are accurately linked to the different needs of the two diverse populations. Nevertheless, the new definition does not contain any specific recommended criteria for intelligence testing or for the determination of a discrepancy. Any such recommendations will be found in the screening and assessment protocol that has been developed to support this definition the more appropriate location for diagnostic directions. Recently, several researchers have begun to oppose the use of the discrepancy formula for the diagnosis of learning disabilities. Various reasons for this are explored below.

1.4 Learning disabilities versus reading disabilities

A great deal of recent research has focused specifically and exclusively on reading disabilities. Reading disabilities are now believed to be predominantly the result of phonological processing deficits.

Some researchers have observed that phonological processing deficits (problems with awareness and understanding of the underlying sound structure of spoken words) are not directly linked to intelligence. In other words, severe phonological processing problems have been observed in some poor readers at all ages, independent of their level of measured intelligence. In contrast, some children with very limited measured IQs have been found to be able to rapidly and accurately decode unfamiliar printed words (albeit with limited comprehension) at a level considerably in advance of their age or grade. Based on such findings, these researchers have questioned the contribution of any measure of intellectual functioning towards the diagnosis of individuals with reading disabilities and have argued for the abandonment of the use of any discrepancy formula as a diagnostic criterion.

The Definition Working Group endorsed the importance of recognizing phonological processing as one of the key factors in an individual's ability to learn to read. However, it also recognized that reading consists of more than just decoding and learning consists of more than just reading. Therefore, in its work to develop a new definition of learning disabilities, the Working Group was not prepared to apply research findings related primarily to reading disabilities to the whole field of learning disabilities. In fact, it strongly supports the importance of recognizing that the term "specific learning disabilities" includes both academic and non-academic areas of difficulty.

1.5 Psychometric issues

Another problem associated with the use of discrepancy formulae involves problems with the psychometric properties of the tests used, as well as influences from statistical phenomena such as regression to the mean. The use of tests that measure ability and achievement based upon different normative samples greatly complicates the degree to which scores from these tests can be meaningfully compared. In addition, due to their imprecision, the use of age equivalents and grade-equivalents as a basis for comparison between tests is indefensible and contributes to serious under- as well as over-estimation of students with learning problems.

The best practice would be to compare **standard scores** from co-normed tests of ability and achievement (i.e., tests based upon the same normative sample), using proper statistical procedures and tables for comparing the degree of discrepancy and the frequency of such a discrepancy among the normative sample.

Some of these psychometric issues are especially problematic in the assessment of students who are outside the traditional “average ability range” as far as standardized scores are concerned. It is generally accepted that the diagnosis of learning disabilities calls for greater diagnostic and clinical judgment when the student’s full scale measured I.Q. is more than one standard deviation from the mean, i.e., is below 85 (16th percentile) and above 115 (84th percentile). The members of the Working Group contend that a good assessment, based on information collected from a variety of test and non-test sources and augmented by clinical judgement will focus on identifying the primary causes of the difficulties and the most appropriate forms of intervention. A key requirement is to ensure that in the diagnosticians’ judgement, the manifestations of academic and other difficulties are related to the observed psychological processing deficits and that neither can be more accurately ascribed to another condition.

1.6 Discrepancy related to delay in service provision

Recently, any use of the discrepancy formula has been opposed by those who think it will result in lengthy delays before students are deemed eligible for assessment, identification and access to special education programming. This has sometimes been described as “waiting to fail”, resulting from an inappropriate application of the concept of discrepancy.

School systems often focus on grade or age equivalents rather than measured ability and performance. The problem occurs when an ability-achievement discrepancy is rigidly and arbitrarily set using grade levels. For example, demanding that in order to be deemed eligible for assessment for learning disabilities and special education help, a student must be academically at least two or more years behind his or her age-appropriate peers. As a result of this misguided approach, students often cannot receive help, however obvious their needs might be, until they are in at least grade 3 and at the same time are functioning at a kindergarten level or below in areas such as reading, writing or math. Instead, educational policies should recognize the value of non-categorical screening and the availability of early intervention, without having to make a

formal diagnosis in the early primary grades. Policies based on this understanding will assist professionals in developing informed diagnoses later based upon the student's response to early intervention and the demonstrated capacity to learn with appropriate accommodations.

In other cases, where the student is receiving significant help at home or has above average ability, he or she may not reach the school's discrepancy threshold for special education until about grade 6 or later. In such cases the student is often really struggling and may end up failing or repeating a grade before being referred for help, making intervention more difficult and less effective. While it is never too late to offer appropriate special education help to a student with learning disabilities, early identification and intervention are particularly important.

1.7 Memory and attention

Short-term memory: is the passive storage of a small amount of information (i.e., about 5 to 10 items) for a limited amount of time, usually no more than about 15 seconds. The information is quickly lost if it is not rehearsed or organized in some way.

Working memory: is the ability to hold information in short term memory while actively performing other mental operations which use this information (e.g., solving mental arithmetic problems).

Long-term memory: is the permanent storage of a seemingly infinite amount of information including knowledge of procedures, experiences and factual information. Long-term storage requires the activation of multiple cognitive abilities such as perception, thought, language, prior memories and, in particular, the use of strategies to process and organize the information meaningfully.

Retrieval: involves the use of cognitive strategies to efficiently and quickly access information stored in memory.

Attention: refers to the ability to selectively focus on some activities while ignoring others, to sustain concentration, to resist distraction and to shift attention among tasks. Attention is a complex and non-unitary activity. Attention may be passive (reflexive, non-voluntary and

effortless) or active and voluntary. It is a necessary but not sufficient condition for any kind of learning activity.

1.8 Processing speed

Processing speed is the ability to perform simple cognitive or perceptual tasks rapidly and efficiently. Tests of processing speed typically require the individual to quickly carry out a sequence of simple mental operations (e.g., scanning visual items and marking those which are identical or which are different) with the stimuli being presented sequentially or randomly in either a visual or auditory mode.

Rapid automatized naming (also referred to a *speed of lexical access*) is a particular way in which processing speed may be measured. Recent research literature on the development of basic reading skills suggests that deficits in this skill may contribute to reading difficulties in some individuals. The nature of this type of disorder is not understood. Some researchers argue that slower performance on naming tasks (e.g., rapidly naming pictures, letters, numbers, etc.) reflects impaired phonological processing. Others claim that such a rapid naming problem is a separate deficit from phonological processing. Individuals who have both phonological processing and rapid naming deficits appear to be the most severely impaired readers.

1.9 Language processing

Receptive language processing refers to an individual's understanding of oral and written language. A student with difficulties in this area may have trouble understanding meaning conveyed by vocabulary (including multiple meanings and figurative language), word structure (such as suffixes and prefixes), sentence structure (syntax/grammar) and meaning conveyed across sentences (such as in stories or a conversation). *Expressive language processing* refers to a student's ability to express ideas in oral and written language. This may include difficulties with recalling and using vocabulary, word structure, sentence structure and the conveyance of ideas across sentences.

Receptive and expressive language processing primarily relate to the *semantics* of the language. People with learning disabilities may also have difficulties with the *pragmatics* of language, which involves communicating in practical ways, using both verbal and non-verbal channels. A

student who has problems with pragmatics may have trouble understanding social contexts (such as an informal meeting with friends versus a formal meeting) and social cues (such as body language) influence meaning or in modifying language to suit the specific social context and/or using non-verbal communication cues, such as body language and how they influence the meaning of any communication.

1.10 Perceptual-motor processing

Perceptual-motor processing refers to an individual's ability to use sensory feedback to guide physical movements, i.e., linking perceptual input to motor output. This type of functioning involves reasoning and judgment as it relates to the processing and elaboration of complex perceptual or sensory inputs. Perceptual-motor processing relies on the integration of the senses (vision, hearing and touch) with co-ordination of the eyes, hands and both sides of the body. Problems may arise when the child's perceptual systems are immature (for example, resulting in a faulty visual interpretation of a design or faulty auditory processing of directions given), when motor skills are impaired (for example, resulting in difficulties in having the hand copy what the visual system recognizes), or when required information from the sensory perception system is unavailable to the motor system. Effective perceptual-motor processing calls for short term memory storage of the original sensory stimulus. This allows time for recognition of the stimulus, processing it e.g., organizing or integrating it and making the appropriate motor response.

- **Executive functions**

Executive functions is a term used to describe specific proactive mental (frontal lobe) control processes, the use of higher level cognitive functions or strategies, that reflect an “anticipatory, goal-oriented preparedness to act” in various cognitive processing, problem-solving and social situations. (Denckla 1994)

The key functions usually included under the heading of *executive functions* include planning, monitoring, regulation, organization and met cognition.

Planning is defined as “a dynamic, transactional process involving the conscious or deliberate specification of a sequence of actions, aimed at achieving some problem goal”. (Herbert, 1994)

Monitoring is described as observing and evaluating one's own performance in problem-solving situations that require goal-oriented intentions and the application of strategies in relation to achieving a desired outcome.

Regulation (or self-regulation) is defined as having three distinct components:

- Motivation (setting goals and expectations),
- Cognition (perception, including the observation and evaluation of the application of strategies and skills)
- and affective skills (the use of active feedback to modify one's performance to achieve a desired outcome). (Mithaug, 1993)

Organization is usually described as the development and implementation of reasoned and logical plans of action that anticipate the consequences of alternative solutions.

Metacognition refers to an awareness and understanding of skills and strategies, including what one knows, how one learns, how one applies knowledge and understanding to the learning process, including the application of coping and learning strategies and the patterns of utilizing prior successes and failures. Flavell et al. (1993) defined metacognition as "cognition about cognition", while Torgesen (1994) recommends the inclusion of motor functions under this heading thereby covering both metacognitive knowledge and metacognitive behaviour.

1.11 The impact of learning disabilities on skill acquisition and execution

Learning disabilities are often first noted when a child enters school and is expected to learn to perform specific academic skills at an age appropriate level.

However, it is important to recognize that, in addition to difficulties with school-based skills, learning disabilities may impact work and life skills. There has been significant debate in the learning disabilities literature concerning the relationship between deficits responsible for academic difficulties and those which contribute to social perception and social interaction skill deficits, problems with self-esteem and difficulties in certain non-academic intelligence areas, such as kinesthetic or emotional intelligence. The evidence suggests that social skill problems

can be the result of processing deficits which occur in learning disabilities, especially non-verbal learning disabilities. The new definition refers to problems in non-academic areas to reinforce their importance in the lives of people with learning disabilities. However, social skill deficits should not be viewed as specific learning disabilities, unless they are accompanied by one or more of the other more traditionally recognized information processing deficit areas.

- **The lifelong impact of learning disabilities**

While it is not usual to diagnose learning disabilities during the pre-school years, some impairments may be noted during that period - for example, language processing difficulties or the failure to meet certain developmental milestones. Prior to school entry, it is often reported by parents, that their children demonstrated uneven or delayed development in the acquisition of basic skills. These may include early language skills (e.g. listening and oral language), motor skills that are initially noted as difficulties with laterality - using pencils or crayons or learning to tie shoelaces, but may later affect writing or printing, as well as more traditional academic readiness skills (e.g. pre-reading skills such as rhyming, or sound-symbol recognition). Parents often describe these difficulties as “**unexpected**” and not in keeping with their children’s level of general development which appeared average or above in other areas, such as problem solving or intellectual functioning.

The work of the Screening and Assessment Working Group, in developing the screening and assessment protocol will elaborate on these observable factors. However, it is important to note that there is ongoing debate about the validity and reliability of psychometric testing, when a child is less than seven years of age. At such a young age, non-categorical screening and intervention may be more appropriate than the use of psychometric tests. During the early primary grades, these same types of unexpected difficulties have often been noted through classroom observation and screening programs offered by the school system.

For example, the development of phonological processing skills, usually acquired by the age of 6 or 7 by most children may be delayed or compromised for some students with learning disabilities. Other children with learning disabilities may have difficulty acquiring basic writing and math skills. This is not due to poor teaching or lack of instruction, nor is it due to lack of effort on the part of the student. In fact, many parents report spending an inordinate amount of

time with their child, completing homework assignments or providing tutors in an effort to remediate or reinforce basic academic skills to have their children achieve. Teachers, while not necessarily aware of the time and effort expended by students and/or their parents in order to enhance school achievement, often comment on the difference they observe between some students' skills in structured academic tasks and their apparent cognitive development.

These problems persist beyond elementary and secondary school into post-secondary and even into the workplace. As a result, individuals with learning disabilities will require ongoing accommodations, supports and services in order to be successful.

Since learning disabilities relate directly to the way that the human brain processes information, the condition does not disappear over time. However, its manifestations will change both in expression (the type of observable difficulties that the learning disability produces) and severity at different life stages, as both the individual and the environmental demands change. In order for adults with learning disabilities to overcome barriers, it is crucial to create a positive match between the demands of the learning, working and living environment and the individual's strengths and needs. Adults with specific learning disabilities can learn to identify such a positive match and to advocate for the accommodations that will ensure their potential future success.

1.12 How learning disabilities relate to other conditions which affect learning

It is very important to differentiate specific learning disabilities from more global intellectual or developmental disabilities. The primary purpose of such differentiation is to ensure that individuals are provided with services, supports and accommodations that are appropriate for them and that meet their specific needs. Mild to moderate intellectual disabilities are sometimes designated "general learning disabilities" within the educational system, which results in confusion and the delivery of inappropriate special education services. This is misleading and should not be done. In some jurisdictions, especially those countries whose educational system is linked to or modelled on the British system, the term "learning disabilities" is used to describe individuals with below average intelligence. At the same time, "dyslexia" is often the sole learning disability recognized for educational purposes. As a result, good readers are sometimes called dyslexic in spite of the fact that they have some other specific learning disabilities. In developing this definition, it was decided to limit the use of related terms and not to include any

of the terminology that is sometimes used to describe learning disabilities, such as dyslexia, dysgraphia, dyscalculia, etc.

It is useful to consider both intrinsic and extrinsic factors that are involved in the development of learning disabilities and about other conditions which may co-exist with learning disabilities. If these factors are fully understood and utilized in developing and delivering programs, services and supports to persons with learning disabilities, then many of the potential problems and barriers faced by this population may be reduced or even eliminated.

- **Congenital versus acquired causes of learning disabilities**

It is obvious that anything which affects the brain will affect learning. It is now well substantiated that factors within the brain itself, genetic influences as well as in the environment can have an impact on learning and consequently on learning disabilities. In the area of reading disabilities, for example, careful research has estimated that about half of the individual differences in these conditions are related to genetic factors.

Other studies have emphasized that learning disabilities can be due to the effects of a number of different genes, acting in combination with environmental influences. In other words, there is no single gene that can be identified as responsible for all of the variation in learning strengths and deficits observed in individuals with learning disabilities. In fact, a number of different chromosomal disorders (for instance, Fragile X and XYY syndrome in boys and Turner's syndrome in girls) can be expressed in forms which include learning disabilities.

Environmental factors which contribute to learning disabilities are also broad. These can roughly be divided into factors that influence the development and integrity of the brain during pregnancy, during the birth process, and after birth. During pregnancy, it is well established that both prescription and non-prescription drugs (especially alcohol and nicotine) can contribute to disorders which may include learning disabilities. Fetal Alcohol Syndrome and Fetal Alcohol Effects are the best known syndromes in this group. Infections of the mother during pregnancy (such as rubella or measles) can also negatively affect the fetal brain, leading to different types of learning difficulties, depending on the nature of the infection and the gestational period during which it occurs.

- **Considering coexisting conditions or comorbidity**

Comorbidity is described as a situation where two or more conditions that are diagnostically distinguishable from one another tend to occur together. The exact nature of the relationship between comorbid conditions is a matter of some debate in the research literature (Martini,

Heath & Missiuna, 1999; Clarkin & Kendall, 1992; Goff, 1992). It is particularly difficult to determine whether one condition is in fact a symptom of the other - causality versus correlation. These important debates aside, research provides support for a number of conditions co-occurring with learning disabilities more often than expected “just by chance”. The largest body of studies supports a comorbid relationship between learning disabilities and attention deficit disorder (with or without hyperactivity). This extensive research, featuring comorbidity estimates as high as 70%, was summarized recently by Riccio, Gonzalez & Hynd (1994) and Maynard, Tyler & Arnold (1999).

In fact, learning disabilities are sometimes confused with Attention Deficit Hyperactivity Disorder (ADHD). It is important to note that these are **two distinct conditions**, in spite of the significant level of co-morbidity. ADHD is **not** a specific learning disability. The distinguishing characteristics of students with ADHD include being more easily distracted, failing to finish assignments, weaker persistence of effort, day dreaming, looking away more often from activities they are requested to do and demonstrating less persistence of effort when completing boring activities (Barkley, Dupaul & McMurray, 1990). As well, children with ADHD have been distinguished from those with LD based on their higher levels of activity and impulsiveness. As mentioned above, a very large percentage of those who have ADHD also have accompanying learning disabilities, while approximately 30% of those who have learning disabilities also have ADHD. Nevertheless, the interventions that benefit people with ADHD and those who have learning disabilities are not the same. Therefore, it is important to diagnose these conditions accurately, before developing an Individual Education Plan for the student.

What do people with learning disabilities need in order to be successful in school and in life?

This is an important part of LDAO’s new definition of learning disabilities and it differentiates it from many others. Most do not include references to treatment and management issues, such as

how individual do persons with learning disabilities may be helped to cope with and compensate for the impacts of their condition.

Research indicates that all of the following four components need to be an integral part of the services and supports that are available to people with learning disabilities, in order to help them achieve their goals and overcome any barriers resulting from the condition. “*Specific skill instruction*” describes appropriate teaching and training that is built on an individual’s identified strengths. It focuses on the development of compensatory strategies in those weaker skill areas where the learning disability interferes with the learning process. Specific skill instruction must be individualized. The teaching/ training process must be adjusted to match the individual’s learning style, rather than assuming that the individual will eventually learn, no matter what the teaching process is, provided that “he/she tries harder”.

Traditional remedial techniques of teaching, testing and re-teaching in essentially the same way frequently do not work for students with learning disabilities. Examples of specific skill instruction include differentiated teaching strategies, for example, reducing the number of tasks without reducing the standard or expected quality; allowing for an extended learning period to achieve mastery; retouching a particular skill in a substantially different way than that used to instruct the rest of the class; and emphasizing the importance of acquiring learning and compensatory strategies.

“*Compensatory strategies*” are ways in which individuals who have learning disabilities can apply coping skills to help themselves overcome the impacts of their learning disabilities, without necessarily having to rely on the assistance of other people or draw particular attention to their needs. Examples of successful compensatory strategies include using colour-coding, applying visual cues such as highlighting, drawing arrows, using a notepad or a handheld tape recorder to ensure that directions are not forgotten, learning a format for approaching certain complex tasks, etc.

Accommodations are defined as alterations and changes in the way individuals with disabilities are enabled to function to demonstrate and apply their skills and knowledge. Accommodations are aimed at eliminating or ameliorating a disadvantage without altering the validity of the work in doing so. Examples of successful accommodations may include using adaptive technology,

getting assistance from another person such as a note taker or scribe or having extra time to carry out certain tasks. It is particularly important that any identified and recommended accommodations are directly linked to the strengths and needs of the person with a learning disability. The obligation to provide accommodations is mandated in the Canadian Charter of Rights and Freedoms and the Ontario Human Rights Code. This obligation applies throughout the individual's life span.

Differentiating between diagnosis and identification

For the purposes of “operationalizing” the new definition of learning disabilities, a clear distinction must be made between “**learning disability**” as a psychological **diagnosis** made by an appropriately qualified professional under the Regulated Health Professions Act (RHPA), 1993, and as a formal **identification** through the Identification, Placement and Review Committee (IPRC) process, governed by the *Education Act*, and Regulation 181/98, the regulation which sets out the IPRC process. Such appropriately qualified professionals include registered psychologists and psychological associates. School board personnel working in the special education field must familiarize themselves with the differentiation between these two processes. This is necessary so that they can accurately communicate the information expected and needed by parents, students and those who develop programming for students with learning disabilities, based on the results of assessments.

Learning Disability as Identification at an IPRC

Identification at the IPRC occurs through reference to the definitions provided by the Ministry of Education and contained in the Special Education Information Handbook. These definitions are also included in every school board's special education plan. The latest version of the definitions of exceptionalities was circulated to school boards on January 15, 1999 and school boards were advised that they were expected to utilize these specific definitions for identification purposes. It is important to note that identification in this context centres on a **need**, while a diagnosis centres on a **cause**. This differentiation follows from the definition of an Exceptional Pupil under Section 1 of the *Education Act*:

“A pupil whose behavioural, communicational, intellectual, physical or multiple exceptionalities are such that he is considered to need placement in a special education program.” (Emphasis added)

Identification involves consideration of a variety of information in order to determine whether a pupil meets the criteria for the Ministry of Education’s definition of “learning disability”, as an area of exceptionality. Information used to make an identification includes reports from teachers, parents, and regulated health professionals including, but not limited to, members of the College of Psychologists. Unlike diagnosis, which involves an individual professional’s formal opinion concerning the cause of an individual’s symptoms, identification is accomplished through a school board committee and is carried out solely for the purpose of planning how best to meet the pupil’s strengths and needs. Under the Regulated Health Professions Act, “communication of a diagnosis” is listed as one of the Controlled Acts, the performance of which is legally restricted to members of certain professional colleges, including the College of Physicians and Surgeons of Ontario and the College of Psychologists of Ontario. Since the Controlled Act is **only** performed when information is communicated to a client or his/her personal representative, communicating such information to other individuals (e.g., to other professionals, in multi-disciplinary teams, or at I.P.R.C. meetings where the client or his/her representative is not present) does **not** constitute performance of the Controlled Act. However, the policy of the College of Psychologists also stipulates:

“Normally, the outcomes of psychological assessments, including any psychological diagnoses, have been communicated to the pupil or his or her parents, guardians or personal representatives prior to the IPRC meeting because of the implications of potential harm and the right of the client to have direct access to the regulated professional who is accountable to the public. Unless a member of the IPRC team is authorized under the law to perform the controlled act, a diagnosis should not be conveyed to the client or his or her personal representative at, or following the meeting, unless this has already been done by a legally authorized health care professional.”

1.13 Learning Disability as a Diagnosis

Section 27(1) of the *Regulated Health Professions Act* defines the Controlled Act of communicating a diagnosis as:

All three elements underlined must be present for the Controlled Act to be considered to have been performed. The following statements further clarify the legal and procedural requirements surrounding the controlled act of diagnosis:

Considering all of the above points, the term “learning disability” constitutes a diagnosis when it is used to provide an explanation for a learning problem through a classification, formulation or causal statement linking it to a neuropsychological disorder and when this information is communicated to the individual, or to his or her personal representatives under circumstances in which he/she or they could be expected to rely upon the diagnosis (i.e., generally in a face-to-face meeting or through a written report).

In the overall interest of public protection (which is the fundamental basis of the RHPA), the only persons legally entitled to communicate such a diagnosis in Ontario are members of a Regulated Health Profession with access to the controlled act. There are substantial penalties under RHPA for individuals who perform the controlled act of diagnosis without authorization, as well as penalties for their employers. Restrictions imposed on the performance of controlled acts are not uniquely the policy of the College of Psychologists of Ontario. They are legislated under the RHPA and apply to all regulated health professionals, unregulated service providers, and the public generally.

Diagnosis versus Identification of Learning Disabilities

Several exceptionality groupings, including “learning disability”, describe conditions which fall within the meaning of the controlled act as defined in the Psychology Act. The policy of the College of Psychologists of Ontario states:

“If a pupil is included in one of these categories due to a determination of the cause of the disorder, this would require a psychological diagnosis. This is in contrast with the use of similar terminology by the IPRC to refer to the nature of services provided by the educational system. Care must be taken in making this distinction clear to the client so as to avoid the unauthorized communication of a diagnosis.”

While the College of Psychologists recognizes the distinction between the diagnosis and identification of learning disabilities, the College has stated to the Ministry of Education that

communication of a diagnosis of a learning disability is a controlled act within the meaning of the *Regulated Health Professions Act* and should be carried out only by qualified health practitioners. This position has been formally acknowledged by the Minister of Education and Training:

The Registrar of the College of Psychologists has made the following recommendation to the Minister of Education and Training:

1.14 Characteristics of Children with Learning Disabilities

Children with learning disabilities are a heterogeneous group. These children are a diverse group of individuals, exhibiting potential difficulties in many different areas. For example, one child with a learning disability may experience significant reading problems, while another may experience no reading problems whatsoever, but has significant difficulties with written expression.

Learning disabilities may also be mild, moderate, or severe. Students differ too, in their coping skills. According to Bowe (2005), “some learn to adjust to LD so well that they ‘pass’ as not having a disability, while others struggle throughout their lives to even do ‘simple’ things. Despite these differences, LD always begins in childhood and always is a life-long condition” (p. 71). Over the years, parents, educators, and other professionals have identified a wide variety of characteristics associated with learning disabilities (Gargiulo, 2004). One of the earliest profiles, developed by Clements (1966), includes the following ten frequently cited attributes:

- Hyperactivity
- Impulsivity
- Perceptual-motor impairments
- Disorders of memory and thinking
- Emotional lability
- Academic difficulties

- Coordination problems
- Language deficits
- Disorders of attention
- Equivocal neurological signs

Almost 35 years later, Lerner (2000) identified nine learning and behavioral characteristics of individuals with learning disabilities:

- Disorders of attention
- Reading difficulties
- Poor motor abilities
- Written language difficulties
- Oral language difficulties
- Social skills deficits
- Psychological process deficits
- Quantitative disorders
- Information processing problems

According to Gargiulo (2004), not all students with learning disabilities will exhibit these characteristics, and many pupils who demonstrate these same behaviors are quite successful in the classroom. As Smith (1979) observes, it is the quantity, intensity, and duration of the behaviors that lead to the problems in school and elsewhere. The focus of this LD Report will be to discuss the most commonly seen characteristics of children with learning disabilities. In almost all cases, a single student will not have deficits in all areas. Understanding the

characteristics of children with learning disabilities is absolutely essential as a future educator in developing prereferral interventions, in making appropriate referrals, and in identifying effective adaptations and intervention strategies (Smith et al., 2004).

1.15 Academic Achievement Deficits

Children with learning disabilities often struggle with various areas of academic performance. During the elementary school years, a discrepancy between ability and achievement begins to emerge in students with learning disabilities. Often puzzling to teachers, these students seem to have strengths similar to their peers in several areas, but their rate of learning is unexpectedly slower (Smith et al., 2004). These problems usually persist from the primary grades through the end of formal schooling, including college (Bradshaw, 2001). Academic deficits for children with learning disabilities normally fall into the following areas: reading, mathematics, and written expression. Some children have problems in only one select academic area, while others may experience difficulties in all three.

1.16 Reading Deficits

Reading provides a fundamental way for individuals to exchange information. It is also a means by which much of the information presented in school is learned. As a result, reading is the academic area most often associated with academic failure. Reading is a complex process that requires numerous skills for its mastery. Consequently, identifying the skills that lead to success in reading is extremely important.

Reading difficulties are observed among students with learning disabilities more than any other problem area of academic performance. It is the most prevalent type of academic difficulty for students with learning disabilities. It is estimated that as many as 90% of students with learning disabilities have reading difficulties, and even the low estimates are approximately 60% (Bender, 2001). Most authorities believe that this problem is related to deficient language skills, especially phonological awareness—the ability to understand that speech flow can be broken into smaller sound units such as words, syllables, and phonemes.

According to Hallahan and Kauffman (2003), it is easy to see why problems with phonology would be at the heart of many reading difficulties. A person who has problems breaking words

into their component sounds will have trouble learning to read. And there is suggestive evidence that readers of English are more susceptible than readers of some other languages to problems with phonological awareness. Some have speculated that this is why reading disabilities are more prevalent in English-speaking countries than in some other countries (p. 162).

Becoming a skilled reader is so important in our culture that an unskilled reader is at a great disadvantage in school and in the workplace. The following problems may prevent a child with learning disabilities from learning to read (Kirk, Gallagher, & Anastasiow, 2003, p. 224).

- Faulty auditory perception without hearing impairment
- Slow auditory or visual processing
- Inability to distinguish or separate the sounds of spoken words
- Lack of knowledge of the purpose of reading
- Failure to attend to critical aspects of the word, sentence, or paragraph
- Failure to understand that letters represent units of speech

1.17 Oral Reading Difficulties

Many students with learning disabilities have difficulties with reading fluency (Mercer, Campbell, Miller, Mercer, & Lane, 2000). Reading fluency, most frequently defined as the rate of accurate reading (correct words per minute), is more than a status symbol for children; it is an important indicator of reading ability (Hunt & Marshall, 2005). Students with fluency problems may read aloud in a word-by-word manner without appropriate inflection or rhythm, unable to relate the patterns of spoken language to the printed word. Students with weakness in this area often dread being asked to read in class (Friend, 2005). According to Salvia and Ysseldyke (1998), common oral reading problems include the following:

- **Omissions.** The student skips individual words or groups of words. **Insertion.** The student inserts one or more words into the sentence being orally read.

- **Substitution.** The student replaces one or more words in the passage by one or more meaningful words.
- **Gross mispronunciation of a word.** The student's pronunciation of a word bears little resemblance to the proper pronunciation.
- **Hesitation.** The student hesitates for two or more seconds before pronouncing a word.
- **Inversion.** The student changes the order of words appearing in a sentence.

Disregard of punctuation. The student fails to observe punctuation; for example, may not pause for a comma, stop for a period, or indicate a vocal inflection, a question mark, or an exclamation point.

Through miscue analysis, teachers can determine the extent to which the reader uses and coordinates graphic, sound, syntactic, and semantic information from the text. According to Goodman and Burke (1972), to analyze miscues you should ask at least four crucial questions:

- **Does the miscue change meaning?** If it does not, then it is semantically acceptable within the context of the sentence or passage.
- **Does the miscue sound like language?** If it does, then it is grammatically acceptable within the context. Miscues are grammatically acceptable if they sound like language and serve as the same parts of speech as the text words.
- **Do the miscue and the text word look and sound alike?** Substitution and mispronunciation miscues should be analyzed to determine how similar they are in approximating the graphic and pronunciation features of the text words.
- **Was an attempt made to self-correct the miscue?** Self-corrections are revealing because they demonstrate that the reader is attending to meaning and is aware that the initial miscuing did not make sense.

Oral reading problems often cause tremendous embarrassment for children. Children with oral reading problems may read in a strained voice with poor phrasing, ignore punctuation, and grope for words like a much younger child would do when reading. Pollaway, Patton, and Serna (2001) confirm that a student's self-image and feelings of confidence are greatly affected by unhappy

1.18 Reading Comprehension Deficits

Students with learning disabilities often have difficulties with reading comprehension (Gersten, Williams, Fuchs, & Baker, 1998). These children often lack the skills required for understanding text and have poor word-analysis skills (Hunt & Marshall, 2005). Reading comprehension refers to a student's ability to understand what he or she is reading. Some students with reading comprehension difficulties are able to read a passage so fluently that you might assume they were highly proficient readers. However, when they are asked questions about what they have read, they have little or no understanding of the words. Students with this problem sometimes are referred to as word callers (Friend, 2005).

It is always necessary to assess not only decoding but also the ability to understand what is being decoded. According to Salvia and Ysseldyke (1998), there are six different types of reading comprehension skills:

Literal comprehension. The student reads the paragraph or story and is then asked questions based on it.

Inferential comprehension. The student reads a paragraph or story and must interpret what has been read.

Listening comprehension. The student is read a paragraph or story by the examiner and is then asked questions about what the examiner has read.

Critical comprehension. The student reads a paragraph or story and then analyzes, evaluates, or makes judgments about what he or she has read.

Affective comprehension. The student reads a paragraph or story, and the examiner evaluates his or her emotional responses to the text.

Lexical comprehension. The student reads a paragraph or story, and the examiner assesses his or her knowledge of vocabulary words.

Here are some common reading comprehension problems of children with LD:

- Difficulties recalling basic facts (unable to answer specific questions about a passage, such as what was the dog's name in the story?)
- Difficulties recalling sequence (unable to tell the sequence of the story that was read)
- Difficulties recalling the main theme (unable to recall the main topic of the story) when evaluating a child for reading comprehension, it is important to ask the following questions:

1.19 Problems with Word Recognition

Students with learning disabilities often have difficulties with word recognition, which relates to the student's ability with respect to sight vocabulary. According to Salvia and Ysseldyke (1998): *A student learns the correct pronunciation of letters and words through a variety of experiences. The more exposure a student has to specific words and the more familiar those words become, the more readily he or she recognizes those words and is able to pronounce them correctly.* (p. 464) In order to identify written words, we use a number of different skills. Here are some of the most important word analysis skills:

- The ability to associate sounds with the various letters and letter combinations used to write them (phonic analysis)
- Immediately recognizing and remembering words (sight-word reading)
- Using the surrounding text to help figure out a specific word (using context) The skills listed above rely heavily on perception, selective attention, memory, and metacognitive skills. Thus, word recognition depends almost entirely on the cognitive skills that are most problematic for individuals with disabilities (Hunt & Marshall, 2005).

According to Gargiulo (2004), here are common word recognition errors:

- **Omissions.** Omitting a word (Tom saw [a] cat.)
- **Insertions.** Inserting words (The dog ran [fast] after the cat.)
- **Substitutions.** Reversing letters in a word (no for on, was for saw)
- **Mispronunciations.** (Mister for miser)
- **Transpositions.** Reading words in the wrong order (She away ran instead of she ran away.)
- **Unknown words.** Hesitating for 5 seconds at words they cannot pronounce
- **Slow choppy reading.** Not recognizing words quickly enough (20 to 30 words per minute)

Poor Reading Habits

Children with reading difficulties often have poor reading habits. As a teacher, it is critical that you be aware of these actions when watching your students read on a daily basis. Gargiulo (2004) lists some behaviors that are exhibited by children with poor reading habits.

- **Tension movements.** Frowning, fidgeting, using a high-pitched tone of voice
- **Insecurity.** Refusing to read, crying, attempting to distract the teacher
- **Loses place.** Losing place frequently
- **Lateral head movements.** Jerking head
- **Holds material close.** Deviating extremely from 15 to 18 inches

Dyslexia

Dyslexia is one of several distinct learning disabilities. It is a specific language-based disorder characterized by difficulties in single word decoding, usually reflecting insufficient phonological processing abilities. The difficulties in single word decoding are often unexpected in relation to age and other cognitive and academic abilities; they are not the result of generalized

developmental disability or sensory impairment. Dyslexia is manifested by variable difficulty with different forms of language, often including, in addition to problems reading, a conspicuous problem with acquiring proficiency in writing and spelling.

Simply stated, dyslexia is a type of reading disorder in which the student fails to recognize and comprehend written words. Dyslexia is a severe impairment in the ability to read, despite normal intelligence, normal opportunities to read, and an adequate home environment. Although the precise organic cause of dyslexia is unknown, it is generally thought that this problem results from difficulties with phonological awareness—a lack of understanding of the rules that govern the correspondence between specific sounds and certain letters that make up words (Lyon & Moats, 1997; cited in Gargiulo, 2004, p. 216). In other words, letter-sound recognition is impaired.

Various types of reading disorders have been recently cited by the American Academy of Special Education Professionals' Educator's Diagnostic Manual of Disabilities and Disorders (2007). Listed below are the reading disorders most frequently seen in children with dyslexia:

- **Direct Dyslexia.** Direct dyslexia refers to the ability of the individual to read words aloud correctly, yet not comprehend what he or she has just read.
- **Dyseidesia Dyslexia.** Such an affected individuals will have poor sight-word vocabularies and will rely on using time consuming word attack skills (a phonetic approach) to decode many words. As a result, students with this condition will read laboriously. Decoding becomes inaccurate for many phonetically irregular words, log for laugh. Characteristic spelling errors include phonetic equivalents for irregular words, such as rede for ready.
- **Dyseidetic Dyslexia.** Children with the dyseidetic type of dyslexia are able to sound out individual letters phonetically but have trouble identifying patterns of letters in groups. Their spelling tends to be phonetic even when incorrect (laf for laugh). Children in this group have deficits in vision and memory of letters and word shapes, making it difficult for them to develop a sight vocabulary. However, they have the ability to acquire adequate phonetic skills.

- **Dyslexia with Dysgraphia** (*Deep Dyslexia*). With this condition, a person has a problem in writing letters and words, grasping word-meanings, integrating the sounds of letters, and in pronouncing unfamiliar and, sometimes, even familiar words. People in this category face the biggest challenge and need our closest attention for educational and career planning.
- **Dyslexia without Dysgraphia** (*Pure Dyslexia*). This disorder occurs when a person has problems reading but not writing. Some students with pure dyslexia have trouble doing written arithmetic because they have to read the text and the numbers, but may not have any problem doing spoken arithmetic. Dyslexia without dysgraphia may never be identified, because, to confuse matters, a person may have nearly normal oral language and his or her writing and oral spelling may be virtually unimpaired.

Math Deficits

Although disorders of reading have traditionally received more emphasis than problems with mathematics, the latter are gaining a great deal of attention (Hunt & Marshall, 2005). Authorities now recognize that math difficulties are second only to reading disabilities as an academic problem area for students with LD (Hallahan & Kauffman, 2003). Researchers estimate that about one out of every four pupils with LD receives assistance because of difficulties with mathematics (Rivera, 1997). According to Lerner (2003), each student with mathematical difficulties is unique; not all children exhibit the same deficiency or impairment.

Students with learning disabilities may have problems in both math calculations and math reasoning (USOE, 1977). These students often have a number of problems in mathematical thinking (Hunt & Marshall, 2005). Mathematical thinking is a process that begins early in most children. Even before formal education begins, children are exposed to various situations that involve the application of mathematical concepts. As they enter formal schooling, they take the knowledge of what they had previously learned and begin to apply it in a more formal manner. It is necessary to understand that mathematics and arithmetic are actually two different terms. Although most people use them interchangeably, they each have distinct meanings. According to Merriam-Webster (m-w.com), **arithmetic is:** a branch of mathematics that deals usually with the nonnegative real numbers including sometimes the transfinite cardinals and with the application of the operations of addition, subtraction, multiplication, and division to them **while**

mathematics is: the science of numbers and their operations, interrelations, combinations, generalizations, and abstractions and of space, configurations and their structure, measurement, transformations, and generalizations.

Mathematics involves many different abilities, as suggested by the list below:

- Estimating
- Doing computational skills
- Solving problems
- Understanding measurement
- Using mathematics for prediction
- Creating and reading graphs and charts

Schools begin the process of learning math skills in kindergarten and proceed throughout the child's formal education. Even at the college level, mathematics is often a core requirement in many liberal arts schools. In general, next to reading, mathematics is probably the area most frequently assessed in school systems.

- **Written Expression Deficits**

Many individuals with LD exhibit deficits in written language (Hallahan, Kauffman, & Lloyd, 1999). Learning disabilities in the area of written expression are beginning to receive more recognition as a serious problem (Smith et al., 2004). Writing is a highly complex method of expression involving the integration of eye-hand, linguistic, and conceptual abilities. As a result, it is usually the last skill children master. Whereas reading is usually considered the receptive form of a graphic symbol system, writing is considered the expressive form of that system. The primary concern in the assessment of composition skills is the content of the student's writing, not its form. The term written language refers to a variety of interrelated graphic skills.

Composition. The ability to generate ideas and to express them in an acceptable grammar, while adhering to certain stylistic conventions

Spelling. The ability to use letters to construct words in accordance with accepted usage

Handwriting. The ability to execute physically the graphic marks necessary to produce legible compositions or messages (Hallahan et al., 1999)

The impact of written language problems increases with a student's age because so many school assignments require a written product. Students with written language problems often exhibit the following characteristics:

- Feel overwhelmed by the idea of getting started
- Struggle to organize and use the mechanics of writing
- Struggle to develop their fluency
- Have difficulties spelling and constructing written products in a legible fashion
- Submit written work that is too brief

Many students with difficulties with written language use a “retrieve-and-write” approach, in which they retrieve from immediate memory whatever seems appropriate and write it down. They seldom use the self-regulation and self-assessment strategies of competent writers: setting a goal or plan to guide their writing, organizing their ideas, drafting, self-assessing, and rewriting. As a result, they produce poorly organized compositions containing a few poorly developed ideas (Sexton, Harris, & Graham, 1998; cited in Heward, 2003).

- **Handwriting Difficulties**

Handwriting refers to the actual motor activity that is involved in writing. Most students are taught manuscript (printing) initially and then move to cursive writing (script) in later grades. Some educators advocate that only manuscript or only cursive should be taught. In truth, problems may appear among students using either system. Children's writing changes as they mature. The focus of a youngster's writing shifts from the process of writing (handwriting and

spelling), to the written product (having written something), to communication with readers (getting across one's message) (Hallahan et al., 1999, p. 396).

Gargiulo (2004) notes that early on, pupils focus on becoming competent in mastering the mechanical aspects of composition—spelling and handwriting; in the later grades, they learn to organize and present their ideas in a lucid and logical fashion. Children with learning disabilities, however, lag behind their nondisabled peers. Investigators have observed that individuals with LD use less complex sentence structure, incorporate fewer ideas, produce poorly organized paragraphs, and write less complex stories (p. 219).

Dysgraphia. Dysgraphia, the learning disability associated with written expression, entails writing skills that fall substantially below those expected given the individual's age, IQ, and education, such that academic achievement or activities of daily living are significantly impaired. Dysgraphia is the inability to perform motor movement, in other words, extremely poor handwriting. It is associated with a neurological dysfunction. Agraphia is an acquired disorder in which the ability to write and make patterns is impaired (Birsch, 1999; cited in Kirk et al., 2003). Students' handwriting problems can rise from any of the following conditions:

- A lack of fine motor coordination
- Failure to attend to task
- Inability to perceive and/or remember visual images accurately
- Inadequate handwriting instruction in the classroom

Dyslexic Dysgraphia. With this disorder, spontaneously written text is illegible, especially when the text is complex. Oral spelling is poor, but drawing and copying of written text are relatively normal. Finger-tapping speed (a measure of fine-motor speed) is normal.

Motor Dysgraphia. With this disorder, both spontaneously written and copied text may be illegible, oral spelling is normal, and drawing is usually problematic. Finger-tapping speed is abnormal.

Spatial Dysgraphia. Individuals with this disorder display illegible writing, whether spontaneously produced or copied. Oral spelling is normal. Finger-tapping speed is normal, but drawing is very problematic.

- **Spelling Problems**

Spelling is the ability to use letters to construct words in accordance with accepted usage. Spelling ability is viewed by some teachers and school administrators equally with other academic skills. Being a poor speller does not necessarily mean that a child has a learning disorder. However, when poor spelling occurs with poor reading and/or arithmetic, then there is reason for concern. It appears that many of the learning skills required for good spelling are the same ones that enable students to become good readers. Learning to spell is a developmental process, and young children go through a number of stages as they begin to acquire written language skills. Writing begins in the preschool years as young children observe and begin to imitate the act of writing.

Dysorthographia is the learning disability associated with spelling. Individuals with this disorder have difficulties utilizing clues from several sources that aid in deciding on the correct spelling of a word. Marshall and Hunt (2005) note that many students with learning disabilities spell a word as if it were being approached for the first time, without reference to an image of the word held in memory. The difficulties students with LD have in learning and applying the rules of phonics, visualizing the word correctly, and evaluating spellings result in frequent misspellings, even as they become more adept at reading. It is not uncommon to find the same word spelled five or six different ways on the same paper, regardless of whether the student is in the fifth grade or college (e.g., ther, there, thare, and they're for their).

Analysis of Spelling Skills. Several questions should be addressed before one begins to analyze a child's spelling abilities (Pierangelo & Giuliani, 2005).

- **Language Deficits**

Students with learning disabilities often have difficulties with the mechanical and social uses of language (Hallahan & Kauffman, 2003). Specific mechanical deficits difficulties are often present in the three different areas (Gargiulo, 2004).

Syntax. Rule systems that determine how words are organized into sentences

Semantics. Word meanings

Phonology. The study of how individual sounds make up words

Language deficits are found in the areas of oral expression and listening comprehension. These two areas control our ability to communicate with others, and therefore a deficit in either or both can have a major impact on the quality of life of a child with a learning disability, as well as his or her life in education (Smith et al., 2004). Studies have found that more than 60% of students with LD have some type of language disorder (Bryan, Bay, Lopez-Reyna, & Donahue, 1991).

- **Oral Language Problems**

Students with LD frequently experience difficulties with oral expression—a problem that can affect both academic and social interactions. Common problems associated with oral language include the following:

- Choosing the appropriate word. Children with LD will often use a less appropriate word because the right word will not come to them.
- Understanding complex sentence structures
- Responding to questions
- Difficulties in retrieving words. The response rate of children with learning disabilities may be slower than that of their nondisabled peers, and they may speak more slowly.

1.20 Listening Comprehension Problems

Listening problems can also be misinterpreted. A child with a disability in listening demonstrates that disability in a negative way, for example, by failing to follow directions or by appearing oppositional or unmotivated. A teacher's careful observation and assessment of a student's language ability is important for ensuring the student's success (Smith et al., 2004).

- **Disorders of Attention**

- Attention is a critical skill in learning. Conte (1991) suggests that to be effective learners, children must be able to initiate attention, direct their attention appropriately, sustain their attention according to the task demands, and shift attention when appropriate. Deficits in these areas can have an impact on all areas of school. When children are not paying attention, they cannot respond appropriately to questions, follow directions, or take notes during a lecture. Social problems occur when the student interrupts others and does not listen to peers. Estimates of the number of students with LD that have attention problems range from 41% to 80% (Smith et al., 2004).

Attention problems for children with learning disabilities are often characterized as short attention span. A short attention span is defined as an inability to focus one's attention on a task for more than a few seconds or minutes. Parents and teachers note that many children with LD share the following characteristics.

- Cannot sustain attention for more than a short period of time
- Exhibit excessive daydreaming
- Are highly distractible

- **Achievement Discrepancy**

Although students who receive special education services under the learning disabilities category are an extremely heterogeneous group, it is important to remember that the fundamental defining characteristic of students with learning disabilities is specific and significant achievement deficits in the presence of adequate overall intelligence (Heward, 2003). Students with LD perform below expectations based on their measured potential, in addition to scoring below their peers in overall achievement. Attempts to quantify the discrepancy between academic achievement and academic potential for students with LD have appeared in the literature for some time, but the field still lacks a broadly accepted explanation of the phenomenon (Roderiques, 2002; cited in Hardman et al., 2005). Early in the school years, youngsters with LD

may find themselves two to four years behind their peers in level of academic achievement, and many fall even further behind as they continue in the educational system. This discouraging pattern often results in students dropping out of high school or graduating without proficiency in basic reading, writing, or math skills (U.S. Department of Education, 2002; cited in Hardman et al., 2005, p. 178).

Memory Deficits

It is well documented that children and adolescents with LD have significant difficulties remembering academic information and nonacademic information, such as doctors' appointments, homework assignments, multiplication facts, directions, and telephone numbers. Teachers frequently comment that, with these students, it seems to be "in one ear and out the other," which can be highly aggravating for teachers as well as parents (Gargiulo, 2004). Teachers and parents also report that memory skills are inconsistent. For example, a student may know the multiplication facts on Thursday and then fail the test on Friday (Hardman et al., 2005).

Parents often state that they cannot understand how their children can be so intelligent and forget such simple things. Early research in learning disabilities has documented that students with LD have a real deficit in memory (Hallahan & Kauffman, 2003). Teachers have long complained that children with LD have poor memory.

Several studies have suggested that students with LD have more deficits in memory than students without LD except in the area of long-term memory (Swanson, 1994). Students with memory deficits have difficulty retaining learned information, repeating information read or heard, following multiple directions, and performing tasks in the right sequence (Smith et al., 2004).

The memory difficulties faced by students are normally either in short-term memory (STM) or working memory (WM). STM involves the ability to recall information after a short period of time. Short-term memory tasks involve the recall, in correct order, of either aurally or visually presented information (such as a list of digits, letters, or pictures) shortly after hearing or seeing the items several times (Hallahan, 1999). Working memory requires that the individual retain information while simultaneously engaging in another cognitive activity. According to Silver (2001), people with LD are more likely to have concerns with short-term rather than long-term

memory. He explained that children and youth with these limitations need to concentrate on new information, and to repeat it continually, in order to keep it in short-term memory. If their attention is disrupted, the information may be lost (Bowe, 2005). Working memory is involved, for example, when we try to remember a person's address while also listening to directions on how to arrive there (Swanson, 1994). Deficits in memory, particularly working memory, often translate into difficulties in the classroom. Success with reading and math seems to depend more on working memory than short-term memory. Working memory also appears to be crucial for word recognition and reading comprehension (Ashbaker & Swanson, 1996).

- **Cognition Deficits**

Students with learning disabilities will often demonstrate problems in cognition. Cognition is a broad term covering many different aspects of thinking and problem solving. Students with learning disabilities often exhibit disorganized thinking that results in problems with planning and organizing their lives at home (Hallahan & Kauffman, 2003). Research suggests that children with LD have differing, rather than uniformly deficient, cognitive abilities (Henry, 2001). This finding has led to the development of specific, highly focused instruction for individuals with learning disabilities to replace generic curricula, reflecting the assumption that their cognitive skills are generally poor (Hardman et al., 2005). According to Smith and colleagues (2004), students with problems in cognition may share the following characteristics:

- Make poor decisions
- Have trouble adjusting to change
- Make frequent errors
- Require concrete demonstrations
- Have delayed verbal responses
- Have difficulties understanding social expectations
- Require more supervision

- Have trouble getting started on a task
- Have trouble using previously learned information in a new situation

- **Metacognition Deficits**

Students with learning disabilities often have problems with metacognition. Metacognition is defined as one's understanding of the strategies available for learning a task and the regulatory mechanisms needed to complete the task. Metacognition has at least three components:

Recognize task requirements. Students with LD frequently have problems judging how difficult tasks can be. For example, they may approach the reading of highly technical information with the same level of intensity as reading for pleasure.

Select and implement appropriate strategies. Students with LD often do not come up with strategies to help themselves in and outside of school. For example, if asked to name ways in which they can help themselves remember to bring their homework into school the next day, they may not have any ideas, whereas the nondisabled peers will suggest writing a note to themselves, putting the homework by the front door, and so on.

Monitor and adjust performance. Students with LD often have problems with comprehension monitoring. Comprehension monitoring is the ability to keep track of one's own comprehension of reading material and to make adjustments to comprehend better while reading. For example, a student with LD may not have a good sense he does not understand what he is reading. Good readers are often able to make the necessary adjustments, such as slowing down or rereading difficult passages. Students with reading problems are also likely to have problems picking out the main ideas of paragraphs. Hallahan and colleagues (1999) refer to metacognition as "thinking about thinking." Students with problems in this area might have difficulty focusing on listening, purposefully remembering important information, connecting that information to prior knowledge, making sense out of the new information, and using what they know to solve a problem. They often lack strategies for planning and organizing, setting priorities, and predicting and solving problems. An important component of metacognition is the ability to evaluate one's own behavior and behave differently when identifying inappropriate behavior or mistakes (Smith

et al., 2004). Competency as a learner requires that students exhibit these metacognitive skills (Kluwe, 1987).

- **Social–Emotional Problems**

The literature suggests that to be socially accepted, students should be cooperative, share, offer pleasant greetings, have positive interactions with peers, ask for and give information, and make conversation (Gresham, 1982). Some children with LD have a real strength in the area of social skills. However, several characteristics of learning disabilities, such as those noted concerning language, can create difficulties in social and emotional life (Smith et al., 2004).

Although not all children with LD have social–emotional problems, they do run a greater risk than their nondisabled peers of having these types of problems. In the early years they are often rejected by their peers and have poor self-concepts (Sridhar & Vaughn, 2001). As adults, the scars from years of rejection can be painful and not easily forgotten (McGrady, Lerner, & Boscardin, 2001). A possible reason for these social–emotional problems is that students with LD often have deficits in social cognition. They may have the following characteristics:

- Misread social cues
- Misinterpret the feelings of others
- Not know when they are bothering others
- Be unaware of the effect of their behavior on someone else
- Be unable to take the perspective of others or put themselves in someone else’s shoes

Research has consistently found a higher-than-normal rate of behavioral problems in the classroom among students with learning disabilities (Cullinan, 2002). In a study of 790 students enrolled in K–12 LD programs in Indiana, the percentage of students with behavioral problems (19%) remained consistent across grade levels. However, it should be noted that the relationships between students’ behavioral problems and academic difficulties are not known. In other words,

we do not know whether the academic deficits or the behavioral problems cause the other difficulty. Furthermore, many children with LD exhibit no behavioral problems at all (Heward, 2003). Research further suggests that social interaction problems for students with LD seem to be more evident in those who have problems in math, visual-spatial tasks, tactual tasks, selfregulation, and organization (Worling, Humphries, & Tannock, 1999).

1.21 Nonverbal Learning Disorders (NLD)

Nonverbal learning disorders (NLD) are a neurological syndrome consisting of specific assets and deficits. The assets include early speech and vocabulary development, remarkable rote memory skills, attention to detail, early reading skills development, and excellent spelling skills. In addition, these individuals have the verbal ability to express themselves eloquently. Moreover, persons with NLD have strong auditory retention. Four major categories of deficits and dysfunction also present themselves.

- **Motoric dysfunction.** Lack of coordination, severe balance problems, and difficulties with graphomotor skills
- **Visual-spatial-organizational dysfunction.** Lack of image, poor visual recall, faulty spatial perceptions, difficulties with executive functioning (the brain's ability to absorb information, interpret this information, and make decisions based on this information), and problems with spatial relations
- **Social dysfunction.** Lack of ability to comprehend nonverbal communication, difficulties adjusting to transitions and novel situations, and deficits in social judgment and social interaction
- **Sensory dysfunction.** Sensitivity in any of the sensory modes: visual, auditory, tactile, taste, or factory Foss (2004) reports that statements like the following are often true of individuals with a nonverbal learning disability:
 - They talk a lot but really say very little.
 - They see the trees, not the forest.

- They focus on details, but do not apprehend the main idea.
- They do not see the whole picture.
- They do not read facial expressions, gestures, nor other nonverbal aspects of communication; they miss the subtleties, nuances.
- They may be inappropriate in their social interactions.
- They have few friends; friendships tend to be with older or younger persons rather than peers.
- They tend to process information in a linear, sequential fashion, not seeing multiple dimensions.
- In spite of relative strength in sequencing or recalling sequences, they may confuse abstract temporal concepts; they have significant difficulty recognizing cause–effect relationships. (NLD Online, 2004)

1.22 Summary

In conclusion, it should be evident that children with LD are truly a heterogeneous group. The characteristics exhibited by one child with a learning disability may be quite different than another one with a learning disability. As a future educator, it is essential that you understand all of the possible characteristics that may be seen in these children. By knowing what to look for and being able to identify the common characteristics, you may be able to help in the identification, diagnosis, and assessment of a child with a suspected learning disability. Ultimately, depending on where a student's problems lie, understanding these characteristics or learning styles can lead to significant improvement in the academic performance, social awareness, and overall self-esteem of a child with a learning disability in your classroom.

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Unit 2

Tools and Areas of Assessment

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2.1 Introduction

One of the outstanding features of studies of assessment in recent years has been the shift in the focus of attention, towards greater interest in the interactions between assessment and classroom learning and away from concentration on the properties of restricted forms of test which are only weakly linked to the learning experiences of students. This shift has been coupled with many expressions of hope that improvement in classroom assessment will make a strong contribution to the improvement of learning. So one main purpose of this review is to survey the evidence which might show whether or not such hope is justified. A second purpose is to see whether the theoretical and practical issues associated with assessment for learning can be illuminated by a synthesis of the insights arising amongst the diverse studies that have been reported.

The purpose of this Introduction is to clarify some of the key terminology that we use, to discuss some earlier reviews which define the baseline from which our study set out, to discuss some aspects of the methods used in our work, and finally to introduce the structure and rationale for the subsequent sections.

Our primary focus is the evidence about formative assessment by teachers in their school or college classrooms. As will be explained below, the boundary for the research reports and reviews that have been included has been loosely rather than tightly drawn. The principal reason for this is that the term formative assessment does not have a tightly defined and widely accepted meaning. In this review, it is to be interpreted as encompassing all those activities undertaken by teachers, and/or by their students, which provide information to be used as feedback to modify the teaching and learning activities in which they are engaged.

Two substantial review articles, one by Natriello (1987) and the other by Crooks (1988) in this same field serve as baselines for this review. Therefore, with a few exceptions, all of the articles covered here were published during or after 1988. The literature search was conducted by several means. One was through a citation search on the articles by Natriello and Crooks, followed by a similar search on later and relevant reviews of component issues published by one of us (Black, 1993b), and by Bangert-Drowns and the Kuliks (Kulik et al., 1990; Bangert-Drowns et al., 1991a,b). A second approach was to search by key-words in the ERIC data-base; this was an inefficient approach because of a lack of terms used in a uniform way which define our field of interest. The third approach was the 'snowball' approach of following up the reference lists of articles found. Finally, for 76 of the most likely journals, the contents of all issues were scanned, from 1988 to the present in some cases, from 1992 for others because the work had already been done for the 1993 review by Black (see Appendix for a list of the journals scanned).

Natriello's review covered a broader field than our own. The paper spanned a full range of assessment purposes, which he categorised as certification, selection, direction and motivation. Only the last two of these are covered here. Crooks used the term 'classroom evaluation' with the same meaning as we propose for 'formative assessment'. These two articles gave reference lists containing 91 and 241 items respectively, but only 9 items appear in both lists. This illustrates the twin and related difficulties of defining the field and of searching the literature.

The problems of composing a framework for a review are also illustrated by the differences between the Natriello and the Crooks articles. Natriello reviews the issues within a framework provided by a model of the assessment cycle, which starts from purposes, then moves to the setting of tasks, criteria and standards, then through to appraising performance and providing feedback and outcomes. He then discusses research on the impact of these evaluation processes on students. Perhaps his most significant point, however, is that in his view, the vast majority of the research into the effects of evaluation processes is irrelevant because key distinctions are conflated (for example by not controlling for the quality as well as the quantity of feedback). He concludes by suggesting how the weaknesses in the existing research-base might be addressed in future research.

Crooks' paper has a narrower focus--the impact of evaluation practices on students--and divides the field into three main areas--the impact of normal classroom testing practices, the impact of a

range of other instructional practices which bear on evaluation, and finally the motivational aspects which relate to classroom evaluation. He concludes that the summative function of evaluation--grading--has been too dominant and that more emphasis should be given to the potential of classroom assessments to assist learning. Feedback to students should focus on the task, should be given regularly and while still relevant, and should be specific to the task. However, in Crooks' view the 'most vital of all the messages emerging from this review' (p. 470) is that the assessments must emphasise the skills, knowledge and attitudes perceived to be most important, however difficult the technical problems that this may cause.

Like Natriello's review, the research cited by Crooks covers a range of styles and contexts, from curriculum-related studies involving work in normal classrooms by the students' own teachers, to experiments in laboratory settings by researchers. The relevance of work that is not carried out in normal classrooms by teachers can be called in question (Lundeberg & Fox, 1991), but if all such work were excluded, not only would the field be rather sparsely populated, but one would also be overlooking many important clues and pointers towards the difficult goal of reaching an adequately complex and complete understanding of formative assessment. Thus this review, like that of Natriello and more particularly that of Crooks, is eclectic. In consequence, decisions about what to include have been somewhat arbitrary, so that we now have some sympathetic understanding of the lack of overlap between the literature sources used in the two earlier reviews.

2.2 Examples in Evidence

- **Classroom Experience**

In this section we present brief accounts of pieces of research which, between and across them, illustrate some of the main issues involved in research which aims to secure evidence about the effects of formative assessment.

The first is a project in which 25 Portuguese teachers of mathematics were trained in self-assessment methods on a 20-week part-time course, methods which they put into practice as the

course progressed with 246 students of ages 8 and 9 and with 108 older students with ages between 10 and 14 (Fontana & Fernandes, 1994). The students of a further 20 Portuguese teachers who were taking another course in education at the time served as a control group. Both experimental and control groups were given pre- and post- tests of mathematics achievement, and both spent the same times in class on mathematics. Both groups showed significant gains over the period, but the experimental group's mean gain was about twice that of the control group's for the 8 and 9-year-old students--a clearly significant difference. Similar effects were obtained for the older students, but with a less clear outcome statistically because the pre-test, being too easy, could not identify any possible initial difference between the two groups. The focus of the assessment work was on regular--mainly daily--self-assessment by the pupils. This involved teaching them to understand both the learning objectives and the assessment criteria, giving them opportunity to choose learning tasks and using tasks which gave them scope to assess their own learning outcomes.

This research has ecological validity, and gives rigorously constructed evidence of learning gains. The authors point out that more work is required to look for long-term outcomes and to explore the relative effectiveness amongst the several techniques employed in concert. However, the work also illustrates that an initiative can involve far more than simply adding some assessment exercises to existing teaching--in this case the two outstanding elements are the focus on self-assessment and the implementation of this assessment in the context of a constructivist classroom. On the one hand it could be said that one or other of these features, or the combination of the two, is responsible for the gains, on the other it could be argued that it is not possible to introduce formative assessment without some radical change in classroom pedagogy because, of its nature, it is an essential component of the pedagogic process.

The second example is reported by Whiting et al. (1995), the first author being the teacher and the co-authors university and school district staff. The account is a review of the teacher's experience and records, with about 7000 students over a period equivalent to 18 years, of using mastery learning with his classes. This involved regular testing and feedback to students, with a requirement that they either achieve a high test score--at least 90%--before they were allowed to proceed to the next task, or, if the score were lower, they study the topic further until they could satisfy the mastery criterion. Whiting's final test scores and the grade point averages of his

students were consistently high, and higher than those of students in the same course not taught by him. 'Me students' learning styles were changed as a result of the method of teaching, so that the time taken for successive units was decreased and the numbers having to retake tests decreased. In addition, tests of their attitudes towards school and towards learning showed positive changes.

Like the previous study, this work has ecological validity--it is a report of work in real classrooms about what has become the normal method used by a teacher over many years. The gains reported are substantial; although the comparisons with the control are not documented in detail, it is reported that the teacher has had difficulty explaining his high success rate to colleagues. It is conceded that the success could be due to the personal excellence of the teacher, although he believes that the approach has made him a better teacher. In particular he has come to believe that all pupils can succeed, a belief which he regards as an important part of the approach. 'Me result shows two characteristic and related features--the first being that the teaching change involves a completely new learning regime for the students, not just the addition of a few tests, the second being that precisely because of this, it is not easy to say to what extent the effectiveness depends specifically upon the quality and communication of the assessment feedback. It differs from the first example in arising from a particular movement aimed at a radical change in learning provision, and in that it is based on different assumptions about the nature of learning.

2.3 Assessment by Teachers

Current Practice

Teachers' practices in formative assessment were reviewed in the articles by Crooks (1988) and Black (1993b). Several common features emerged from these surveys. The overall picture was one of weak practice. Key weaknesses were:

Classroom evaluation practices generally encourage superficial and rote learning, concentrating on recall of isolated details, usually items of knowledge which pupils soon forget.

Teachers do not generally review the assessment questions that they use and do not discuss them critically with peers, so there is little reflection on what is being assessed.

The grading function is over-emphasised and the learning function under-emphasised.

There is a tendency to use a normative rather than a criterion approach, which emphasises competition between pupils rather than personal improvement of each. The evidence is that with such practices the effect of feedback is to teach the weaker pupils that they lack ability, so that they are de-motivated and lose confidence in their own capacity to learn.

More recent research has confirmed this general picture. Teachers appear to be unaware of the assessment work of colleagues and do not trust or use their assessment results (Cizek et al., 1995; Hall et al., 1997). Both in questioning and written work, teachers' assessment focuses on low-level aims, mainly recall. There is little focus on such outcomes as speculation and critical reflection (Stiggins et al., 1989; Schilling et al., 1990; Pijl, 1992; Bol & Strage, 1996; Senk et al., 1997), and students focus on getting through the tasks and resist attempts to engage in risky cognitive activities (Duschl & Gitomer, 1997). Although teachers can predict the performance of their pupils on external tests--albeit tests reflecting low-level aims--their own assessments do not tell them what they need to know about their students' learning (Lorsbach et al., 1992; Rudman, 1987).

Reviews of primary school practices in England and in Greece have reported that teachers' records tend to emphasise the quantity of students' work rather than its quality, and that whilst tasks are often framed in cognitive terms, the assessments are in affective terms, with emphasis on social and managerial functions (Bennett et al., 1992; Pollard et al., 1994; Mavromattis, 1996). There are some striking comments by those who have researched these issues--one report on science practices sees formative and diagnostic assessment as 'being seriously in need of development' (Russell et al., 1995, p. 489), another closes with a puzzled question 'Why is the extent and nature of formative assessment in science so impoverished?' (Daws & Singh, 1996, p. 99), whilst a survey of teachers in Quebec Province, Canada, reports that for formative assessment 'Indeed they pay lip service to it but consider that its practice is unrealistic in the present educational context' (quoted by Dassa et al., 1993, p. 116). The conclusion of a survey about practice in Belgian primary schools was that the criteria used by teachers were 'virtually invalid by external standards' (Grisay, 1991, p. 104). A study which used interviews and so produced a richer picture of the perceptions of US teachers concludes as follows:

Most of the teachers in this study were caught in conflicts among belief systems, and institutional structures, agendas, and values. The point of friction among these conflicts was assessment, which was associated with very powerful feelings of being overwhelmed, and of insecurity, guilt, frustration, and anger. These teachers expressed difficulty in keeping track of and having the language to talk about children's literate development. They also described pressure from external accountability testing. They differed in their assessment strategies and in the language they used to describe students' literacy development. Those who worked in highly controlling situations were inclined to use blaming language and tended to provide global, negative descriptive assessments in impersonal language. Their assessments were likely to be based on a simple, linear notion of literacy. The less controlling the situation the less this was likely to occur. This study suggests that assessment, as it occurs in schools, is far from a merely technical problem. Rather, it is deeply social and personal. (Johnston et al., 1995, p. 359)

This last quotation also draws attention to the dominance of external summative testing. The effects here run deep, witness the evidence in Britain that when teachers were required to undertake their own assessments they imitated the external tests (Bennett et al., 1992), and seemed to be able to think only in terms of frequent summative tests with no feedback action (Ratcliffe, 1992; Harlen & Malcolm, 1996). A similar effect was encountered in assessment reforms in Queensland (Butler & Beasley, 1987). A different tension between formative and summative assessment arises when teachers are responsible for both functions: there has been debate between those who draw attention to the difficulties of combining the two roles (Simpson, 1990; Scott, 1991; Harlen et al., 1992) and those who argue that it can be done and indeed must be done to escape the dominance of external summative testing (Black, 1993a; William & Black, 1996). The requirement in Scotland, that teachers use external tests when they think their pupils are ready, and mainly for moderation purposes (i.e. checking for consistency of standards between schools), does not seem to have resolved these tensions (Harlen et al., 1995).

Assessment, Pedagogy and Innovation

Given these problems, it is not surprising that when national or local assessment policies are changed, teachers become confused. Several of the reports quoted above give evidence of this. A

patchy implementation is reported for reforms of teacher assessment in France (Broadfoot et al., 1996) and in French Canada (Dassa, 1990), whilst in the UK such changes have produced a diversity of practices, some of which may be counter-productive and in conflict with the stated aims of the changes which triggered them (McCallum et al., 1993; Gipps et al., 1997). Where changes have been introduced with substantial training or as an intrinsic part of a project in which teachers have been closely involved, the pace of change is slow because it is very difficult for teachers to change practices which are closely embedded within their whole pattern of pedagogy (Torrie, 1989; Shepard et al., 1994, 1996; Shepard, 1995) and many lack the interpretive frameworks that they need to co-ordinate the many separate bits of assessment information in the light of broad learning purposes (Bachor & Anderson, 1994). Indeed, some such work fails to produce its effect. A project with teachers in the creative arts, which tried to train them to communicate with students in order to appreciate the students' view of their own work, found that despite the training, many teachers stuck to their own agenda and failed to respond to cues or clues from the students which could have re-oriented that agenda (Radnor, 1994).

The issue that emerges here, as it did in the section above on Classroom experience, is the close link of formative assessment practice both with other components of a teacher's own pedagogy, and with a teacher's conception of his or her role. In a project aimed at enhancing the power of science teachers to observe their students at work, teachers could not find time for observing because they were not prepared to change classroom practices in order to give students more free responsibility and give themselves a less closely demanding control. The authors interpreted this as a reluctance to break the existing symbiosis of mutual dependency between teachers and students (Cavendish et al., 1990). In research with special education teachers, Allinder (1995) found that teachers with a strong belief in their high personal and teaching efficacy made better use of formative assessment than their less confident peers.

2.4 Students and Formative Assessment

The core of the activity of formative assessment lies in the sequence of two actions. The first is the perception by the learner of a gap between a desired goal and his or her present state (of

knowledge, and/or understanding, and/or skill). The second is the action taken by the learner to close that gap in order to attain the desired goal (Ramaprasad, 1983; Sadler, 1989). For the first action, the prime responsibility for generating the information may lie with the student in self-assessment, or with another person, notably the teacher, who discerns and interprets the gap and communicates a message about it to the student. Whatever the procedures by which the assessment message is generated, in relation to action taken by the learner it would be a mistake to regard the student as the passive recipient of a call to action. There are complex links between the way in which the message is received, the way in which that perception motivates a selection amongst different courses of action, and the learning activity which may or may not follow.

For the purposes of this review, the involvement of students in formative assessment will be considered by division into two broad topics, as follows:

(1) The first of these will focus on those factors which influence the reception of the message and the personal decisions about how to respond to it. The concern will be with the effects of beliefs about the goals of learning, about one's capacity to respond, about the risks involved in responding in various ways, and about what learning work should be like: all of these affect the motivation to take action, the selection of a line of action and the nature of one's commitment to it.

(2) The second will focus on the different ways in which positive action may be taken and the regimes and working contexts in which that action may be carried out. The focus here will be on study methods, study skills, collaboration with peers, and on the possibilities of peer and self-assessment.

There is clearly a strong interaction between the two areas. In particular, if self and peer-assessment are promoted in a classroom, this affects the initial generation of the message about a gap as well as the way in which a learner may work to close it. However, the over-arching sets of beliefs to be considered within the first focus bear on the perception of and response to feedback messages, albeit in different ways, whether they are generated by the self or by others. In the studies reported within the first topic, both sources of feedback have been considered.

Reception and Response

In his analysis of formative assessment by teachers in France, Perrenoud comments that:

A number of pupils do not aspire to learn as much as possible, but are content to 'get by', to get through the period, the day or the year without any major disaster, having made time for activities other than school work [...] Formative assessment invariably presupposes a shift in this equilibrium point towards more school work, a more serious attitude to learning [...] Every teacher who wants to practise formative assessment must reconstruct the teaching contracts so as to counteract the habits acquired by his pupils. Moreover, some of the children and adolescents with whom he is dealing are imprisoned in the identity of a bad pupil and an opponent. (Perrenoud, 1991, p. 92 (author's italics))

This rather pessimistic view is supported, but modified, by the finding of Swain 1991) that some secondary students working on teacher assessed science projects in England would respond to serious difficulties by working on subsidiary aspects of the task, so avoiding the main problem, and would be 'insatiable' in their search for cues for the 'right answer' from teachers. These symptoms of insecurity were accompanied by frequent moves to secure the esteem of the teacher. Similarly, Blumenfeld reports (1992) that some US students will try to avoid the risks involved in tackling a challenging assignment.

Thus whilst reluctance to be drawn into a more serious engagement with learning work may arise from a wish merely to minimise effort, there can be other influences. One problem may be fear--the extra personal commitment required can carry with it an enhanced penalty for failure in terms of one's self-esteem. Another problem may be that students can fail to recognise formative feedback as a helpful signal and guide (Tunstall & Gipps, 1996a). Purdie & Hattie's (1996) comparative study of the responses of Japanese and Australian students, which aimed to explore their self--regulation strategies, shows that response can be culturally determined. Many researches report that positive learning gains secured by formative feedback are associated with more positive attitudes to learning--notably in mastery learning regimes where the use to be made of the feedback is clearly planned (Kulik et al., 1990; Whiting et al., 1995), but there can also be negative affects and the notions of attitude and motivation have to be explored in more detail if the origin of such effects is to be understood.

Goal Orientation

This effect of goal orientation on learning has been extensively studied. The study of Ames & Archer (1988) involved only enquiry into the goals that students already held. They found that their sample of 176 students ranging over grades 8 to 11 could be divided into two groups--those with mastery orientation and those with performance orientation. The former spoke of the importance of learning, believed in the value of effort to achieve mastery, and had a generally positive attitude to learning. The latter attributed failure to lack of ability, spoke more in terms of their relative ability, about learning with relatively little effort if able, and focused on the significance of out-performing others. A similar distinction was made in the intervention study by Butler (1988) already described in the section on Classroom experience above in which the terms 'ego-involving feedback' and 'task-involving feedback' were used. The surprising result of this study, that the giving of grades could undermine the positive help given by task comments, illustrates the sensitivity of the issues raised here. In a later study, Butler & Neuman (1995) showed that those in task mode were more likely to seek help and to explain help-avoidance in terms of seeking independent mastery, whilst those in an ego mode sought help less and explained their avoidance in terms of masking their incapacity. Two general reviews of this field both stress that feedback which draws attention away from the task and towards self-esteem can have a negative effect on attitudes and performance (Cameron & Pierce, 1994; Kluger & DeNisi, 1996). It is even the case that giving praise can have bad effects, particularly when it is not linked to objective feedback about the work. Lepper & Hodell (1989) argue that reward systems can undermine both interest and motivation, whilst a detailed study by Pryor & Torrance (1996) shows how a teacher can concentrate on protective care for a child at the expense of helping the child to learn.

Self-Perception

In a more general review of the literature in this field, Ames (1992) started from the evidence about the advantages that 'mastery' (i.e. task-related) goals can secure and reviews the salient features of the learning environments that can help to secure these advantages. She concludes that evaluation to students should focus on individual improvement and mastery, but before this

the tasks proposed should help students to establish their own self-referenced goals by offering a meaningful, interesting and reasonably demanding challenge. She also recommends that feedback should be private, must be linked to opportunities for improvement, and should encourage the view that mistakes are a part of learning. The self-perception of students is all-important here, and this will be strongly influenced by teachers' beliefs about the relative importance of 'effort' as against 'ability' in their views of learning. In particular, it is important that motivation is seen to involve changes in students' qualitative beliefs about themselves, which the setting of goals and the style of feedback should both be designed to secure. The use of extrinsic rewards can be counter-productive if they focus attention on 'ability' rather than on the belief that one's effort can produce success. Of course, the beliefs of peers and of parents can also affect the ways in which the self-concepts of students are developed, as is pointed out in Blumenfeld's analysis (1992), which draws general conclusions similar to those of Ames.

Studies by Skaalvik (1990), Siero & van Oudenhoven (1995) and Vispoel & Austin (1995) all show that the reasons students gave for the results of their learning differ between low achievers, who attribute failure to low ability, and high achievers who tend to attribute success to effort. Vispoel & Austin urge that teachers should help students to overcome attributions to ability, and should encourage them to regard ability as a collection of skills that they can master over time.

Craven's work in mathematics and reading with students in grades 3 to 6 (Craven et al., 1991), showed that students' self-concept could be enhanced by feedback designed to this end and that whilst those whose self-concept was initially low showed large gains, those with initially high self-concept showed no gains. In addition, the students' attribution of success in the work to effort increased whilst attributions to ability did not. However, in this short intervention, the results obtained by the researcher could not be replicated by the teacher and there were no significant differences in achievement between experiment and control groups. A final and further perspective is added by the review of Butler & Winne (1995), who, in addition to covering the evidence that many of the factors mentioned above can have on learning achievement, also draw attention to the importance of learners' beliefs about the importance of effort, about the amount of effort that successful learning can demand, about the nature of learning, and about the--immature--expectation that all learning should lead to simple and unambiguous answers to all the questions that can be raised.

Assessment by Students

The focus of this section is to discuss one aspect of the learning activity which may follow from the student acceptance and understanding of the need to close a gap between present achievement and desirable goals. In formative assessment, any teacher has a choice between two options. The first is to aim to develop the capacity of the student to recognise and appraise any gaps and leave to the student the responsibility for planning and carrying out any remedial action that may be needed. This first option implies the development within students of the capacity to assess themselves, and perhaps to collaborate in assessing one another. The second option is for teachers to take responsibility themselves for generating the stimulus information and directing the activity which follows. The first of these two will be the subject of this section, whilst the second will be discussed in the sections titled Strategies and tactics for teachers and Systems below. The two options overlap in that it is possible to combine the two approaches: the boundary between this section and the section on Strategies and tactics for teachers will therefore be arbitrary, as is the boundary between this section and the section on Classroom experience.

The focus on self-assessment by students is not common practice, even amongst those teachers who take assessment seriously. Daws & Singh (1996) found that only about a third of the UK science teachers whom they sampled involved pupils directly in their own assessment in any way, and both Parkin & Richards (in Fairbrother et al., 1994, pp. 15-28) and the account of Norwegian initiative by Jernquist (reported in Black & Atkin, 1996, pp. 92-119) describe the introduction of self-assessment, respectively in secondary school science in the UK and in secondary mathematics in Norway, as innovations. In the general literature on classroom assessment, the topic is frequently overlooked--for example, the otherwise comprehensive collection by Phye (1997) contains no piece which focuses explicitly on self- and peer-assessment.

The motives for introducing this practice are diverse. Parkin & Richards started because of the practical impossibility of appraising the level of need of each individual in a class of about 30 students engaged in practical laboratory work--if they could do it for themselves the teacher could deploy his/her effort more efficiently. In his review of the literature on student self-

evaluation in professional training courses in the health sciences, Arthur (1995) reported that the requisite skills are not purposefully taught in most programmes, but also described new research to develop these skills in nursing education. The motive given here is that the future professional will need all of the skills necessary for life-long learning, and self-evaluation must be one of these.

The Norwegian initiative started from a more fundamental motive, which was to see self- and peer-assessment as an intrinsic part of any programme which aims to help students to take more responsibility for their own learning. A different slant on this aspect is provided in the study by James of recorded dialogues between teachers and students (1990). This study showed that in such dialogues, the teacher's power easily overwhelms the student's contribution, the latter being too modestly tentative. The effect is that inquiry into the reasons for a student's difficulty is not pursued. Some of the research discussed in the section on Classroom experience above involved experiments where work on goals was pursued both with and without training in self-evaluation; an example is the research by Schunk (1996) which showed that, if combined with performance goals, self-evaluation practice improved persistence, self-efficacy and achievement.

Studies of Self-assessment

Research studies of self- and peer-assessment can be broadly divided into two categories--those involving experimental work yielding quantitative data on achievement and those for which the evidence is qualitative. These will now be discussed in turn. Two quantitative examples have already been described in some detail in the section on Classroom experience (Fontana & Fernandes, 1994; Frederiksen & White, 1997). Both of these have in common an emphasis on the need for students to understand the learning goals, to understand the assessment criteria, and to have the opportunity to reflect on their work. Peer evaluation played a part only in the Frederiksen & White study.

Two studies have worked with children who have learning difficulties. In the first of these (McCurdy & Shapiro, 1992), the oral reading rates of elementary school students were improved by giving them verbal and visual performance feedback, either by the teacher only, or through peer-monitoring, or self-monitoring. The largest gains, measured by comparison of pre- and

post-test scores over the programme's period of nine weeks, were achieved by the self-monitoring group, whilst all three did better than a control group who had no formative feedback. Both on the grounds of acceptability to the teachers involved and on the reliability of their own appraisal of their work, the peer- and self-monitoring methods were preferred and one benefit of both was that they reduced the amount of time that the special education teachers had to spend on measurement in their classrooms. In the second research (Sawyer et al., 1992) the focus was on the writing composition skills of 4th and 5th grade students. Here, a group who were taught self-regulated strategies with explicit attention to goals did better than a similar group without the goal emphasis and a group without self-monitoring instruction. The first group were better overall on generalisation of the writing skills taught, but all groups with feedback did better, after the particular experiment was over, than other learning disability students without any experience of such feedback.

In research to investigate the most effective way of using a problem-solving software programme (Delclos & Harrington, 1991), two groups of 5th and 6th grade students were both given training in their pro-active use of the programme, but one of them also had to take part in monitoring exercises, described by the authors as meta-cognitive training. There was also a matched control group who used the programme without the training. The monitoring exercises were provided by a booklet of questions with which students monitored their results on a set of practice problem-solving exercises selected from the software. Both trained groups achieved greater success with the programme than the control group, but those with the monitoring training were also significantly better than those without it. They were more successful with the more complex problems, they succeeded more quickly, and overall they were seen to be employing more effective strategies. They seemed to do better, not because they could use the particular strategies more effectively, but because they started by reflecting on a problem and considering the possibilities of using different strategies before proceeding--an outcome which seemed to link with the meta-cognitive emphasis underlying the self-monitoring training.

A focus on self-directed learning was seen, in the review by Thomas (1993), to be a necessary concomitant to the moves to develop practical work, study skills, and responsibility for learning amongst students. He distinguished course features that discourage independent learning, such as test review handouts, from those that encourage it, including extensive performance feedback,

and reviewed evidence which established that such activities can improve student achievement. In a review of the practice of writing, Zimmerman & Risemberg (1997) discussed the different forms of the practice of self-regulation employed by several well-known authors and linked this to research evidence on the effectiveness of supporting students by encouraging self-monitoring (Schunk & Swartz, 1993b; Zimmerman & Bandura, 1994). A closely related set of studies by King (1994) on students' questioning strategies will be reviewed in the section on Questions below.

Peer-assessment

Several of the accounts described in this section involve both self-assessment and peer-assessment, Peer-assessment as such is included in several accounts of the development of group collaboration as a part of classroom learning activity. In an experimental study by Koch & Shulamith (1991), college students were taught to generate their own questions about topics in physics, and achieved better learning gains than those who used only teacher's questions; amongst those generating their own questions, some also used peer feedback to answer and discuss their efforts, and this group showed even greater learning gains than the rest. Higgins et al. (1994) also used collaborative work, in their work with 1st and 2nd grade school-children developing assessment skills in their integrated project work. The children generated their own criteria, and the quality of these rose during the study. Good agreement with teachers' assessments was achieved, with children tending to under-assess. However, groups were not accurate in their assessments of other groups. The reliabilities of self-and peer-assessments were also investigated, in work with college biology students, by Stefani (1994). He found correlations with teachers' assessments of 0.71 for self-assessments and 0.89 for peer-assessments. All of the students said that the self- and peer-assessment work made them think more, and 85% said that it made them learn more. Hughes & Large (1993) also investigated peer-assessment of final year undergraduates in pharmacology and found a correlation coefficient of 0.83 between the mean ratings of peers and those of a group of staff.

It is often difficult to disentangle the peer-assessment activity from other novel activities in work of this kind, and impossible in general to ascribe any reported gains to the assessment component. General reviews are given by Slavin (1991) and by Webb (1995). The second of these does focus on assessment practices in group work and it stresses the importance of training

in group processes and of the setting of clear goals and clear achievement criteria. In such groups, a clear choice has to be made, and shared in the group, between a goal of the best performance from the group as a group, and a goal of improving individuals' performances through group collaboration. The question of the optimum group composition is a complex one; where a group goal has priority, then for well defined tasks, established high achievers are the most productive, but for more open tasks a range of types of students is an advantage. Where individuals' performance has priority, then the high achievers are little affected by the mix, but the low achievers benefit more from a mixed group provided that the group training emphasises methods for drawing out, rather than overwhelming, their contribution. The need for such care is emphasised in a study of group discussions in science education by Solomon (1991).

Links to Theories of Learning

The arguments given by Zessoules & Gardner (1991) show how any assessment changes of the types described above might be expected to enhance learning if they help students to develop reflective habits of mind. They further argue that such development should be an essential component in programmes for the implementation of authentic assessment in classroom practice. Assessment is to be seen as a moment of learning, and students have to be active in their own assessment and to picture their own learning in the light of an understanding of what it means to get better.

In summary, it can be seen that these various approaches to developing self-assessment by pupils hold promise of success. However, their interpretation in relation to more general theories of learning raises fundamental problems, as illustrated by the analysis of Tittle (1994). Full discussion of this and similar work will be deferred until the last part of this article. A few points can be introduced here. In a review of European research in this field, Elshout-Mohr (1994) points out both that students are often unwilling to give up misunderstandings--they need to be convinced through discussion which promotes their own reflection on their thinking--and also that if a student cannot plan and carry out systematic remedial learning work for himself, he or she will not be able to make use of good formative feedback. Both of these indicate that self-assessment is essential. Similarly, Hattie et al. (1996) argue that direct teaching of study skills to students without attention to reflective, meta-cognitive, development may well be pointless. One reason for the need to look for radical change is that students bring to their work models of

learning which may be an obstacle to their own learning. That pupils do have such models that are to a degree culturally determined is illustrated by a comparison of the approaches to learning of Australian and Japanese students (Purdie & Hattie, 1996), whilst the finding that the most able students in either country are more alike than their peers in having developed similar effective habits of learning shows that such constraining traditions can be overcome.

2.5 Strategies and Tactics for Teachers

Overview

The various aspects of a teacher's work in formative assessment can be organised in relation to the temporal sequence of decisions and actions that are entailed. This approach will be used here as a framework for describing the work reported in the literature. Thus, the sub-sections below will deal in turn with the choice of tasks, with classroom discourse, with several aspects of the use of questions, with tests and then with feedback from tests. A closing section will then look at strategies overall, including work that looks more deeply at the assumptions and rationales that might underlie the articulation of tactics.

Choice of Task

It is obvious that formative assessment which guides learners towards valued learning goals can only be generated with tasks that both work to those goals and that are open in their structure to the generation and display of relevant evidence, both from student to teacher and to students themselves. In their detailed qualitative study of the classroom characteristics of two outstandingly successful high-school science teachers, Garnett & Tobin (1989) concluded that the key to their success was the way they were able to monitor for understanding. A common feature was the diversity of class activities-with an emphasis on frequent questioning in which 60% of the questions were asked by the students. In a more general review of classroom environments, Ames (1992) selects three main features which characterize successful 'mastery' (as opposed to 'performance'-see section on Goal orientation above) classrooms. The first of these is the nature of the tasks set, which should be novel and varied in interest, offer reasonable challenge, help students develop short-term self-referenced goals, focus on meaningful aspects of learning and support the development and use of effective learning strategies. Blumenfeld (1992)

explores some of these issues, pointing out that such notions as 'challenging' and 'meaningful' are problematic. A task where the challenge goes too far can lead to student avoidance of the risks involved, and for students who are far behind it is difficult to encourage their efforts without at the same time making them aware of how far behind they are. Similarly, tasks can be meaningful for a variety of reasons and it is important to emphasise those meanings which might be productive for learning.

In an earlier review about science teaching, Dumas-Carre & Larcher (1987) were more ambitious. They emphasised the need to shift current pedagogy to give more emphasis to procedural aspects of knowledge and less to the declarative aspects. They outlined a scheme for the comparative analysis of tasks which could be deployed by teachers to produce a descriptive analysis of the tasks they were using. This scheme distinguished tasks which (a) presented a specific situation identical to the one studied, or (b) presented a 'typical' problem but not one identical to the one studied, requiring identification of the appropriate algorithm and its use, rather than exact replication of an earlier procedure as in (a), and (c) a quite new problem requiring new reasoning and construction of a new approach, deploying established knowledge in a new way. Students would need special and explicit training for tackling tasks of type (c). They recommended that all three types of task are needed, but that teachers do not currently plan or analyse the tasks that they set by any scheme of this type. Such foresight is an essential condition for planning the incorporation of formative assessment, both for provision of feedback and for planning how to respond to it.

Discourse

That the quality of the discourse between teacher and students can be analysed at several different levels is evident from the extensive literature on classroom interactions and discourse analysis. In a review of questioning in classrooms, Carlsen (1991) contrasts the process-product approach with the socio-linguistic paradigm, and argues that inconsistency of research results on the cognitive level of questioning may be due to neglect of the fact that the meaning of a question cannot be inferred from its surface features alone. As both he and Filer (1995) argue, the meaning behind any discourse depends also on the context, on the ways in which questions in a particular classroom have come to signify the patterns of relationships between those involved that have been built up over time. Pryor & Torrance (1996) give an example of how a habitual

pattern can be unhelpful for learning. Newmann (1992) makes a plea, on similar grounds, for assessment in social studies to focus on discourse, defined by him as language produced by the student with the intention of giving narrative, argument, explanation or analysis. The plea is based on an argument that current methods, in which students are constrained to use the language of others, undermine the constructive use of discourse and so trivialise social knowledge. A striking example of such effects is reported in a paper by Filer (1993). In the same vein, Quicke & Winter (1994) report success in work with low achieving students in Year 8, where they aimed to develop a social framework for dialogue about learning. The work of Ross et al. (1993) in aesthetic assessment in the arts can be seen as a response to this plea, and the difficulties reported by Radnor (1994) are evidence of the inadequacy of established practice.

Radnor's paper uses the phrase 'qualitative formative assessment' in its title, and this may help to explain why quantitative evidence for the learning effects of discourse variations is hard to find. An exception is the research by Clarke (1988) on classroom dialogue in science classrooms. He analysed the discourse of three teachers in four classrooms, grading the quality of the discourse by summation over four criteria. These included the numbers of interpretable themes, the numbers of cross-correlations (an indicator of coherence) and proportions of themes explicitly related to the content of the lessons. This discourse variable was included with three other measures, of scholastic aptitude, locus of control and Piagetian level respectively, as independent variables, and an achievement post-test as dependent variable. With the class as the unit of analysis, the discourse variable accounted for 63% of the variance, with the three others accounting respectively for under 4%, 22% and 14%.

Johnson & Johnson (1990) present a meta-analysis to show that collaborative discourse can produce significant gains in learning. Rodrigues & Bell (1995), Cosgrove & Schaverien (1996) and Duschl & Gitomer (1997) all report work with science teachers to promote such discourse. The concern in all three cases was to help students to move, in talk about their work, from a focus in everyday and content based terms towards a deeper discussion of conceptual learning. Roth & Roychoudhury (1994) recommend the use of concept maps as an aid in such discussions; such maps, drawn by the students, serve to provide useful points of reference in clarifying the points under discussion and enable the teacher to engage in 'dynamic assessment'.

Questions

Some of the relevant aspects of questioning by teachers have already been introduced above. The quality of classroom questioning is a matter for concern, as expressed in the work of Stiggins et al. (1989) who studied 36 teachers over a range of subjects and over grades 2 to 12, by observation of classroom work, study of their documentation, and interviews. At all levels the questioning was dominated by recall questions, and whilst those trained to teach higher-order thinking skills asked more relevant questions, their use of higher-order questions was still infrequent. An example of the overall result was that in science classrooms, 65% of the questions were for recall, with only 17% on inferential and deductive reasoning. The patterns for written work were similar to those for the oral work. Bromme & Steinberg (1994) studied the classroom strategies of novice teachers in mathematics and found that they tended to treat students' questions as being from individual learners, whereas the responses of expert teachers tended to be directed more to a 'collective student'.

Several authors report work focused on question generation by students, and as pointed out in the section on Self-assessment above, this may be seen as an extension of work on students' self-assessment. With college students, King (1990,1992a, b; 1994) found that training which prompted students to generate specific thought-provoking questions and then attempt to answer them is more effective than training in other study techniques, which she interprets in terms of the strategy underlying the training which aimed to develop learner autonomy and learners' control over their own work. Similar results, which also showed that students' own questions produced better results than adjunct questions from the teacher, were reported for Israeli college students by Koch & Shulamith (1991). In work with 5th grade school students, a similar approach was used in training students with problem-solving on computer administered tasks (King, 1991). With a sample of 46 students, one group was given no extra instruction, another was trained to ask and answer questions with student partners, whilst a third group were also trained in questioning one another in pairs but directed to use strategic questions for guidance in cognitive and meta-cognitive activity. The latter training focused on the use of 'generic questions' such as 'How are X and Y alike?' and 'What would happen if .. ?'. The outcome was measured by a post-test of written problems and a novel computer task. The group trained to ask strategic questions of one another out-performed the others. Foos et al. (1994) have reported similar success on outcome measures when students were trained to prepare for examinations by several techniques, the most successful being the generation of their own study questions

followed by attempts to answer them. This work, and similar work in school science classes (King & Rosenshine, 1993) can be seen as part of a larger strategy to promote inquiry-based critical thinking (King, 1995).

The Use of Tests

One study giving evidence that frequent testing can lead to improved learning has already been cited in the section on Classroom experience above (Martinez & Martinez, 1992). Bangert-Drowns, et al. (1991b) reviewed the evidence of the effects of frequent class testing. Their meta-analysis of 40 relevant studies showed that performance improved with frequent testing and increased with increased frequency up to a certain level, but that beyond that (somewhere beyond 1 and 2 tests per week) it could decline again. The evidence also indicated that several short tests were more effective than fewer longer ones. Similar evidence is quoted by Dempster (1991,1992,see the section on Task motivation processes, below).

In a later investigation with college students in psychology, Iverson et al. (1994) found that the addition of frequent ungraded tests produced no significant improvement in performance, even though the students in the experiment said that they would like to have such tests in other courses also. A similarly negative result was reported by Strawitz (1989), but a contrary, positive effect was found by Schloss et al. (1990) working with graduate students in teacher training for special education. When given a short formative quiz after each lecture students performed significantly better than they did when no quiz was provided on three measures post-tests of familiar items, post-tests of unfamiliar items and a survey of satisfaction with the instruction.

In some subject areas, teachers are reluctant to use tests for fear of inhibiting creativity. Gilbert (1996) attempted to tackle this problem with primary school teachers in the assessment of art. The project worked with both experienced and trainee teachers to develop a framework and a language for assessing art, and from this the group were able to formulate guidance about the feedback appropriate to children according to the classification in the framework to which their work was judged to correspond. Developments of new assessment methods appropriate to specific subjects are also described by Adelman et al. (1990) for visual and performance arts with older secondary students and by Harnett (1993) for history in primary schools.

The aim and structure of the tests employed are not explained in most studies. Crooks speculated that low level aims might benefit from more frequent testing, but that higher level aims would benefit from a lower frequency. Khalaf & Hanna (1992) conducted a more searching review and disagreed with Crooks on this point. They selected 20 studies, of which 18 gave positive effects. They point out that the nature of the criterion test could distort results. In any such study, the criterion test might be more important to a control group than to the treatment group. On the other hand, if the final summative test were to contain questions similar to those in the class tests, there could be a distortion in the reverse direction. They concluded that only four of the studies they reviewed were free from this type of flaw. The results of these were all positive with a mean effect size of 0.37, but they were all with college students. Following their critical review, these authors describe the results of an investigation with 2000 students in 93, 10th grade, classes in Saudi Arabia. Control classes were given the normal monthly quizzes, whilst the others were given bi-monthly quizzes. The criteria were tests set by an investigator who had not seen any of the quizzes constructed and used by the teachers, and included a test at the end of the course and a delayed test three months later. Treatment and control classes were set up in pairs having the same teacher for the two members of each pair. There were significant differences between the two groups on both test occasions, with an effect size of about 0.3 in favour of the more frequently tested group. The effect was much greater for high achievers than for the medium and low achievers. However, the tests used were composed only of true-false or multiple choice items, so that the learning at issue was relatively superficial in nature. The care taken with this experiment shows how studies cannot be accepted as relevant without careful scrutiny of the experiment design, and of the quality of the questions used for both the treatment and the criterion tests.

The Quality of Feedback

Both of the previous sub-sections lead to the almost obvious point that the quality of the feedback provided is a key feature in any procedure for formative assessment. The instructional effect of feedback from tests was reviewed by Bangert-Drowns et al. (1991a) using a meta-analysis of 58 experiments taken from 40 reports. Effects of feedback were reduced if students had access to the answers before the feedback was conveyed. When this effect had been allowed

for, it was then the quality of the feedback which was the largest influence on performance. Programmed instruction and simple completion assessment items were associated with the smallest effects. Feedback was most effective when it was designed to stimulate correction of errors through a thoughtful approach to them in relation to the original learning relevant to the task.

The feedback provided by teachers' written responses to students' homework was studied in an experiment with over 500 Venezuelan students involving 18 mathematics teachers in three schools (Elawar & Como, 1985). They trained the teachers to give written feedback which concentrated on specific errors and on poor strategy, with suggestions about how to improve, the whole being guided by a focus on deep rather than superficial learning. A control group followed the normal practice of marking the homework without comments. In order to check whether effects of the feedback training on their teaching could account for any results, a third group of the trained teachers marked half of their classes with full feedback and the other half with marks only. All were given a pre-test and one of three parallel forms of post-test. Analysis of variance of the results showed a big effect associated with the feedback treatment, which accounted for 24% of the variance in final achievement (with another 24% associated with prior achievement). The treatment also reduced the initial superiority of boys over girls and had a large positive effect on attitudes towards mathematics.

In a quite different sphere of learning, Tenenbaum & Goldring (1989) carried out a meta-analysis with 16 studies of the effects of 'enhanced instruction', involving emphasis on cues, participation, reinforcement, feedback and correctives, on motor skill learning in physical education. Exposure to these forms of enhancement produced gains with a mean effect size of 0.66, and they also enhanced students' time on task.

Formulation of Strategy

The sub-sections above can be regarded as treatments of the various components of a kit of parts that can be assembled to compose a complete strategy. The research studies described can be judged valuable, in that they explore a complex situation by treating one variable at a time, or as flawed because any one tactic will vary in its effect with the holistic context within which it

operates. It has also emerged that at least some of the accounts are incomplete in that the quality of the procedures or instruments that evoke the feedback, and of the assumptions which inform the interpretation of that feedback, cannot be judged. At the same time it is clear that the fundamental assumptions about learning on which the procedures, instruments and interpretations are based are all important.

Several authors have written about the larger strategic picture, and reference has already been made to some of their arguments. The analysis of Thomas et al. (1993) stands out because they have attempted a quantitative study to encompass the many variables involved. They applied hierarchical linear modelling to data collected from 12 high school biology courses, focusing on the features of the courses which placed demands on and gave the support to the students. At the student level, the results indicated a positive link between achievement and both their self-concept of academic ability and their study activities; these last two were also linked with one another. Students' engagement in active study work was positively associated with the provision of challenging activities and with extensive feedback on their work in the course. Such feedback was also directly related with high achievement. Amongst the other relationships that were teased out was the finding that instructor support which reduced course demands strengthened the relationship between self-concept and achievement. This work constitutes an ambitious effort, but, almost of necessity, only very general information is provided about the underlying quality of the work being studied.

Weston et al. (1995) have argued that if the literature on formative assessment is to inform instructional design, then a common language is needed. They identify four components--who participates, what roles can be taken, what techniques can be used and in what situations these can occur, and argue that instructional design should be based on explicit decisions about these four, to be taken in the light of the goals of the instruction. The model was used to analyse 11 instructional tests and revealed that there were many assumptions about these four issues which were embedded in the language about formative evaluation.

Thus task selection, and the type of feedback that a task might generate, require a cognitive theory which can inform the link between learners' understanding and their interactions with assessment tasks, in the light of which assessment activities can be designed and interpreted. Such an approach will of course interact strongly with the pedagogy adopted, and may have to

temper an a priori theoretical position with a readiness to adapt and develop by an inductive approach as formative feedback challenges the rationale of the work (Fuchs & Fuchs, 1986). The conclusion of the analysis is that a very substantial effort, involving collaboration between psychometricians, cognitive scientists and subject experts is needed.

All of these discussions point to the need for very far-reaching changes if formative evaluation is to realise its potential. Some large-scale changes in pedagogy have attempted to meet such targets, and are distinguished from what has been discussed in this section by their comprehensive and strategic approach. These will be the subject of the next section.

2.6 Systems

2.6.1 General Strategies

Good assessment feedback is either explicitly mentioned or strongly implied in reports of a range of studies and initiatives in which such feedback is one component of a broader strategy. Thus, for example, in summarising a study of school effectiveness Mortimore et al. (1988) point out that feedback and good record-keeping are key aspects of effectiveness. In the initiative in Britain to develop a holistic 'Record of Achievement' to cover all aspects of a student's work and contribution within a school, the student is to be involved in negotiating an agreed record. Thus self-assessment is reported to be an important feature, but it has been incorporated in a variety of ways and is sometimes superficial (Broadfoot, 1992; Broadfoot et al., 1990). Enhanced attention to diagnosis and remediation is a feature of many other schemes, for example reading recovery and Slavin's Success for All scheme (Slavin et al., 1992, 1996).

Assessment and feedback are also an important feature of mastery learning programmes, discussed in more detail below, but with many (if not all) of these teaching systems, even identifying the precise nature of the formative feedback used, let alone its contribution to the global improvements in attainment generated, is difficult. For this reason, these systems are reviewed only briefly in what follows.

2.6.2 Studies of Mastery Learning

Mastery learning originated as a practical implementation of the learning theories of John B. Carroll. He proposed that success in learning was a function solely of the ratio of the time actually spent learning to the time needed for learning—in other words, any student could learn anything if they studied it long enough. The time spent learning depended both on the time allowed for learning and the learner's perseverance while the time needed to learn depended on the learner's aptitude, the quality of the teaching, and the learner's ability to understand this teaching (see Block. & Bums, 1976, p. 6). Two main approaches to mastery learning were developed in the 1960s. One, developed by Benjamin Bloom, using teacher-paced group-based teaching approaches, was called Learning for Mastery (LFM) and the other was Keller's individual-based, student-paced Personalized System of Instruction (PSI). The vast majority of the research undertaken into mastery learning has been centred on Bloom's, rather than Keller's model, and that which has been done on PSI is largely confined to further and higher education.

A key consequence of Bloom's LFM model is that students of differing aptitude will differ in their achievements unless those with less aptitude are given either a greater opportunity to learn or better quality teaching. For most proponents of mastery learning, this is not to be achieved by targeting teaching resources at students of lower aptitude, but by improving the quality of teaching for all students, the underlying assumption being that students with higher aptitude are better able to make sense of incomplete or poor instruction (Milkent & Roth, 1989).

The key elements in this strategy, according to McNeil (1969) were:

The learner must understand the nature of the task to be learned and the procedure to be followed in learning it.

The specific instructional objectives relating to the learning task must be formulated.

It is useful to break a course or subject into small units of learning and to test at the end of each unit.

The teacher should provide feedback about each learner's particular errors and difficulties after each test.

The teacher must find ways to alter the time some students have available to learn.

It may be profitable to provide alternative learning opportunities.

Student effort is increased when small groups of two or three students meet regularly for as long as an hour to review their test results and to help one another overcome the difficulties identified by means of the test.

Therefore, although the principles of mastery learning cover all aspects of learning and teaching, effective formative assessment is a key component of effective mastery learning. Three major reviews of the research into the effectiveness of mastery learning use the technique of 'meta-analysis' to combine the results from a variety of different studies. The review of Block & Bums (1976) covers work done in the first half of the 1970s while Guskey & Gates (1986) and Kulik et al. (1990) cover the subsequent decade.

2.6.3 The Relevance of Mastery Learning

The only clear messages emerging from the mastery learning literature are that mastery learning appears to be effective in raising students' scores on teacher-produced tests, is more effective in teacher-paced programmes than in self-paced programmes, and is more effective for younger students.

However, while establishing that under certain circumstances, mastery learning is effective in raising achievement, the literature gives very little evidence as to which aspects of mastery learning programmes are effective. For example, while most of the research concentrates on the effects of mastery learning on students, the explanation of the effects could be that preparing for teaching for mastery provides professional development for the teacher (Whiting et al., 1995).

Indeed, one of the criticisms voiced by the reviewers cited above is that too often it is impossible to establish from the research reports which features of mastery learning were implemented in the research being reported, let alone which were effective (Guskey & Pigott, 1988). There are at least five aspects of 'typical' mastery learning programmes that are relevant to the purposes of the present review:

- that students are given feedback;

- that students are given feedback on their current achievement against some expected level of achievement (i.e. the 'mastery' level);
- that such feedback is given rapidly;
- that such feedback is (or is at least intended to be) diagnostic;
- that students are given the opportunity to discuss with their peers how to remedy any weaknesses.

However, it is by no means clear that all of these are necessary to achieving the gains claimed for mastery learning. For example, Kulik & Kulik (1987) argue that mastery testing is a crucial component in the success of mastery learning, and that its omission leads to a substantial drop in a programme's effectiveness. On the other hand, programmes without formalised 'mastery' testing, but with many of the other features of 'mastery' programmes can show significant effects.

For example, in a study by Brown et al. (1996) a group of low-achieving second grade students were given a year of 'transactional strategies instruction' (TSI), in which teachers explained and modelled strategies, gave additional coaching as needed, and encouraged students to explain to each other how they used the strategies. At the end of the year, this group outperformed by a considerable margin a similar group taught by highly regarded teachers using more traditional methods (exact effect sizes are not given, but they range between 1 and 2 standard deviations).

Assessment Driven Models

2.6.4 Portfolios

The portfolio movement is more closely associated with efforts to change the impact of high-stakes, often standardised, testing of school learning. There is a vast literature associated with the portfolio movement in the USA. Much of it is reviewed, by Collins (1992), in the edited collections of Belanoff & Dickson (1991) and-for assessment of writing--by Calfee & Perfumo (1996a), whilst Courts & McInerney (1993), set out some of the issues in higher education. Mills (1996) gives an account of the origins of the innovation, describing the work as an attempt in Vermont to satisfy demands of accountability whilst avoiding the pressures of standardised tests.

A portfolio is a collection of a student's work, usually constructed by selection from a larger corpus and often presented with a reflective piece written by the student to justify the selection.

The involvement of the student in reviewing and selecting is seen as central--as Mills says 'Inventing ways to promote that kind of reflection on a wide scale has been at the heart of the Vermont assessment from the beginning' (Mills, 1996, p. 192) and, speaking of the response of students 'What was striking was their ability to reflect on their own work in relation to a set of internalised standards--standards that they shared with many others' (Mills, 1996, p. 194). Similarly, writing about a different, national, project, Daro (1996) reports on the innovators' enthusiasm, both for the power of portfolios to focus student attention on their own learning efforts and accomplishments, and for the evidence that teachers believe the work changes the ways in which they teach and increases their expectations for their students. Calfee & Freedman (1996) see portfolios as offering a technology for helping the slogan of 'student-centred learning' to become a reality. Others (Herman et al., 1996) emphasise that it is valuable for students to understand the assessment criteria for themselves, whilst Yancey (1996), in a more subtle analysis of the concept of reflection as learning, points out that the practice of helping students to reflect on their work has made teachers more reflective for themselves.

However, there is little by way of research evidence, that goes beyond the reports of teachers, to establish the learning advantages. Attention has focused rather on the reliability of teachers' scoring of portfolios because of the motive to make them satisfy concerns for accountability, and so to serve summative purposes as well as the formative. In this regard, the tension between the purposes plays out both in the selection and in the scoring of tasks (Benoit & Yang, 1996). Daro (1996) describes scoring approaches based on a multi-dimensional approach, with the criterion that each dimension reflect an aspect of learning which can be understood by students and which reflects an important aspect of learning. However, he identifies the problem thus 'But it does not necessarily follow that it will be practical to bring national standards into the focus of self-assessing students and their teachers' (Daro, 1996, p. 241).

Calfee & Perfumo (1996b) report on research with teachers into their experience of using portfolios. The results showed a disturbing gap between the general rhetoric and actual practice, for they showed that many teachers were paying little attention to external standards and were producing little evidence of any engagement of students in understanding why they are doing this work. Their conclusion was that the future fate of the movement hung in the balance, either

between three negative possibilities--anarchy, disappearance, or becoming too standardised--or the positive possibility of promoting a profound revolution in learning.

2.7 Prospects for the Theory and Practice of Formative Assessment

2.7.1 No Meta-analysis

It might be seen desirable, and indeed might be anticipated as conventional, for a review of this type to attempt a meta-analysis of the quantitative studies that have been reported. The fact that this hardly seems possible prompts a reflection on this field of research. Several studies which are based on meta-analyses have provided useful material for this review. However, these have been focussed on rather narrow aspects of formative work, for example the frequency of questioning. The value of their generalisations is also in question because key aspects of the various studies that they synthesise, for example the quality of the questions being provided at the different frequencies, is ignored because most of the researchers provide no evidence about these aspects.

Individual quantitative studies which look at formative assessment as a whole do exist, and some have been discussed above, although the number with adequate and comparable quantitative rigour would be of the order of 20 at most. However, whilst these are rigorous within their own frameworks and purposes, and whilst they show some coherence and reinforcement in relation to the learning gains associated with classroom assessment initiatives, the underlying differences between the studies are such that any amalgamations of their results would have little meaning.

At one level, these differences are obvious on casual inspection, because each study is associated with a particular pedagogy, with its attendant assumptions about learning: one that in many cases has been constructed as the main element of the innovation under study. There are however deeper differences: even where the research studies appear to be similar in the procedures involved, they differ in the nature of the data which may have been collected--or ignored. The fact that important determining features are often given no attention is one sign of the inadequate conceptualisation of the issues involved, indicating a need for further theory building. From the evidence presented in this review, it is clear that a great deal of theory building still needs to take

place in the area of formative assessment, and we shall make suggestions below about a basis for this development.

An underlying problem, which we have already noted in an earlier paper (William & Black, 1996), is that the term 'formative assessment' is not common in the assessment literature. Such meaning as we have attached to the term here is also represented for others by such terms as 'classroom evaluation', 'curriculum-based assessment', 'feedback', 'formative evaluation' and so on.

Taking further the argument in the section on feedback, we propose, for the sake of simplicity, that the term feedback be used in its least restrictive sense, to refer to any information that is provided to the performer of any action about that performance. This need not necessarily be from an external source (as, for example, would be required by Kluger & DeNisi, 1996), nor need there necessarily be some reference standard against which the performance is measured, let alone some method of comparing the two. The actual performance can be evaluated either in its own terms, or by comparing it with a reference standard. The comparison can either be in terms of equality (i.e. these are the same or different?), as a distance (how far short of--or indeed beyond--the standard was it?) or as diagnosis (what do I need to do to get there?). Adopting the definition (although not the term) proposed by Sadler (1989), we would argue that the feedback in any assessment serves a formative function only in the latter case. In other words, assessment is formative only when comparison of actual and reference levels yields information which is then used to alter the gap. As Sadler remarks, 'If the information is simply recorded, passed to a third party who lacks either the knowledge or the power to change the outcome, or is too deeply coded (for example, as a summary grade given by the teacher) to lead to appropriate action, the control loop cannot be closed' (Sadler, 1989, p. 121). In such a case, while the assessment might be formative in purpose, it would not be formative in function and in our view this suggests a basis for distinguishing formative and summative functions of assessment.

Gipps (1994, chapter 9) draws attention to a paradigm shift from a testing culture to an assessment culture, associated with a shift from psychometrics to the assessment of learning. Similarly, Shinn & Good III (1993) argue that there needs to be a 'paradigm shift' in assessment, from what they call the current assessment paradigm (and what we have here called summative functions of assessment) to what they call the 'problem-solving paradigm' (broadly equivalent to

what we are here calling the formative functions of assessment). They illustrate the distinction by the differences in the way that questions are posed in the two paradigms along various dimensions (see Table 1--from Shinn & Hubbard, 1992). Summative functions of assessment are concerned with consistency of decisions across (relatively) large groups of students, so that the over-riding imperative is that meanings are shared by different users of assessment results. A particular problem for the constructors of summative assessments is that exactly who will be making use of the assessment results is likely to be undetermined. In contrast, formative functions of assessment prioritize desirable consequences either for (relatively) small groups of students (such as a teaching group) or for particular individuals.

2.7.2 The Theoretical Basis

All that can be set out here are a few 'notes towards a theory of formative assessment', which are offered partly because they may be a helpful aid to reflection on the work surveyed and partly because they may be helpful in looking ahead to the implications of this work.

Two key contributions, to which reference has already been made, are those of Sadler (1989) and Tittle (1994). Sadler built upon Ramaprasad's notion of the gap between the state revealed by feedback and the desired state, emphasising that action will be inhibited if this gap is seen as impracticably wide. He further argued that ultimately, the action to close that gap must be taken by the student--a student who automatically follows the diagnostic prescription of a teacher without understanding of its purpose or orientation will not learn. Thus self-assessment by the student is not an interesting option or luxury; it has to be seen as essential. Given this, the orientation by a student of his or her work can only be productive if that student comes to share the teacher's vision of the subject matter. Some (e.g. Klenowski, 1995) argue that this can be done by clarifying objectives, but others (e.g. Claxton, 1995; Wiliam, 1994) argue that these definitions must remain implicit if they are not to distort learning.

A development of this theory seems to call for links to compatible learning theories and to theories of the meta-cognition and locus of control of the learner.

Tittle's (1994) framework emphasises three dimensions. The first, the epistemology and theories involved, can relate both to positions held in relation to learning in general, and to the particular epistemology relevant to the subject matter concerned. The nature of the epistemology, and so of

the meta-cognition involved, in (say) aesthetic appreciation of poetry will be very different from that for (say) physics, and hence many features of formative assessment will differ between these two fields of learning. The second dimension is the more evident one of the assessment characteristics; it can be remarked here that in several of the studies reported here, little is said about the detail of these, or about the distinctive effects of the particular subject matter involved.

Tittle's third dimension brings in the interpreter and user, and she particularly stresses the importance of these. In relation to students, this emphasis is reinforced and developed by Sadler's arguments, but the teacher's beliefs, about the subject matter, about learning, and about the students and the class, must also be important components in any model, if only because it is on the basis of these that appraisals of Sadler's 'gap' must be formulated. Tittle also makes the important point that while modern conceptions of validity theory (e.g. Messick, 1989) stress the value-laden nature of assessment processes, the actual nature of those values is excluded, creating the impression that one (presumably reasonably coherent) set of values is as good as any other. Thus current conceptions of validity provide no guide as to what 'ought' to be going on, merely a theoretical framework for discussing what is going on.

2.7.3 Expectations and the Social Setting

These last two analyses bring out a feature which in our view has been absent from a great deal of the research we have reviewed. This is that all the assessment processes are, at heart, social processes, taking place in social settings, conducted by, on and for social actors. Guy Brousseau (1984) has used the term 'didactical contract' to describe the network of (largely implicit) expectations and agreements that are evolved between students and teachers. A particular feature of such contracts is that they serve to delimit 'legitimate' activity by the teacher. For example, in a classroom where the teacher's questioning has always been restricted to 'lower-order' skills, such as the production of correct procedures, students may well see questions about 'understanding' or 'application' as unfair, illegitimate or even meaningless (Schoenfeld, 1985).

As Tittle's (1994) approach emphasises, the 'opening moves' of teachers and students in the negotiation of such classroom contracts will be determined by their epistemological, psychological and pedagogical beliefs. For example, when a teacher questions a student, the

teacher's beliefs will influence both the questions asked and the way that answers are interpreted. An important principle here is the distinction between 'fit' and 'match' (Von Glasersfeld, 1987, p. 13). For example, a teacher may set student problems in solving systems of simple equations. If students answer all the questions correctly, the teacher may well conclude that the students have 'understood' the topic, i.e. they assume that the students' understanding matches theirs. However, this is frequently not the case. For example, when asked to solve the following two equations

$$3a = 24$$

$$a + b = 16$$

many students believe that it is impossible, saying things like 'I keep getting b is 8, but it can't be because a is 8'. This is because in the examples encountered in most textbooks, each letter stands for a different number. The students' understanding is therefore not a match but only a 'fit' with the teacher's. The relationship between fit and match depends critically on the richness of the questions used by the teacher, and this, in turn will depend on the teacher's subject knowledge, their theories of learning, and their experience of learners.

A study of seven experienced elementary school teachers examined the implicit criteria that teachers used to determine whether students had 'understood' something (Reynolds et al., 1995). After studying and discussing video extracts and transcripts of lessons, seven 'indicators of understanding' emerged which were agreed by all seven teachers, although they were regarded not as a static check-list, but rather as a series of potential clues to the level of the student's understanding:

- (1) changes in demeanour: students who had understood were 'bright-eyed' while those who had not appeared half-hearted;
- (2) extension of a concept: students who have understood something often take the idea further on their own initiative;
- (3) making modifications to a pattern: students who understand, spontaneously start making their own modifications, while those who don't understand imitate or follow rules;

- (4) using processes in a different context: students who have understood a particular idea often start seeing the same patterns elsewhere;
- (5) using shortcuts: only students who are sure of the 'big picture' can short-cut a procedure so that thinking up or using a short-cut is taken as evidence of understanding;
- (6) ability to explain: students who have understood something are usually able to explain it;
- (7) ability to focus attention: persistence on a task is taken as a sign of understanding.

It may be that some teachers are content with 'fits' rather than 'matches' because they are unaware of the possibilities for students' conceptions that are different from their own. However, it seems likely that most teachers are aware of the benefits of richer questioning styles, but find that such approaches are difficult to implement in 'real classrooms' (Dassa, 1990). In this respect, computer software that enables teachers to provide formative and diagnostic feedback may have a role to play (Dassa et al., 1993; Wiliam, 1997), although there is little evidence so far about the actual benefits of such software.

In turn, the student's responses to questioning will depend on a host of factors. Whether the student believes ability to be incremental or fixed will have a strong influence on how the student sees a question-as an opportunity to learn or as a threat to self-esteem (Dweck, 1986). Even where the student has a 'learning' as opposed to 'performance' orientation, the student's belief about what counts as 'academic work' (Doyle, 1988) will have a profound impact on the 'mindfulness' with which that student responds. The study of two middle-school teachers by Lorsch et al. (1992) cited in the earlier section on Current practice found that a major threat to the validity of test-result interpretation was the extent to which students could construct meanings for the tasks they were set, and the extent to which teachers could construct meanings for the students' responses. They also found that teachers used assessment results as if they gave information on what students knew, whereas, in fact, they were better indicators of motivation and task completion.

More specifically, the actual context of the assessment can also influence what students believe is required. An example is a study of a grade 5 geometry class (Hall et al., 1995) where performance was assessed in two ways-via a multiple choice test and with an assignment in

which students had to design a HyperCard geometry tutorial. In the multiple choice test, the students focused on the grades awarded, while in the tutorial task, students engaged in much more presentation and qualitative discussion of their work. Perhaps most significantly, discussion amongst students of the different tutorials focused much more directly on the subject matter (i.e. geometry) than did the (intense) comparison of grades on the multiple choice test.

The actions of teachers and students are also 'enframed' (Mitchell, 1991) by the structures of schools and society and typically knowledge is closely tied to the situation in which it is learnt (Boaler, 1997). Spaces in schools are designated for specified activities, and given the importance attached to 'orderliness' in most classrooms, teachers' actions are as often concerned with establishing routines, order and student satisfaction as they are with developing the student's capabilities (Torrance & Pryor, 1995; Pryor & Torrance, 1996). A review by Rismark (1996) shows that students are frequently marginalised and their work undervalued if they use frames of reference from their personal experiences outside school and Filer (1993) found that children learning handwriting and spelling in English primary school classrooms were constrained by the teacher to develop these skills in standard contexts, so that their own personal experiences were 'blocked out'. In this way, formal, purportedly 'objective' assessments made by teachers may be little more than the result of successive sedimentation of previous 'informal' assessments-in extreme cases the self-fulfilling prophecy of teachers' labelling of students (Filer, 1995).

2.7.4 Research--prospects and needs

The above discussion has clear implications for the design of research investigations. It draws attention to the range of important features which will combine to determine the effects of any classroom regime. In the light of such a specification, it is clear that most of the studies in the literature have not attended to some of the important aspects of the situations being researched. A full list of important and relevant aspects would include the following:

the assumptions about learning underlying the curriculum and pedagogy;

the rationale underlying the composition and presentation of the learning work;

the precise nature of the various types of assessment evidence revealed by the learner's responses;

the interpretative framework used by both teachers and learners in responding to this evidence;

the learning work used in acting on the interpretations so derived;

the divisions of responsibility between learners and teachers in these processes;

the perceptions and beliefs held by the learners about themselves as learners about their own learning work, and about the aims and methods for their studies;

the perceptions and beliefs of teachers about learning, about the 'abilities' and prospects of their students, and about their roles as assessors;

the nature of the social setting in the classroom, as created by the learning and teaching members and by the constraints of the wider school system as they perceive and evaluate them;

issues relating to race, class and gender, which appear to have received little attention in research studies of formative assessment;

the extent to which the context of any study is artificial and the possible effects of this feature on the generalisability of the results.

To make adequate report of all of these, let alone control them in any classical quantitative design, would seem very difficult. This is not to imply that reliable measures of outcomes, both of learning and of attitudes to the subjects learnt, are not to be sought--although one of the problems evident in many of the studies seems to be that although they are serving learning aims that the established methods ignore or play down, they have to justify themselves in relation to tests which are adapted to the established methods only. There is clearly a need for a combination of such measures with richer qualitative studies of processes and interactions within the classroom. If, as we believe, there is a need to evolve new approaches as quickly as possible, such studies might well focus on the problems of change and attendant disorientations.

Particular attention ought to be paid to two specific problems. The first is the evidence in many studies that new emphasis on formative assessment is of particular benefit to the disadvantaged

and low-attaining learners--evidence which is not supported in the results of other studies. The apparent contradictions here probably arise because there are some important features of the classrooms that have yet to be recorded and understood. If it is true that the ranges of school achievement might be narrowed by the enhancement of the achievement of those hitherto seen as slow learners, then there are very strong social and educational reasons for giving high priority to sensitive research and development work to see how to understand and tackle the issues involved.

2.7.5 Are there Implications for Policy?

Table I could be read alongside the section on Strategies and tactics for teachers as helping to determine the essential elements of any strategy to improve learning through thorough implementation of formative assessment. These elements would be the setting of clear goals, the choice, framing and articulation of appropriate learning tasks, the deployment of these with appropriate pedagogy to evoke feedback, noting the arguments in the section on Students and formative assessment, and the appropriate interpretation and use of that feedback to guide the learning trajectory of students. Within and running through any such plan should be a commitment to involving students in the processes of self- and peer-assessment as emphasised in the above section, underpinned by a constructivist approach to learning.

There are clearly many different ways in which such guidelines could be incorporated into classroom practice, and whilst the various experiments and schemes described throughout this review, and the particular strategies explored in the section on Systems, give useful examples, there is clearly no single royal road. In particular, a careful reading of the earlier sections on Reception and response, on Goal orientation and Self-perception on students' response to feedback, and of parts of sections on The quality of feedback, Feedback and Expectations and the social setting, should show that in framing the feedback that they give to students teachers have to keep in mind several important and delicate considerations which are neither widely known nor understood.

For public policy towards schools, the case to be made here is firstly that significant learning gains lie within our grasp. The research reported here shows conclusively that formative

assessment does improve learning. The gains in achievement appear to be quite considerable, and as noted earlier, amongst the largest ever reported for educational interventions. As an illustration of just how big these gains are, an effect size of 0.7, if it could be achieved on a nationwide scale, would be equivalent to raising the mathematics attainment score of an 'average' country like England, New Zealand or the United States into the 'top five' after the Pacific rim countries of Singapore, Korea, Japan and Hong Kong (Beaton et al., 1996).

If this first point is accepted, then the second move is for teachers in schools to be provoked and supported in trying to establish new practices in formative assessment, there being extensive evidence to show that the present levels of practice in this aspect of teaching are low (Black, 1993b; McCallum et al., 1993), and that the level of resources devoted to its support, at least in the UK since 1988, has been almost negligible (Daugherty, 1995).

There is no doubt that, whilst building coherent theories, adequate descriptions, and firmly grounded guides to practice, for formative assessment is a formidable undertaking, there is enough evidence in place for giving helpful guidance to practical action (for an account of a major state-wide assessment system which incorporates formative and summative functions of assessment, see, for example, Rowe & Hill, 1996). Furthermore, despite the existence of some marginal and even negative results, the range of conditions and contexts under which studies have shown that gains can be achieved must indicate that the principles that underlie achievement of substantial improvements in learning are robust. Significant gains can be achieved by many different routes, and initiatives here are not likely to fail through neglect of delicate and subtle features.

This last point is very important because there does not emerge, from this present review, any one optimum model on which such a policy might be based. What does emerge is a set of guiding principles, with the general caveat that the changes in classroom practice that are needed are central rather than marginal, and have to be incorporated by each teacher into his or her practice in his or her own way (Broadfoot et al., 1996). That is to say, reform in this dimension will inevitably take a long time, and need continuing support from both practitioners and researchers.

2.8 Summary

The Graded Assessment schemes in England were comprehensive provisions designed to replace terminal examinations for public certificates by a series of graded assessments, conducted in schools but moderated (i.e. checked for consistency of standards between schools) by an examining authority. In that they replaced the terminal examinations by frequent tests within each school, and enhanced the importance of components of assessed coursework as contributors to the summative results, they influenced the ways in which assessment was operated within schools and provided a distinctive scenario for the working out of formative-summative tensions. Whilst general accounts of these schemes have been published (Pennycuick & Murphy, 1986; Lock & Ferriman, 1988; Swain, 1988, 1989; Ferriman & Lock, 1989; Iredale, 1990; Lock & Wheatley, 1990) there does not appear to be any published research which could identify the particular developments of the formative functions within these schemes. A similar scheme in sciences, in that the summative function was linked to frequent assessment over extended periods within classroom work, has been described by Ratcliffe (1992). In all of these accounts, one of the problems that stands out is the difficulties that teachers and developers met in trying to establish a criterion-referenced approach to assessment. In several such schemes, an important feature has been the provision of a central bank of assessment questions from which teachers can draw according to their particular needs--but these have generally been designed with summative needs in mind. In Canada, Dassa et al. (1993) describe the setting up of a bank of diagnostic items organised in a three-dimensional scheme: diagnostic context, notional content and cognitive ability, the items being derived from a study of common errors so that they could provide a basis for causal diagnosis. 'The overall purpose was to help teachers to provide formative personal feedback within the constraints of normal classrooms. Trials in five classrooms showed that by comparison with a control set of five more classrooms, those using the item bank had superior gains, the mean effect size being 0.7.

The problems of developing criterion-referenced assessment beset the far more radical reforms in Queensland (Withers, 1987; Butler, 1995). This Australian state abolished external examinations for secondary schools in 1971, but subsequently encountered problems in the quality and the norm-referencing of school-based assessments. Butler's article chronicles the development of a criterion-referenced approach, with teachers having to learn the skills and the state having to

develop systems to ensure comparability of interpretation of the criterion standards. Greater emphasis on assessment spread over two years within classroom work, on feedback to students on successive assessment results, and on the production of student portfolios as evidence for the moderation procedures, have been required in these developments. However, the impact of these on the formative role of assessment has yet to be researched.

The systems described in this category should indeed have implications for formative assessment and could approach the problem of the formative-summative relationship from a different direction from most other studies, where the high-stakes pressures are either ignored, or accepted in that achievements on the existing measures are used (problematically) as the criteria of success. However, there is little evidence that the formative-summative relationship has been thought out in their design, and little substantial evidence about how it has worked out in practice (but see Rowe & Hill, 1996 and section titled Are there implications for policy? below).

2.9 References

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Unit 3

Strategies for reading, Writing and Math's

Content

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3.1 Introduction

A key principle of the *K–10 Curriculum Framework*, which guides K–10 syllabus implementation, is that the curriculum is inclusive of the learning needs of all students in New South Wales government schools (NSW Board of Studies, 2002). The rationale, aim, objectives, outcomes and content of each syllabus have been designed to accommodate teaching approaches that support the learning needs of all students. This includes students experiencing difficulties in learning and those with a disability enrolled in regular classes, support classes and special schools. The Stage Statements and the Continuum of Learning in each K–10 Languages syllabus can help teachers identify the starting points for instruction for the students in their class.

Most students experiencing difficulties in learning or who have a disability will participate fully in learning experiences and assessment activities provided by the outcomes and content within the syllabuses. Some students will require additional support, including adjustments to teaching and learning activities and/or assessment. For a small percentage of students (approximately 2%), the Life Skills outcomes and content within each 7–10 syllabus can provide an accessible and meaningful curriculum option. If a student is unable to access any regular outcomes or content after adjustments have been made, only then will the student access the Life Skills outcomes of a particular syllabus?

By denying some students access to regular outcomes and content, we may be providing reduced opportunities for these students to learn, reducing their chances in life (Ainscow and Muncey, 1990). Refer to Appendix A for more information about accessing Life Skills outcomes.

Who are students with special needs?

These may include:

- students with sensory or physical disabilities
- students with intellectual disabilities
- students with emotional and/or behavioural disorders
- students with communication and/or language disorders
- students experiencing difficulties in learning.

Students classified as having a learning difficulty are a heterogeneous group and have a wide variety of characteristics, ranging from academic difficulties to cognitive and social-emotional problems (Kraayenord and Elkins, 1990). The term ‘learning difficulties’ is a general one used widely and without much precision. Usually the term applies to approximately 10 to 16% of the school population and refers to those students who have general problems in learning (*Mapping the Territory*, DETYA, 2000).

Very seldom is there a single cause found for a particular student's problems with learning and sometimes it is impossible to determine any predisposing factor that has led to a student experiencing difficulties.

3.2 What are some common characteristics of students who experience difficulties learning?

Some general learning characteristics of these students are that they may:

- experience difficulties with reading, writing, spelling and mathematics – sometimes in all areas and sometimes in just one or two
- have difficulty following instructions or directions to complete a task
- have poor fine motor skills which can affect legibility of handwriting
- pronounce or read multi-syllabic words with difficulty
- have reduced or limited vocabulary and word knowledge in comparison to their peers
- have poor retrieval of information perhaps due to problems with memory
- have difficulty copying from the board or overhead projector
- have poor literacy in their first language. Students may also demonstrate the following behaviors.

They may:

- be introverted or exhibit attention seeking behavior
- be easily distracted
- have poor organizational skills
- show reluctance to learn or get involved in classroom activities
- be dependent on the teacher or a friend
- work slowly or work too quickly and inaccurately
- have poor or inappropriate social skills
- be frequently absent from class and/or school
- demonstrate a lack of motivation
- have poor self-esteem.

Teachers should be aware that students who display some or all of these characteristics and behaviors may not necessarily be experiencing difficulties in learning. Further investigation by the teacher may be necessary.

3.3 The importance of effective instruction

Students experiencing difficulties learning benefit most from explicit teaching of new concepts and skills. Effective teaching practices are those which provide all students with maximum opportunities to learn. Effective lessons contain the following elements:

- o Daily review of previous work and how the lesson will build on previous lessons
- o Clear presentation of new skills and concepts with much modelling by the teacher
- o guided student practice, with high success rates and with specific feedback to individual students
- o Independent student practice, applying new knowledge and skills appropriately
- o Systematic cumulative revision of work previously covered.
- o The importance of effective teaching
- o Effective teachers of Languages:
 - o demonstrate a clear understanding of the structure and substance of the core content being taught
 - o are specific about lesson outcomes
 - o Successfully break lessons into manageable and logical sequences
 - o Judge accurately the time needed to accomplish these outcomes
 - o use instruction and vocabulary appropriate for different abilities
 - o adjust the level of questions for different abilities
 - o provide meaningful, purposeful opportunities for the students to communicate in the target language, both in and out of the classroom
 - o devote time to the four macro skills appropriate to the course objectives
 - o use a variety of groupings, including pair work and larger group work
 - o use mainly student-centered activities
 - o use explicit error correction in activities focusing on accuracy and uses implicit or no error correction in activities focusing on communication
 - o use a variety of activities and approaches to address student learning styles and needs
 - o connect language learning to real situations and other KLAs
 - o encourage students to assess their own progress
 - o provide many opportunities for student success
 - o establish a climate in which students feel comfortable taking risks
 - o use the textbook as a tool, not as the curriculum
 - o use a variety of print and non-print tools, including authentic material
 - o facilitate teaching and learning through technology.

3.4 Planning teaching and learning activities

There are four areas to consider when planning teaching and learning activities which are inclusive of all students in the class.

3.4.1 Content

Teachers need to consider the core content to be covered from the syllabus and to sequence a particular unit of work or series of lessons to ensure all students will learn and be successful. For example, if the goal of a lesson is to have students draw up their own timetable in the target language, the prerequisite knowledge students would need to bring to the task would include:

- o Days of the week
- o Time
- o Ordinal numbers
- o School subjects
- o Other relevant vocabulary, for example recess, lunch, sport
- o understanding a timetable in English
- o Socio-cultural knowledge, for example that the school day in Germany may finish at 1:00pm. Teachers need to work with students to brainstorm this prerequisite knowledge before completing the task. When choosing teaching and learning activities teachers need to:
 - o ensure the activities chosen link to prior student knowledge by designing tasks which are similar to those already mastered by the student, but which take them one step further
 - o ensure there is no prerequisite knowledge required in the target language that the student does not possess
 - o ensure that there are no prerequisite skills required that the student does not possess, for example computation skills in a lesson on numbers
 - o lead all students to the same outcome, but provide students experiencing difficulties with smaller steps to reach that outcome, maximizing their chance of success
 - o relate learning as much as possible back to the student's own world or to a real situation.

For assessment tasks, the teacher should plan for students experiencing learning difficulties to access and become familiar with the appropriate support that will allow them to complete assessment tasks. This may include:

- o The provision of a scribe and/or reader
- o Additional time
- o Rest breaks
- o Additional assessment opportunities, for example performing a written task orally.

3.4.2. Instruction

Instruction should be explicit and systematic. Explicit instruction refers to the active and deliberate development of all aspects of students' learning rather than leaving anything to chance. Systematic instruction refers to the selection and sequencing of activities in order to ensure that students develop comprehensive knowledge, skills and understanding which are relevant to the identified needs of the student and to the task at hand (*Focus on Literacy: Writing*, 2000).

The way teachers instruct students varies from teacher to teacher. Many students experiencing difficulties learning have problems recognising and adapting to different teaching styles in a school setting. Teachers need to be aware of their instructional style and consider the needs of particular students. To be explicit and systematic teachers need to:

- o define the purpose of every task and explain to students what is required in fulfilling the purpose of a lesson
- o Clearly state the criteria which will be used to assess student achievement
- o provide a range of learning and responding modes to cater for the different learning needs of the students, for example, supporting reading and writing skills with oral and aural skills
- o think about how they interact with the students, for example time spent by only the teacher talking, involving students in discussions, varying the levels and variety of questions and ensuring all students contribute ideas and feel their contributions are valued
- o consider their questioning techniques, including rephrasing questions, simplifying language or providing alternative formats for information to be provided, for example more use of visual cues
- o be aware which students require additional instruction and/or support
- o be aware that, for some students, the teaching of organizational strategies (e.g. diaries, notebooks) for assignments may be necessary to help keep students on task and on schedule
- o ensure modeling occurs frequently, followed by guided or peer practice and then independent practice to provide an effective scaffold for success
- o provide additional support, for example through group work, peer tutoring, direct assistance and frequent feedback
- o rephrase or adjust questions using fewer words or less complex language when necessary
- o wait a few seconds longer before expecting an answer
- o be descriptive when praising students.

3.4.3 The classroom environment

Teachers may need to consider the classroom environment and perhaps:

- o have rules clearly displayed and clarified regularly, ensuring all students know the procedures and routines of the class
- o consider arranging the room in a way which maximizes the student's participation and/or comprehension, for example small group work, peer work, quiet individual work area
- o use grouping and cooperative learning as a strategy
- o think of other ways for information to be relayed to students if copying from the board or overhead projector is difficult
- o allow additional time for practice and/or task completion for those students who need it
- o be discreet when assisting students and providing them with adjusted material and/or activities
- o Reward students, via incentive charts or card systems.

3.4.4 Materials and tasks

When selecting or developing resources and materials used in a lesson the teacher should:

- be aware that some students may require more prompts or scaffolds to complete an activity
- design worksheets which use the same font throughout and have a predictable layout
- consider using multiple choice answers in place of sentence completion
- include mnemonic devices or graphic organizers that assist students to remember new material, for example visual webs, organizational charts, first letter acronyms and visual prompts and icons
- provide visual cues (both gesture and illustrations), being as consistent as possible. For example if one gesture represents "open your book", always use that same gesture, if one illustration represents "happy", always use that illustration
- provide concrete learning aids which facilitate learning by using a 'hands on' approach
- remove unnecessary detail from textbooks or worksheets
- highlight critical features that are the most important for students to learn
- enable students to access alternate formats, such as large print, computers, and simplified texts.

3.5 Effective teaching strategies and adjustments for students experiencing difficulties with learning: the four macro skills promoting success in listening and responding

When listening to a language tape the teacher can:

- set the scene – explain what the recording is about, for example title, purpose, monologue/dialogue, who is talking, key language items to listen for
- Model how to complete the listening task by playing the tape to the whole class while the teacher or a volunteer fills in the beginning of an OHT of the exercise
- provide more guided support by:
 - having the teacher repeats the key points
 - breaking the listening text down (by repeating the tape or reading the transcript) into smaller chunks
 - providing students with a transcript of what is on the tape, combining their listening and reading skills
- allow students to play the tape at their own pace, for example by using a Discman with headphones
- organize students to work in smaller groups, working out answers together or replaying tape as necessary – students could even provide oral answers to the questions. Vary the groupings when adopting this strategy
- ask the students to listen for specific information on the tape – cued listening and reporting, one item at a time. For example, in a text where teenagers are talking about which subjects they like and dislike, ask students in the first listening to identify the subjects, in the second listening to identify which subjects the teenagers like and in the third listening to identify which subjects the teenagers dislike
- have students put a series of statements from the activity into the correct order by numbering or reordering them in accordance to the order they appear on the tape
- provide a range of ways in which students can respond, for example true/false, multiple choice, tick-a-box activities, short answers
- allow multiple exposures to the tape by writing a new activity or adapt an old one while still using the same tape
- teach the vocabulary the students will encounter explicitly and classify by function or meaning.

3.6 Promoting success in speaking skills

When engaging students in speaking skills teachers can:

- use games such as Snap, Battleships, Go Fish that may provide a common context for understanding and at times may be more effective than just pair work

- use whole class repetition games (for example saying words as a chain, word substitution), varying the vocabulary or by varying volume, tone and mood of the sentence
- provide questions in pair work activities that are simple and clearly scaffold. For example, when completing an information gap activity on timetables, use a visual scaffold to prompt students (see Appendix B)
- accept different levels of responses, from one word to complete sentences
- provide opportunities for students to read role-plays together and then act out for the class
- make oral assessments accessible and relevant – for example is there any point asking a student experiencing difficulties to discuss likes and dislikes of school subjects if they cannot name the school subjects themselves?
- provide students with the opportunity to do speaking skills on to a tape, to allow them to listen to and refine what they say.

3.7 Promoting success in reading and responding skills

Students experiencing difficulties with learning can be overwhelmed by the amount of text on the average page of a textbook. To support students to read text teachers can:

- model how to identify what is relevant on a textbook page, e.g. by showing the text on an OHT, by asking students to tag the relevant passage in their books with a *Post-it* note, or by having students cover irrelevant information on a page
- identify the key concepts for the students before reading to ensure they know what they are reading for
- use illustrations as much as possible to support understanding
- read the passage aloud to the students or provide the passage on tape
- have students work in smaller groups and work out answers cooperatively, either in written or oral form
- have students match pictures with text to show understanding or allow students to draw their own pictures to illustrate what they have understood
- provide a range of ways in which students can respond, for example true/false, multiple choice, tick-a-box activities, short answers
- teach the vocabulary the students will encounter explicitly and classify by function or meaning.

3.8 Promoting success in writingskills

Writing is often the skill that students may need the greatest support in. To support students with writing teachers can:

- ensure writing tasks are clearly scaffolded (see Appendix C) and explicitly modelled by the teacher
- encourage the use of ICT that may assist some students to write or take notes especially if fine motor problems are an issue
- ensure there is a range of opportunities for the student to succeed with writing tasks, including cutting and pasting, copying words/sentences, comic strips, sequencing, choosing from a list of words, filling in missing letters or words, rearranging items according to criteria not disability!

3.9 Historical Perspective

Morgan, a general practitioner in Sussex, England, published the first case of what is now known as dyslexia, a word derived from the Latin word “dys”, which translates to ‘difficult’, and the Greek word “lexia”, which translates to ‘words’; it literally means, “*difficulty with words*”. On 7th November, 1896. Morgan wrote in the *British Medical Journal*, about Percy F., a 14-year old, who was intelligent, bright, quick with learning games, and the intellectual equal of his peers, but fell behind, in his inability to learn how to read.

Today, as in 1896, most people associate intelligence with the ability to read, but Percy F. and the experience of millions of people with dyslexia breaks down the relationship between reading and intelligence. Researchers were left with the question, “What causes dyslexia if intelligence is not the marker?” (Snowling MJ, 1996). Morgan and Hinshelwood, an ophthalmologist also writing at the turn of the Century, speculated that such difficulties with reading and writing were due to “congenital word blindness”, and for many years, the dominant view was that dyslexia was caused by *visual processing deficiencies*. There is still interest in the role of visual factors in the etiology of dyslexia, especially in low level impairments of the visual system. However, the most widely accepted view today is that dyslexia is a verbal deficit and can be considered part of the continuum of language disorders. Indeed, converging evidence supports a specific theory, that dyslexic readers have phonological (speech) processing deficits (Snowling MJ, 1996).

The identification and description of Learning Disabilities as being deficient general learning processes centering mostly on what we today call distractibility, hyperactivity and visualperceptual and perceptual-motor problems began in the Western world in the 1950s and 1960s (The Nalanda Institute, 2002). The major developments of the LD movement during this period centered on children who appeared normal in many intellectual skills, but who also displayed a variety of cognitive limitations that seemed to interfere with their ability to read, write and learn in the classroom. LD was seen primarily as a processing disorder with difficulty

in cross-modal integration (Karanth, 2002). Dyslexia at this stage was a term coined to describe right brained thinkers who have difficulty in reading, think in pictures and are very imaginative and multidimensional (Eklavya School). Famous personalities, Walt Disney and Albert Einstein were cited as examples. It was a unanimous thought even at this time that these children needed to be accommodated in the mainstream class and rather than expecting them to mould themselves to the system, *the system needed to become flexible to adapt to their needs*. Gardner's theory (1983) of *Multiple Intelligences* talked of different ways to teach these children. In addition, detailed assessments in various processing areas such as auditory or visual sequencing, auditory/ visual memory and discrimination (which are still included in most test batteries for LD) resulted in specific remedial measures to deal with a deficient processing pathway.

The 1980s, however, witnessed a renewed emphasis on the association of language disturbances with Learning Disabilities (Karanth, 2002). Today it is accepted that LD is a *language based disorder*. In the years following the report on the first case of dyslexia, different types of specific learning disabilities were defined: dyslexia (difficulty in reading), dysgraphia (difficulty in writing), dyscalculia (difficulty in numbers and mathematical concepts) and dysnomia (difficulty in naming). Simultaneously dysphasia (expressive language difficulty) was also being noted together with receptive language difficulties (Karanth, 2002).

Today all these are included under the umbrella of Specific Learning Disability (SLD). Hence using the word dyslexia interchangeably with LD is technically incorrect. It is important to remember that a Specific Learning Disability, as the name suggests, includes difficulties in specific processing areas as opposed to global difficulties in children with compromised intelligence (Karanth, 2002).

- **Federal Definition of LD**

The following *Federal definition* by the U.S. Government in Public Law 94-142 of Learning Disabilities *has been adopted in India*. "Specific Learning Disabilities means a disorder in one or more of the basic psychological processes involved in understanding or in using language, spoken or written, which may manifest itself in an imperfect ability to listen, speak, read, spell or to do mathematical calculations. The term includes such conditions as perceptual handicaps, brain injury, minimal brain dysfunction, dyslexia and developmental aphasia. The term does not include children who have learning problems which are primarily the result of visual, hearing or motor handicaps, or mental retardation, emotional disturbance or environmental, cultural or economic disadvantages." (*Federal Register*, 1977, p. 65083) (Karanth, 2002).

The LD movement in India is of more recent origin and comparable today with that of the western LD movement of nearly half a century ago. In the eastern world, LD was earlier considered a problem of English speaking countries. The apparent lower incidence of these types of difficulties resulted in a relative lack of concern about LD in Asian countries such as India and China. Reports of lower incidences of LD in the eastern world were attributed by Western

scholars to the general lack of awareness and sensitivity among educationists. The specific difficulties faced by children learning to read were attributed to the overcrowded classrooms. At the same time, reports of the high incidence of problems associated with the acquisition of reading in Western countries was attributed by easterners to the vagaries and complex nature of alphabetic writing systems such as English (Karanth, 2002). During the last decade or two, however, there has been an increasing awareness and identification of children with LD in India. Despite this growing interest, we still have no clear idea about the incidence and prevalence of LD in India. Epidemiological studies of LD are fraught with difficulties ranging from the very definition of LD, identification and assessment, to socio cultural factors unique to India. The Federal definition implies key factors: adequate intelligence, appropriate instruction and socio cultural factors. The implications of these terms for identification of children with LD in a pluralistic society such as ours are immense and cannot be easily handled (Karanth, 2002). The inherent complexities of the notion of LD or proficiency in reading and writing (Karanth, 2002). The examples given demonstrate the difficulty these children have with processing and using language, are it spoken or written.

- **Difficulties with Language of Print**

Reading Difficulties: Unable to read, poor tracking, reading errors. *Spellings of a 9 year old boy in Std IV; not only are the formation of letters reverses, he is unable to differentiate the sounds in words and associate them with correct letters. He is a very bright child and can give all answers orally. The fox jumped over the fence. The eagle flew over the high mountain top. 11 year old Hindi speaking girl in Std V has difficulty with sounds, predominantly vowels; good at orals but problems with reading and writing. Sample of an 8 year old boy in Std II showing difficulty in writing due to fine motor difficulties commonly referred to as Dysgraphia Sample of a 11 year old in Std VI showing difficulty with vowels*

Net

Can

Fun

Top

Rag

Eat

Hit

Lid

Cap

Had

Whilst narrating a story

- **Difficulties with Oral Language**

Expressive Language

Children with expressive language difficulties exhibit slow vocabulary growth, pronunciation difficulties, difficulty in *expressing* (single words, poor/wrong retrieval of words, poor answering and narrative and conversational skills) and Grammatical difficulties.

*Difficulty with **word retrieval***

4.5 year old boy

Teacher: "What do you want Karim?"

K: (pointing to car)"that"

T: What is "that"

K: (still pointing to car) That, want that

- **Receptive Language Difficulties**

Difficulties with processing sounds affects understanding which in turn Difficulty with sequencing, linking thoughts, difficulty with concepts. Difficulty with usage 9 year old English speaking boy asked to write 10 - 15 lines for an essay.

Tells mother

"Why 10 - 15 lines, it should be 1-15 lines"

"We start from 1st line to 15, not from 10 to 15"

- **Diagnostic and Statistical Manual– IV Definition**

The definition of Learning Disability as per the *Diagnostic and Statistical Manual – IV (DSM-IV)* (American Psychiatric Association, 1994): Learning Disorders are diagnosed when the individual's achievement on individually administered, standardized tests in reading, mathematics or written expression is substantially below that expected for age, schooling and level of intelligence. The learning problems significantly interfere with academic achievement or activities of daily living. Learning disorders are of four types: Disorders of Reading, Disorders of Mathematics, Disorders of written expression and Learning disorder not otherwise specified (NOS) (American Psychiatric Association, 1994).

In fact, many institutions use *a discrepancy of 2 years between ability and performance as a marker of LD*, provided that other factors, emotional disturbance or environmental, cultural or economic disadvantages have been excluded. Various terms are used to describe specific

learning disabilities. A person may exhibit one or more of them. Some of them are as follows (codes provided are ICD-10 and DSM-IV, respectively). (F80.0- F80.2/315.31)
Dysphasia/Aphasia - Speech and language disorders

- Difficulty producing speech sounds (articulation disorder)
- Difficulty putting ideas into spoken form (expressive disorder)
- Difficulty perceiving or understanding what other people say (receptive disorder). (F81.0/315.02)

Dyslexia

The general term for reading disability which involves difficulty in phonetic mapping, where sufferers have difficulty with matching various orthographic representations to specific sounds. Some claim that dyslexia involves a difficulty with sequential ordering such that a person can see a combination of letters but not perceive them in the correct order. (F81.1/315.2)

Dysgraphia

The general term for a disability in physical writing, usually linked to problems with visualmotor integration or fine motor skills.

Dyscalculia

The general term for disability in mathematics. (American Psychiatric Association, 1994).

• The ICD-10 1999 Definition of LD

The ICD DSR (International Statistical Classification of Diseases and Health Related Problems) is a detailed description of known diseases and injuries. Published by the World Health Organization, it is used world-wide for morbidity and mortality statistics, reimbursement systems and automated decision support in medicine (GEON, 2005). Learning disorders (LD) refer to a significant deficit in learning due to a person's inability to interpret what is seen and heard, or to link information from different parts of the brain (GEON, 2005). Learning disorders can be classified into three major types: disorder of written expression (DWE); reading disorder (RD); and mathematics disorder (MD).

The description of learning disorders corresponds to the educational legal designation of learning disabilities. Learning disabilities are legally defined in a law called the Individuals with Disabilities Education Act, or IDEA. The rules and related laws of IDEA stipulate that children with LD are entitled to free education and special services. A fourth category of LD has also been established for an LD that does not completely fit into any category (called an LD not otherwise specified) (GEON, 2005). The difficulty with LD is that:

(a) The DSM and ICD definitions of LD are medical and used as a marker for eligibility for special education facilities; however LD primarily requires educational interventions.

(b) The federal definition adopted in India is based more on processing difficulties, however without a national policy for LD and standardized assessment procedures, this adoption is almost redundant.

3.10 Incidence and Magnitude of the Problem

With controversies raging regarding the diagnostic criteria for LD, the estimates on the prevalence of LD in the U.S. show marked variation. However, it is a fact that the LD category now accounts for 52% of all students with disabilities served in special education under the Individuals with Disabilities Education Act (IDEA). Between 1976–77 and 1996–97, the number of students serving as LD increased from 797,213 to 2,259,000, a 283% increase (Macmillan, et al., 1996).

The spectrum of difficulties and their severity makes diagnosis of SLD extremely difficult especially when they are confounded by environmental, cultural and economic disadvantages. In India these factors namely, the poor exposure of many of these children to education, knowledge and language makes diagnosis even more problematic.

Developed countries today face some of the same difficulties due to large number of children from immigrant populations; the diagnosis of SLD is certainly not easy. An important landmark in the history of LD was the Washington Summit on Learning Disabilities in 1994. At the summit, the prevalence of LD was quoted as 15% of the population. Among the array of learning disabilities, deficits in basic reading skills were the most prevalent (Reid L, et al., 1994).

Some of the areas of concern were:

1. The most recent scientific discoveries about learning disabilities have exciting implications for helping children, but they have not been quickly translated into appropriate interventions for students with learning disabilities, particularly in the education system.
2. The field of learning disabilities is somewhat fragmented across a number of academic and professional disciplines. Each of these groups has focused on different aspects of LD, which has led to disagreements and differences in priorities; there are few formal communication channels.
3. Despite substantial gains that have been made via federal legislation for those with learning disabilities since the passage of Public Law 94- 142, known now as the Individuals with Disabilities Education Act (IDEA), and the Americans with Disabilities Act, the uneven and uninformed implementation of the law has led to many tragic failures.
4. *Statistics:* The numbers and the tragedies:

- o 50% of all students in special education in the public schools have learning difficulties(US Dept of Education, 1992).
- o 75-80% of special education students identified as LD have their basic deficits in language and reading (National Institute of Health).
- o 35% of students identified with learning disabilities drop out of high school. This is twice the rate of their nondisabled peers. (This does not include the students who are not identified and drop out). (National Longitudinal Transition Study Wagner, 1991).
- o 60% of adults with severe literacy problems have undetected or untreated learning disabilities. (National Adult Literacy and Learning Disabilities Centre, 1994).
- o 50% of the juvenile delinquents tested were found to have learning disabilities. (National Centre for State Courts and Educational Testing Service, 1977).
- o Up to 60% of adolescents in treatment for substance abuse have learning disabilities. (Hazelton Foundation, Minnesota, 1992).
- o 62% of learning disabled students were unemployed one year after graduation. (National Longitudinal Transition Study Wagner, 1991).
- o Learning Disabilities and substance abuse are the most common impediments to keeping welfare clients from becoming and remaining unemployed according to the 1992 report of the Inspector General (Reid L, et al., 1994). As a follow up to the Learning Disabilities Summit: Building a Foundation for the Future, the Office of Special Education Programs (OSEP) in the U.S. Department of Education brought together a group of researchers for a meeting on November 29-30, 2001.

They concluded as follows: *It is difficult to know the true prevalence rate of specific learning disabilities.* However, based on reading research, conducted largely in the elementary grades, we know that high quality classroom instruction is a way to meet many of the educational needs of individuals with learning difficulties. Also known as supplemental intensive small group instruction can reduce the prevalence of learning difficulties. *Even with these interventions, approximately 6 percent of students may exhibit specific learning disabilities* and will need special education interventions. Prevalence rates of students with specific learning disabilities involving math and written expression are difficult to estimate given the current lack of research evidence (Cook L, et al., 2001).

The Situation in India

We need to learn from these experiences. At present, in India, LD is considered the prerogative of a few in the big cities. Even Directors of State Education are known to express doubts at the existence of any such disability. Unfortunately, the confounding factors of English as a foreign language and lack of proper education and exposure whilst aggravating the academic difficulties for the children, also play a major part in masking the processing problems and hence make LD an elusive entity. Teachers attribute the learning difficulties to a “language problem”, not realizing that LD too is a language based disorder.

Most of the (research and intervention) work in the area of LD is being done by private organizations and the NGOs. There is little communication between these organizations and the state educational authorities. Adding further to the problems, there is a divide between the personnel in the health and the educational fields, be they private or government. LD as all other developmental problems is both a health and an educational issue, but regrettably, the meeting point between the two is few and far between. The multilingual social context in India, where children often have to learn to study through a medium other than their mother tongue is a complexity that makes not only diagnosis extremely difficult but also, estimation of prevalence next to impossible.

The language issue is further compounded by factors such as age of enrolment in school, preschool exposure and literacy support available in their respective homes during the school years. Consequently, relating “adequate instruction” and “social opportunity” as is required by definition of SLD to children from varied backgrounds (from an urban child enrolled in pre-school at age 2½ years with early and sustained support to a rural child attending school for the first time at age 6½ years with no additional literacy support of any kind is a tremendous challenge (Karanth, 2002). If this is true of identification and assessment, the challenges faced with respect to remediation and management are no less daunting. Our educational system with its overwhelming emphasis on knowing rather than learning, theory rather than application, is ill-suited for the child with LD. The overwhelming influence of Western thought with lack of indigenous research has led to a situation where even ones strengths are turned into liabilities, an example being the ‘phonemecisation’ of the Indian scripts under the influence of the phonic method of the West. The near total lack of alternate systems of education and the social premium for a handful of vocational courses with an utter disregard for all other vocational training are other major hurdles in the ‘education’ of the child with LD. These are but some of the issues faced by the individual and the family of the learning disabled, to date in India (Karanth, 2002).

An epidemiological study (1995-2000) of child and adolescent psychiatric disorders in urban and rural areas of Bangalore, was done by the Dept of Psychiatry, Epidemiology and Biostatistics, National Institute of Mental Health and Neuro Sciences, Bangalore to determine prevalence rates of child and adolescence psychiatric disorders for the Indian Council of Medical Research. The total prevalence rate in 4-16 year old children in urban middle class, slum and rural areas was 12%. However the children with *SLD were eventually excluded from this study* as most of them lacked adequate schooling as per the ICD-10-DCR criteria for SLD. In addition, many of the *assessments were incomplete* due to lack of cooperation for the lengthy testing for Specific Learning Disabilities (Srinath S, et al., 2005).

The prevalence study on Learning Disability conducted at the L.T.M.G. Hospital, Sion, Mumbai reveals that of the total number of 2,225 children visiting the hospital for certification of any kind of disability, 640 were diagnosed as having a Specific Learning Disability. These children came from the lower, middle and upper middle socioeconomic strata of society. Referral was due to their poor school performance (LTMG, 2006). Studies conducted by the Sree Chithira

Thirunal Institute of Medical Sciences and Technology in Kerala in 1997 revealed that nearly 10% of the childhood population has developmental language disorders of one type or the other and 8-10% of the school population has learning disability of one form or the other. The Institute for Communicative and Cognitive Neurosciences (ICCONS), Kerala, has been conducting research programs in child language disorders and developing research and rehabilitation programs for learning disabilities. Screening for LDs for Classes I to VII in schools with follow up assessments by experts in 10 panchayats in Kerala revealed that 16% of these school children have a learning disability (Suresh, 1998). Other studies have been done at childguidance clinics in India (Khurana, 1980; John & Kapur, 1986) where 20% children attending the clinic were diagnosed to be scholastically backward. However, variables such as the socio-economic class, exposure to language act as confounding variables in such clinic-based studies (GEON, 2005).

Dyslexia is not a transient lag in development (Bruck, 1992; Francis, et al., 1996; Shaywitz, 2003). It is a persistent condition that will not fade away with brain maturation. It is critical that children at risk for dyslexia begin treatment as early as possible. More significant is the successful outcome of early intervention than the one that is initiated later. (Lyytinen, et al., 2005; Torgesen, 1998; Hinton 2006). *If LD is etiologically related to heredity and abnormalities in the brain, is treatment futile?* The structure of the brain is a function of a synergistic interaction of genetics and experience. Neural circuitry is continually constructed and reconstructed in response to experience. Consequently, while genetic predispositions influence the architecture of the dyslexic brain, there is considerable potential for functionally significant structural modification (Shaywitz, 2003). Therefore, it is crucial that individuals at risk for dyslexia receive effective treatment (Hinton 2006).

This implies that though LD is not curable, there is much scope for using compensatory mechanisms to alter functional gaps which are to be initiated early to ensure that the disability is not aggravated further. The child must be able to develop and learn to the best of his/her potential. Early intervention presupposes early identification. At present, there is no universally standardized screening procedure to guide referrals from schools.

3.11 Early Identification and Assessment Procedures

The Schwab Foundation for Learning has a grade specific checklist to help identify “at risk for LD” children. This list is comprehensive and usually followed by organizations working in the field of LD (Schwab Learning, 2002). The checklist for LD in the Sarva Shiksha Abhiyan Manual (SSA, 2003) is also a helpful tool for initial screening by teachers in the schools. However, at present, the assessment itself is being used as a screening/identification procedure. The children are referred for assessment by the school/teacher for reasons of failure, underachievement or behavioral problems. For the same reasons, parents may take the child directly, and avail of examination concessions that exist in some states. In the rural areas, there is near zero awareness of LD and practically no assessment facilities.

3.12 Assessment

Before a specialized evaluation of a student is conducted, pre-referral discussions by teachers regarding the nature of the problem, and what possible modifications to instructions in the classroom might be made are important. The child must be assessed in all areas related to the suspected disability such as health, vision, hearing, social and emotional status, general intelligence, academic performance, communicative status, and motor abilities. (National Information Centre for Children and Youth with Disabilities, 2000).

An ideal assessment for LD is a long process requiring several sessions with a qualified educational psychologist. Apart from administering a battery of tests, the psychologist also gathers relevant information about the child from the teachers and school records. The assessment procedure for LD involves the following steps:

Parental Consent and Parent Interview

- o Parents' consent must be obtained before evaluating the child. The academic, developmental and medical history along with the linguistic usage and communications patterns of the child must be obtained from the parents.
- o The parent must be involved in the planning of the intervention program such as attending a resource room, provision of accommodation and modifications to the child.

• **Gathering Information from the Teachers/School**

The psychologist must also observe the child in his/her school setting to know about the child's performance and behavior in the class, and gain insights from the teacher. Review of previous grades will show the pattern of academic progress. These may throw light into the problem areas of the child. A student's current classroom performance can be compared to Test scores.

• **Looking at Student Workbooks**

Regrettably, in the present educational set up, very often the notebooks don't reflect the learning difficulties faced by the child due to rote learning especially when the child can easily copy from the blackboard. The examination papers may give a clearer picture of the specific nature of difficulty.

Only through collecting data through a variety of approaches (observations, interviews, tests, curriculum-based assessment, etc.) and from various sources such as parents, teachers, peers, adequate picture be obtained of the child's strengths and weaknesses. Synthesized, this information can be used to determine the specific nature of the child's special needs, whether the child needs special services and if so, to design an appropriate program. (National Information Centre for Children and Youth with Disabilities, 2000).

A number of approaches being used recently include *curriculum-based assessment*, *task analysis*, *dynamic assessment*, and *assessment of learning style*. These approaches yield rich

information about students and are especially important when assessing students from *culturally or linguistically diverse backgrounds*, and therefore, are critical methods in the overall approach to assessment. (National Information Centre for Children and Youth with Disabilities, 2000).

- **Interview with the Child**

“An Interview should be a conversation with a purpose” (Wallace, Larsen, & Elksnin, 1992, p. 16), with questions designed to collect information that “relates to the observed or suspected disability of the child”. (National Information Centre for Children and Youth with Disabilities, 2000). A careful review of the student’s school records or work samples help the assessment team identify patterns or areas of specific concern which may be focused on at the time of interview. The student too, may have much to say to illuminate the problem (Hoy & Gregg, 1994, p. 44). (National Information Centre for Children and Youth with Disabilities, 2000).

- **Testing**

Though increasingly controversial, most assessments for LD include standardized tests. There are two types of tests.

(i) *Criterion-referenced tests* are scored

according to a standard, or criterion decided by the teacher, the school, or the test publisher. An example of a criterion referenced test might be a teacher-made spelling test where there are 20 words to be spelled and where the teacher has defined an “acceptable level of mastery” as 16 correct (or 80%). (National Information Centre for Children and Youth with Disabilities, 2000).

(ii) *Norm-referenced tests*: Scores on these tests

are not interpreted according to an absolute standard or criterion (i.e., 8 out of 10 correct, etc.) but, on how the student’s performance compares with that of the norm group (a large number of representatives of that age group). This helps evaluators determine whether the child is performing at a typical level, below, or above that expected of a given ethnicity, socio-economic status, age, or grade. (National Information Centre for Children and Youth with Disabilities, 2000).

The drawback of this type of test is that the norms in different regions of a country will vary and too, the norms of the same region will change over a period of time. Hence in a diverse country like India, each area would have to develop its own norms which would need to be reviewed periodically. Essentially, the tests for LD have two major components:

1. Testing for Potential: Performance Discrepancy.
2. Testing Processing Abilities.

A two-year discrepancy between *potential* and *performance* is an indicator of a possible LD. Validity of a significant discrepancy will be evaluated on a case by case basis (Hirisave U, et al., 2002). The recommended Psycho-educational tests are discussed below under various heads:

1. *Intellectual Assessment*: Weschler Adult Intelligence Scale Third Edition (WAISIII), Woodcock Johnson Tests of Cognitive Ability.
2. *Achievement*: Recommended tests include: Woodcock Johnson Psycho166 Educational Battery-Revised, Nelson Denny Reading Test, SATA.
3. *Cognitive Processing Abilities*: Woodcock Johnson Psycho-Educational Battery- Revised (Part 1 - Tests of Cognitive Ability), Weschler Memory Scales- Revised, Benton Visual Retention Test, Berry Visuo-Motor Integration Test, Raven Colored Progressive Matrices, Rex Auditory-Verbal Learning Test, Bender Visual Motor Gestalt Test, Halstead-Reitan Neuropsychological Test Battery, Memory-For-Designs Test, Nimhans Index (Hirisave U, et al., 2002).

These tests would have to be modified and norms created for children who come from culturally and linguistically diverse backgrounds. *Exclusion of other disabilities* as the primary cause of learning difficulties is essential. Such disabilities include:

- o Mental retardation.
- o Sensory deficits. Example: Visual and/ or hearing impairment.
- o Physical impairment.
- o History of multiple education settings.
- o Poor educational background or lack of prior learning.
- o Cultural differences or lack of experience with the English language (Office of Disability Services).

However, a learning disability may co-exist with the above. *SLD being a language based disorder, it is imperative that tests for both receptive and expressive language be included in the assessment procedures.*

• **Co-Morbidity with ADHD**

Many children with LD develop secondary inattention and behavioral difficulties; Attention Deficit Hyperactivity Disorder (ADHD), which is characterized by developmentally-inappropriate inattention, hyperactivity and/or impulsivity, is often co-morbid with dyslexia. (Kadesjö & Gillberg, 2001). The two disorders occur simultaneously in 12% to 24% of individuals with dyslexia (Shaywitz, 2003). However, they do not appear to share a common cause (Doyle, 2001; Shaywitz, 2003). Under these circumstances, it becomes difficult to differentiate LD from a Primary ADHD. (National Information Centre for Children and Youth with Disabilities, 2000).

- **Other Assessment Procedures**

Curriculum Based Assessment

Direct assessment of academic skills (Curriculum Based Assessment) is one alternative that has recently gained popularity. “Tests” of performance in this case come directly from the curriculum. For example, a child may be asked to read from his or her reading book for one minute. Information on the accuracy and the speed of reading can then be compared with other students in the class. CBA is quick and offers specific information about how a student may differ from his peers. (National Information Centre for Children and Youth with Disabilities, 2000). Because the assessment is tied to curriculum content, it allows the teacher to match instruction to a student’s current abilities and pinpoints areas where curriculum adaptations or modifications are needed.

CBA provides information that is immediately relevant to instructional programming. (National Information Centre for Children and Youth with Disabilities, 2000). The merits of a CBA are lost in a system with a rigid curriculum based mainly on memorization as is true in India where CBA may not be the right option.

Dynamic Assessment

The goal “is to explore the nature of learning, with the objective of collecting information to bring about cognitive change and to enhance instruction” (Sewell, 1987, p. 436). (National Information Centre for Children and Youth with Disabilities, 2000). Dynamic assessment includes a dialogue or interaction between the examiner and the student. This interaction may include modeling the task for the student, giving the student prompts or cues as he/she tries to solve a given problem, asking what a student is thinking while working on the problem and giving praise or encouragement (Hoy & Gregg, 1994).

The interaction allows the examiner to draw conclusions about the student’s thinking processes and his/her response to a learning situation. The “teaching” phase is followed by a retesting of the student with a similar task but without assistance from the examiner. (National Information Centre for Children and Youth with Disabilities, 2000). Dynamic Assessment Tools (LPAD) have been developed by Prof. Reuven Feurenstein at the International Centre for Enhancement of Learning Potential (ICELP), Jerusalem. Of course, dynamic assessment is not without its limitations or critics. One particular concern is the amount of training needed by the examiner to conduct both the assessment and interpret results. Another is a lack of operational procedures or “instruments” for assessing a student’s performance or ability in the different content areas (Jitendra & Kameenui, 1993). Even with these limitations, it is a promising addition to current evaluation techniques because it incorporates a teaching component into the assessment process. (National Information Centre for Children and Youth with Disabilities, 2000). Given, the difficulties in diagnosis due to environmental deprivation developing simple dynamic assessment tools would greatly benefit children with learning difficulties.

Learning Styles

We know that all children have different learning styles. A learning style assessment, attempts to determine the elements that has an impact on a child's learning. Some of the common elements that may be included here would be the way in which the material is presented (i.e., visually, auditorily, tactilely) in the classroom, the environmental conditions of the classroom (hot, cold, noisy, light, dark) the child's personality characteristics, the expectations for success that are held by the child and others, the response the child receives (for example, praise or criticism) and the type of thinking the child generally utilizes in solving problems (for example, trial and error, analyzing). Identifying the factors that positively impact the child's learning are very valuable in developing effective intervention strategies. (National Information Centre for Children and Youth with Disabilities, 2000).

Outcome-based Assessment

Outcome-based assessment involves considering, teaching and evaluating the skills that are important in real-life situations. Assessment, from this point of view, starts by identifying what outcomes are desired for the student (for example, being able to use public transportation). The team then determines what competencies are necessary for the outcomes (for example, the steps or sub skills the student needs to have mastered to achieve the outcome desired) and identifies which sub skills the student has mastered and which he/she needs to learn. (National Information Centre for Children and Youth with Disabilities, 2000). This type of assessment though generally used for the mentally challenged or autistic, may also be used for children in the general classroom with severe behavioral difficulties.

Assessment of the Culturally and Linguistically-Diverse

Because culture and language affect learning and behavior (Franklin, 1992) the school system may misinterpret what students know, how they behave, or how they learn. Students may appear less competent than they are, leading educators to inappropriately refer them for assessment. Once referred, inappropriate methods may then be used to assess the students, finally leading to inappropriate conclusions and placement into special education. There is also a great deal of research and numerous court decisions (for example, Larry P. V. Riles, 1979; Guadalupe V. Tempe Elementary District, 1972) to support the fact that *standardized tests (particularly intelligence and achievement tests) are often culturally and linguistically biased against students from backgrounds different from the majority culture*. Most cognitive, language, and academic measures are developed using standards met by the English-speaking majority. It is, therefore, imperative that the evaluation team collect the information about the student through interviews, observations, and approaches such as dynamic assessment (Lidz, 1987). (National Information Centre for Children and Youth with Disabilities, 2000).

Assessments in India

The National Institute of Mental Health and Neurosciences (NIMHANS), Bangalore has developed the index to assess children with LD (Hirisave U, et al., 2002). There are two levels of this index. They are: Level I for children 5-7 years and Level II for 8-12 years. The index comprises of the following tests:

- a. Attention test (Number cancellation).
- b. Visuo-motor skills (the Bender Gestalt test and the Developmental test of Visuo – Motor integration).
- c. Auditory and Visual Processing (discrimination and memory).
- d. Reading, writing, spelling and comprehension.
- e. Speech and Language including Auditory behaviour (Receptive Language) and Verbal expression.
- f. Arithmetic (Addition, subtraction, multiplication, division and fraction) (Hirisave U, et al., 2002).

At the Lokamanya Tilak M.G. Hospital, Sion, Mumbai, the procedure for assessment of Specific Learning Disability involves the following:

- a. Neurological assessment.
- b. Vision and Hearing tests.
- c. Analysis of school progress report.
- d. I.Q. test.
- e. Educational assessment.
- f. Psychiatric assessment.
- g. Case conference.
- h. Counseling.

Most private institutions in India follow some, if not all of these procedures. In our country where numbers often determine procedures, it would be beneficial to provide basic facilities for assessments within the educational setting. The reasons are overwhelming:

- o Children experiencing delays or learning problems may be screened at the first level, provided with timely help and only those requiring further assessment would need to undergo further testing.

- o Ideal assessment procedures being very elaborate cannot be completed in a single session.
- o Attending clinics and hospitals would be difficult for the parents from a lower socioeconomic background.
- o Information can be easily gathered from within the school. Observation of the child in the educational setting would be preferable to those made in a clinic.
- o The assessment team could include a psychologist, special teacher/educator, class teacher which, with input from the parent and child, would facilitate a comprehensive assessment of the child.
- o Assessment procedures would include instructional planning, placement and development of an Individualized Education Program (IEP) appropriate to the child's special needs with a follow up evaluation of student progress.
- o Eligibility for special education services/ classroom and accommodations/ modifications is best determined by a knowledgeable school team. Given the lengthy assessment procedures, it is vital that proper pre-referral procedures are formulated for implementation. Teacher-training would avoid over referral.

The exact cause of SLD is not yet known. Genetic factors and brain insult in the antenatal, natal and postnatal periods are possible etiologies. There is no cure for SLD, either medical or through other measures. However, the difficulties due to LD in the areas of reading, spelling, expression, etc., can be managed effectively with timely and appropriate intervention. Processing information in the brain is a multi path process; hence, strengthening alternate modes of processing, building language and cognition are good compensatory strategies.

3.13 Prevention and Intervention

Approaches that use a phonics drill may seem effective in the short term, but unless they are embedded within meaningful and purposeful texts and reading activities, they may well remain to be viewed as exercises for school and not as reading 'for real'. Wray and Medwell (1999) reported that the most effective teachers of literacy put into context, the skills needed for decoding using meaningful texts for a real purpose (Special Focus, 2005).

• Interventions

Several components constitute interventions for individuals with LD. First, they need intensive, targeted treatment aimed at developing phonemic awareness, phonics and fluency. Also needed is instruction in vocabulary, background knowledge, and comprehension strategies (Hinton, 2006). Reading programs through library activities, for example, story telling and individualized remedial reading sessions are essential. The reading programs developed in the Western world are not suitable in the Indian context. Most of our children will need input in language development and enhancement of vocabulary alongside the phonemic awareness activities. For example, asking an Indian child to list words rhyming with ball will not work, as words like call,

hall, and mall are not in his/her vocabulary. *Development of indigenous programs and tools would be extremely beneficial.* All reading programs must be linked with language development programs. Many children with LD have difficulty with expressive language. Though it is important to encourage usage, it is crucial that these children be evaluated for non-language subjects through more multiple choice questions, drawings, etc., to avoid repeated failures. As *linking and retrieval of information* is also a major difficulty, strategies to help them are: use of cues, prompts and development of pictorial mapping techniques. Finally, intervention should focus on *preservation of self-esteem*. Individuals must be supported and praised as they learn; motivation being an important ingredient of success (Hinton, Dyslexia Primer).

One must remember that *many children with LD are very bright and may show brilliance in other areas.* Hence they should be *encouraged to nurture the development of their talent*, be it in athletics, music, art, science, writing, or mathematics.

- **Pre-school Intervention**

At present there is no standard developmentally appropriate pre-school curriculum followed in our country. This in itself creates difficulties and for children at risk for LD, aggravates the problem. A diagnosis of LD is not made before a child is 7 years old which is the time when the brain matures. However, there are many warning signs (Schwab Learning, 2002) and interventions to help the LD children must be initiated at this level. Pre-school intervention should focus on (a) Language development, (b) development of fine motor and visual motor skills. Adaptations of the Developmental Program in Visual Perception (Frostig, et al., 1972) can be included as a part of the regular pre-school curriculum, and (c) Synthetic and Analytic Phonics as mentioned earlier. Children “at risk” for LD receiving timely help are able to cope much better as they move into the next level of schooling.

- **Interventions at Primary School**

Interventions should focus on developing and strengthening language and basic skills of reading, writing and arithmetic. In addition ensuring that children are allowed to “think” for themselves, to develop higher cognitive functioning is vital. A reading strategy developed by Das based on the *PASS (Planning - Attention - Simultaneous - Successive) theory of cognitive development* (Das, JP, 1998) may be used. This program is being used at the Maharashtra Dyslexia Association in conjunction with other remedial measures. Emotional development must be incorporated into language-teaching with the ability to express emotion, both positive and negative, appropriately, a vital aspect of education.

- **Interventions at Middle School**

Middle school is the time when the foundation for Sciences and Social Sciences are laid. Children with LD have great difficulty in memorizing, retrieval and linking of information. If they also have difficulties with learning English as a second language, failure is likely in every

aspect of learning. Interventions at this stage, in addition to continuing language development and basic skills, must focus on teaching of concepts, critical thinking, and problem solving whilst encouraging creativity and divergent thinking.

3.14 Accommodations and Modifications: Adjusting the Classroom Experience

Accommodations

Accommodations provide different ways for children to take information or communicate their knowledge back. The changes don't alter or lower the standards or expectations for a subject or test. A child with delayed reading skills can participate in class discussions about a novel if she/he has listened to the audio tape version of the book. A child with poor writing and spelling skills may use assistive technology (tape recorder or word processor) rather than struggle with pencil and paper to do her report. Accommodations would include classroom alterations such as seating the child in front, alterations in class work and homework such as individualizing assignments, regarding length, number, due date, topic and alterations in examinations such as multiple choice questions, oral examinations, reading of question paper, allowance for spelling errors (Schwab Foundation, 2006).

Modifications

Modifications are changes in the delivery, content, or instructional level of subject matter or tests. They result in changing or lowering expectations and create a different standard for kids with disabilities than for those without disabilities. Modifications mean that the curriculum is changed quite a bit (Schwab Foundation, 2006). A fifth grade child with a severe math disability who isn't ready to learn fractions and decimals may still be working on addition and subtraction. This means that his instructional level has changed significantly (second, not fifth, grade instruction) (Schwab Foundation, 2006).

Present accommodations and modifications, commonly called concessions, being offered by some secondary boards in India are provision of a writer, reader, extra time, exemption from second and/or third languages, etc.

3.15 Education

- **The Quality of Education Imparted**

As stated at the OSEP meet in November, 2001, The burden of learning that has crept into and taken gargantuan proportions in our education system has been the subject of debate for a considerable period of time now. Decreasing the "school bag" load, making learning more joyful and experiential have been strongly recommended but not implemented to the extent desired. "Rote learning" is the result of examination-oriented teaching, with far reaching detrimental effects especially in the development of language. Language is a vehicle of communication,

expression of needs and thought; it is a medium of learning, a tool for social interaction. Language can only be developed through a period of time with allowance for errors during development. Children, especially from less privileged socio economic homes, do not have much exposure to language; their language develops mainly through the school experience. The school insists on perfectly correct answering skills in rote fashion, right from the entry stage. The child does not develop any independent expressive skills, language which should be a joy is nothing short of drudgery and without language, all other learning is restricted.

The linguistic diversity in India further compounds the problem. A headmaster from a rural school in Goa acknowledges that the children have not developed language skills even, in their own mother tongue. Specific Learning Disability being language based, this has disastrous effects. Hence children with even mild processing deficiencies have great difficulty in learning and achieving. If these children are allowed free expression with less stress on errors, there is a tremendous motivation to express and write. In addition their grades show appreciable improvement.

A study by Pratham (NGO) covering 3,000 children in Delhi revealed that 37% of children in the age group 7-10 years in government schools cannot read words and in private schools 16% of these children can read only letters or nothing at all (NGO Forum, 2006). This serious gap needs to be filled.

Also important for all children, is developing higher brain functions like understanding of concepts, and cognitive development through critical thinking, problem solving and encouraging creativity. Some children develop these on their own, for other children with LD, extra efforts are needed.

At their training level, the teachers are trained in “teaching methodology”. However, they do have difficulty in identifying the concepts to be taught. This results in teaching verbatim from the text book. In teaching language, the stress is on grammar, rather than communication. Hence, a child may score high marks in a language, but may still not be able to communicate in it. In addition, there is no talk of allowing the child to develop his/her own ideas, deduce meanings from context, and no mention of the encouragement of free expression and linking to emotional growth and development.

In India due to over-crowded classes and the stress on exam oriented learning and large curriculum, multi sensory teaching, experiential learning and encouragement of thinking has not found a place on a regular basis. This has led to a large number of under- or non-achievers and eventually school drop outs. Ironically, if teachers were to spend time and energy teaching children with LD, teaching would automatically attain a high quality. In other words, children with LD would make better teachers of us all!

3.16 Social Issues, Socio Economic Rehabilitation, Human Resource Development

Schools play a crucial and formative role in the spheres of cognitive, language, emotional, social and moral development of a child (Kapur, 1995) (Johnson B, 2002) Academic skills such as reading, writing and mathematics form the foundation upon which a student's performance at school is assessed. A learning problem may therefore engender feelings of anxiety, inadequacy and shame, leading to behavioral disturbances in children of school age. Any negative feedback from school is likely to have an impact on the emotional, social and family functioning of a child. A review article by Johnson(2002), has thrown light into the significant association between learning disabilities and behavior problems. Many other studies including a retrospective study at Child and Adolescent unit at the National Institute of Mental Health and Neurosciences (NIMHANS), Bangalore found that 79% of children with learning disabilities had co-morbid psychological disorders, in which 32% had internalizing disorders such as anxiety, depression, 28% had externalizing disorders such as Attention Deficit Hyperactivity Disorder (ADHD), Oppositional Defiant Disorder (ODD) and Conduct Disorder (CD) and 19% had other disorders.

The co-morbidity of developmental dyslexia with both internalizing and externalizing disorders as well as other learning disabilities underscores the need for cognitive and behavioral approaches in the remediation programs offered to children with LD (Muthukumar, et al., 1999). Early diagnosis and intervention in children with learning disorders makes a substantial improvement in self-confidence and social competency.

Major Difficulties

Labeling

Many of these children are branded as “stupid”, “lazy” or/and “slow learners”. The presenting symptom being behavioral, the children get labeled as “trouble makers”, a label that sticks because the performance and the behavior continue without appropriate help and interventions. The child may come to believe the label, and may act upto his/her reputation.

Failure and Non-achievement

Repeated failures lead to very poor self esteem, low motivation, further failure and rejection. In today's world where academic achievements are the gateway to success in life, the children are under tremendous pressure to excel in the examination, their parents under great stress to see their child find their place in the world, and the teachers are equally pressurized to produce results in the examinations. The burden of academic (examination) achievement affects families and teachers, takes away the joy of not just learning and teaching but the joy of teacher/child interaction as well.

Emotional Development

Both failure and labeling will not only affect the emotional development of these children; they also lack the ability to express their feelings due to the difficulties in expressive language which compounds the problem.

Social Stigma

Due to lack of awareness there is no real social stigma associated with SLD, the stigmatization is more due to the effects of LD, i.e., the behavioral issues, the labeling and non achievement. At this point, certification without awareness is likely to result in discrimination rather than inclusion.

Parental Difficulties

Since LD is not an obvious disability, it generally takes time for the parents of a child with LD to accept that the child has different needs and requires certain accommodations. “My child is quite okay, why can’t he learn and achieve like other children. Why am I being constantly called by the school to be told how my child is difficult and will not study, is badly behaved and inattentive?” Parents find it hard to understand. Once the parents recognize the difficulty, they have to take the child for assessments (often stigmatizing) in order to avail of concessions during exams. Many parents are known to spend months, even years rushing from hospital to government offices to even courts of justice to fight for the rights of their child. Concessions, when they are granted are a relief, but the parent and child still have to face the accusations of teachers and other parents who frown upon these as being unfair. In the Western world, parent support groups including on-line ones are a great help to parents of children with LD. They provide emotional support as well as practical advice and information. In India, many NGOs have attempted to facilitate parent support groups with little success; parent involvement is still at the “creating awareness” stage.

Parental Involvement and Community Mobilization

“Perhaps most important, parent groups can actually help the whole village” (Woodward L, et al., 2002). Parents being the greatest advocates, empowering them can bring about major changes in processes and policies that would be beneficial for all children with LD. The involvement of parents can mobilize entire communities to be part of this support movement. However, many children are first generation learners. The parents are happy just to have their child in school; failure means the child has not put in any effort and a good beating should solve the problem! Tuitions are now a norm rather than an exception. Almost all children have been for afterschool tuitions including children in Classes I and

II. Much stress has been laid on parents giving “quality time” to their children. Parents of LD children need to give more than that, they need to understand and be involved with the learning

of their child. Again, this is all very well for some children, but for others like first generation learners, the parents feel equally helpless and abandoned. The entire burden of looking into the needs of these children has to be shared by the Government, the school system and the parent. The community must feel the need to be involved in the development of children. Though they are aware of the consequences of academic failure, awareness regarding LD is almost nil. Community endeavors like libraries, where children with reading and language difficulties can be helped, involvement of Parent-Teacher bodies in sponsoring special teachers, corporate bodies adopting schools will all go a long way in rehabilitating LD children. Corporate bodies may also offer apprenticeship/vocational training for the needy. Panchayats can also contribute towards creating awareness and organize support services/ measures through the community.

3.17 Government and Policies

Newspaper Article 2: Policies on LD as both a health (disability) and education(education for all) issue. The legal definitions view disability strictly from the medical and/or psychometric perspective. This ends up reinforcing a medical model of intervention rather than the much-needed community-based rehabilitation. In the case of LD, the diagnosis has remained in many instances medical; any scheme for children with LD has had to include identification in a hospital. This could lead to inappropriate referral, with the added danger of labeling and stigmatization, it is more discriminatory than inclusive. In addition “there is currently no universally accepted test, test battery, or standard for identifying children with LD” (Reid L., et al., 1994).

The Disability Focus

In India, we are going through a crucial phase in the historical development of LD. As awareness about LD is just in its infancy, there is no stigmatization attached. This is a great opportunity to create non-stigmatizing processes to address the problem. For example, sending children to Hospitals and Psychiatric Institutions (to the lay person a “mental hospital”) is more likely to lead to stigmatization whereas assessment in schools after following thorough pre-referral processes is more child friendly. In fact, present research (Chapter 8) is looking towards non-threatening ways of helping children with LD.

Legislative Actions

The last decade of the 20th Century saw the enactment of three legislations for the rehabilitation and welfare of people with disabilities. All the three legislations, namely, The Persons with Disabilities Act, 1995; The Rehabilitation Council of India Act, 1992; and The National Trust Act, 1999 are comprehensive in spirit, and together deal with all aspects pertaining to rehabilitation, from prevention, training, employment, long-term settlement, human resource development and research, and documentation (SSA, 2003). However, Specific LD is not included in any of these acts. The difficulty in diagnosis of Specific LD has made the

formulation of a National Policy for these children no easy task. As a result, selected states have tried to provide for children with LD by bringing in their own legislations.

Maharashtra Government

Maharashtra Government has provided concessions for children with LD from Standards I to XII. However in the absence of a National policy, schools in Mumbai have been reluctant to follow these guidelines. There is a general feeling amongst those in the field of education that these concessions will (a) dilute the standards of achievement, and (b) create difficulties with parents of other children. For example a child with LD will not have marks cut for spelling errors and may get higher grades than another, who has produced better work.

These fears whilst being valid emphasize what is wrong with our education system; the focus is not on the child and his/her potential. The system is not designed to ensure all children learn and get educated; rather the child has to fit into the system. Any attempt at flexibility is challenged. Every child has the right to basic school education. No system has the right to tell a developing child that he/she is a failure. It is not the child who is a failure; it is the rigid system that has failed the child. Recently, cases have been cited where children with LD have been denied concessions at examinations. In response to the parents' petitions, the Mumbai High Court directed all schools in Maharashtra to abide by the guidelines for students with LD (*Times of India*, 22nd July, 2006).

Other states too that are aware of the problem, are struggling to take decisions that will provide for children with LD. The Delhi High Court, for example, recognizing dyslexia as a form of disability, has ordered Delhi University to grant admission to dyslexic students under a three percent quota for people with disabilities (*The Tribune*, June 22, 2004).

Such efforts are, however, few and far between in the absence of a National Policy for Children with LD.

Goa Govt. Incentive Scheme for Special Children

In February 2005, the Goa Government introduced a novel scheme in promoting education among children with special needs, slow learners and children with LD, in the state (Govt of Goa, 2005). The families of the children with special needs, slow learners and children with LD, are given financial benefits, such as transport, uniform allowance and allowance for aids and appliances. Institutions are also given incentive money in order to promote inclusion. Though a laudable initiative, children with LD primarily require infrastructure that facilitates their learning, such as trained teachers, resource rooms, educational aids, a flexible curriculum and evaluation methods. Here again the absence of proper guidelines has resulted in unfocused initiatives. There is an urgent need for a National Policy on Learning Disability.

3.18 Summary

“Although some learning disabilities are now known to be biological in origin, the treatments are more often educational” (Reid L., et al., 1994). There have been various initiatives in the field of education such as DPEP, National Policy of Education, Total Literacy Campaign and establishment of DIETs, but none of these initiatives address the problems of LD and its related issues.

In recent years, the Government of India has launched Sarva Shiksha Abhiyan for Universalization of Elementary Education. The objective of UEE cannot be achieved without including children with special needs including LD under the ambit of elementary education. Experiences of programmes like DPEP and various research findings have shown that the number of children with special needs in every district is by no means small. The Ministry of Social Justice and Empowerment, which is a nodal Ministry for disability issues, estimates the number of children having special needs as 5% (SSA, 2003). Given the difficulty in diagnosis of LD, this figure may not be accurate as for children with LD is concerned. At the same time, Sarva Shiksha Abhiyan has included LD in their categorization of children with special needs. “The term children with special educational needs, refers to all those children who are challenged with various problems such as that of vision, hearing, movement, learning, cerebral palsy or mental retardation” (SSA, 2003). Education of children with special needs being a relatively new concept requiring a great deal of technical expertise, a manual has been developed under SSA to initiate, implement and monitor the program of educating children with special needs. The SSA manual, though comprehensive in many respects, has not given clear cut directives regarding children with LD. Apart from a check list for LD, the structuring of the resource room, aids, teacher training, use of funds, curriculum alterations (accommodations and modifications) have not been defined. As a result, the state governments are still unsure of the steps to be taken with CWSN under the LD category.

The National Curriculum Framework, 2005 also recommends an *inclusive curriculum* that ensures full participation of all children including first generation learners, learning disabled, slow learners and children from SC and ST (NCERT, 2005).

3.19 Resources

Ainscow, S. and Muncey, J. (1988), *Meeting Individual Needs in the Primary School* Asch, A. (1989) *Has the law made a difference?*, in D. Lipsky and A. Gartner (eds) *Beyond separate education*, Baltimore:

Brookes. Board of Studies NSW (2002), *K–10 Curriculum Framework* Board of Studies NSW (2004), *English and Mathematics Years 7–10 Life Skills: Advice on Planning, Programming and Assessment*

Louden, W., Chan, L.S., Elkins, J., Greaves, D., House, H., Milton, N., Nicols, N., Rivalland, N., Rohl, N., Van Kraayenord. (2000) *Mapping the territory – Primary students with learning difficulties: Literacy and Numeracy*, Department of Education, Training and Youth Affairs, Commonwealth of Australia Kraayenoord, C.V. and Elkins, J. (1990), *Learning difficulties* in A. Ashman and J. Elkins (eds) *Education children with special needs*, New York: Prentice Hall NSW Department of Education and Training (2000), *Focus on Literacy: Writing* Westwood, P. (1993), *Commonsense Methods for Children with Special Needs*

Unit 4

Curricular Adaptation, IEP, Further Education

4.1 Introduction

This resource guide is an updated version of the 1996 Ministry of Education resource guide *Individual Education Planning for Students with Special Needs*. It provides information and strategies that align with professional practice and current legislation. The guide is designed for teachers involved in developing and implementing Individual Education Plans (IEPs). While the content in this resource is organized sequentially, the circular diagram at left represents the cyclical process of IEP development, which includes stages of PLANNING, IMPLEMENTING and EVALUATING. Within these stages are seven essential components: Assessment, Collaboration, Writing, Introducing, Monitoring, Reviewing and Reporting. Ideally, schools engage in these stages to develop and use the written IEP as an effective tool for the individualized program for a student. The information in this resource is presented using the following format:

- The Big Idea, which describes the overarching theme or concept
- Key Points that should be considered in the process
- Practices that have proven to be effective

Definitions of terms, links to legislation, Ministerial Orders, clarifications and additional resources may be found directly by clicking on the highlighted text within the document. Appendices include an array of useful, reproducible tools to assist teachers in planning, implementing and evaluating IEPs.

4.2 Determining Which Students Require an IEP

The Individual Education Plan order (M638-95) provides direction about who should and who should not receive an IEP. It states that a board must ensure that an IEP is designed for a student with special needs as soon as practical after the student is so identified by the board. The order

indicates three circumstances where an IEP for a student with special needs is not necessary. Those instances are where:

- the student with special needs requires no adaptation or only minor adaptations to educational materials, or instructional or assessment methods
- the expected learning outcomes established by the applicable educational program guide have not been modified for the student with special needs
- the student with special needs requires in a school year 25 hours or less remedial instruction by a person other than the classroom teacher, in order for the student to meet the expected learning outcomes

(To view the order see www.bced.gov.bc.ca/legislation/schoollaw/e/m638-95.pdf) Looking at the inverse circumstances of this may help clarify which students are entitled to an IEP. A student with special needs is entitled to an IEP if:

- the student with special needs requires more than just minor adaptations to educational materials, or instructional or assessment methods, or
- the student with special needs is working on outcomes other than the prescribed outcomes of the curriculum, or
- the student with special needs is working on the regular outcomes with little or no adaptations, but receives 25 hours or more of remedial help from someone other than the classroom teacher to meet the expected learning outcomes

Some students, although not identified as having a special need, may require adaptations and remedial help from someone other than the classroom teacher. It is not a requirement that these students have an IEP. In this case, it is good practice to put in place a learning plan to document specific needs of these students. Local school boards may have established practices and templates for learning plans.

4.3 Purpose and Process

An IEP is a documented plan developed for a student with special needs that summarizes and records the individualization of a student's education program.

Key Points

- An IEP provides accountability for:
 - individualized goals
 - the means to achieve these goals
 - additional services and how they will be delivered
 - responsibility for various aspects of the IEP
- An IEP assists teachers in monitoring student growth and progress.
- An IEP provides an ongoing record to assist with continuity in programming and transition planning.
- An IEP guides the implementation of learning support services inside or outside the classroom to align the educational program with the needs of the student.
- Development of an IEP may be undertaken by one teacher in consultation with parents, a small group or an expanded team, depending on the complexity of the student's needs.
- An IEP team might include any of the following participants:
 - teacher(s)
 - case manager
 - specialist staff
 - school counselor
 - community specialists
 - parent(s) or legal guardian
 - the student, if appropriate
- The IEP process, when it works well, involves a dynamic cycle of planning, monitoring and evaluating.

4.4 Planning

Assessment and collaboration should precede the actual writing of an IEP document.

4.4.1 Assessment

Key Points

- The purposes of educational assessment are twofold. Assessment *for* learning provides information that directs further instruction. Assessment *of* learning monitors student achievement.
- Assessments exist in a variety of forms: observation, reviewing historical data, interviewing, and informal and formal testing.
- A variety of assessments should be used to inform the development of appropriate IEP goals and planning of services.
- Assessment not only considers a student's current level of performance and learning style but also methods of instruction being used, features of the learning environment, and the appropriateness of the curriculum for that student.

Practices

Teachers will improve students' assessment results and access to the curriculum when they accommodate different learning styles. There are three main learning styles. A student may prefer one or a combination of all.

- Visual - learns best by seeing (e.g., watching demonstrations or videos)
- Auditory - learns best by listening (e.g., through verbal interactions with others or with self)
- Kinesthetic - learns best by doing

Most informal and formal assessment methods can be grouped within four general categories: Reviewing, Interviewing, Observing and Testing (RIOT).

Reviewing

Understanding a student's needs can be informed by reading various records with both current and historical information:

- permanent record files and special services files
- school documentation (e.g., school-based team notes, discipline records)
- teacher marks/records
- performance logs

Interviewing

Valuable insight can be gained from talking to a variety of people:

- o parents, who have rich insights and knowledge about their children
- o current and past teachers and other relevant service providers
- o students themselves (e.g., questionnaires, self-assessments, response journals or learning logs)

Observing

Watching, listening and examining student work and monitoring responses to instruction are all important sources of assessment information. Various tools can be used to document observations:

- o checklists of student performance and/or behaviour
- o rubrics and rating scales
- o product/work samples
- o portfolios
- o audio or video recordings
- o functional assessments

Testing

Testing, both formal and informal, is an important part of the assessment process and can include a variety of tests:

- o quizzes, probes and diagnostic skill inventories
- o criterion-referenced tests
- o norm referenced tests
- o alternative test formats – oral, graphic, electronic

4.4.2 Collaboration

Key Points

- Collaboration is a process in which people work together to solve a common problem or address a common concern. Collaboration is founded on clear communication and a willingness to work toward a common purpose.
- Through collaboration a commitment to action can be built.
- The planning of an IEP can draw on the knowledge and expertise of several people, depending on the needs of the student. Consultation is a process of seeking expert information and advice. It is foundational to the collaborative process.
- By authority of the Individual Education Plan Order schools are required to offer parents, and where appropriate, the student, an opportunity to be consulted on IEPs.
- Collaboration on an IEP should begin as soon as practically possible after a student is identified as having special needs.

4.5 Terms Related to IEP Development

Accommodations

Accommodations are the means to facilitate a variety of forms of representation, expression, and/or engagement that may be necessary to help each student in the classroom succeed. Both differentiations and universal design provide varied approaches to setting goals, providing choice in use of materials and media, and flexibility in gathering assessment data. Adaptations and/or modifications are each a form of accommodation. Use of accommodations allows a teacher to use a personally responsive approach to meeting the needs of the student.

Adaptations

Adaptations are teaching and assessment strategies especially designed to accommodate a student's needs so he or she can achieve the learning outcomes of the subject or course and demonstrate mastery of concepts. Essentially, adaptations are "best practice" in teaching. A student working on learning outcomes of any grade or course level may be supported through use of adaptations.

Adaptations do not represent unfair advantages to students. In fact, the opposite could be true. If appropriate adaptations are not used, students could be unfairly penalized for having learning

differences, creating serious negative impacts to their achievement and self-concept. An education plan with adaptations retains the learning outcomes of the regular curriculum and is provided so that the student can participate. Students with education programs that include adaptations are assessed using the standards for the course/program and receive credit toward a Dogwood Diploma for their work in the Graduation Program. Adaptation can be made to:

- o the environment in which learning and teaching will occur
- o the processes to be used (level of support, instructional method, mode of learning and assessment)
- o the depth of the content to be learned
- o the products that will record the student's individual achievement

Assessment *for* Learning

Assessment *for* Learning is a term used to describe the collection of a wide range of data that highlights a student's strengths and weaknesses, and provides insight into a student's learning. The purpose of assessment for learning is formative – to provide feedback about a student's learning that can be used to advise the next stage of learning/remediation.

Assessment *of* Learning

Assessment *of* Learning is a term used to describe an evaluation, often an exam, that includes questions drawn from the material studied during a set period of time. It is usually done at the end of a unit, course, grade or program. The purpose of assessment of learning is summative, intended to measure learning and report to parents and students about the student's progress in school.

Board Authorized Authority (BAA)

Board Authorized Authority is the district level authority to officially adopt a Grade 10, Grade 11 or Grade 12 course that meets the requirements set by the Minister in the Course Information Booklet (www.bced.gov.bc.ca/graduation/courseinfo/). For more information see www.bced.gov.bc.ca/legislation/schoollaw/e/m285-04.pdf.

Case Manager

A case manager is the person assigned to coordinate the collaborative process involved in developing, writing, and introducing and evaluating an IEP. A case manager promotes quality and effective interventions and outcomes. Case management is characterized by advocacy, communication and resource management.

Collaboration

Collaboration is a process in which people work together to solve a common problem or address a common concern. A successful collaborative process is characterized by the following features: it is voluntary, there is mutual trust and open communication among the people involved, identification/clarification of the problem to be addressed is a shared task, the goal is shared by all participants, each participant's contribution is valued equally, all participants' skills are employed in identifying and selecting problem-solving strategies, and there is shared responsibility for the program or strategy initiated.

Conditions

Conditions are the circumstances or setting in which a student may work. There are a wide variety of conditions for learning. Two key learning conditions to consider include "in context" and "in isolation." For example, a student may need to learn to read consonant-vowel-consonant words in isolation first and then practise them in context of appropriately levelled text.

Consultation

Consultation is a process of seeking information and advice. It is founded on clear communication and facilitates collaboration and problem solving. As such, it is not a one-time process. Consultation may include the family or families, the student, other professionals and outside agencies.

Criteria

Criteria are a set of elements that describe what constitutes success with a given skill or acquisition of knowledge. Criteria will state:

- o The content of what the student is to learn
- o The observable behaviour – what the student does to indicate that she/he has learned

- o The level at which the student needs to perform or exhibit the behaviour, knowledge, etc. (e.g., 8 out of every 10 times observed)
- o The conditions for the behaviour to occur – the circumstances under which the student will work

Criterion-Referenced Test

Criterion-referenced evaluation is based on comparing a student's performance to a specified criterion such as a performance standard. Its purpose is to gauge individual mastery and to determine if the student can or cannot perform at a certain standard. How well others perform on the same standard is not considered.

Curriculum

Curriculum is a logically organized, developmentally appropriate set of learning outcomes categorized by subject or grade. View the prescribed curriculum for BC schools at www.bced.gov.bc.ca/irp/irp.htm.

Differentiated Instruction

Differentiated instruction employs strategies to accommodate a range of abilities and learning styles. Differentiation can address content, process, procedures, presentation strategies, or assignment tools.

Evaluation

Evaluation involves interpreting, analyzing and reflecting on observation and assessment data for the purpose of reporting progress and making decisions about logical next steps for the student. The goal of evaluation is to determine student progress, celebrate the student's success, and address the student's needs.

Functional Assessment

Functional assessment is an assessment of a student's functioning within a context.

Functional assessment is a two-part process:

- 1) Observing the student's functioning while taking note of environmental conditions such as physical factors (e.g., heat, light, noise) and instructional factors (e.g., pace, teacher expectations,

peer interaction). During observation, the teacher or other qualified professional develops a working hypothesis about the reasons for the student's responses and the possible impact of various environmental conditions.

2) Identify replacement skills and behaviours that can be taught and that would meet the same function or need expressed by the student, but that enable him or her to improve his or her independence, adjustment and learning. At the same time, staff identifies possible changes to environmental conditions (see above) that would support the student's progress.

4.6 IEP Goals

An IEP goal is a statement of what a student will be able to accomplish academically, developmentally and/or functionally, usually over a year's time. Goals should be specific, measurable, achievable, relevant and time-framed (SMART).

Individual Education Plan (IEP)

An Individual Education Plan is a documented plan developed for a student with special needs that describes individualized goals, adaptations and/or modifications and the support services to be provided, and includes measures for tracking achievement. More information is contained in the BC Ministry of Education's Special Needs Student Order (www.bced.gov.bc.ca/legislation/schoollaw/e/m150-89.pdf).

4.7 A GUIDE TO ADAPTATIONS AND MODIFICATIONS

In British Columbia, three principles of learning guide practice in the development of Integrated Resource Packages (IRPs), which contain the provincially prescribed learning outcomes for grades and subjects. These are:

- Learning requires the active participation of the student.
- Students learn in a variety of ways and at different rates.
- Learning is both an individual and group process.

These same three principles should guide the differentiation of instruction, assessment methods, and/or materials--particularly the principle that people learn in a variety of ways and at different rates. Today's classrooms are diverse and inclusive by nature. Differentiation of instruction and assessment and the principles of universal design are now recognized practices for teachers.

Both differentiation and universal design provide systematic approaches to setting goals, choosing or creating flexible materials and media, and assessment. To undertake differentiation and universal design, teachers need to be aware of a range of accommodations (multiple means of representation, of expression, and/or of engagement) that may be necessary to help each student in the classroom succeed. These accommodations may take the form of adaptations and/or modifications. Many students with special needs and significant learning challenges will be able to achieve the learning outcomes for subjects or courses with no or minor adaptations. Some may be able to achieve the learning outcomes of some subjects or courses with adaptations. A small proportion will need to work on individualized outcomes, goals different than the curriculum; this is referred to as modification.

4.8 Adaptations

In BC policy, all students should have equitable access to learning, opportunities for achievement and the pursuit of excellence in all aspects of their educational programs. (Policy Document: Special Education: http://www.bced.gov.bc.ca/policy/policies/special_ed.htm.) Adaptations are teaching and assessment strategies especially designed to accommodate a student's needs so he or she can achieve the learning outcomes of the subject or course and to demonstrate mastery of concepts. Essentially, adaptations are "best practice" in teaching. A student working on learning outcomes of any grade or course level may be supported through use of adaptations. Adaptations do not represent unfair advantages to students. In fact, the opposite could be true. If appropriate adaptations are not used, students could be unfairly penalized for having learning differences, creating serious negative impacts to their achievement and self-concept.

Examples of Adaptations

Accommodations in the form of adaptations occur when teachers differentiate instruction, assessment and materials in order to create a flexible learning environment. For example, a

student could be working on below grade level learning outcomes in Language Arts and at grade level in all other subjects or courses, some of which require reading materials at the lower reading level.

Adaptations include, but are not limited to:

- o Audio tapes, electronic texts, or a peer helper to assist with assigned readings
- o •Access to a computer for written assignments (e.g. use of word prediction software, spell-checker, idea generator)
- o Alternatives to written assignments to demonstrate knowledge and understanding
- o advance organizers/graphic organizers to assist with following classroom presentations
- o extended time to complete assignments or tests
- o Support to develop and practice study skills; for example, in a learning assistance block
- o Use of computer software which provides text to speech/speech to text capabilities
- o Pre-teaching key vocabulary or concepts; multiple exposure to materials
- o working on provincial learning outcomes from a lower grade level

Best practice in teaching suggests that a record of successful adaptations for any student should be kept within a student's file to both document current practice and support future instructional needs. In the case of a student with special needs who has an Individual Educational Plan (IEP) or English as a Second Language students who have Annual Instructional Plans, successful adaptations are recorded in these plans to document how the student is being supported currently and also so other teachers will know what works well for that student.

For students with special needs, adaptations that are used for tests and exams must be included in their IEPs in order for them to be considered for adaptations to the provincial exam conditions (adjudication). Students whose education programs include adaptations will generally be working toward graduating with a Dogwood Diploma.

Grading and Reporting When There Are Adaptations

Grading for students who have been provided with adaptations should be in relation to the outcomes of the curriculum. If the learning outcomes that a student is working toward are from the curriculum of a grade level lower than the current grade placement, this should be indicated

in the IEP or learning plan and in the body of the student's progress report. Further information on this subject is available in the Ministry document:

4.9 Modifications

This section may not apply to students in ESL programs unless they are also identified as a student with special needs as determined by Ministry and district processes.

Accommodations in the form of modifications are instructional and assessment-related decisions made to accommodate a student's educational needs that consist of individualized learning goals and outcomes which are different than learning outcomes of a course or subject.

When To Use Modifications

The decision to use modifications should be based on the same principle as adaptations—that all students must have equitable access to learning, opportunities for achievement, and the pursuit of excellence in all aspects of their educational programs. Before modifying the outcomes for a student, schools should review all instructional interventions tried and consider assessment information, utilizing a process that is ongoing and consultative—similar to IEP development practices overall.

Modifications should be considered for those students whose special needs are such that they are unable to access the curriculum (i.e., students with limited awareness of their surroundings, students with fragile mental/physical health, students medically and cognitively/multiply challenged.) Using the strategy of modifications for students not identified as special needs should be a rare practice.

In many cases, modifications need only form part of an educational program for a student with special needs, and they need not be a permanent or long term solution. Whether to use modifications should be reviewed on a regular basis. Decisions about modifications should be subject or course specific wherever possible. For example, a student with an intellectual disability may require modifications to a specific subject area such as mathematics; however, modifications may not be required to meet the provincial outcomes in physical education. Although decisions about modifications to a student's courses or subjects may take place in grades earlier than Grade 10, a formal decision that an overall program is modified does not need

to occur until Grade 10. The decision to provide modifications, particularly at the secondary school level, will result in students earning a School Completion Certificate upon leaving school rather than credits toward graduation or a Dogwood Diploma. Therefore, the critical decision of whether a students' education program should include modifications should not be made in isolation by a single classroom teacher. The decision should be carefully and thoughtfully made, in consultation with parents, school administration, and/or instructional support personnel. This decision should address longer term educational, career and life goals of students and encompass plans for attaining these goals.

Examples of Modifications

An educational program for a student might include a combination of accommodations which includes modifications. For example, a student could be working on grade level learning outcomes in Physical Education

and Health and Career Education and below grade level learning outcomes in Mathematics, all with adaptations while at the same time working on individualized learning outcomes that meet the student's IEP goals in all other subjects. The individualized outcomes address functional life skills and foundational academic skills. For students with special needs, modifications that consist of individualized learning outcomes or goals must be included in the IEP. Some further types of modifications include:

- Content and evaluation related to the course or subject but at a lower level of conceptual difficulty that is based on a student's individualized outcomes or goals. When students do well on this especially designed material, they have a chance to feel successful. For example, while students in a Grade 3 class are researching for presentations on the solar system, a student with special needs in this class uses a computer to drag and click planets into a template of the solar system and learns to say the names of each planet. At the secondary level, a Grade 9 student with special needs learns how to count change and manage a personal budget while other students are introduced to algebraic expressions.
- Only portions of the learning outcomes are addressed so that a student may participate in the classroom and feel success even though they are working at a conceptual level significantly

different from the other students. For example, in a science class a student with special needs learns to identify safe and dangerous chemicals used in the lab, while other students carry out a chemistry experiment.

- Although related to the outcomes of the curriculum, the goals for a student with special needs are significantly different. For example, while other students are learning how to read and respond to text in a Grade 4 classroom, a student with special needs is learning how to listen to stories at a pre-primary level and when to turn the page at the appropriate time using assistive technology.

Grading and Reporting When there are Modifications

If schools are using BCeSIS or Student Achievement Data Exchange (SADE) to record progress for students in Grades 4 to 12, a value is required to be entered to maintain student records over time. For more information about BCeSIS, please contact your local school district. For more information about SADE, please see the following link:
<http://www.bced.gov.bc.ca/datacollections/sade/>

Progress reports to parents for students with special needs who are working toward individualized outcomes or goals in an IEP rather than the outcomes of the curriculum for that subject or course may be done using structured written comments or letter grades. The most appropriate form of reporting for the student should be determined collaboratively at the school level. If letter grades are used when modifications have been made, the body of the student progress report should state that the evaluation is in relation to the individualized outcomes or goals and not in relation to learning outcomes for the subject or course. The specific IEP outcomes or goals evaluated should be included in the student progress report. Further information on this subject is available in the Ministry document: *Reporting Student Progress: Policy and Practice*:
http://www.bced.gov.bc.ca/classroom_assessment/09_report_student_prog.pdf

Is the student struggling with some course or subject? Continue with differentiation of instruction using adaptations and universal design for learning concepts noting strategies that work particularly well for the student. Try different strategies and monitor. Did the student

demonstrate improvement? Consult with others, review assessment information, initiate further assessments or make referrals, gather ideas for other

4.10 Special Education Support Service: Information on Mild General Learning Disability

Definition:

1. From Circular 08/02:

Such children have been assessed by a psychologist as having a mild general learning disability.

2. From Report of the Special Education Review Committee¹ (1993):

Pupils with mild general learning disability² have significantly below-average general intellectual functioning, associated with impairment in adaptive behaviour. This may be reflected in a slow rate of maturation, reduced learning capacity and inadequate social adjustment ... Insofar as an Intelligence Quotient may be used as an indicator of mild general learning disability, such pupils would lie within the IQ range 50 to 70.

Common Characteristics:

The SERC Report notes that a person with a mild general learning disability may display some of the following characteristics:

- o Delayed conceptual development;
- o Slow speech and language development;
- o Limited ability to abstract and generalize;
- o Limited attention-span;
- o Poor retention ability;
- o Poor adaptive behavior;
- o Inappropriate or immature personal behavior;
- o Low self-esteem;
- o Emotional disturbance;
- o General clumsiness and lack of co-ordination and of gross and fine motor skills.

The National Council for Curriculum and Assessment (NCCA) in their Guidelines for Teachers of Students with General Learning Disabilities³ also refer to:

- o Limited retention
- o Difficulties in expressing ideas and feelings in words
- o Difficulties adapting to new situations

These all appear to be negative traits, however, the SERC Report also notes “that they are capable of achieving vocational and social adequacy with proper education and training but will need support and

What can I do as a teacher?

The SERC Report sees the learning needs of the student with mild learning disability as requiring:

- o A curriculum adapted to their learning need and capacities;
- o A variety of appropriate teaching approaches and methods;
- o Specialist assistance and teaching equipment;
- o Individual or small-group instruction;
- o Concrete materials and aids to assist in learning;
- o Special attention paid to the development of language and communication skills;
- o Specific programmes for social and personal development;
- o An emphasis on functional literacy and social mathematics;
- o An emphasis on pre-vocational skills and life-skills’;
- o Regular diagnostic assessment and review of progress.

The Guidelines for Teachers of Students with General Learning Disabilities⁵ note that students with mild general learning disabilities are not easily distinguishable from their peers. Their learning needs are often difficult to define, due to the multiplicity of factors that can contribute to their learning disability. Their difficulties are often only identified after starting school, when they show a slower pace of learning progress than others of their age group. Like their peers who do not have learning disabilities, they have a wide range of learning styles, varied interests and the same need for affirmation and success. Practical suggestions from the Guidelines for Teachers of

Students with General Learning Disabilities in reference to students with mild general learning disability include:

- Using objectives that is realistic for the students ensuring that the learning task is compatible with prior learning
- Providing opportunities for interacting and working with other students in small groups
- Spending more time on tasks
- Organising the learning task into small stages
- Ensuring that language used is pitched at the students' level of understanding and does not hinder understanding the activity
- Using task analysis outlining the steps to be learned/completed in any given task
- Posing key questions to guide students through the stages/processes and to assist in self-direction and correction
- Using graphic symbols as reminders to assist in understanding the sequence/steps in any given task/problem
- Modelling task analysis by talking through the steps of a task as it is being done
- Having short and varied tasks
- Creating a congenial learning environment by using concrete and, where possible, everyday materials, and by displaying word lists and laminated charts with pictures.

To attain success, learning experiences, as noted in the Guidelines for Teachers of Students with General

Learning Disabilities need to be characterised by:

- A range of activities that will interest and motivate the student
- Differentiated teaching approaches, methods, materials, resources and learning tasks

- Clarity in relation to the beginning and end points of tasks, ambiguity to be avoided
- consider using a model or picture of the final goal or end product, so that the student knows what is expected
- The use of visual cues to highlight meaning; present tasks using symbols, or give instructions with a written/pictorial list or prompt cards
- Tasks that are presented in small, manageable, clearly identifiable steps
- Skills that are introduced one at a time
- A hierarchy of sub-goals and sequence activities that are developed towards the learning goal
- Modification of tasks that harness and build on the student's strengths; tasks can be graded so that they make increasing demands on the student
- Instructions that are explicit; do not assume that the context will help to make the meaning clear
- Explicit connections to previous skills or knowledge
- Opportunities for the student to generalize knowledge and skills
- The gradual introduction of choice, thereby encouraging decision-making
- Routines that are taught through a structured approach
- Teacher attitudes - be positive, patient and aware of how and what you communicate to the student

4.11 Differentiation

This is the concept of varying the pace, content, level and amount of material presented to students so that each student receives the appropriate learning experiences. Westwood⁶ defines differentiation as 'learning things differently according to observed differences among learners'. All effective teaching requires a degree of differentiation, but it is particularly important in

supporting students with general learning disabilities. How can this be incorporated successfully into teaching? Differentiation can occur:

- o **By level and pace:** students work on a similar topic, at a level and pace appropriate to their ability; e.g. Topic – Money: some students work on coin recognition, others on shopping / change activity, etc.
- o **By interest:** students select a topic of interest to themselves, learning can be enhanced; e.g. Topic – Bar Charts: groups / students select from: favourite singer, TV programme, football team,
- o **By access and response:** students respond in different ways to the material presented to them; e.g. Topic – Bar Charts: some draw, some use ICT, some use different formats for presenting data.
- o **By structure:** students guided step-by-step or in whole blocks of integrated curricular content; e.g. Topic – Regrouping in subtraction: some practice skill independently, others monitored / supported.
- o **By sequence:** different students access different parts of a subject at different times; e.g. Subject – LCA Leisure & Recreation: modular course, students select the elements they wish to do;
- o **By teaching style:** teacher varies approach / methodology in the classroom; e.g. ‘chalk and talk’, using ICT or video, research using the Internet, class discussion, small group work.
- o **By time:** teacher adapts use of time with class to meet the needs of all children; e.g. Topic – Bar Charts: some students work independently, others closely monitored by teacher / SNA.
- o **By classroom structure:** teacher structures groups in class to maximise learning e.g. some students work in pairs, others in groups, others individually, each with a purpose and role.

The NCCA in the Guidelines for Teachers of Students with General Learning Disabilities note that effective planning for differentiation depends on an accurate assessment of the student’s previous level of achievement in order to plan learning objectives appropriate to future learning needs. The teacher begins by establishing the student’s prior learning experiences (what he/she already knows or can do); his/her interests (what the student likes, what the student is good at) and his/her learning profile (more personal information about the student and the factors that affect how the student learns). The NCCA observe that in planning for differentiation teachers need to be guided by three key considerations:

1. What the student could know and do

2. What the student should know and do

3. What the student must know and do

Remember the needs of individual students may be different even if their learning disability, in this instance mild general learning disability, is the same.

4.12 Use of ICT to support learning for students with Mild General Learning Disabilities

The effective use of ICT has benefits for all learners, including students with mild learning disabilities. For the teacher, it offers a different way of presenting material; for the learner, it offers a different way of interacting with the curriculum. Both software and hardware can be utilized to support students with mild general learning disabilities.

Software: A range of software is now available for a variety of different settings, needs and subject areas. This can be either content-rich or content-free software. While some is specifically categorized as ‘Special Needs’ Software, many of the regular software packages are also suitable for use by students with mild learning disabilities, including office applications such as Word Processing, Spreadsheets and Desktop Publishing packages.

Hardware: Most students with mild general learning disabilities will not need high-tech assistive technology, such as touch monitors, switches, screen readers, etc. However, they may benefit from adaptations such as lower-case keyboards or keyboard overlays, optical mice, rollerballs, etc. Peripheral hardware items, in particular digital cameras and scanners, have proven to be of huge benefit in the classroom. Use of the digital camera allows the student to relate their work directly to their own experience, whether it is photos of family members or of a field trip in the local environment. The Guidelines for Teachers of Students with General Learning Disabilities state that ICT can benefit students with mild general learning disabilities in the following ways:

- ICT can provide a non-threatening environment in which the level and pace of instruction can be differentiated for individual students.
- Software programs can provide exciting and stimulating repetition that is often required for students to master skills.

- o ICT can offer graphics, sound effects and immediate rewards to help encourage the learner.
- o ICT may provide the needed support to effectively accomplish tasks in a variety of contexts and settings that the student may otherwise find difficult and stressful.
- o ICT can provide a means for some students with general learning disabilities to accomplish tasks independently, and therefore not have to rely continually upon others.
- o ICT promotes independence and communication, since for some students; technology may be the only method they have to communicate with the world around them.
- o ICT provides a multi-sensory approach to the teaching of essential skills and concepts in a variety of contexts and settings.
- o ICT reinforces the concept of 'cause-and-effect' and pre-reading and pre-number concepts.
- o ICT can be used to introduce and reinforce simple concepts, such as matching and sorting, as well as basic literacy and numeracy skills.
- o ICT facilitates the development of motor skills, eye tracking and hand-eye co-ordination.
- o ICT facilitates communicative situations and language development.
- o ICT facilitates social interaction and gives students an opportunity to experience turn taking and co-operative situations.
- o ICT provides motivating and stimulating learning experiences and gives instant feedback to students' responses.
- o ICT is non-judgmental and allows students to work at their own pace.

However, while ICT can be powerful as a tool for engaging the student, for motivation, and for self-directed learning it ought to be appropriate to the student's functional and cognitive abilities, temperament, attitudes and motivation and that the ICT used should be interesting and motivating to the student. In the Guidelines for Teachers of Students with General Learning Disabilities it is suggested that that the following questions may help in selecting the most appropriate use(s) of ICT:

- o Does the student have difficulty with social interaction?
- o Does the student have a communication difficulty in speaking?
- o Does the student have a communication difficulty in writing?

- o Does the student have difficulty in planning, organizing ideas, and formulating language?
- o Does the student have sensory or physical needs?
- o Does the student have social, behavioral and emotional needs?
- o Does the student have more complex multiple support needs?

Resources provided by the National Centre for Technology in Education(NCTE) for Special Educational Needs and ICT include a set of critical factors to consider when matching technology. These are available on the NCTE website at <http://www.ncte.ie/SpecialNeedsICT/ResourcesAdvice/MatchingTechnology/> For more information on the benefits of ICT in Special Education, go to the relevant sections in the SESS website: http://www.sess.ie/sess/Main/Resources_ICT.htm or http://www.sess.ie/sess/Main/Links_SEN&ICT.htm

Adolescents with Mild General Learning Disabilities – Issues at Second Level

The challenges faced by students who have mild general learning disabilities may be significantly exacerbated by the onset of adolescence. The physical, social and emotional changes, which characterize the transition from childhood to adulthood, are significant for all young people. These changes can pose particular challenges for a young person with mild general learning disabilities. The emergence of the peer group as an important social context is one of the characteristics of adolescence. A student with a learning disability can easily become isolated from a peer group, which thinks, communicates and reacts at a quicker rate. The classroom and school can offer not simply a context for learning, but an important focus for the development of good peer relationships and a secure foundation for self-esteem. This is true, of course, for all young people in school, but particularly so for those with mild general learning disabilities. The process of transfer from primary to post-primary school poses further challenges. For example, the difficulties students with mild general learning disabilities may have with retention can be highlighted when a student has to cope with more than one teacher, with different classrooms, and the variety of books, equipment and homework required for each lesson. The learning experiences and organizational support offered to students with mild general learning disabilities should not contribute to even further isolation by being ‘different’. Rather, in a dynamic and differentiated learning environment, the variety of learning experiences available

to all students should include activities designed to meet the particular needs of students with mild general learning disabilities.

Students with mild general learning disabilities, attending either mainstream second level schools or special schools, may need an adapted curriculum relative to their needs and abilities. There is now a range of options for these students. These include:

- Junior Certificate School Programme / Selected Junior Certificate courses

Within the Junior Certificate programme a range of subjects are available but the subjects most commonly taken by students with mild general learning disabilities tend to include: English, Mathematics and Gaelge to Ordinary or Foundation Level; Art, Craft & Design, Materials Technology (Wood), Home Economics and History to Ordinary Level.

Leaving Certificate Applied Programme

Many students with special educational needs are following this excellent programme - the modular and task-based nature of the work is very 'student-friendly', rewarding those who put in the effort. The final exams take place at the same time as the traditional Leaving Certificate, but 66% of the overall marks are available for satisfactory completion of Modules and Tasks prior to the terminal exams.

FETAC (NCVA) Foundation Level Courses

The FETAC courses are suitable for many students over 15 who may not be in a position to follow either the Junior Cert or LCA course. Foundation Level courses may be suitable for students with mild general learning disabilities. There are a range of Foundation Level courses available.

School-designed programmes

Schools are coming up with their own solutions to assist students whose needs are not being met by any of the traditional or alternative curriculum models. Taking the local environment, the resources of the school and the interests and abilities of the student(s) into consideration is a good starting point.

The NCCA Guidelines for Teachers of Students with General Learning will also be of benefit to teachers delivering programmes to pupils with mild general learning disabilities at second level. Subject areas covered for teachers of pupils with mild general learning disabilities are English, geography, history, science, home economics, religious education, physical education, social personal health education, art craft and design, music, technology education and mathematics. It is intended that these guidelines be utilised in tandem with other post-primary curricular options. Copies of these subject guidelines (PDF format) are available on the NCCA website, the exact link is <http://www.ncca.ie/eng/index.asp?docID=293>

4.13 Summary

Curriculum modification is necessary for learners with special needs even when an inclusive curriculum is being followed. The suggestions posed by Sear are applicable in the case of Botswana. Botswana has an enabling environment for such modifications to be done given that government policies do recognise the need to educate equally all citizens. Accommodation, adaptation, parallel curriculum outcomes and overlapping curriculum as modification strategies would require that teachers have training in handling learners of diverse abilities and be committed to teaching. Special education teachers and the regular teachers would need to work together and this will be facilitated by having regular teachers also having some training on handling special needs learners. The support structures need to be strengthened to facilitate this cooperation. The assessment of such a modified curriculum will also benefit from the preparedness of teachers and the schools as well as the national curriculum unit. The suggested modifications of assessment procedures and materials need coordinated efforts on both the government side and that of service providers (teachers). Parents also have to be brought on board even more than for regular learners as they need to play a role to ensure that the special need child does not fall behind with their school work.

4.14 References

1. Department of Education and Science (1993) Report of the Special Education Review Committee Available from Government Publications 01-6476834 €12.70 Catalogue No. E/109
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Unit 5

Transition Education, Life Long Education

Content

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5.1 Introduction

When we think about current adult education in the context of the uneven and contradictory social and economic impact of globalization, it necessarily implies thinking about the transfer from the paradigm of lifelong education to the paradigm of lifelong learning. We shall examine the essential quality involved in the social significance and the political dimension of each of these paradigms, because, since the post-war period, both became innovative educational policy strategies capable of mobilizing and transforming society. We would like to stress the importance of rethinking the role of adult education today in the light of the responsibilities shifting from the state to individuals, arising from the implications of this transition of paradigms: we do this by framing it in the context of the socio-productive restructuring movement, which speeded up the move from the so-called model of qualification, associated to social emancipation, to what is known as the model of competence (later competences), which is associated with individual empowerment. Therefore in this article we intend to question this new policy direction, which is associated with a conceptual and methodological shift in adult education practices, by using the prism of a critical sociology of education.

First thoughts

This article addresses the establishment of an international discourse. We therefore briefly introduce definitions of basic concepts that will be discussed throughout the paper. The first main concept is *adult education*. We accept that this concept is pluralist: it is ideological as well as technical. The concept of adult education is mostly understood as a social and human right (Gadotti, 2011), approximating the UNESCO agenda. We follow those critical educators (Lima, 2003; Torres, 2008) who perceive the concept of *adult learning* as an expression that opposes the humanist-critical roots of the popular tradition of *adult education*. There has been a shift in paradigms from [120] Rosanna Barros *lifelong education*, where the concept of *education* is seen as a collective entity and a state obligation, to *lifelong learning*, where the concept of *learning* is seen as an individual entity and a personal duty. This has led to a loss in the broader view of adult education as a key issue related to social themes such as inequalities, power relations and political struggle for a better world for all. The research question guiding these reflections is: who gains most and who loses most in this conceptual and paradigmatic shift in the public policy agenda of adult education, especially in the European context? We shall now look at this issue in greater depth.

5.2 The main characteristics of adult education under the paradigm of lifelong Education

In the history of modern systems of education the so-called world crisis in education which took place at the end of the 1960's would mark the beginning of a gradual shift in the understanding

given to adult education, and in more general terms to the actual nature of education itself. In this way, the *lifelong education movement* that emerges at this time represents a real turning point in thinking about education, due both to its worldwide dimension as well as its socio-political stance. The instability of our economic, political and social reality, in the context of the crisis in the welfare state, and the central role played on the international stage by UNESCO in the area of education, is the basis for the context in which the *concept of lifelong education* is reintroduced in debates about education. We say re-introduction due to the fact that some of the key theoretical ideas about lifelong education actually date back to the early 20th century, based mainly on the theories of Dewey, Smith, Lindeman and Yeaxlee. Their work represents an indispensable legacy, even though they referred to another socio-political context and a different age, when central assumptions about adult education were forged, namely: the understanding that education does not end when a person leaves school, insofar as education fulfils the role of ensuring the viability of life in society, irrespective of age (Dewey, 1916); the idea that adult education, as a basic necessity to a country, must be expanded, insofar as it helps to build citizenship, and this is then perceived as universal and desirable throughout the life of its citizens (Smith, 1919); the growing importance given to experience in the learning process amongst adults, as well as the importance attached to discussion groups as an appropriate methodology for adult education (Lindeman, 1926); also the idea that human needs are part of the social fabric of a powerful socio-political system that is contained within a context of pedagogical action in adult education (Yeaxlee, 1929).

An interesting outcome of our research on the genealogy of concepts in adult education (Barros, 2011a) is that despite the intensity of these pioneering debates we should point out that there was a kind of gap between the 30's and the 50's as far as adult education's conceptual heritage is concerned, and this has helped to gauge the effective use of the *concept of lifelong education* in public debates on adult education. However, when it resurfaced in the sixties, the dominant understanding given to the *concept of lifelong education* comprises three basic dimensions: one is a severe criticism of the school model of formal education¹; a second dimension is related to the need to ensure a form of 'lifelong' education which keeps knowledge up to date and allows adults to keep up with technological developments in society (UNESCO, 1960); and a third dimension promotes equal educational opportunities and access to permanent and effective social promotion². These three dimensions create a concept and an ideal out of *lifelong education*, and indeed two of these fundamental dimensions lead us to think in humanistic terms about educational policies for adult education. Looking at the work and effects of policies, we should underline that the main international body to have an effective role in the promotion of the lifelong education paradigm was UNESCO. Under its patronage several documents were produced, among which stands out the influential Report coordinated by Edgar Faure et al. (1972)

Learning to Be – the world of education today and tomorrow. The concept of lifelong education that was developed has implications for the entire educational process, and includes all

its forms, expressions and moments in order to ensure the implementation of an innovative strategy in the global education of children and young people, and to prepare adults to pursue their autonomy and freedom with a sense of social justice. Lifelong education is understood, in this context, as an educational project that is continuously inter-relating with the individual as well as the social dimension of education, and is aimed at the construction of a 'new man', and the offer of a humanist system of collective values. This vision of lifelong education contains explicitly political choices and represents a project which has an ideological nature, a utopia (Faure et al., 1972, p. 143). Since then the understanding given to the concept of lifelong education would become more and more clearly linked with the political dimension of education (Lengrand, 1989; Lima, 2003). The concept is now commonly associated with, on the one hand the restructuring of school systems, and on the other hand, the drafting of social transformation projects. In other words, associated with lifelong education is a new vision and interpretation, both with respect to the educational process at the various levels of education, as well as to the perception of a common destiny for mankind.

Given that the *lifelong education movement* has developed within a global framework of strong criticism and opposition to the school model, it is no wonder that the repercussions in the field of the discursive and pedagogical practices of formal adult education would be profound. Besides, this period corresponded to an expansion, unprecedented in the history of adult education, particularly with regard to its non formal and informal characteristics: in fact it went against the two structural axes that characterize the whole school model - the spatial and the temporal axis. In this way it reached beyond the public space of the school with regard to its educational practices, and stepped outside the temporal constraints of the inflexible logic of a school education by introducing the possibility of negotiating schedules and timetables with a degree of flexibility. The aim was to allow education to be present at all stages of life (UNESCO, 1976). In the Faure Report it says in relation to this that 'the educational enterprise will only become efficient, just and human by undergoing radical changes affecting the essence of educational action, as well as the time and place for education' (Faure et al., 1972, p. 142). This alternative understanding of the concept of education assumes, in the same way, a break with what this report calls 'preconceived ideas about instruction –it was for the young and took place in schools– prevented people generally from conceiving of lifelong education in normal educational terms' (ibid., p. 142). The role of the school with regard to *lifelong education* changes completely, to the extent that the basic education that is acquired there is now perceived as only a prelude, and whose aim is as follows: to provide adults with the best means to communicate with each other; to promote the ability to obtain information independently; and to create a more cohesive society where individuals can communicate, work and live cooperatively with each other. So, *lifelong education* represents a lever to change the entire understanding of the modern concept of education. We can safely emphasize that the *lifelong education approach*, with regard to the restructuring of school systems and the entire educational process, aims to reunite what the school system of education has locked up. For this to be achieved, a restructuring of this nature would involve at least [122] Rosanna Barros

two different consequences: first, the separation between the idea of education and the idea of there being a right age to learn; and second, the total loss of significance for the notion of success and failure at school, with a focus now only on personal growth and the creation of a path towards 'learning to be'. The political dimension of education, which has been recognized and debated vigorously with regard to *lifelong education*, now represents its most significant contribution in the light of a critical sociology of education (Giroux & McLaren, 1997; Mészáros, 2005; Zinn & Macedo, 2007), by proposing a theoretical path as well as a plan of action, in this case related to the vision of building a new society, a *learning society* and a new type of city, the *educational city*. We believe this vision is in fact the essence of the *paradigm of lifelong education* and also of what is understood as adult education in the Faure Report. The established political and educational agenda is based on a theory of action with clear objectives of social transformation, and these are progressively examined throughout the report. This new society, the society of learning, will only make sense in the context of this new understanding given to education, which is seen not only as school education but as *lifelong education*. Twenty-one principles and recommendations are presented in this document, and these show the way forward to make the 'today and tomorrow's' reality as close as possible to the idealized utopia. Thus, in the report Faure defends his point of view from a firm political-ideological positioning that leaves no doubts about the nature of social transformation behind this vision, that 'it is out of the question for education to be confined, as in the past, to training the leaders of tomorrow's society (...) education is no longer the privilege of an elite' (Faure et al., 1972, p. 160). The effect of such policies on adult education and learning clearly means social emancipation for all. According to *lifelong education*, the idea of the educational city, which is part of the underlying vision of the learning society, is based on the principle that when we accept that education will increasingly become a primal need of each individual we will have to invest all our efforts to broaden and expand its scope mostly in two necessary directions: first to develop other types of schools, polytechnics and universities which benefit from other forms of teaching, as well as from other types of educational relationships, and which are constructed from existing models; and secondly, the creation of other types of institutions in the city, both public and private, which will be able to embrace an educational role in the various institutional dimensions of city life by maximising the existing educational potential in local communities. It is in this sense that the city can also be understood as being educational. This is a deliberate comparison with the Athenian educational ideal of a *paideia*, and in this way the educational city represents 'a school for civic sentiment and fellow-feeling' (Faure et al., 1972, p. 162).

In Faure's report the vision of the educational city and a learning society, which is a vehicle for social transformation whose fundamental purpose is to democratize education and democracy itself, is as important today as it was then. According to this political-philosophical perspective a truly democratic education is the basis for guaranteeing 'man's right to be' (ibid., p. 162). But the goal of democratizing education implies changing the traditional bases of the relationship between society, the state and education. All layers of society, and particularly those referring to the political, economic and family spheres are called on to make a real contribution, because, in

this perspective, a society of learning, besides being democratic, also needs to be mutually supportive, and in this context it is assumed that ‘the fight against ignorance is as important as the fight against hunger’ (Faure et al., 1972, p. 235). Lastly, this learning society is not only a democratic and supportive movement but is also pacifist in nature. Its proposed path of action will reverse mankind’s warlike propensity. The plan is manifestly utopian but it doesn’t diminish its power of action. We think the utopian vision is simply reinforcing the underlying principle of today’s educational debate (still). That is to say, as Paulo Freire put it, it is the torch which lights the way towards the creation of a more ‘human’ society.

However, mostly on account of its somewhat ideological slant, the *concept of lifelong education* has created controversies, divisions, schools of thought, and counterclaims, and disagreements. This has led to a general reflection about education on an international scale, as well as to the very particular interpretation given to literacy and adult education as part of a liberating project (Freire, 1975; Gadotti, 2001; Gelpi, 1983). Nevertheless, and despite its utopian branding, *lifelong education* has actually made its way into the public educational policy-making process. It did so on the back of UNESCO (1985, 1997), which in some ways helped to put it on the global stage and partly explains the sociological and historical importance that the *lifelong education perspective* has attained, as Lima has stated in this respect, ‘we need only to remember that lifelong education as a *continuum* between the education of children, young people and adults, is seen in certain countries as cornerstone work of the welfare state, and which goes hand in hand with other social policies and also redistributive policies’ (Lima, 2003, p. 129).

As so many of its minor failings have been pointed out and given that forty years have passed since Faure’s Report, allowing us time to step back and look critically, it is not difficult to understand that many of the promises of *lifelong education* have not been kept: they have remained dormant, with much of their transforming power untapped, both in terms of national systems of education, as well as with regard to society itself (Comissão Europeia, 1995; UNESCO, 1996, 2009a, 2009b; Organisation for Economic Cooperation and Development [OECD], 2009). In the words of one of lifelong education’s foremost advocates: ‘from the theoretical point of view the principle [of lifelong education] has progressed considerably, but in practice the situation is less impressive (...) it does not appear that the set of traditional structures has in fact been substantially modified’ (Lengrand, 1989, p. 9). Yet despite these promising discursive practices, with regard to the actual educational practices a fundamentally school-based logic has persisted.

There is no single reason behind a change in the world of education since those times, especially since it has been the humanistic approach to *lifelong education* that has surrounded the issue. But the growing *fin-de-siècle* dominance of a public discourse about a pragmatic approach to social and educational policies, inevitably linked to private groups interested in taking advantage of the crisis in the welfare state in core countries in order to expand the market, would constitute a powerful force for change (Sousa Santos, 2005). What we have here is for the most part an ideologically constructed crisis which draws attention to the difficulty of putting the political

principles of *lifelong education* into practice, and invariably points to the economic unsustainability of a *lifelong education for all*. Alongside the transformation in the socio-economic climate caused by the oil crisis, a new global swing to the neo-liberal right at the end of 20th century had a significant impact on what inspired the U-turn with regard to adult education. [124] Rosanna Barros

5.3 The main characteristics of adult education under the paradigm of lifelong learning

In the second half of the 20th century, it was the *lifelong education* approach that was mostly behind the educational policies of the welfare state and social thinking in social sciences, but in the 21st century the concept of *lifelong learning* quickly replaced it. This change in emphasis is a consequence of the domination of neoliberal globalization (Barros, 2009). It concerns a change in the socio-educational panorama which in turn reflects the effects of a broader change that took place in the international political and economic context (Falk, 1999; Bauman, 1999; Bourdieu, 1998, 2001). What took place was that the *Keynesian consensus* of post-war international politics, according to which education is perceived primarily as a citizen's right and which the state should provide, was replaced by a new agreement, the *Washington consensus* of post-cold war world international politics, under which education was to be seen first and foremost as a service to the consumer that the state should liberalise (Field, 2002). To understand the impact of the new neoliberal political consensus we have to be aware of at least three other levels of consensus, intended to reverse the political-ideological assumptions of the Keynesian consensus: the first one concerns the future of the economy (World Bank, 2000); the second is about development policies (McMichael, 1996); and the third refers to the role of the state in the economy (World Bank, 1997). Out of all this arises in the first instance a new global economy associated with a growing global prevalence of a financial and investment way of thinking over the real economy (Chossudovsky, 1997), as well as the emergence of a new international division of labour (Tilly, 1995; Olesen, 2008, 2010).

The result has been a profound impact on the social structures that shape the socialization of individuals, with the result that, on the one hand, there has been a huge increase in the social divide between classes within national societies, even changing the norms of social stratification which had prevailed since the post-war period; and on the other hand, it has caused a break with the collective cultural values of the construction of the sense of belonging and social identity, by putting constraints on and personalizing their own symbolic classification criteria of social and cultural relationships. There is therefore a new economy, a new type of politics and a new social stratification emerging in our contemporary history. And it is in this broader scenario of social transition that we witness the shift from the hegemony of the perspective of *lifelong education* to that of *lifelong learning*.

The concept of *lifelong learning* was the dominant political outlook at the beginning of this century. However, this is not strictly speaking a 'new' concept, as Lima points out, 'it is

important to understand the centrality assigned to lifelong learning starting with its secondary or rather peripheral emergence in the 1970, as the satellite idea of the then key concepts of lifelong education or continuing education' (Lima, 2003, p. 130). It is important to remember that many buzz words in the field of education received official recognition within the context of the main international political bodies. Thus, the promotion of the *lifelong learning* paradigm is especially linked to the OECD and the EU, as both produced several policy documents, such as the OECD report published in 1973, *Recurrent Education – a strategy for lifelong learning*, (OECD, 1973) and after that, the document entitled *Memorandum on Lifelong Learning* published by the European Commission in 2000, one of the most important and disseminated policy documents in 21st century. In the widely distributed *Memorandum* the concept of *lifelong learning* is understood as: 'any learning activity with an objective, undertaken on a continuous basis and aimed at improving knowledge, skills and competences' (Comissão das Comunidades Europeias, 2000, p. 3). The distinction is clearly made between two of the dimensions associated with the concept: one of them is expressed in the term 'lifelong', where 'the emphasis is laid on time: learning during a lifetime, continuously or periodically' (ibid., p. 3); and a second dimension is expressed in the term 'lifewide' in all areas of life, which:

draws attention to the dissemination of learning, which can take place in all aspects of our lives (...) reminding us that a useful and pleasant acquisition of knowledge can take place, and indeed does, within the bosom of the family, during our leisure time and in our continuing social and professional life. (ibid., p. 3)

It is widely known that the concept of *lifelong learning* has become quite dominant today in transnational policies concerning adult education, and has enjoyed a degree of recognition in particular in the context of European policies since the 1995 White Paper (Comissão Europeia, 1995) and 1996, the *European year of lifelong learning*. This event heralded the launching pad for European guidelines concerning political agendas in the field of national mandates for adult education. Since then, within this 'silent enrolment explosion' taking place in the field of adult education (Tuijnman, 1996, p. 26), it is possible to pick out the most frequent interpretations presented in public debate about the concept of *lifelong learning*, and which form the basis of its widespread adoption in the new canonical educational *ethos* present in many of the contexts of today's adult education, namely: one in which it is presented as being a key tool for adaptation to change, both by individuals as well as organizations and society in general, to the extent that the concept is perceived as being the best educational tool to increase flexibility and economic competitiveness; in another sense it is seen as a policy of social cohesion and for combating exclusion through educational programs intended for adult audiences considered problematic; another interpretation presents it as a factor of employability and professional promotion; and a final one that shows it as a strategy to develop consumer-citizen participation in the social, cultural and political spheres of their societies. It is significant that since the turn of the century, all European governments have been implementing adult education policies referring to the *lifelong learning* paradigm that are eminently vocational and technocratic in nature, as we found

during the course of a recent research project on European countries' national reports to CONFINTEA VI (Barros, Guimarães & Lima, 2012).

This widespread adoption has been behind the latest turning point to take place in the history of this sector, and which Jarvis sums up as follows: 'the commoditization of education for adults' (Jarvis, 1995, p. 242). In public debate in this field there was a conceptual and theoretical reconstruction in adult education, based on sources from management, especially on the theory of human capital and the theory of development of human resources, whose ideological basis is neoliberal (Milana, 2012). The main consequence of this is that the meaning and purpose of education is now reinterpreted in terms of productivity and competitiveness. This political-ideological trend is reflected in the majority of adult educational practices on offer in the capitalist centres and southern European developing countries³. This can be illustrated by the Portuguese case, where we find a predominance of educational political discourses oriented towards adaptability, employability and the production of competitive advantages in the global market (Barros, 2009, forthcoming).

In this way, the economic, political and cultural bases that underline the adoption of *lifelong learning* in a political sense have contributed to the spread of a new mission statement for adult education: as a management tool of the work force; as a means to prevent forms of social conflict; and as a tool of adaptability. The first statement sees adult education as a central instrument for the management of the workforce (European

Round Table of Industrialists [ERTI], 1995). The premise for this interpretation is based on changes in the economic situation recorded in the last decades, which are presented as inevitable, in particular with regard to the dramatic transformations that have occurred in the world of work. These changes have led to the systematic erosion of outward signs of well-being that marked the so-called 'thirty glorious years', which followed World War II, namely: full employment, labour rights, universal social rights, the gradual tertiarisation of business, the exponential growth of consumption, increased leisure time, the expansion of access to education (Mishra, 1995, 1998). The oil crisis of the 70's is the milestone that marks the end of this cycle and the beginning of another more austere age, which has been labelled the period of 'thirty disastrous years' by the most critical sectors (Forrester, 1997; Chomsky, 2000), or as the 'new economy' by the more conservative sectors (Taleb, 2007; Krugman, 2008). The most representative characteristics of this second cycle are: the emergence and maintaining of mass structural unemployment; international competition with regard to work; and rapid and constant technological innovations. In this context, adult education, now restructured for a market context which also favours a *lifelong learning* approach, is there to provide a range of services, which, from the point of view of human resources management, are seen as the key to increasing efficient productivity (Comissão Europeia, 2002; Consejo Europeo, 2011). These educational offers are presented as a required investment for the entire manufacturing sector that will represent gains in competitiveness in the national economies in the global market, with the promise of future returns and benefits for the population.

The second statement sees adult education as a means to prevent forms of social conflict (OECD, 2003, 2005). This interpretation is based on the promotion of social cohesion and the fight against social exclusion, which are essential in a context where the result of social options and policies of the new economy, namely the neoliberal consensus, is the systematic churning out of the unemployed, the spread of poverty, the generalization of insecure employment and a sharp drop in real wages, as well as an unprecedented creation and concentration of wealth, while at the same time there is more social inequality and a rising rate of bankruptcy (Martin & Schumann, 1996). The result of this is an imminent threat of civil violence on the part of a growing section of the population that is marginalized by an uncaring political class. In Europe, the social consequences of the erosion of the welfare state (which reflected the political commitment that permitted the articulation of democracy and capitalism on behalf of social justice) gave rise to an unprecedented increase in the scale of long-term structural unemployment, creating social framework that is harmful from a social point of view, but not necessarily from an economic perspective (Boyer, 1999). The massive amount of existing offers of education for young people and adults, both in the public and private sector, which are presented as new opportunities for employability (European Association for the Education of Adults [EAEA], 2006), can be interpreted as a way of curbing violence, a sort of safety valve or palliative factor in the social management of unemployment, inspired by the classic social control mechanisms, and well-known among conservative social policies. It is all about maintaining social order, despite injustice, to ensure the proper functioning of the economy (George & Taylor-Gooby, 1996). The third statement sees adult education as a tool of adaptability for the benefit of the working population. This way of seeing it is based on the key assumption that it is employees themselves who are responsible for maintaining their jobs, and this necessarily involves self-empowerment and the constant updating of skills. Here the future is presented as inexorable with technological change and development seen as something inevitable and unstoppable (Beck, 1998, 2000). In this context, it is argued that employees must learn to learn throughout their life, in an aggressive logic, in which, along with the current systematic appeal for training and *lifelong learning* there is a renewed highlighting of individual responsibilities for the acquisition of technical knowledge (OECD, 2000) and ‘competitive skills’ (Lima, 2003, p. 129). In short, the current understanding of the adult education mission is founded on these three central political statements. From this it follows that the adoption of the perspective of *lifelong learning* represents, essentially, a partial and instrumental appropriation of the field of adult education, whose potential for social transformation and social emancipation, which was developed in the framework of a philosophical tradition with critical and radical roots, has been strategically confined to a state of latency. In this way its use has been strictly limited to what best suits the market, that is, it is confined to the scope of a philosophical tradition with technocratic and vocational roots.

5.4 Mapping the effects of policy changes: from collective qualifications to individual competences

As we have seen, *lifelong education* and *lifelong learning* each have a different approach and a firmly based way of understanding a definition of adult education, especially in terms of how intentionality or mission is attributed. In other words, each of these models involves the adoption of specific underlying principles that imply very different practices.

This discussion (occurring on capitalist centres) could be situated within the socio productive restructuring movement which began in the last quarter of the twentieth century. This shift of contexts has implied the transfer of the *model of qualification* to what is known as the *model of competence*, which gave the concept of competence (later competences) and empowerment such a strong central position within the framework of today's adult education. This is so both at the level of its discursive practices, as well as formal, non-formal and informal educational practices. From this perspective particular attention is drawn to the fact that contemporary educational practices carried out with adults which appeal to the concept of competence, particularly those with a non-formal character, turn out to reveal an *ethos* whose main characteristic is to put education at the service of business interests (Boshier, 1998). Most of the current adult education discursive practices dominant in European policy documents leave us with little doubts about this. In fact, the *lifelong learning* paradigm provides an understanding of the concept of competence that appears to be deeply instrumental and applies overwhelmingly to key benchmark skills that underpin a logic of educational results (meaning productivity gains for companies, and competitive empowerment for employees). Despite theoretical work to accomplish an acceptable educational definition of competence (Gillet, 1998; Perrenoud, 1999; Le Boterf, 2000), there is not yet a consensus on this. In mapping the effects of changes in terms of this conceptual domain, we see that the shift in thinking from competence to competences was mainly the rolling out of a new European Union policy concept.

The understanding of the evolution of the *lifelong education* approach towards that of *lifelong learning* requires an understanding of some continuities (these include the critique component of the school system in the Faure Report, which allowed a more inclusive view of formats and innovative contexts for adult education and learning), as well as an understanding of some tensions and ambivalences stemming from an accelerated process of socio-productive restructuring. There has been a change in models, from the Taylor-Ford production paradigm to a new flexible or post-Ford production model. In the present context of hybrid structural transitions, we can observe that the concept of competences is winning the central ground. The current hegemony of the concept of competences (particularly the prevailing understanding which reduces it to the concept of professional competence) is firmly bound to the interpretation that has traditionally been given to the concept of qualification. The spread of the concept of qualification, which came to the fore in the late 1940s, was based on two main axes of socio-political structuring, which together would help to sustain a model of social organization that would dominate throughout the Western world. These two axes were: the system of collective agreements, aimed at classifying and ranking jobs; and the systems of education and vocational training, with the task of classifying and organizing knowledge around diplomas and certificates.

It is because of this that there is now a triple meaning attached to the current understanding of the concept of qualification. First, in terms of employees, referring to the knowledge and skills which stem from their vocational training; next in terms of the job, indicating the standard requirements required by the job; thirdly in terms of classifying the employees in the hierarchy of professional categories, each with their own salary and legitimate status, and which then serve as a basis for the idea of promotion and social emancipation (Kovacs, 1994; Schwartz, 1995; Bellier, 2001).

This qualification-based model, because it was so highly structured, allowed for a society in which individuals *a priori* believed that a certain level of qualification would correspond to a certain level of social status. This *status quo* acquired by qualifications can imply two things. First, it allows a collective feeling of belonging, which is important for the construction of professional and social identity among workers (Olesen, 2008), since different qualifications had a certain secure market value, that is, they would be the holders of a particular value that is unchanged by the context. Second, it allows for the creation of concrete social promotion channels resulting from a legitimate and permanent collective bargaining, based on collective agreement pay scales, out of which came, for example, the statutes of civil service careers.

However, this sort of social structuring came to a crisis at the end of the twentieth century, giving way to a time of uncertainty in which new modes of organization and social stratification emerged, whose rules are not always easily understood. This is a paradoxical situation in society, since, within the framework of changes in the relationship between the economy, the workplace and between education and training systems, some rather ambiguous situations have arisen (Bron, Kurantowicz, Olesen & West, 2005). For example, alongside a trend towards the devaluation of certificates and diplomas, and an ending to meritocratic opportunities for social advancement and access to jobs, there co-exist different ways of social, symbolic and professional classification, not unlike the transition in hegemony from the qualifications model to the new model of competences. In this way a new set of discursive practices emerge that point to a growing inadequacy of the qualification model. These appear to be based on changes in the employment system, which now requires technical skills to deal with the innovations introduced into these organizations by new information and communication technologies (OECD, 2006). Therefore the model of competences is now presented as a response to the inadequacies diagnosed in the model of qualifications. And, as is to be expected, this response springs promptly from the theories of human resources management, and in particular from the schools of thought of the so-called forward From thinking and predictive school of management (considered as key elements of the new economy).

The model of competences, on which the recent models and provisions for non formal education for adults are founded, presupposes two things: first, that there is a connection between competence and action, in which experience is the central concept; and second, that competences refers exclusively to the person, by which competence implies a new assumption of responsibility at the individual level, although previously it was at the collective level, as well as

the social recognition and encouragement of this assumption of responsibility through the concept of empowerment.

The influence of this shift in responsibilities is quite profound for adult education, both in the political discourse about the field and in the underlying logic of educational practices regarded as innovative, especially in formal and non-formal methods with singular emphasis on the recognition of prior learning (Barros, 2011a). These new practices based on competences are, of course, the ones which are given the most support today by the supranational financing policies based on the *lifelong learning* approach, which, in the context of the European Union in turn mark the agenda both of educational policies carried out by the Ministries of Education, as well as the social policies carried out by the Ministries of Labour of the various Member States. Thus there appears a new ideological-political rhetoric that articulates the field of adult education with the world of training and employment by means of various educational neoliberal governance mechanisms (Dale, 2005; Barros, 2009). This new way of thinking comes on the back of the idea of an inevitable technological shock (Fukuyama, 1992), requiring people as well as organizations and institutions to constantly adapt to evolving technologies, at the risk of countries losing their global economic competitiveness.

Considering the close relationship today between the new economy, flexible work practices and educational practices aimed at competences and skills, it is not surprising that the terms, approaches and concepts used in the context of management theories often appear in discursive practices on education in general and about adult education in particular. Moreover, in turn, and in a similar way, some terms, approaches and concepts in the field of education are appropriated and re-conceptualized under the auspices of management and organizational flexibility (Tuijnman, 1996; Lima, 2003; Barros, 2011b).

These much used flexibility strategies are the most visible aspect of an ongoing neoliberal globalization (Soros, 1987; Greenspan, 2007), and are based on four axes of action: in numerical flexibility, by which businesses quickly reduce or increase their workforce, basing this on the concept of additional human resources, which is used in the form of temporary work, short term contracts, part-time, and occasional work. The result is that this kind of flexibility, which is very popular at the moment, leads increasingly to the casual nature of employment; a second axis, which supports the first, is flexibility in salaries which, in essence, breaks wage agreements standardized by the model of qualifications, reducing expenditure on employees and increasing profits; a third strategic axis is called distancing which functions by subcontracting both goods and services; and finally a fourth axis is functional flexibility, which appears closely associated, on the one hand, with the general idea of multi-competences, and on the other hand, with the idea of transferability of these same competences, by applying these in such a way the desired results for the organization will be achieved. The result is the change from a model of salaried fixed employment to a job model which is becoming increasingly autonomous in that it transfers the responsibility for job management to the one who carries it out. This model of competences, which has come to take precedence in the discursive practices of adult education, is widely

advocated by management theorists. For example Senge (1990) defends, under the new information and management systems, the appliance of five key aspects, designed with the explicit purpose, it should be noted, to create new models of thought among the workforce that may contribute towards the liberation of their collective aspirations. All these transitions are a reflection of the latest evolution in capitalism, which has been transformed into a wild capitalism, which leaves its marks on the whole of today's social life, making it de-standardized and individualistic and creating a constant climate of uncertainty and risk. We live in times of growing insecurity arising from the instability of our professional situation, which is also reflected in the physical and psychological health of workers, who are becoming increasingly isolated and socially unprotected. This is exacerbated by increasing social inequality, exclusion and violence.

Seen as a whole, this situation gives society a general sense of disenchantment and alienation, allied to a paradoxical situation in which the adult is incited to action while at the same time is discouraged from acting, that is, people are encouraged to be compulsively active while receiving education. Furthermore it is to be noted that they are engaged in *lifelong learning* in all aspects of their lives. These individuals are told not only that their employability is in their own hands but also that the possibility of them becoming competent workers is also their own responsibility. But a competent worker at the beginning of the 21st century is a well-adapted one, whose personal action is reduced to the sphere of their positive professional commitment, and their positive performance as a consumer. Therefore, a citizen that is competent to work and consume cannot be too active in asking questions and in querying unfair social realities. In a way, the competences model of lifelong learning seems today to be actively advocating the consolidation of a 'collective free aspirations world'. We believe that this neoliberal utopia can be upset by another kind of commitment in educational work.

5.5 Final thoughts on interpreting shifts in adult education policies

The impact of these transformations in adult education is having a profound effect, and is generating a paradoxical situation. In fact this has traditionally always been a field of knowledge in which critical perspectives have played a part, and this has given rise to most of its inheritance and heritage in terms of theoretical and pedagogical conceptualisations developed by a broad set of humanist educators, many of whom agree with the general assumptions of the paradigm of *lifelong education*. Despite this, the phenomenon of education and training for competences invaded the field by means of its polyvalent rhetoric, which comes from the normative, political-philosophical discourses which are in essence administrative and managerial (Fragoso & Guimarães, 2010).

We can safely conclude that one of the main differences between *lifelong education* and *lifelong learning* comes down to the role and mission that is assigned to adult education itself. Thus, as far as lifelong education is concerned adult education political mandate was essentially a social transformation plan, based on a humanist ideology, seen as a vision of building a learning

society, made up of interdependent institutions and committed to the safeguarding of social justice and the *res publica*. But with respect to *lifelong learning*, the political mandate applied to adult education is seen as a social adaptation policy, following a neoliberal and individualistic ideology, represented in the vision of building a learning society, made up of autonomous organisations that provide qualifications and which are committed to safeguarding private interests. They are two opposite poles of a *continuum* with contextual characteristics to which researchers should apply even greater tools of analysis.

It is clear to us that any debate about adult education will have to be conducted in the domain of political principles and values (Barros, 2012). We disagree here with discursive practices of mainstream European Union and OECD policy documents. We take the line of some critical educators and researchers who see the evolutionary path of *lifelong education* towards *lifelong learning* mainly as a division rather than a continuity. Indeed, the concept of *lifelong learning* may seem on the surface to follow in the footsteps of the underlying principles of the concept of *lifelong education*. However, as we have highlighted in this article, the way many of these concepts was understood underwent drastic change, and some of them re-appeared dressed up in technical and de-politicized language. In other words, we believe that the current interest in *lifelong learning*, far from implying a renewed interest in the political social ideals of the seventies, as at first glance the discourse appears to suggest, is in fact based on exactly the opposite: its deliberate and continuous erosion. Boshier goes even further and is quite candid in his criticism when he says that ‘if lifelong education was an instrument for democracy, lifelong learning is almost entirely preoccupied with the cash register’ (Boshier, 1998, p. 5). Basing our arguments on the principles of critical social thought, we can interpret these changes and observe that the original socio-political vision of *lifelong education* that advocates social justice and a ‘co-naissance’ (Wildemeersch, 2010), has been subverted; it has been replaced with *lifelong learning* that advocates social peace, which turns it into a new form of oppression (lifelong and lifewide). Faced with this, many critical educators and researchers have sought to revive or at least give visibility to educational practices in the field of critical and engaged adult education within their spheres of influence, albeit on a somewhat small-scale. They carry this out on a more informal level, potentially still adhering to a tradition inherited from the principles of a socio-educational and socio-cultural emancipatory perspective (Ackland, 2011), or based on popular education (Martin & Shaw, 2006), and community interventions for local development (Wildemeersch & Kurantowicz, 2011).

These are in some way strongholds, normally seen as radical or critical (and usually not subject to EU financial programmes), and represent small pockets of political resistance to the dominant management-based *ethos* currently associated with adult education and learning in both public and private institutions (which appears to have succumbed to a market-based rhetoric). It is urgent that counter arguments of a critical and irreverent nature should be applied to these overwhelmingly dominant discourses, so that at least one clear lesson concerning the historical heritage of adult education should be drawn: it is a field with a long conceptual tradition linked

to the safeguarding of the democratisation of society, the promotion of human rights and social transformation. It will, thus, help to create more enlightened communities and consequently societies where there is a greater sense of justice and solidarity. This is what the participants in the WSF (World Social Forum) discussions have been comprehensively demanding for the benefits of all.

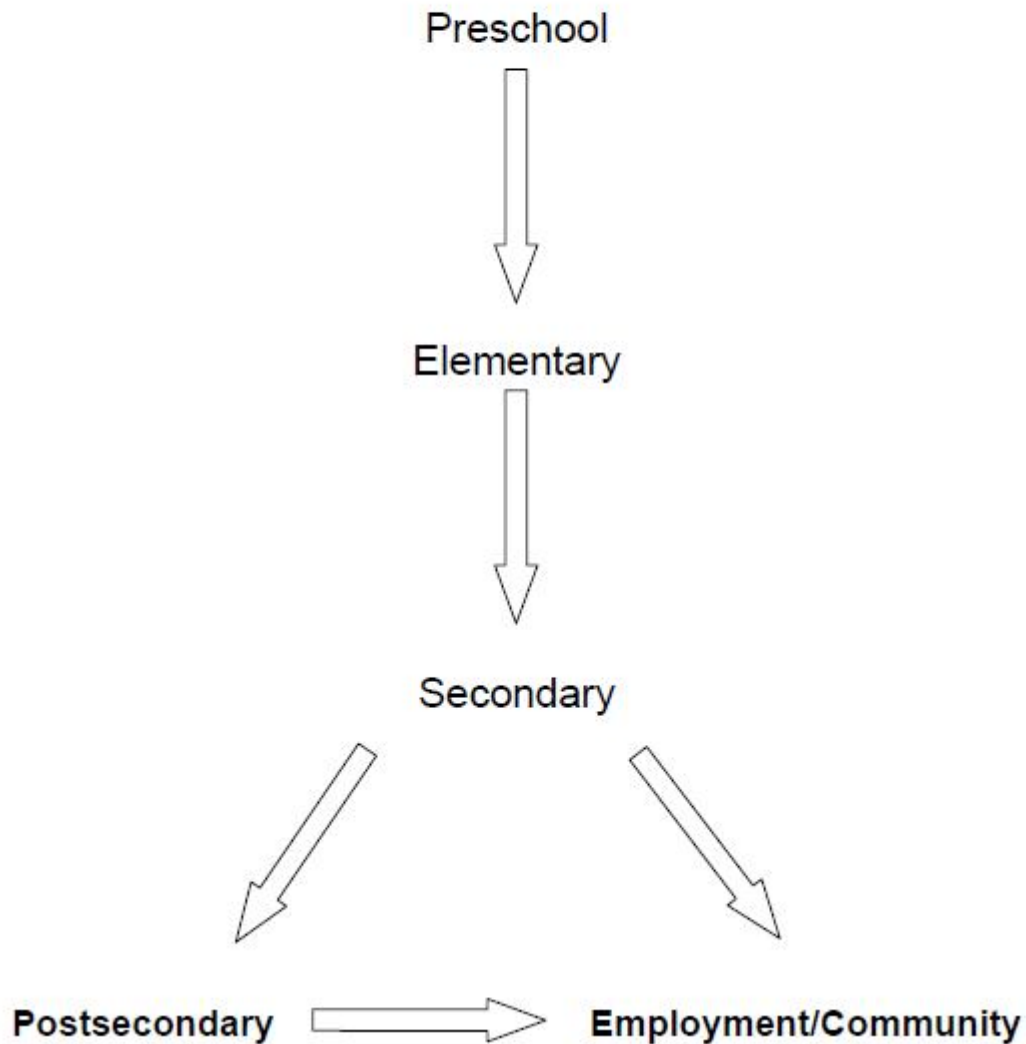
5.6 What is Transition?

Transition is a passage from one place or time to another. In education, it is used to describe the passage from school to adult services and full community participation. It is a bridge between the security and structure offered by school and the opportunities and risks of adult life (Will, 1989). For the student with a learning disability, transition offers challenges that require guidance and assistance from families and others to help him or her effectively deal with these challenges.

5.7 Where Do We Begin?

Transition should be considered an outcome-oriented process. This means that the process first identifies future-oriented visions and life goals of your adolescent and then provides needed instruction or services to fulfill those goals. The process begins with identifying life goals developed by a "transition team" ideally involving school and community professionals and you and your adolescent. The plan then identifies services which assist your adolescent in living successfully in the community. The transition process should include ways to facilitate overall adult adjustment by addressing family, emotional, social and daily living issues as well as educational and vocational programming. The final step in the plan involves developing transition goals and action statements written into the IEP.

TRANSITIONS



What is an IEP Meeting that focuses on Transition Planning?

This involves meeting(s) where the IEP is planned to specifically focus on: 1) long-term life skills and employment goals; 2) short term objectives; and 3) actions to accomplish these goals. It includes services to be provided; programs and personnel to be implemented to help accomplish the goals; and objectives designed for your adolescent.

When does it take place?

Kansas law requires that transition planning in the IEP begins when your adolescent reaches the age of 14. However thinking and dreaming about the future occurs throughout your child's life.

Who attends?

The "transition team" consists of several or all of the following, dependent upon your Adolescent's needs:

- You;
- Your adolescent;
- Special, regular, and/or vocational education personnel;
- School psychologist;
- Counselor;
- Adult service providers;
- Employers;
- Case manager*; and
- Liaison from college (if possible).

The following is a description of the roles of key individuals involved in the transition planning meeting:

Student 's

1. Identifies career interests.
2. Discusses views about various postsecondary options.
3. Describes academic strengths and weaknesses and learning styles.
4. Describes different accommodations needed and prioritizes the importance of each.

5.8 Postsecondary Educational Options and Support Services

As you and your adolescent with learning disabilities begin to sort out educational options, you need to determine what kinds and how much support your adolescent needs. Levels of support vary greatly from school to school. Because each school fashions and implements its own services and because there are no accrediting devices for LD programs, levels of service should be compared and weighed before you and your adolescent make his/her selection. These levels can be thought of as minimal, moderate, or intensive--and are not related to size or type of school (Scheiber & Talpers, p.41).

Minimal support services

These services are found in colleges where the student adapts on his or her own. *No special consideration is offered during the admissions process, nor are there special programs available after the students have enrolled.* Academic support services include tutoring, writing labs and study skills courses, and are available to all students. The school may also offer basic, remedial, or developmental classes but they may not always be appropriate for students with learning disabilities. Students must be their own advocates and self-identify with individual instructors. (Students need to be familiar with their rights under the law to do this.) Students who do well on such campuses are usually confident of their abilities or talents, bright, well-motivated and able to cope with their disabilities.

Moderate support services

These services are found in colleges where the campus helps the students adjust. Students with learning disabilities are admitted as part of the regular student body and are enrolled in mainstream courses. These colleges have a range of services that are available to help students with learning disabilities compensate for special difficulties. Classroom and testing accommodations, advocacy and referral can be obtained through Student Support Services. (You should check with the dean of students or admissions to find out who coordinates these services.) Students can schedule meetings with counselors who coordinate services to meet academic and personal needs. Schools using this model intend to work with students to offer them the best opportunity to develop academic and personal skills. Students who do well on such campuses come prepared to access services and staff, to identify their needs and to be assertive. They know how to compensate for their disability.

Intensive support services

These services are found in colleges where the program is adapted to students. The college offers a special program for students with learning disabilities with the goal of helping them recognize and use their strengths, to manage their disabilities and to participate and succeed in the academic mainstream. Most of these programs are separate entities within a 2 or 4 year college. Trained personnel assess the learning styles and needs of students and individualized courses of study are developed. Counseling and student support groups assist with personal adjustment and help strengthen personal skills. This type of program provides a period of intensive support to prepare students to function independently in regular classes. In addition to taking regular courses, students get special instruction to improve academic and study skills and to develop strategies to compensate for special learning problems. There is ongoing communication between program staff and college faculty to increase understanding of students with learning disabilities. These colleges usually charge a fee in addition to tuition to cover the costs of special instruction and other services provided by the program.

5.9 Summary

1. The services and supports needed by your adolescent in school as a result of his/her learning disability will continue to be needed as he or she pursues post-secondary education, job training and/or work experiences.
2. You and your adolescent must understand and accept his or her disability. Only with this knowledge and acceptance can there be successful transition for your adolescent.
3. It is equally important for you and your adolescent to recognize and build on his or her strengths rather than magnify weaknesses.
4. Transition focuses on fostering personal growth and individual competence (Reiff & deFur, 1992).
5. Student involvement in transition planning can act as a powerful process that fosters empowerment, ownership and responsibility. Student participation in setting goals fosters gaining control of his or her life, a factor considered critical in vocational success for adults with learning disabilities (Reiff & deFur, 1992).
6. Utilize all of the resources that are available to you. This includes seeking the assistance of supportive people--family, friends, at school and in the community.
7. Listen, listen, listen.

8. Your adolescent has options and opportunities. Help him or her by working as a vital partner in the transition process to make informed decisions. Be assertive. No one is as concerned about your adolescent as you are. Ask questions. Insist on answers. Work with schools, services, counselors, to get the help your adolescent needs and deserves.

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Block 2: Intellectual Disability: Nature, Needs and Intervention

Unit 1: Definition, Types and Characteristics

Unit 2: Tools and Areas of Assessment

Unit 3: Strategies for Functional Academics and Social Skills

Unit 4: Assistive Devices, Adaptations, Individualized Education Plan, Person Centered Plan, Life Skill Education

Unit 5: Vocational Training and Independent Living

Unit 1

Definition, Types and Characteristics

Content

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1.1 Introduction

Intellectual disability (ID) is an abnormality that has enormous social effects; it not only affects the people who suffer from it but also the family and society as a group. Millions of people worldwide have intellectual disability and prevalence is calculated to be 1 to 3% in developed countries.¹ In Mexico, according to INDESOL (National Social Development Institute) calculations, there are 2 925 000 persons with some type of intellectual disability.² In spite of this fact, knowledge of its causes is very limited and the etiological diagnosis in developed countries is almost always established as being a little less than half of the subjects affected.³ The frequency with which exogenous and genetic causes are identified varies; some authors have reported proportions ranging from 17 to 47%.^{4,5} Of course these differences are attributed to the selection of the population studied, the degree of mental delay, the heterogeneity of study protocols, technological advances and case definitions. This article outlines basic concepts related to intellectual disability as a function of diagnosis, treatment and prognosis.

According to the tenth revision of the WHO (World Health Organization): Intellectual disability (ID) is a disorder defined by the presence of incomplete or arrested mental development, principally characterized by the deterioration of concrete functions at each stage of development and that contribute to the overall level of intelligence, such as cognitive, language, motor and socialization functions; in this anomaly, adaptation to the environment is always affected. For ID, scores for intellectual development levels must be determined based on all of the available information, including clinical signs, adaptive behavior in the cultural medium of the individual and psychometric findings.

On the other hand, the American Association on Intellectual and Developmental Disabilities (AAIDD) indicates that in addition to a significantly sub-average intellectual functioning, concomitant limitations are observed in two or more areas of adaptive skills, described in table I, and the disorder presents itself before the age of 18.

In summary, intellectual disability is characterized by evident limitations in intellectual functioning and adaptive conduct, the latter expressed as conceptual, social and practical adaptive skills. Therefore, for the study of ID, according to the AAIDD, five dimensions must be considered (table II), one of which was recently included: the dimension composed of participation, interaction and social networks. This definition not only agrees with the International Association for the Scientific Study of Intellectual Disability (IASSID), but also with the recent World Health Organization's International Classification of Functioning, Disability and Health (ICF) of subjects' age and expectations for the future, with the only goal being to provide the support necessary for each one of the dimensions or areas in which the person's life is expressed and exposed. Illnesses can be easily diagnosed when intra-hospital births are involved, during which neonatal metabolic screening can be conducted; however, when the latter is not conducted a considerable risk is presented since such causal factors for mental delay are not identified.

People have different abilities and develop at different rates. Some people find learning new skills or information difficult. This could be because they have an intellectual disability. A person has an intellectual disability if they have both the following before they are 18 years of age: An IQ below 70 (average IQ is 100) Significant difficulty with daily living skills including looking after themselves, communicating and taking part in activities with others.

About two to three per cent of the population has an intellectual disability. This is more than 100,000 Victorians. People with an intellectual disability have significant health disadvantage with a life expectancy up to twenty years less than the general population, and many health conditions not identified or inadequately managed.

Intellectual disability can be mild, moderate or severe and factors such as personality, coping strategies and the presence of other disabilities (motor, social or sensory) will influence a person's requirement for support with daily living.

1.2 Common characteristics

Every person is unique, regardless of their IQ score. Everyone has their own personality and areas of ability and areas of difficulty. Generally a person with an intellectual disability has difficulty: learning and processing information as quickly as people without an intellectual disability, grasping abstract concepts such as money and time, understanding the subtleties of interpersonal interactions (and so may sometimes behave awkwardly or inappropriately in social situations), manipulating the ideas and concepts required for planning and organisation.

1.2.1 Needs depend on individual factors

Arbitrary categories of mild, moderate, severe and profound levels of intellectual disability are defined on the basis of IQ scores. These levels give some guide to the level of support someone might need but the way a person functions in their life also depends on other factors including:

personality, coping skills, other disabilities – for example, physical, social or sensory, the amount of support offered by family, friends and the community, what is demanded of them in different situations.

1.2.2 People with a mild intellectual disability

A mild intellectual disability is defined as an IQ between 50 and 70. Generally speaking, a person with a mild intellectual disability: participates in and contributes to their families and their communities, has important relationships in his/her life, works in either open or supported employment, may live and travel independently but will need support and help to handle money and to plan and organise their daily life, may marry and raise children with the support of family, friends and the service system, may learn to read and write.

1.2.3 People with a moderate intellectual disability

A moderate intellectual disability is defined as an IQ between 35 and 50. Generally speaking, a person with a moderate intellectual disability: has important relationships in his/her life, enjoys a range of activities with their families, friends and acquaintances, understands daily schedules or future events if provided with pictorial visual prompts such as daily timetables and pictures, makes choices about what s/he would like to do, eat, drink etc may learn to recognise some words in context, such as common signs including 'Ladies', 'Gents' and 'Exit', may develop independence in personal care, will need lifelong support in the planning and organisation of their lives and activities.

1.2.4 People with a severe or profound intellectual disability

A severe or profound intellectual disability is defined as an IQ below 35. Generally speaking, a person with a severe or profound intellectual disability: recognises familiar people and may have strong relationships with key people in their lives, has little or no speech and relies on gestures, facial expression and body language to communicate, requires lifelong help with personal care tasks, communication and accessing and participating in community facilities, services and activities.

1.3 What is intellectual disability?

- a disability that affects the way you learn
- you can be born with an intellectual disability or acquire one under the age of 18
- it can affect a person in the following ways:
- it may take longer to learn things
- the person may have difficulty reading and writing
- it can affect how the person's communication
- it will affect the person's understanding
- the person may find it difficult to maintain eye contact
- the person may have difficulty understanding abstract concepts
- it may affect the person's ability to plan and to problem solve
- it may affect the person's ability to adapt to new or unfamiliar situations
- it can involve the person presenting well □ "cloak of competence"

1.4 Clinical definition of intellectual disability

Intellectual disability is a disability which occurs in the developmental period of life (i.e. before the age of 18) and is characterised by below average intellectual functioning. Most people with intellectual disability are born with the disability.

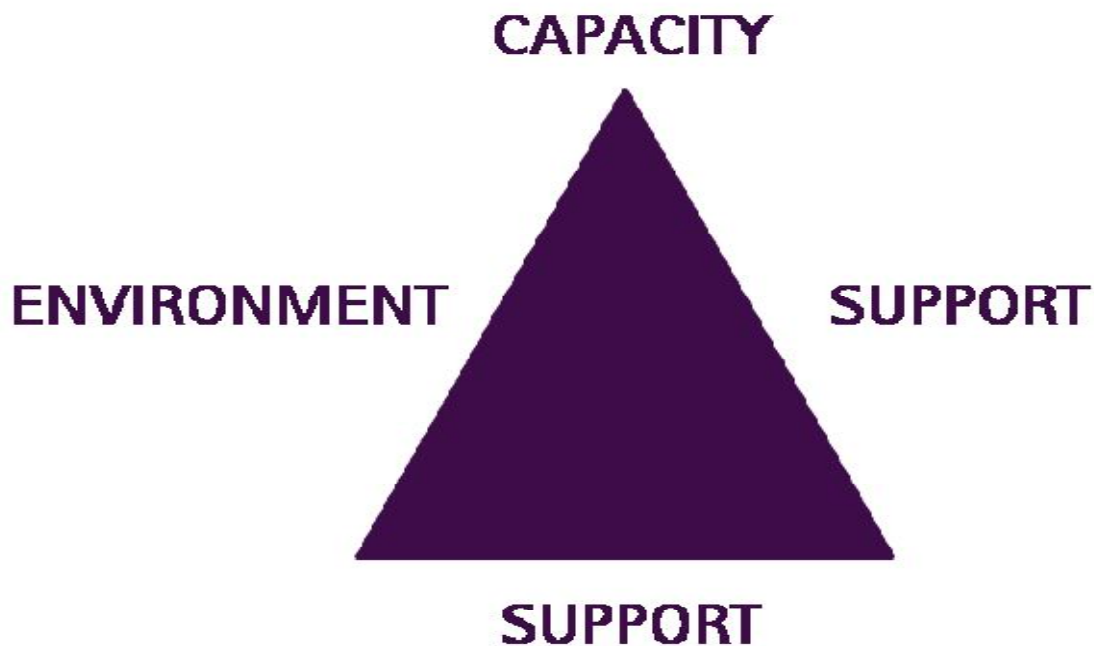
Clinically, and for the purposes of proving in a court that a person has an intellectual disability, intellectual disability is best assessed by a psychologist as:

- an IQ of 70 or under, PLUS
- deficits in at least 2 areas of adaptive behaviour, ie:
 - o communication
 - o self-care
 - o home living
 - o social skills
 - o self direction
 - o leisure and work
 - o Learning.

2-3% of the New South Wales population has an intellectual disability

1.5 Sociological definition of intellectual disability

A more constructive and pragmatic definition is to define intellectual disability in terms of the support needs of an individual. This approach sees the effect of the disability as something that will vary and can be increased or decreased by external factors. It does not view intellectual disability as an unchangeable characteristic of the individual. This definition does not rely on the capacity of the person being set in stone, but also on the environment and the support that they receive.



1.6 Intellectual disability is different from other cognitive disabilities

☐Mental illness

- Often episodic
- Affects perception and mood
- E.g. Schizophrenia, Depression, Bipolar Affective Disorder (used to be known as manic depression).

☐Brain injury

- can be caused by stroke, accident or drug/alcohol
- Can impair memory, thinking, perception, attention and emotions. The brain injury can result in disinherited behavior or poor impulse control.
- People rehabilitate over time i.e. years

☐Dementia

- loss of short-term memory
- progresses into confusion/disorientation
- Alzheimer's Disease is one type of dementia

☐Autism

- Lifelong developmental disability
- Affects ability to relate to people and the world around them
- Impairs social interaction, communication and behavior (eg may be compulsive)
- Often experience overwhelming anxiety, frustration, confusion
- Most people with autism have some degree of intellectual disability but not all. People with Asperger's syndrome generally have average to above average intelligence

1.7 The Power of Language

The use of appropriate language is not just an exercise in political correctness. It is exercising respect for the people that you are speaking about or to.

Language is not fixed and changes with time to reflect changes in government policy, public perception and, for many groups, a move towards more equal opportunity.

Up to the 1980's policies of institutionalisation for "handicapped" people were the norm. People lived in cramped wards where their basic physical needs were met and little else. These people were seen as numbers in many cases. They were seen to have no value in society; they could not work or contribute in any way. Words like "mongol", "retard", "spastic" and "handicapped" were socially acceptable.

Today, due to new models of disability and a move towards people with intellectual disability living and working in the community, we have come to realise that people with intellectual disability have a place in society and make valuable contributions in daily life. People with intellectual disability have families, get married, get divorced, get emotional and get in trouble the same as everyone else!!!

Disability is not an illness. It is part of a person's identity but it is not all that a person is. People with intellectual disability do not suffer, are not afflicted with and are not victims of their disability.

When thinking about language, always think "person first".

Do say	Do not say
Person who has/ person with/ person who has experienced	Victim of/ crippled/ suffering from/ afflicted by. Do not use words that invite pity or reinforce impressions of frailty or dependence.
Wheelchair user/ person who uses a wheelchair	Wheelchair bound or confined to a wheelchair. Remember that a wheelchair can represent freedom to its user.
Person with a disability	Invalid. Don't equate illness with disability.
Person who has epilepsy	Epileptic. Remember this is a person first not a medical condition.
Person who has cerebral palsy	Spastic
People with disability/ with autism	The Disabled/ The Autistics. This makes people with disabilities seem like a group who are separate from the rest of society

1.8 Disability Etiquette

- Offer assistance to a person with a disability if you feel it is appropriate but wait for an answer before you help. Remember it is their right to say no. Do not assume you know the best way of helping - listen to instructions you are given by the person.
- Treat adults like adults. Do not use gestures more suitable for children.
- Do not refer to an adult as "cute" or "adorable". Would you use these kinds of terms when speaking about a person who did not have a disability?
- If someone is a wheelchair user, do not lean on their chair. The chair is part of the personal space of the person using it.
- Make appropriate physical contact with people with disabilities according to the situation, the same as anyone else (e.g. handshake on introduction). Some people with intellectual disability may for various reasons have inappropriate social responses. (e.g. hugging or kissing new people or in work situations). It is important to establish appropriate boundaries that both you and the person are comfortable with. On the other side of this, a person who has autism may be uncomfortable with any form of physical contact but it is important to offer.
- Talk directly to the person not through a support worker or companion. Make eye contact.

- When speaking to a wheelchair user, position yourself at eye level if possible.
- Do not be embarrassed about using common expressions such as “I’ll see you later” when talking to someone with a visual impairment.
- Do not make assumptions about the severity of disability. Some people hide their disability better than others.
- Just because a person does not communicate verbally, doesn’t mean they do not understand.
- In the same way, just because a person speaks very well does not mean they DO understand. Ask questions to check understanding if you are unsure.
- Do not shout at a person if you think they cannot hear you or they do not understand. If they cannot hear you this will not help. You may need to write it down.
- Facial gestures and expressions help people who have hearing problems to understand you. Do not hide your mouth or look away when speaking.
- To attract someone’s attention, a light touch on the shoulder is okay.
- If someone has speech problems, give them your whole attention. Be patient. Do not correct or speak for the person. Resist the temptation to finish the sentence.
- If you have difficulty understanding, don’t pretend. Repeat what you do understand and let the person’s reaction guide you.

1.9 Improving Communication with People with an Intellectual Disability

Tips	Blockers
Allow plenty of time	Being rushed
Build rapport, start with something easy and of interest to the person	Not taking time to put the person at ease
Invite the person to say if they aren’t sure what you mean	Assuming the person will tell you if they don’t understand
Let the person tell the story in their own words and in their own time first. Then clarify	Doing all the talking. Firing question after question. Closed questions or questions that suggest an answer
Deal with one piece of information or question at a time	Too much information or double barrelled questions
Check understanding but make sure the person knows this is to make sure you’ve explained well enough	Giving more information without making sure that the person has understood
Ask open questions □ encourage the person to tell their story at their own pace	Asking closed questions or leading questions that suggest an answer
Allow time for a response even if this is slightly uncomfortable. Wait longer than you normally would	Rushing / not waiting for the answer. Needing to fill the silence. Coming in with more information when the person is still thinking about the last
Use plain everyday language	Jargon, technical language, long sentences, double negatives
Relaxed body language	Impatient or worried body language
Checking for understanding	Problem solving instead of listening

Easy Read

Having an accessible environment is about being proactive not reactive. This means thinking about how we present all of our information. Are letters that are written to clients accessible? Can clients access all the information that you would expect any other client to access?

To make written information more accessible the following tips are important...

- Size of print should be 14 or 16 point
- Font should be easy to read (the nearest to handwriting is the easiest to read. The font in this document is Comic Sans and is considered the most accessible)
- Use no more than one or two fonts
- Line spacing should be 1.5 □ 2
- Use headings when you change subject or have a lot of text
- Use short sentences. (Think of a world where the only punctuation is a full stop!)
- Have one idea per sentence
- Try not to use abstract concepts
- Illustrations are important to help you get your point across if possible
- Black writing on yellow paper is the easiest to read
- Test your information if possible with a person with an intellectual disability.

1.10 NSW Disability Services Act 1993 & Disability Services Standards

All NSW government funded services for people with disabilities have to meet the Disability Service Standards.

These standards have been developed from the Disability Services Act 1993 (NSW). The basis of the Disability Services Act (DSA) is that people with disabilities have the same basic human rights as other members of Australian society. They also have the rights needed to ensure that their specific needs are met. There are also Commonwealth Disability Standards which are very similar.

Through regulating services to people with disabilities, the DSA aims to:

- maximise potential for people with disabilities;
- allow for further integration of people with disabilities;
- achieve positive outcomes such as increased independence, employment opportunities, etc;
- promote positive image.

There are 10 Disability Standards, they are:

1. Service Access
2. Individual Needs
3. Decision Making and Choice
4. Privacy, Dignity and Confidentiality
5. Participation and Integration
6. Valued status
7. Complaints and Disputes
8. Service Management
9. Family Relationships
10. Protection of Human Rights and Freedom of Abuse

1.11 Acquired factors Congenital

Can be grouped as follows:

1. Metabolic: neonatal hypothyroidism; 2. Toxic: lead poisoning,²⁰ fetal alcohol syndrome,²¹ prenatal exposure to substances;²² and 3. Infectious: rubella,²³ Cytomegalic Inclusion Body Disease,²⁴ syphilis,²⁵ toxoplasmosis,²⁶ simple herpes (genital type II).²⁷

1.12 Developmental

During the prenatal period, possible pregnancy complications exist, such as toxemia²⁸ and uncontrolled diabetes, ²⁹ intrauterine malnutrition,³⁰ vaginal hemorrhages,³¹ placenta previa³² and umbilical cord prolapse.³³ During the perinatal period, there are common birth complications: prolonged fetal suffering with neonatal anoxia,³⁴ asphyxia related with suffocation,³⁵ inadequate application of high forceps³⁶ or a poorly applied Kristeller maneuver.³⁷ During the postnatal period, complications are observed such as encephalopathy from hyperbilirubinemia (kernicterus),³⁸ encephalic traumatism³⁹ and infections⁴⁰ (encephalitis and meningitis).

1.13 Environmental and socio cultural factors

Epidemiological studies have consistently reported a notable link between poverty and intellectual disability. The available evidence suggests that this connection reflects two distinct processes. The first establishes that a relation exists between poverty and exposure to a wide range of environmental and psychosocial factors;⁴¹ the second indicates that families with members who suffer from intellectual disability have an increased risk of catastrophic expenses that considerably affect poverty levels. These factors are direct causes of the disproportionate

increase in the incidence of intellectual disability in developing countries.⁴² Interactions have been reported between scarcity and poor prenatal, perinatal and postnatal health care, adolescent maternity, family instability, poor natal health care due to multiple and inadequate caregivers and health professionals, low

- **Etiopathogeny**

Causal factors related with cognitive disability are described in table III and can be classified as follows: genetic, acquired (congenital and developmental), environmental and socio cultural. *Genetic factors* Chromosomal or hereditary disorders⁶ The greatest number of cases are patients with Down syndrome,⁷ an anomaly that occurs in 15 of every 10 000 births and is due to chromosome 21 trisomy or the translocation of chromosomes 21 and 15. Other much less frequent chromosomal abnormalities are those of the fragile X chromosome syndrome,⁸ Prader-Willi syndrome,⁹ Rett syndrome, neurofibromatosis,¹¹ tuberous sclerosis,¹² Lesch-Nyhan syndrome,¹³ adrenoleukodystrophy¹⁴ and other very rarely occurring related conditions.

- **Classification**

Cognitive disability has as a common denominator a subnormal intellectual functioning level; nevertheless, the extent to which an individual is unable to face the demands established by society for the individual's age group has brought about four degrees of severity:⁴³ mild, moderate, severe and profound. The characteristics of individuals in these age groups have little in common with those of other age groups and, therefore, distinguishing the limitations of each one in order to establish intervention guidelines, are fundamental. Cognitive disability is classified according to the characteristics outlined in table IV.

- **Diagnosis**

Subjects with intellectual disability and co morbidities, especially those that are psychiatric in origin, have less social capital than individuals with ID who do not have psychiatric disorders.⁴⁴ In addition, the absence of legislation in developing countries, the lack of knowledge about ID, the stigmatization and discrimination as well as the lack of training and infrastructure for establishing a diagnosis significantly delays possibilities for intervention and the utilization of specific services.⁴⁵ The cultural context establishes denial on the part of the parents who are faced with the possible diagnosis of intellectual disability. It is indisputable that parents take on the diagnosis of mental delay with great difficulty, to the extent that the doctor frequently chooses not to confront this reality in the attempt to avoid the intense pain that it represents. It is unquestionable that the acceptance of a condition that signifies a life of constant difficulties is not easy for anyone. All parents expect to be gratified by their children and hope that they will be intelligent, brilliant students, triumphant and, subconsciously, expect then to become "a second, corrected and better version of themselves". Nevertheless, upon confirmation of the diagnosis,

their expectations are destroyed becoming overwhelmed by a sense of loss, not to mention the obligation of having an unwanted child. This news often provokes feelings of guilt in the parents and outright aggression between them, in addition to hostile reactions toward the doctor; consequently, it is necessary to be prepared and to provide the support needed.

The initial reaction is denial. Then, during the first years, the parents justify the discrepancies during the critical development periods while becoming more demanding as the child grows up, without results. The impossibility of attaining the expected goals generates a great deal of frustration and can lead to serious errors in childrearing, which without doubt, significantly affects the psychological development of these children. As a result, parents must face the correct diagnosis so as to minimize the denial mechanism (among other issues that will be described later). A great deal of empathy is required on the part of the doctor when reporting the diagnosis. Parents need to perceive security and confidence from health professionals; know that they will not face the problem alone and that they will receive guidance throughout the long and arduous path. If the deficit is notable, the diagnosis can be made earlier, but the effect is more devastating. It is necessary, therefore, to provide support for accepting the diagnosis; only in this way can the search for “magic cures” brought on by pain and denial be avoided. This not only provokes catastrophic expenses but also extremely traumatic experiences; each time, the failures become evident.

When there is organic evidence, the initial diagnosis and the need to confront the parents almost always falls on the gynecologists or pediatricians (neonatologists). If a moderate or mild mental disability is involved, the diagnosis is generally established by personnel in the field of education. A referral to a doctor is recommended only if an incorrect diagnosis is established, if behavioral problems are present or if the family requires guidance. On occasion, when a diagnosis is determined by the educational institution, a doctor will be consulted to corroborate the diagnosis or to advise the parents as to treatment for behavioral disorders that prevent the child’s normal development in school or at home. In these cases, the use of instruments for the diagnosis is essential, such as psychometric tests, developmental scales, abilities and adaptive behavior tests or psychopedagogical tests. These instruments are highly useful for psychologists. Once the diagnosis is corroborated, the parents should be advised as to the existing abilities and the expectations on the part of the school or family, since an incompatibility could result in aberrant behavior.

In Down syndrome, establishing a diagnosis at birth is indispensable given that the characteristic features of this disorder are evident; consequently, once the grief is overcome, there is a greater tendency for acceptance on the part of the parents and society, which allows for less to be demanded of them. As for their intellectual capacity, there is a large discrepancy because it can vary from profound to mild disability, and in the case of mosaicism (partial trisomy), even an average intelligence is possible. Currently, some still believe that people with this syndrome live for just a few years however. Thanks to medical progress, the life expectancy is roughly 50-60 years of age, at which point many develop Alzheimer’s disease.

Another important group to mention is individuals who possess an intelligence quotient (IQ) between 70 and 85 points range (borderline) placing them in what is known today as limitrophe. Until the beginning of the 1980s they were considered to be subjects with mental delay, but since they did not present adaptation problems they were not considered to have a disability. Nevertheless countries like Mexico, this population faces serious problems, especially in urban areas and mostly when they belong to socio culturally and economically competitive classes; school and work programs are designed for people with normal average IQ, which makes it very difficult to satisfy such expectations. Furthermore, given their functionality level, they can recognize their own limitations and generate a good deal of suffering with notable tendencies toward depression. This group's highest prevalence is found in rural areas with the most important causal factor being family heredity; therefore, the disorder in these cases does not result in serious conflicts because social and family expectations adapt to their abilities.

- **Clinical diagnosis**

In the figure 1, a proposed algorithm for a diagnostic protocol is described.⁴⁶ The clinical history must put an emphasis on health care during the prenatal, perinatal and postnatal period and include the results of all previous studies, including a genealogical tree for at least three generations and an intentional search for family antecedents of mental delay, psychiatric illnesses and congenital abnormalities.

The physical exam should focus on secondary abnormalities and congenital malformations, somatometric measurements and neurological and behavioral phenotype evaluations. If, as a result, signs suggesting a particular etiological diagnosis are recognized, specific analyses are requested; in the case of Rett syndrome, an MeCP2 gene mutation analysis is recommended.

1.14 Search for genetic abnormalities

If it is not feasible to establish a clinical diagnosis, it is necessary to conduct high-resolution cytogenetic studies⁴⁷ in addition to metabolic clinical evaluations.⁴⁸ For patients with normal results from the previously mentioned analyses, it is possible to request fluorescence *in situ* hybridization (FISH)⁴⁹ which is an evaluation of the entire subtelomeric region. In the next step, if no abnormal data are identified, submicroscopic chromosomal disorders are evaluated. Neuroimages can be useful only if the occipitofrontal circumference of the head is abnormal – below the second or above the 98th percentile– or if the neurological exams reveal abnormalities. In addition, they are recommended if a specific neuroanatomical effect exists (tuberous sclerosis) or a clinical history indicates perinatal hypoxia. Metabolic analyses can include urinary tests for amino acids, organic acids, oligosaccharides, mucopolysaccharides and uric acid. Plasma levels 7- and 8- dehydrocholesterol for total cholesterol and diesterol are useful for identifying defects in the distal cholesterol pathways. Other diagnostic tests can be conducted for finding congenital disorders of glycosylation.⁵⁰

- **Differential diagnosis**

Since mental delay is a developmental disorder that is secondary to many etiological factors, there are few nosological units with which they can be confused; among these are generalized developmental disorders⁵¹ (DSM IV 299.00 to 299.80) and, in particular, autism with low-functionality and autistic spectrum disorders.⁵² Nevertheless, it is important to establish a differential diagnosis since, while some therapeutic interventions are similar with respect to social and work inclusion programs, future expectations are very different.

- **The clinical profile**

Based on the gradual division described above, it is possible to infer that a specific clinical profile does not exist; each one of the levels for each group has its own clinical characteristics.

Symptoms are closely related to the specific sensorimotor, linguistic and cognitive processes, among others. The more moderate the disability, the less evident its symptoms; perhaps alterations in language construction shed more light and, in particular, the comprehension level and behavior of the children themselves compared with that of a chronologically younger age. It is useful to ask the parents if they can establish a comparative age in relation to brothers and sisters or close cousins in the family (if they exist) who could serve as a reference. Parents often say that the patient “speaks and understands everything”; in this case it is important to clarify if they are referring to the use of nouns and verbs or the comprehension of orders and concepts. It is essential to give credibility to the parents when they insist that a doctor recognize a developmental disorder. It is true that in some cases they tend to be delay or apprehensive or overprotective, but it is more often the case that their perceptions are based on objective, comparative judgments and are therefore correct. Frequently, by the time they get to a doctor who establishes the diagnosis, they painfully express the great frustration that they experienced when they had to repeatedly tell their doctor that “something was wrong,” especially if that doctor insisted on not making comparisons and affirming that “no two human beings are alike,” that “developing the skills is just a question of time”; such an attitude causes late intervention, among other things. When a diagnosis of slow-learner is established, this creates confusion since it suggests normality rather than disability.

- **Comorbidities**

Individuals who suffer mental delay are more vulnerable for developing a psychiatric abnormality. Statistics exist today that confirm that this population experiences two or three times more mood disorders, anxiety disorders and behavioral problems than persons without intellectual disability.⁵³ Such vulnerability, on occasion, leads to a set of psychotic symptoms that are secondary to the stress experienced when social, family and school demands are excessive.⁵⁴ On the other hand, other illnesses comorbidly exist that are unrelated to this vulnerability factor; the most common being epilepsy,⁵⁵ attention deficit disorder and hyperactivity,⁵⁶ schizophrenia⁵⁷ and infantile cerebral paralysis.

- **Treatment**

Cognitive disability must, evidently, be understood and diagnosed as a developmental disorder, but fundamental for providing adequate professional care; than to take into account the intellectual level of the patient and his or her ability to adapt to the existing environment and, of utmost importance, the reaction of the parents to this situation.

Since cognitive disability is not curable, the treatment objectives must focus on the normalization of behavior in accordance with the norms and rules determined by society. To this end, intervention as early as possible is fundamental since treatment for developmental disorders can reach its maximum scope only through early intervention. It must be remembered that the process of cerebral plasticity has its greatest potential during the first five years of life and this is one of the most important supporting factors in attaining the therapeutic goals. Furthermore, since the behavior disorders which are so frequently observed are secondary –due to deficient childrearing in the majority of cases– the parents must be worked with and advised as to adequate childrearing methods for developing adaptation patterns that lead to optimal social integration. People with mental delay, being limited to a mental age of less than 13 years, cannot aspire to levels of abstract cognitive thought that in normal conditions appear at the beginning of adolescence; for this reason, educational objectives must be focused on the development of skills for achieving a self-sufficient life at an adult age and not on schooling as it happens in many countries. In addition, interaction between peers should be promoted for creating group belonging (relationships with others with approximately the same disability level).

It is evident that the results obtained depend on the age of the subject when the diagnosis is established and when treatment is initiated, the severity of the disability, the affected areas and the methods or techniques used for rehabilitating these areas. Accordingly, treatment is established in the following manner: During the infancy period (zero to two years), disorders in muscular tone and motor development are treated (motor therapy) and signs of sensory disintegration that almost always accompany motor problems are treated through sensory integration therapy. As for the linguistic developmental disorders themselves, symbolization processes are treated (language therapy) as well as muscular tone alterations that affect respiration, phonation and the bucofacial region (articulation therapy). Likewise, cognitive stimulation should be used for these children at the onset of treatment. When children are diagnosed after two years of age and before puberty, the ideal is to use instruments that determine the maturity level for each one of the developmental areas and apply the same therapies (motor, sensory, linguistic, etc.), in addition promoting the development of perceptual abilities, with deficits in, and learning abilities (reading, writing, mathematics, etc.), using techniques similar to those used in children with learning disorders (dyslexia, etc.). It is common to observe behavioral problems in this population, which often motivate parents to seek out professional advice; therefore, behavioral therapy should be included in the health care plan and should be based on humanistic principles (clarification of feelings and positive reinforcement) and not on aversion techniques. Finally, it is essential to work with independent living skills and give the individual the necessary elements for self-sufficiency.

Although it is rare, an adolescent can be seen in the doctor's office for poor academic performance or behavioral problems without having been previously diagnosed with ID. This most likely involves a subject with limitrophe and who, upon entering middle school, does not manage to pass the subjects that require abstract thinking (mathematics, physics, chemistry, etc.). In these cases, as well as for adolescents and adults with intellectual disability, programs oriented toward independent living should be recommended. These programs should cover the areas necessary for achieving a partially or totally self-sufficient life, among which are: the academic-basic skills, community integration programs, developing skills for managing domestic tasks, personal healthcare and sexuality. In addition, a prevocational program should be included for the development of abilities for the workplace and, when possible, for the individual to become integrated into the labor market.

Pharmacological treatment is used for the treatment of comorbidities, and is specific for each one according to the criteria followed for these disorders, both in psychiatry and in neurology. On occasion, when a favorable response is not obtained, a random treatment scheme is established. In the case of attention deficit disorder and hyperactivity, the use of stimulants for the central nervous system (the treatment of choice for this disorder) had been considered not to produce favorable results; nevertheless, recent research shows that elevated doses of methylphenidate (0.60 mg/kg) produce obvious improvements in symptoms related to inattention, impulsivity and hyperactivity. Because of the above, it is evident that treatment for persons with cognitive disability must be multi disciplinary and include medicine (family, pediatrics, neurology and psychiatry), psychology (educational and clinical), education (regular, and special), rehabilitation (physical, occupational and recreational), nursing, social work, etcetera.

Prognosis

As was already established, intellectual disability is not curable; and yet, the prognosis in general terms is good when using the emotional wellbeing of the individual as a parameter. When the parents of these subjects are asked, after they have accepted the diagnosis, they express that their long-term expectation, that "their children be happy" or that "they be self-sufficient"; that is, to see them content and self-sufficient compensates for the lost hope of having a professional child or one that satisfies the parents' unfulfilled dreams. This is a striking contrast with parents who reject the disability and for whom the only acceptable parameter is "normality." Of course, in these cases, the patients not only suffer more but the parents also live in constant agony. Other conditions exist that, when attained, guarantee the achievement of the objectives and, therefore, the prognosis in terms of obtaining that state of happiness is wholly viable. It is feasible to see an adult with mild or moderate cognitive disability or limitrophe maintain a job, travel to the city, take care of his or her apartment, have a social life or, in other words, manage on one's own. It is therefore necessary to fulfill the following requirements. The first and most important is the parents' acceptance of their children's limitations and the impossibility of attaining "normality." After this, the age upon beginning the rehabilitation program should be considered as well as obtaining a level of continual productivity that is consistent with the disability, the existence of a

group to which to belong and, lastly, the unconditional affection of the parents. It has been stated that the above mentioned factors reduce psychiatric comorbidities (depression, anxiety, behavior) and a prevalence equal to that of the non-disabled population is found; or as one patient said who was asked how it feels to attend an independent living center: "I know I can't go to the university like my brothers and sisters but I can do everything that they can do."

1.15 Summary

Intellectual disability should be treated in a comprehensive manner. Nevertheless, currently, the fundamental task and perhaps the only one that applies is the detection of the limitation and abilities as a function of subjects' age and expectations for the future, with the only goal being to provide the support necessary for each one of the dimensions or areas in which the person's life is expressed and exposed. Nevertheless, this perspective tends to be reductionistic and subjects with disabilities who live in developing countries should have multiple social benefits that today are not present, and there are no signs of them on the near horizon. Intellectual disability has been absent in the political, social and economic agenda of the countries in the region. In this sense, Mexico is no exception. Subjects with intellectual disability are not considered in the planning strategies for health services, the predominant educational proposals are not based on scientific evidence and virtually no research has been developed in this knowledge area. Substantive but elemental aspects for a dignified life for individuals with intellectual disabilities must be a priority in the legislative arena. Among these are the possibilities for guaranteeing human development, teaching and education, home life, community life, work, health and security, optimal behavioral development, social integration, as well as protection and defense. The organized social response for confronting the social needs of subjects with intellectual disabilities cannot be delayed indefinitely.

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Unit 2

Tools and Areas of Assessment

Content

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2.1 Introduction

There are a number of difficulties in accurately estimating the number of children who have an intellectual disability, due to differences in the assessments, methodologies (Roeleveld et al., 1997) and terminology used as well as in the populations studied (Fryers, 1997), however, worldwide prevalence rates for mild and severe intellectual disability are estimated to be 3% and 0.38% respectively (Roeleveld et al., 1997). Intellectual disability is defined as: a significant impairment of intellectual functioning (i.e., an IQ of less than 70); a significant impairment in two or more areas of daily living and onset before age 18 (World Health Organisation, 1992). It is diagnosed by a standardised assessment of cognitive and adaptive functioning, as well as a developmental history (American Association of Intellectual and Developmental Disabilities, (AAIDD), 2009).

By definition, all children with an intellectual disability will have significant difficulties with cognitive and adaptive functioning. They also have a number of additional care needs, including increased mental health problems (Emerson and Hatton, 2007), increased rates of severe sleep disorders (Richdale et al., 2000) and challenging behaviours (Baker et al., 2003), physical health problems (Courtman & Mumby, 2008) and higher rates of co-morbidity e.g. Autistic Spectrum Disorder (Croen et al., 2002). Many are thought to need higher levels of care and more educational support than children without intellectual disability (Simonoff et al., 2006). In addition, parents of children with an intellectual disability are more likely to experience mental health difficulties and stress than other parents (Fidler et al., 2000).

There are, therefore, a number of reasons to accurately identify those children who have an intellectual disability and differentiate them from children who may have other cognitive difficulties, such as specific learning difficulties. The identification of intellectual disability in children can, however, be problematic, especially for those with significant intellectual impairment (Hamilton, 2006; Simonoff et al., 2006). This means that the timing and process of diagnosis is extremely variable (Quine & Rutter, 1994) and may not occur until the individual is in his/her teens (Simonoff et al., 2006) or has reached adulthood (Hamilton, 2006). This can result in a delay in meeting the support needs of the child (McGinty & Fish, 1992) and such delays have been found to be one of the main factors related to stress and dissatisfaction with service provision in parents of children with an intellectual

disability (Carmichael et al., 1999). In addition, delays in identification and intervention can have a negative impact on the child's learning, communication and adaptive skills (Herbert, 2006; Chadwick et al., 2005) and can result in others failing to understand the child's difficulties and behaviour (Goodman & Linn, 2003).

The difficulty in identifying children with an intellectual disability at an early stage may be due to a number of factors. Firstly, there is limited knowledge about what an intellectual disability is on the part of education (Rae et al., 2011), health (McKenzie et al., 2000) and social care staff (Williams et al., 2009) and, in the U.K., frequent confusion between the terms 'learning disability' and 'learning difficulty' (Hames & Welsh, 2002). This is likely to be exacerbated by the fact that different countries use different terminology to describe the same group of people. In the UK, the term 'learning disability' is used, while many European countries and until recently, the US adopt the term 'mental retardation' (AAIDD, 2009; Reid, 1997). 'Intellectual disability' has now been adopted by the AAIDD as the term in the U.S. and is used by academics and clinicians in a number of other countries (Reid, 1997). Screening for Intellectual Disability in Children Secondly, the diagnosis of intellectual disability requires an assessment of intellectual and adaptive functioning, as well as a developmental history. These assessments are time-consuming to complete and the former must be carried out by, or under the supervision of, qualified professionals, normally applied psychologists (Goldstein et al., 2004; British Psychological Society (BPS), 2001). There has, therefore, been an increasing impetus to find alternative ways of identifying children who may have an intellectual disability at an early stage. This has led to an interest in the use of screening tools.

Screening tools may be used for a number of different purposes. Clinical services may screen as a way of increasing efficiency through reducing waiting times and guiding appropriate referrals (Bailey et al., 2005; BPS, 2003). Early identification can also inform early interventions (Guralnick, 2005). Education services may wish to predict which children are at risk for educational difficulties and identify appropriate support for them within the school setting (Sonnander, 2000). Researchers may use screening tools to provide a quick and easy means of identifying particular population samples (Charman et al., 2007) while forensic services may wish to identify potentially vulnerable adolescents with an intellectual disability in the criminal justice system (Ford et al., 2008; Talbot & Riley, 2007). At a broader level, screening tools may

be used in an attempt to meet government policies and targets which require early and accurate identification of children with intellectual disabilities (Sonnander, 2000; Glascoe & Byrne, 1993) or to provide estimated prevalence rates of intellectual and other disabilities in certain populations or countries (Mirza et al., 2008; Xie et al., 2008). As screening tools may be used for a range of different purposes, it might be expected no one tool can meet these differing needs (Rafoth, 1997). Screening for Intellectual Disability in Children

For any screening tool to be useful and effective it has to meet a number of basic criteria. As one of the main aims of screening is to reduce the amount of time involved in assessment, it has to be quick and easy to use and appropriate to the group for which it was designed (Glascoe, 2005). In order to identify the target population, it must also have strong psychometric properties including standardisation, reliability, validity, with acceptable values of 70-80% in relation to sensitivity and 80% for specificity (Rafoth, 1997; Sonnander, 2000; Glascoe, 2005). These values may, however, vary depending on the purpose of the screening tool and the relative costs of false negatives and false positives (Charman et al., 2007). Additional calculations of positive and negative predictive powers may also be included (Glascoe, 2005). The psychometric properties of any given tool will depend on the characteristics of the population being screened (Charman et al., 2007), for example the predictive values of a test are influenced by the prevalence of the condition in the population being screened, although these values are seldom reported (Camp, 2006). Table 1 below summarises the psychometric characteristics required of a good screening tool.

The development and use of screening tools is, however, not straightforward. The rapid development of young children's cognitive and social processes makes very early identification of intellectual disability difficult (Bornholt et al., 2004) and screening carried out with infants during the first 24 months of life has low predictive validity in relation to later cognitive functioning (Sonnander, 2000). Developmental changes with age also present challenges for screening in older children. Different skills are acquired at different ages and are influenced by a range of environmental factors, therefore, developmental problems which Screening for Intellectual Disability in Children were not apparent at one age may become obvious at another (Glascoe, 2005), meaning that a screening tool must be developmentally sensitive.

There are also a number of ethical issues that have to be considered when using screening tools. Sonnander (2000) notes that early identification is only advantageous if there are adequate resources to subsequently meet the needs of children who are found to have an intellectual disability. In addition, as a screening tool can only give an indication of whether an individual falls into a particular category or not, there must be the facility to provide proper diagnostic assessment when required. A screening tool that produces a number of false positives may result in services being stretched by requests for diagnostic assessment or to provide input to those who do not fall within their remit. It is also likely to cause unnecessary stress for the families of those children who are incorrectly classified as having an intellectual disability. Conversely, false negatives may result in children who require additional support, failing to receive it.

2.2 Conceptual Definition of Intellectual Disability

IDEA 2004 defines mental retardation as “. . . significantly subaverage general intellectual functioning, existing concurrently with deficits in adaptive behavior and manifested during the developmental period, that adversely affects a child's educational performance.”

The Connecticut State Department of Education has adopted the term *intellectual disability* as synonymous with the federal term, mental retardation, for purposes of IDEA identification. The Department defines intellectual disability as:

Significant limitations both in intellectual functioning and in adaptive behavior as expressed in conceptual, social and practical adaptive skills. Inherent in the definition adopted by the Department and supported in the professional literature is the emphasis on the dual dimensions of general intellectual functioning and adaptive behavior (Luckasson et al. 2002). The American Association on Mental Retardation (AAMR) has defined mental retardation since 1960 as significant impairments in general intellectual functioning and adaptive behavior.

Three essential characteristics of the mental retardation diagnostic construct have been recognized for more than 70 years. These are:

1. Impairment in general intellectual functioning

2. Impairment in a domain variously termed social incompetence, adaptive skills and adaptive behavior

3. Developmental onset, requiring that evidence must be available of significant limitations in intellectual functioning and adaptive behavior by age 18

The continued use of the term *intellectual disability* is supported by a consensus of Connecticut educators and parents. Intellectual disability is believed to more fully capture the essential feature of the disability, specifically, limited intellectual ability leading to or causing adaptive behavior deficits. Moreover, the term has fewer negative connotations and incorrect stereotypes than the term mental retardation. Many nations in Western Europe and in the English speaking world adopted the term intellectual disability in place of mental retardation beginning in the mid-1990s and continuing to the present. It is worth noting that AAMR adopted the term intellectual disability in its 2006 name change to American Association on Intellectual and Developmental Disabilities. It is not unreasonable to think that the term may be more widely adopted as a result.

2.3 Developmental Period

In Connecticut, evidence of significant limitations in intellectual functioning and adaptive behavior must appear by age 18 if intellectual disability is an appropriate classification, even if the student is older than age 18 at the time of eligibility determination. On the other hand, students who do not show such limitations by age 18, but acquire significant limitations in intellectual functioning and adaptive behavior after 18 but before age 21, perhaps due to injury or disease, are NOT classified as having an intellectual disability/mental retardation.

For many decades all conceptions of intellectual disability have included the notion of developmental origin: specifically, evidence of significant limitations must appear prior to a specific age, usually 18. Critical to note, however, are terms like “appear before” or “originates.” The diagnosis of intellectual disability/mental retardation does not have to be made by age 18, only that evidence of significant limitations in the appropriate areas has to be present prior to age 18. For example, a 19-year-old high school student referred for an evaluation might be validly identified with intellectual disability in Connecticut *if* substantial evidence of significant limitations in general intellectual functioning and adaptive behavior appeared before age 18 even

though the actual eligibility determination did not occur until age 18 or later. Applying Criteria Accurately and Conscientiously

Planning and Placement Teams (PPT) may be reluctant to identify mild intellectual disabilities, defined in part by performance on measures of general intellectual functioning as an IQ score in the range of 60 to 75. Today, in some school districts and states, children performing in this range of intellectual ability are identified frequently as specific learning disabled (MacMillan 1996; Reschly et al. 2002).

Failure to apply classification criteria accurately and conscientiously is a matter of professional ethics and, conceivably, possible legal sanction. Misdiagnosing children and youth into what are perceived as more acceptable categories, for example, diagnosing students as specific learning disabled (SLD) who in fact are more properly identified with mild mental retardation, may be a short-term expediency, but will often create long-term problems for the individual and the agency. For example, persons who as adults need Supplemental Social Insurance (SSI) assistance under mild mental retardation are markedly disadvantaged if their school records show a diagnosis of SLD. Many individual cases of this nature have emerged in recent years in SSI cases and in other agencies making decisions about adult services.

2.4 Prerequisites to Appropriate Identification

Appropriate identification of children with intellectual disability involves more than a special education evaluation. It begins in general education with effective instruction that uses scientifically based curriculum, culturally responsive teaching and early intervening services to meet academic and behavior needs of the child in the general education setting. Effective Instruction

Children are entering our schools with more complex educational, medical and social needs than ever before. In its Position Statement on the Education of Students with Disabilities (2001), the Connecticut State Board of Education expressed the following belief: "... all students are unique and are influenced by cultural, linguistic, intellectual, psychological, medical, social and economic factors. These factors create a need for a varied educational environment that provides for, and accommodates, each child's strengths and areas of needed improvement. The Board also believes that a unified and coordinated continuum of educational opportunities and supports,

designed to address individual needs, serves and benefits all students. The Board encourages the implementation of educational models that promote multiple instructional strategies, which encourage and accommodate students in the general environment to the maximum extent appropriate. It is the responsibility and obligation of educators to design and provide teaching strategies, methods and materials that are suitable for each individual learner. As appropriate a continuum of these strategies should be implemented before a child is referred to special education.”

Prior to any identification process, students suspected of having an intellectual disability should be extended the same opportunity to receive an appropriate education as their non-disabled peers; this includes explicitly taught and effective instruction. The practice of lowering standards without evidence of attempts at providing effective explicit instruction in the general education curriculum can lead to negative outcomes such as inappropriate identification and overrepresentation of particular racial and ethnic groups within disability categories, including the category of intellectual disability (see Appendix E, Understanding Disproportionality).

Instruction must first be explicit to be effective. Explicit instruction has been defined in various ways by researchers. Hall (2002) defines explicit instruction as a systematic instructional approach that includes a set of delivery and design procedures derived from effective schools research merged with behavior analysis. There are two essential components to well-designed systematic and explicit instruction: (a) visible delivery features such as group instruction with a high level of teacher and student interactions, and (b) the less observable, instructional design principles and assumptions that make up the content and strategies to be taught. Birsh, 1999, citing a 1988 study by Winograd and Hare, stated that Direct instruction that is both planned and controlled by the teacher has proven to be a useful way to teach academic skills to students with disabilities. The following are critical components of explicit and systematic instruction:

Explicit Instruction

- Explicit means students are not expected to infer knowledge and care is taken to direct students’ attention to what is being taught (Connecticut Blueprint for Reading Achievement 2000).

- Explicit instruction ensures students' attention is drawn to important features of an example or demonstration (National Reading Panel 2000).

Systematic Instruction

- Skills are taught in a planned, logically progressive sequence according to individual needs.
- Lessons focus on clearly defined objectives that are stated in terms of what students will do.
- Multiple practice activities are scheduled purposefully to help students master and retain new skills.
- Students work on carefully designed tasks that give them opportunities to apply what they have been taught.
- Assessments are designed and used in a timely fashion to monitor skill acquisition as well as students' ability to apply new skills, to retain them over time, and to use them independently (National Reading Panel 2000).

Effective instruction and improved student outcomes begin with the teacher. Researchers Jere Brophy and Thomas Good (1986, 370), upon review of numerous studies about teacher impact on student achievement, concluded that "the myth that teachers do not make a difference in student learning has been refuted." More recent studies note that the individual classroom teacher has an even greater effect on student achievement than originally thought (Sanders and Horn 1994; Wright, Horn and Sanders 1997). Douglas Reeves (2006) cites an earlier study by

Rosenthal and Jacobson (1968) that hypothesizes the "Pygmalion Effect," supporting the notion once again that when teachers expect more, they get more. However, high expectations alone are not enough. High expectations must be accompanied by effective, explicit instruction in relevant curricular domains.

Improved student outcomes for *all* students, those with and without disabilities, is the keystone of the No Child Left Behind (NCLB) Act of 2001. The Individuals with Disabilities Education

Improvement Act of 2004 (IDEA 2004) upholds this expectation for students with disabilities, by embracing the specific language used in NCLB regarding the necessity for highly qualified personnel, scientifically based instruction, and student assessment.

IDEA 2004 states that a child must not be determined to be a child with a disability if the determinant factor for that determination is:

- lack of appropriate instruction in reading, including the essential components of reading instruction as defined in section 1209(3) of the Elementary and Secondary Education Act (ESEA);
- Lack of appropriate instruction in math; or
- Limited English proficiency.

The essential components of appropriate reading instruction are identified in NCLB as:

- Phonemic awareness
- Phonics
- Vocabulary development
- reading fluency, including oral reading skills
- reading comprehension

Some professionals and researchers have defined highly effective instruction as simply instruction that places about 80-85 percent of *all* students on trajectories to meet established benchmarks, such as passing state tests at specific ages or grades. Other researchers have defined effective instruction in various ways. A study conducted at Midcontinent Research for Education and Learning (McRel), examined years of research on instruction to synthesize and make useful to the public. Robert J. Marzano (2002) published those results after examining the strategies to

determine those that had a high probability of enhancing student achievement for all students in all subject areas at all grade levels. Marzano's study revealed nine categories of strategies that have a strong effect on student achievement:

1. Identifying similarities and differences
2. Summarizing and note taking
3. Reinforcing effort and providing recognition
4. Homework and practice
5. Nonlinguistic representations
6. Cooperative learning
7. Setting objectives and providing feedback
8. Generating and testing hypotheses and questions
9. Cues and advance organizers

Douglas Reeves (2002, 19) cites that it is a holistic accountability system, "a continuous cycle in which research informs professional practice and professional practice yields evidence of its impact on student achievement," that makes the difference in student achievement. In practice, this is a system of reviewing and analyzing student data and changing instruction as appropriate. While his work is far-reaching, extending beyond the scope of this document, Reeves also indicates that a connection between standards, curriculum and accountability is a critical feature of improved student achievement, an intended outcome of highly effective instruction.

Instructors at Vanderbilt University's Peabody College of Education and Human Development, identify high-quality instruction by indicating that it implies the effective use of curriculum and instructional procedures that have been validated through rigorous research. These instructional procedures are referred to in a variety of ways, including such terms as scientifically based practices, evidence-based instruction, and research-validated instruction (IRIS STAR Legacy Module).

Wiggins and McTigue (2006, 27) have developed a model of nine principles illuminating the process of learning. Principle number nine refers to *instruction*, maintaining that “instruction is most effective when it is personalized—when we sufficiently honor learners’ interest, curiosity, strengths, contributions, and prior knowledge, making learners feel that they are an important part of something larger than themselves.”

2.5 Culturally Responsive Teaching/Pedagogy

A critical piece of effective instruction and part of IDEA 2004 requirements is consideration of the diversity of the student population and providing teaching/pedagogy that recognizes and takes into consideration cultural differences within the classroom.

Gay (2000, 29) is cited by several sources as defining culturally responsive teaching as “using the cultural knowledge, prior experiences, and performance styles of diverse students to make learning more appropriate and effective for them; it teaches to and through the strengths of the students.” Culturally responsive teaching:

- acknowledges students’ cultural heritage as it affects their dispositions, attitudes, and approaches to learning, and recognizes that it contains content worthy to be included in the curriculum;
- builds meaning between students’ home and school experiences as well as “school stuff” and the students’ lived realities;
- uses of a wide variety of instructional strategies;
- teaches an appreciation of the students’ own cultural heritage as well as that of others; and
- incorporates multicultural information, resources, and materials in all subjects and skills routinely taught in schools (Gay 2000).

Nieto (1999, 69) addressed the responses to what she calls “cultural discontinuities” that lead to school failure. One of these discontinuities is in curriculum and instruction. Nieto argues that, in

spite of promising research, a serious problem of perception remains as “culturally responsive pedagogy sometimes is based on the static view of culture that may even verge on the stereotypically.” Attempts to be culturally responsive may be applied or implemented in ways that defeat the purpose and the “result is that entire cultures are identified by a rigid set of characteristics.” The complex nature of culture and the fact that culture is affected by other cultures need to be considered in planning culturally responsive pedagogy. Nieto further states, “any approach to meaning and effective pedagogy needs to take into account how students’ languages, cultures, and other differences exist within, and are influenced by, mainstream U.S. culture as well as by other cultures with which they come into contact.” Nieto (1999, 70) warns that “it is therefore necessary to look beyond cultural responsiveness alone to help explain student academic success” in light of the impact of “structural inequalities...with which so many students, especially those who live in poverty, contend on a daily basis.” Given (a) the focus on student achievement for students with and without disabilities, (b) the research that supports effective practices that are culturally responsive, and (c) the knowledge that the teacher can affect student learning, it is the role of educators and paramount to student success that initial provision of appropriate and effective instruction to all students is provided within the context of the general education curriculum. Effective instruction that is culturally responsive can easily be integrated into the conceptualization and the delivery of early intervening services and nonbiased assessment. Early Intervention-Early Intervening (EI-EI) Services

A proactive preventative approach to address the needs of students is the most effective means of ensuring that students receive appropriate academic and behavioral supports and to identifying appropriate referrals to determine eligibility for special education and related services. It also is a *necessary* first step in the process to reduce assessment bias and to ensure that students suspected of having an intellectual disability are provided appropriate interventions and instruction prior to a referral to determine eligibility for special education and related services. This is especially important for students who may be having significant difficulty due to a lack of appropriate instruction, rather than an intellectual disability.

Although early intervention is sometimes referred to in the literature as focusing on students in kindergarten through Grade 3, early intervention is appropriate for students from preschool through high school. The term is used to indicate the need to identify and address concerns

regarding a student's academic skills or behavior early in the development of an emerging difficulty, before the need for more intensive supports. For example, a high school student may experience a drop in grades that may be related to academic difficulties or changes in home life. The school should have a process in place to identify the focus area of improvement and determine appropriate interventions before the student's failing his or her courses, which could lead to a decision to drop out of school. In this example, early intervention is implemented with an older student for prevention purposes.

All schools should have a data-driven process in place to monitor student progress and support teachers with instructional options in order to meet students' needs. An effective process includes a collaborative approach that involves strategic decision making and ongoing progress monitoring. This ensures that all students have access to and are provided with a continuum of instructional supports and interventions prior to consideration of referral for special education evaluation.

IDEA 2004 regulations strengthen the language of the use of early intervening services. In implementing coordinated, early intervening services in section 34 CFR §300.226 (b) (1) and (2), a local educational agency may carry out activities that include...

- professional development (which may be provided by entities other than local educational agencies) for teachers and other school staff to enable such personnel to deliver scientifically based academic instruction and behavioral interventions, including scientifically based literacy instruction, and, where appropriate, instruction on the use of adaptive and instructional software; and
- providing educational and behavioral evaluations, services, and supports, including scientifically based literacy instruction

Prevention and early intervention are most effective when incorporated into a system of multiple tiers of academic and behavioral intervention that stress scientifically based instruction and empirically validated behavioral interventions.

Primary/universal prevention and intervention begins with universal screening and early intervention procedures for *all* students to provide effective instruction and behavioral supports

that will meet the academic and behavioral needs of most children (e.g., scientifically based core curriculum, school wide positive behavior support). With effective universal practices in place, most students (about 80-90 percent) will not require additional support. If less than 80 percent of the school's population is responding to universal practices, then the universal practices need to be improved. If the general education context is not effective in meeting the needs of most students, more students will perform at low levels, which may lead to increased numbers of inappropriate referrals and special education identification. However, students who do not respond to effective primary/universal prevention and intervention (i.e., effective general education context) will require additional support.

Secondary/targeted prevention and intervention incorporates universal procedures with additional targeted interventions for about 5-15 percent of the school population that doesn't require individualized support but needs support beyond universal practices (e.g., small group tutoring, check and connect, or social skills instruction). Prevention at this level is necessary to reduce the number of students who may need intensive support.

Tertiary/intensive prevention and intervention involves individualized academic and behavioral interventions with frequent progress monitoring and formative evaluation. Students who require individualized interventions at this level (typically 1-5 percent of the school population) have more intensive needs that require a more focused team-based approach for determining what supports the student needs to be successful. Figure 1 illustrates a three-tiered system of early intervening services to address academic and behavioral concerns prior to a referral to special education.

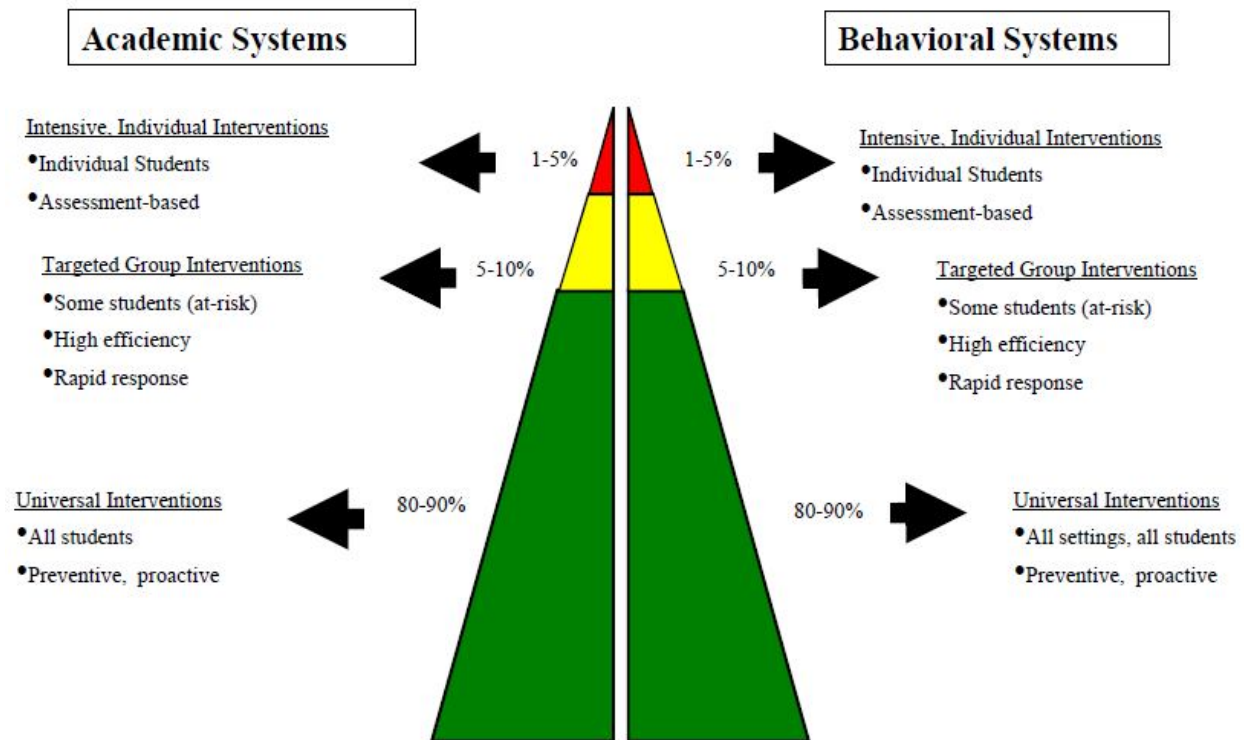


Figure 1. Three-tiered system of early intervening services to address academic and behavioral concerns prior to a referral to special education (Sugai 2001).

A high-quality early intervention process uses “best practices” to ensure the development and implementation integrity of effective interventions that are matched to the level and intensity of student need. Best practices in early intervention include the following:

- A comprehensive review of the student’s attendance, academic history, (e.g., report cards, curriculum exposure), school history, health history, health record (e.g., vision, hearing and current health status), experiential background, cultural issues, and language proficiency
- Observations in a variety of settings
- Developing a comprehensive plan of action that includes clearly stated goals, a reasonable timeline, specific assignment of responsibilities, progress monitoring and a decision-making process to determine if the plan is successful or requires changes, or the development of a new plan to meet the student’s needs

- Curriculum-based assessments
- Ongoing assessment to document growth and to provide a basis for instructional decisions

The promise of an effective early intervention system to ensure appropriate referrals to special education is lost if interventions are not linked to student needs, implemented with fidelity, and evaluated to determine the effectiveness of the intervention. Improved academic and behavior outcomes are associated with interventions that incorporate quality indicators (Flugum and Reschly 1994; Flugum and Tilly 2002; Tilly, Reschly and Grimes 1999). Incorporated in this document is a sample checklist to assist school personnel with intervention design, implementation integrity, and intervention evaluation (see Appendix F, Checklist for Intervention Quality Indicators).

As barriers are being removed, district personnel have the opportunity to unify resources and services to address the needs of *all* students. School district personnel are encouraged to break away from the old service paradigms and create new unified models of service delivery. For example special educators might be very involved in intervention programs involving K-2 children who are at risk for learning problems. Similarly, general education resource specialists (e.g., reading and math specialists, Title I personnel) may include children with and without disabilities in the same instructional group when the learning needs of the children are similar.

Aim

The aim of this paper is to address the question: how well do screening tools specifically identify children with an intellectual disability?

2.6 Method

Search strategy

A literature search was carried out with the keywords 'screening' or 'assessment' tool or test + 'intellectual disability', 'mental retardation' or 'intellectual disability' + 'child*' using the following databases: Ovid, PsycINFO, Global Health, EMBASE, AMED, CINAHL, Medline,

Web of Science, World CAT . A search was also carried out using the Cochrane database.

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Secondary searches were carried out from references of relevant papers. The search was restricted to articles in the English language and from 1990 to early 2009. The exclusion term 'learning difficulty' was used. For the purposes of the study, children were defined as being aged up to 18 years.

The initial search produced over 3000 potential papers. Once duplicates and papers which were clearly irrelevant were excluded, approximately 600 remained. To refine the search further, a number of additional exclusion criteria were applied. Papers were excluded if the screening was based on short forms of intellectual or neuropsychological assessments, as such screening would still require to be carried out by appropriately qualified professionals, such as applied psychologists (Goldstein et al., 2004; BPS, 2001). In addition, research with adults with an intellectual disability shows that many do not have a uniform cognitive profile (Murray et al., 2002), suggesting that short-forms of intellectual assessments may be inaccurate with this group. Using such short-forms with children may have similar limitations. Papers which screened for specific learning difficulties or general school performance and those relating to genetic or medical prenatal or postnatal screening were also excluded. Papers relating to screening for very young children (under age 2) were excluded due to previous reviews which have indicated that such screening is poor at predicting future intellectual functioning (Bornholt et al., 2004, Sonnander, 2000). An additional condition which was applied was that the minimum criterion for concurrent validity was the comparison of the screening tool with a valid, reliable and standardised assessment of intellectual functioning. Finally, papers which reported on screening tools which were designed primarily for adults, but included some adolescents in the standardisation sample were also excluded (e.g. Ford et al., 2008).

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A detailed examination of the remaining 26 papers indicated two main limitations in terms of their applicability to the screening of children with an intellectual disability. One group of studies reported on screening tools for a range of disabilities e.g. physical, intellectual, hearing which were used to provide, amongst other things, estimated prevalence rates of intellectual

disability in certain countries and populations. The classification was subsequently confirmed or otherwise by diagnostic assessment, but no information was provided that allows the psychometric properties of the tool to be calculated specifically in relation to identifying intellectual disability (Xie et al., 2008; Mirza et al., 2008; Bashir et al., 2002; Christianson et al., 2002;). These papers were, therefore subsequently excluded from the review.

A second group of studies reported on the reliability and validity of screening tools in large, heterogeneous populations such as children at risk for low IQ (Russell et al., 2002, McIntosh, 1999: McIntosh et al., 2000; Montgomery et al., 1999, Andrews et al., 1995) and poor educational attainment (Scott et al., 1998; Scarr et al., 1994) or children with general disabilities (Chopra et al., 1999, Thorburn et al., 1992, Zaman et al., 1990), general cognitive impairment (Besson & Labbe, 1997) or developmental delay (Heo et al, 2008, Rydz et al., 2006, Walsh et al., 2007, Billard et al., 2002, Lenkarski et al., 2001, Sonnander & Claesson, 1999, Leppert et al., 1998, Deuel, 1998; Glascoe & Byre, 1993). Some researchers reported on the identification of children with intellectual disabilities within the wider group being screened, however, the screening tools used were not developed specifically for this purpose. It was, therefore, unclear to what extent they could differentiate between children with an intellectual disability and children with other cognitive, behavioural or social difficulties. These papers were also excluded from further detailed review. Screening for Intellectual Disability in Children

A final quality indicator was then applied: that the study had a sufficient sample size to achieve statistical power. Camp (2006) also notes that small sample sizes in screening tools can lead to unreliable results due to the large confidence intervals involved. This led to the exclusion of a small pilot study of the child learning disability screening tool (CLDSQ) (McKenzie et al., 2008) developed from the adult version (LDSQ, McKenzie & Paxton, 2006). The CLDSQ was specifically developed to identify children with probable intellectual disability and had concurrent validity based on a clinical assessment of all three diagnostic criteria for an intellectual disability, however, it was based on a small sample size of 33 children.

The one remaining paper was by Simonoff et al. (2006) and utilised the Cognitive Abilities Test (CAT) (Thorndike & Hagen, 1986) as a screening tool. The CAT was originally designed to assess reasoning abilities linked with academic success, with the aim of identifying talented children or children at risk, however, it was included in this review as Simonoff et al. (2006)

utilised it specifically as a screening tool for mild intellectual disability. The authors note, however, that due to a lack of a standardised short form of CAT, items for their study were selected based on face validity and from the psychometric data provided for the complete test: they included verbal and non-verbal subtests but omitted the mathematical subtests.

The study included 2726 children between 12-13 years old from 15 schools (including schools for children with moderate learning disability, physical disability and emotional and behavioural disorders). The children completed the CAT individually, in small groups or in their class or year group. Scores were age-corrected allowing children to be identified according to risk of mild intellectual disability. A follow-up study of 204 children (80 Screening for Intellectual Disability in Children children categorised as high risk, 60 moderate risk and 64 low risk) involved in depth assessment, including a range of cognitive, psychological and attainment assessments, information from parents and a medical examination. Intellectual functioning was assessed using 9 subtests from the WISC-III (Wechsler, 1992). Data were adjusted in 3 stages to account for non-participants, the sampling design and differences between the number of children who were recorded on the school rolls and those who were estimated as being resident in the area from the local census.

The authors found a correlation of 0.76 between total CAT scores and WISC-III full Scale IQ scores with the highest correlation of 0.77 being found between Verbal IQ and CAT total score. They conclude that CAT is an effective screening tool for intellectual disability, but unfortunately fail to provide sensitivity and specificity values or cut-off scores for the test which would indicate intellectual disability in this sample. The CAT was also only administered to children in the 12-13 age group and its developmental sensitivity as a screening tool for children with an intellectual disability is unknown. The authors do not specifically state the length of time it took to complete the short form of the CAT, however they note that the time period allowed to complete the test was extended, enabling it to be completed in two school class periods. This suggests that the version of the CAT used by Simonoff et al. (2006) may not be much quicker to administer than standardised intellectual assessments.

The paper reports on a comprehensive and interesting study and the authors were thorough in accounting for potential sources of bias in the data, for example in relation to non-participants, however, it is unclear what the impact was of using a non-standardised short form of the CAT as

a screening tool and a shortened form of the WISC III (Wechsler, 1992) Screening for Intellectual Disability in Children to assess intellectual functioning. In addition, both of these assessments have been revised since the study was carried out, meaning that research using the updated tools would be required before firm conclusions could be drawn about the utility of the CAT as a screening tool for children with an intellectual disability.

2.7 Summary

Accurate and timely diagnosis of intellectual disability has clear benefits in terms of accessing support and services, but it is estimated that only 15-20% of professionals such as paediatricians and doctors use formal screening measures (Dobrez et al., 2001, Hamilton, 2006), partly due to a lack of agreement on which screening tool to use (Dobrez et al., 2001). This lack of agreement is understandable, given that no single screening tool which was reviewed was shown to be sufficiently valid or reliable for use with children with an intellectual disability.

The review did, however, highlight a number of areas that future researchers need to address if developing a screening tool. Importantly, the tool must serve the purpose for which it was designed. In some developing countries or areas where service provision is limited, the purpose of the screening tool may be to provide an initial indication of the number of children with a range of different disabilities, rather than focusing on children with intellectual disabilities per se (e.g. Thorburn et al., 1992; Zaman et al., 1990). In countries where service provision for children with an intellectual disability is well established, screening tools are likely to serve a range of different purposes including early identification. In these circumstances, for maximum utility, the screening tool should be designed specifically to identify children with an intellectual disability and differentiate these children from other groups, rather than including them in a large heterogeneous sample such as children with Screening for Intellectual Disability in Children developmental disabilities, low IQ or educational difficulties. The concurrent validity of the tool should be, at minimum, measured against the diagnostic criteria of intellectual disability, which in turn should be assessed by a valid, reliable, individually administered assessment of intellectual functioning. The tool needs to be developmentally sensitive and to have sufficient participants in each age band to ensure that there are not fluctuations in sensitivity and specificity due to children at different ages having scores with vastly different standard deviations (Camp, 2006). Similarly the overall number of participants must be sufficient to ensure the study has

statistical power. Data should be provided that allows the tool to be assessed in terms of the criteria of a good screening tool (Charman et al., 2007; Camp, 2006; Glascoe, 2005) and the likely prevalence of intellectual disability in the population being screened should be accounted for to ensure an accurate measure of the positive predictive power of the tool. Screening for Intellectual Disability in Children

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Unit 3

Strategies for Functional Academics and Social Skills

Content

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3.1 INTRODUCTION

Social skills are defined as the set of skills used to interact and communicate with one another. These skills include daily interaction skills such as sharing, taking turns, and allowing others to talk without interrupting. According to Kratchowill and French (1984) [1] social skills are learned verbal and nonverbal behaviors that are performed within a specific social context. Social skills are necessary to form and sustain relationships with others. These skills may be acquired through gradual learning and are largely influenced by a variety of social agents present in the culture. This process of learning and acquiring in the society is called socialization and when these skills are performed adequately they are referred to as social competency. Social competence includes both social skills and adaptive behavior. Children differ greatly in their social traits because of the influence of family, neighborhood and school environment which are important units of a society. Schools and classrooms are social environments where children function effectively from instructional activities that occur in an interactional context. However children with mental retardation encompass a heterogeneous group of people with varying needs. They learn things more slowly than other children of the same age. They show a delay in development like late to smile, to move, show interest in things, sit, and walk. They have a

deficit in one or more areas like self care, home living, communication and social skills, use of community resources. They have difficulty in intellectual functioning and in the performance of day to day activities expected of a person of similar age. Children with mental retardation are often at risk for difficulties in social and emotional development. These children lack the social and language skills needed to initiate and maintain relationships. The social repertoires of children with mental retardation have been found to be limited compared to those of normal children. One of the early findings by Guralnick&Weinhouse (1984) [2] and Strain (1984) [3] was that children with mental retardation initiate fewer social interactions and demonstrate fewer responses to peers when compared to normal children. As early as preschool, many children with developmental disabilities have difficulty forming friendships. They tend to interact less frequently with peers and engage more frequently in solitary or unoccupied activities.

Children with disabilities experience difficulty in initiating activities and forming peer groups (Guralnick& Groom (1987) [4], Wilson (1999) [5]. Families of children having mental retardation often face challenges because of the special demands placed upon them. The families typically make a strenuous effort to organize the family environment and to reach out to obtain resources that will meet the needs of their child. Strategies like social skills training, sensitivity training for peers, imitation of peers, teacher prompting, group affection strategies, peer mediated intervention, cooperative learning have been developed for training children with mental retardation.

Rationale of the study Social skills are an important aspect of education for all young children with and without disabilities. Inclusive settings provide forum through which children with disabilities can learn through modeling from their typically developing peers as well as from teacher led instruction. Children with disabilities need to learn appropriate social gestures such as smiling, and making eye contact. In India children with mental retardation receive educational services through special schools, integrated set up home based and community based programs. Educational programs for these children focus on functional academics, community living and vocational training. These programs are provided in various setting ranging from specialized educational centers to home based services. Research studies on social skill development of children with mental retardation through these service models are very scant. However there is a need to find out the effectiveness of teaching and the learning of social skill development in the

children. Therefore the current study focuses on investigating the achievement of social skills in children with mental retardation attending home based program and children attending regular special schools. The study will throw light on the effectiveness of home based programs and special schools.

Objectives The objectives of the study are

- 1) To compare social skills of children with intellectual disability who are attending home based programs and those children attending regular special schools.
- 2) To compare the social skills of children with intellectual disability attending home based programs and those children attending regular special schools – Domain wise

3.2 METHODOLOGY

The present study is a Descriptive research study. Survey method was used for collecting the data from children attending home based program and children attending regular special schools.

3.2.1 Participants The participants for the present study consist of children with mild and moderate mental retardation, age ranging from 5 to 10 years. The sample was drawn from special schools situated in the twin cities of Hyderabad and Secunderabad. The sample size for the present study is 150 children, in which 75 children are attending home based program and 75 children are attending regular special schools. Children with mental retardation having associated conditions like cerebral palsy, autism and sensory impairments were not included in the study. Convenience sampling was used by the researcher because this method consists of groups or individuals who are readily available.

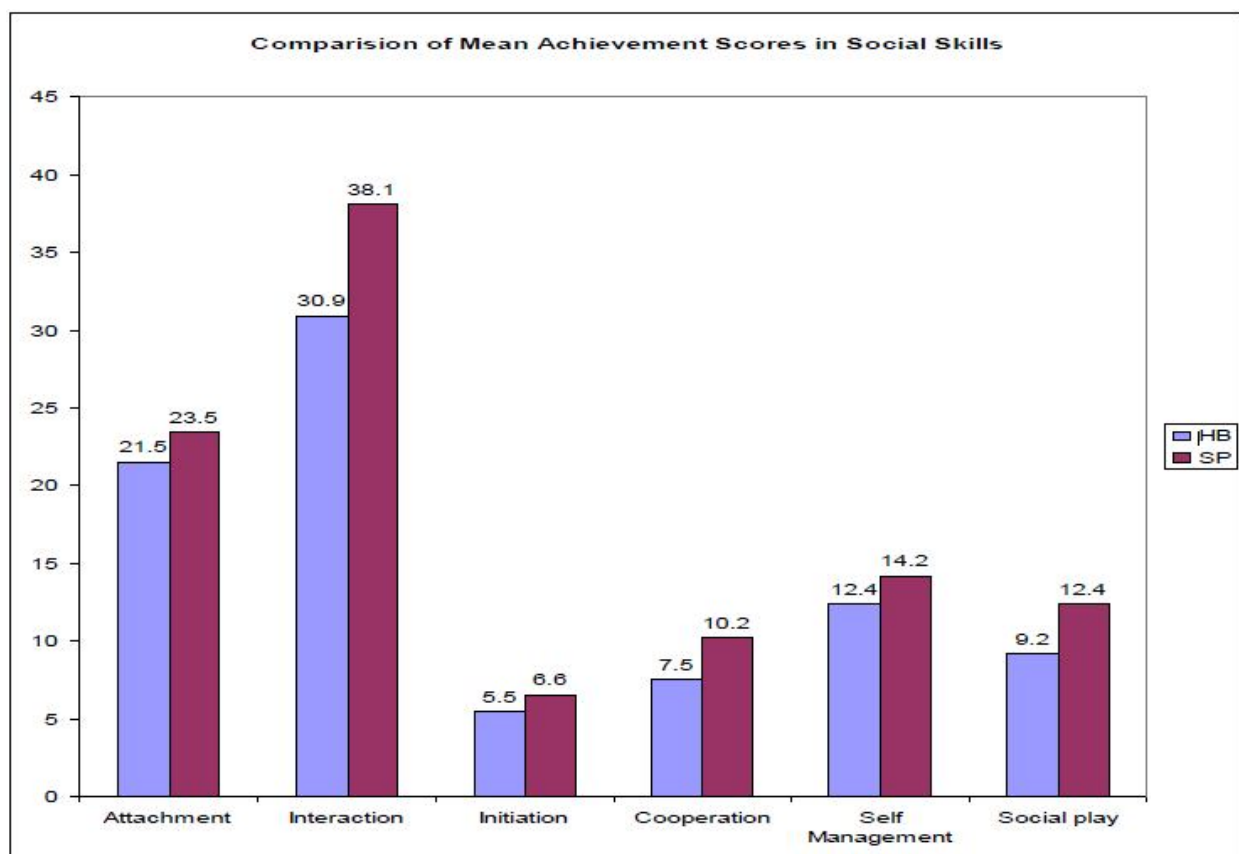
3.2.2 Instrument Social Skills Rating Scale (SSRS) was developed by the researcher for evaluating the social skills of children with mental retardation. The Social Skills rating scale is a 3 point rating scale – Always (2), Sometimes (1) and Never (0). Cumulative model of scoring was used for the scale. There are six domains in the SSRS – they are Attachment (20), Interaction (35), Initiation (9), Cooperation (12), Self-management (13) and Social play domain consists of (13) items. Reliability and Validity of the SSRS was established and the final checklist consisted of 79 items.

3.2.3 Procedure The researcher had taken permission from the principals of the schools. Parents were informed about the study before filling the scale. The SSRS was administered to 150 subjects. The researcher administered the scale to each subject individually, and collected the data through observation and asking information from the parents.

3.3 RESULTS

The data collected was analyzed using appropriate statistical measures. To compare the social skills achievement of the subjects the Mean, SD and independent t-test were computed. The results are discussed below *Home based (HB)*, *Special School (SP)*, $p < 0.05$, $p < 0.01^{**}$ Table 2 indicates the social skill achievement of children attending home based program and children attending special school with respect to different domains. Achievement of social skills in each domain is compared among children of both the groups to see whether there is any difference. From the results it is evident that children in special schools have achieved higher scores in all the domains when compared to children in home based program. Further comparison of results indicated that though there is a difference in the achievement of social skills it was observed that in both the groups' the scores in the Interaction domain is high and low in Initiation domain. Interaction skills are fundamental for functioning in a society. It may occur at verbal or nonverbal level. Children interact with each other when they play in groups, whether at school, home or in the community. Theories of Mead, Freud, Bowlby, Erikson, Sullivan, Piaget and Kohlberg all emphasize the fact that the nature and quality of children's social interactions have a significant impact on their development. Similar findings were reported by Rubin et al. (1998) [8] that a competent social development in early childhood includes the ability to interact with others effectively and develop positive relationships. In schools activities like sports, competitions, and cultural programs are conducted. Participating in these programs provide a wide range of opportunities like sharing, cooperating in social situations, interacting with each other. Howes (1987) [9] and Saarani (1990) [10] have reported that participation in interaction requires the ability to enter into play with others, to coordinate one's actions appropriately during the course of social interaction, should show positive behaviors towards their peers.

Fig- 1 Comparison of mean achievement of social skills of children in home based programs and children in regular special schools with respect to domains.



3.4 The Role of Functional Skills Instruction

While many students transition successfully into adult life, many are at risk for experiencing difficulty during this period (Zarrett & Eccles, 2006). Even greater difficulty has been reported for students with disabilities (Brolhier, Shepard & Markley, 1994; Wagner, Newman, Cameto, Levine, & Garza, 2007). Therefore, students with disabilities, and particularly students identified as having an intellectual disability, should be provided functional skills instruction within the educational environment with the premise of teaching skills necessary for successful transitions into adulthood.

Functional skills, according to Cronin (1996), are the tasks that help individuals become successful and independent adults. Vandercook (1991) stated, “a true functional skill is one that is initiated, used, and maintained under typical circumstances” (p. 320). Functional skills are often taught in conjunction with functional academics. Bouck and Joshi (2012) defined functional academics as an approach to teach “students the skills to help them be productive members of society and support post school outcomes” (p. 140). Functional academics may

include “core subject content, vocational education, community access, daily living, personal finance, independent living, transportation, social skills and relationships, and self-determination” (Bouck & Joshi, 2012, p. 140). This article will discuss functional academics as it relates to successful transitions into adulthood through literacy, social skills, self-determination, and community involvement instruction.

3.5 Transition to Adulthood

In order to create a context for the understanding of functional academics and functional skills, it is imperative to view this notion within the broader scope of the concept of transition. High school transition can be “a period of *floundering* that may occur for at least the first several years after leaving school as adolescents begin to assume a variety of adult roles in their communities” (Halpern, 1992, p. 203). Evidence of the difficulty experienced by students with disabilities during this critical transition was brought to light through a number of follow-up and follow-along studies (Hasazi et al., 1985; Hasazi, Johnson, Hasazi, Gordon, & Hull, 1989; National Commission on Excellence in Education, 1983; Newman, Wagner, Cameto, & Knokey, 2009; Wagner et al., 2007) conducted in the 1980s and 1990s. As a result, schools in the United States are federally mandated to make available to youth with disabilities, techniques, skills, and the guidance necessary to facilitate successful transition to adult life.

A number of recurring areas of need were highlighted in these studies, among which was the need for what Kohler and Field (2003) and IDEIA (2004) refer to as student development. Critical to student development is the need for instruction in functional life skills. In addition, the No Child Left Behind Act (NCLB), (2001) mandated that such instruction be substantiated by scientifically based research. Researchers have identified functional skills that lead to post-school success (Bouck, 2010; Test et al., 2009).

3.6 Functional Literacy

Literacy can be considered a functional skill when viewed in terms of everyday spoken and written communication. Literacy can also be a means for students with disabilities to access other academic areas. Being literate also increases these students’ opportunities to interact with peers and gain entry to community offerings. Literacy as part of a functional skills curriculum

should focus on those skills that increase students' ability to be autonomous and independent (Ruppar, Dymond, & Gaffney, 2011).

An essential component of being literate is acquiring reading skills. Reading can be a struggle for many students with disabilities; therefore, the use of evidence-based reading instruction can help them acquire basic reading skills needed to function in school and community settings (Pullen & Cash, 2011). Although it may seem late for effective reading intervention, secondary students with disabilities benefit from instruction which focuses on word study and text comprehension (Malmgren & Trezek, 2009). Diagnostic, prescriptive reading programs (e.g., The Wilson Reading System and Corrective Reading) are particularly effective in bolstering adolescent students' skills in phonemic awareness, phonics, fluency, vocabulary, and comprehension. For students with disabilities, reading allows them to find work, participate in leisure activities, and pursue interests outside of school and/or work settings (Forts & Luckasson, 2011).

A second essential component of literacy is the ability to write. Writing instruction can allow for participation in class activities as well as a feeling of being a member of the classroom community (Ruppar et al., 2011). In addition, because of the increasing demand for writing in everyday life, students with disabilities must become as proficient as possible in this skill (Graham & Harris, 2011). A writing strategy, such as Cognitive Strategy Instruction, provides students with disabilities with a structure for understanding and approaching writing tasks (Guzel-Ozmen, 2009). Writing, as part of functional literacy, presents an opportunity for students to develop essential skills as well as to develop an additional meaningful means of communication and social connection within their respective communities.

3.7 Social Skills

Students with disabilities are at risk for diminished social skills. As a result of this deficit, their ability to successfully navigate their personal and educational environments may be negatively impacted. Social skills are those skills or behaviors that are perceived and positively reinforced by others as socially acceptable and then therefore lead to the avoidance of socially awkward situations (Johns, Crowley, & Guetzloe, 2005). It has been suggested that while an academic disability in the classroom may not be evident on the playground, difficulties with social skills impact a student wherever they may go (Snider & Battalio, 2011).

According to Johns et al. (2005), teaching social skills can involve various instructional methods. Direct instruction identifies a specific student, a particular skill, and then instruction is delivered, progress is monitored, and opportunities are provided to use that skill across various environments. Another strategy for teaching social skills is taking advantage of the teachable moment where a socially awkward moment is addressed with immediate feedback in a constructive manner. A third strategy is for the teacher to model appropriate social skills for the students, and to check for understanding. A fourth strategy suggested is recognizing or acknowledging when a student displays appropriate social skills and providing positive feedback. A fifth strategy is to have students participate in group projects that emphasize working together on social skills such as giving and receiving a compliment, and how to express thankfulness. Additionally, instruction in conflict resolution will benefit students as they interact in a social world.

3.8 Self-determination

The need to provide students with disabilities the necessary functional skills to become active participants in their education and planning for their future has led to an increase in instruction in self-determination skills over the last two decades. According to Carter, Lane, Crnabori, Bruhn, and Oakes (2011), self-determination is “the capacity to direct one’s life in ways that are personally valued” (p.100). Denny and Davison (2012) suggested that research has demonstrated a relationship between self-determination and optimistic outcomes for students as they transition to adulthood. Self-determination instruction includes teaching skills essential for potential success. These skills include decision-making with regard to education, leisure, vocational training, and independent living (Palmer, 2010). Instruction to increase self-determination may include teaching students how to develop problem solving skills; how to set personal goals, how to make personal choices; how to set goals and develop plans to reach those goals; and finally instruction on self-regulation (Wehmeyer, 2002).

3.9 Community Involvement

According to Carnahan, Hume, Clarke, and Borders (2009), independence is essential in order for individuals to successfully integrate into the community and employment. Independence skills can be taught across disciplines. Teachers actively engage students with disabilities in the

planning of their futures in order to help them recognize, analyze and generalize information. Therefore, educational agencies, through the application of functional skills, teach students with disabilities at an early age to learn to think and function independently (Carnahan et al., 2009).

The Individuals with Disabilities Education Improvement Act (IDEIA) (2004) ensures that all children with disabilities have access to a free and appropriate education, along with special education services that prepare them for further education, employment, and independent living. IDEIA (2004) states that transition planning should help students enter “vocational education, postsecondary education, community participation, adult services, independent living, integrated employment, and/or continuing and adult education” (Sec. 602(34) (A)). Therefore, schools provide transition services in order to prepare students with disabilities for adulthood (IDEIA, 2004). Often these transition services are taught through community involvement. Therefore, community involvement is a key addition to instruction in functional skills, including functional academics. Through community involvement, students are provided with hands-on opportunities to learn skills necessary for a successful adulthood.

Gerhardt (2007) stated that students with disabilities should be exposed to jobs in order to learn the skills necessary to be successful. Individuals with disabilities will need to learn through experiences on a job to know what skills and conditions are needed in order to successfully gain and sustain employment. When students with disabilities are provided vocational opportunities through experiential learning and functional academics, they can learn firsthand the skills that can be generalized into other environments and settings.

Vandercook (1991) concluded that recreation and leisure skills also should be taught within the context of community involvement in functional academics. When recreation skills are taught as a component of functional skills instruction, students are able to effectively generalize those skills into other environments with peers without disabilities. These skills help individuals with disabilities adapt in social situations through community involvement.

3.10 LEARNING DIFFICULTIES FOR PEOPLE WITH SPECIAL NEEDS AND HOW TO HELP

1.1 ATTENTION

At the encoding level, attention is the main learning process vital for receiving incoming information. Research has shown that people with intellectual disability have problems in the level of attention, focusing attention and selective attention.

How can you help?

General suggestions to increase attention level and focus

- 1) Designate an area of work where full attention and focus is required then. Place pictorial symbols to remind your client that he/she needs to pay attention and focus.
- 2) Ensure that the material is of your client's level.
- 3) Make the tasks interesting and attractive (use colourful material and vary the tasks). For instance, while teaching numbers, you can use a stack of UNO cards to play a number game, count marbles etc.
- 4) Divide the tasks into those that are interesting and less interesting and alternate them.
- 5) Use colour cues to direct attention to important information, key words and directions.
- 6) Give your client a choice for an activity. Your client will feel more involved and hence more focused.
- 7) Provide the element of surprise to gain client's continued attention. For instance when teaching puzzle arrangement, give your client each piece one at a time.

Eye contact and focuss some intellectually disabled persons may have difficulties maintaining eye contact and focusing on an object. Here are a few helpful suggestions:-

- 1) Tie bells to the tool your client is using (eg. broom when he is sweeping) to attract his/her attention to the tool.
- 2) Attach an object (eg. ball with bells attached)) to a string and swing it like a pendulum. Encourage your client to follow its movements.

3) Move a noisy object (squeaky toy, bell etc.) at a horizontal line your client's eye level. Start from an extreme left to your client's extreme right. Repeat it with movement of the object up and down and the four principal diagonal directions.

4) Sit in front of your client. Push a bead/roll a ball along a string within your client's view. Encourage him/her to follow its movements.

3.11 Selection and Discrimination

Intellectually disabled people have difficulties differentiating the visual objects presented to them. This concept is important to learn how to acquire concepts like colour, shape, size etc. Some research has suggested that people first acquire the concept of shapes (form) before colour and three dimensional before two dimensional discrimination. The following are a few suggestions to help your client take note of differences in the visual field:-

1) Spot the difference games. Eg. pictures that are similar except for minute details, colour, shape etc.

2) Copying block formations. Help your client build model shapes from an example. Start by using blocks of the same colour before varying the difficulty of the task by adding different colour patterned.

3.12 MEMORY

Memory is important to help your client store new information gained for future use. Research shows that people with intellectual disability have difficulties in short-term memory but have normal abilities for long-term memory. Most short-term memory problems are related to a deficit in the ability to use rehearsal strategies or adequate rehearsal activities. This means that they do not practice or repeat the new information sufficiently or effectively to remember it.

How can you help?

1) Teach your client rehearsal strategies

- Verbal rehearsal –teach your client to verbalise the steps of a task. After becoming expert at the task, verbalization usually ceases to silent verbalizations. For example, when

teaching the a puzzle formation, teach your client to repeat verbally the required steps “look for the triangle, then the blue, then turn to fit together.”

- Image rehearsal-teach your client to associate aspects of the task with a visual image. Eg. Tie shoelaces by making a rabbit ear, another rabbit ear and a big knot together. When teaching your client, remember to use the same instructions and verbal or image prompt.

2) Six common strategies to remember when helping your client to remember:

The Steps Example task: identification of modes of transportation

- 1) Actively label and repeat presented Items -label each picture of the transport mode with the corresponding spelling (eg. car, boat, aeroplane)
- 2) Group common items together -group transportation by air, land and sea
- 3) use category cues -show that these are categorized under modes of transportation
- 4) repeat items presented earlier -eg. repeat ‘car’ and ‘boat’ that was learned the week before
- 5) develop distinct visual images -eg. the aeroplane is like a bird. Teach your client to stretch his arms like a bird/aeroplane
- 6) develop verbal links -eg. the aeroplane flies in the sky

3.13 GENERALIZATION

This is crucial to ensure that your client is able to independently apply available skills in novel situations. Most intellectually disabled people find generalizations difficult as they do not know the appropriate moment to use the information/processing strategy.

How you can help?

- 1) Expose your client to as many different contexts to maximize the ability to transfer. For instance, when teaching time telling, once your client has mastered the basic skills, show him/her as many different clock faces for both digital and analog clocks.

3.14 PROBLEM SOLVING

This involves higher level processing, an analysis of problems, anticipation of outcomes and decisions on how a problem should be solved. With exposure and experience, your client will learn the necessary skills to enable him/her to approach problems.

1) The most direct way is to teach your client to stop and think. For example, ‘Am I doing this the right way? Is this correct?’ This ties in closely with verbalization skills mentioned above where verbalizations remind your client to systematically solve problems.

2) Group the information/material presented to your client in a meaningful way. For instance, in puzzle solving, teach your client to look out for general colour similarities, form and shape. Also remind your client to check the picture against the model he/she is building.

3.15 DO’S & DON’TS WHEN TAKING YOUR TRAINEE

1. Provide your trainee with a positive model of appropriate behaviour and social skills.
2. Be certain that your behaviour mirrors the skills that you are teaching your trainee.
3. Correct your trainee behaviour if it is inappropriate.
4. Before the start of the task, you have to demonstrate the steps to your trainee, explain to him/her the objectives of the task and what you expect from him/her.

5. Do the task with your client. Hold your client’s hands and bring him through the task.

Initially you may end up completing the whole task but once you’ve guided your client three or four times, reduce your physical assistance. Let your client take more responsibility in completing the task.

6. Be willing to interact with your trainee.

7. Be committed.

8. Talk to your trainee more even if he/she doesn’t speak because he/she will understand and may respond through body language. For those who can speak, encourage them to talk.

9. Volunteers to seek help from each other if lost and ask around if not sure.

Don'ts

1. Don't leave your trainee alone.
2. Don't encourage your frustrated trainee to relieve stress via pointless activities.
3. Don't share plates, forks, spoons, cups with your trainee as he/she have lower body resistance and are prone to illness.
4. Don't place your trainee in a highly competitive/dangerous situation.
5. Don't assume that your trainee understood your oral directions or instructions just because he/she did not ask any questions.
6. Don't rush your trainee into completing the tasks, let him/her take his/her time and enjoy the process.
7. Don't treat your trainees as kids, children. Treat them as adults, maybe with mentality years younger than their age.

3.16 HOW TO TRAIN PEOPLE WITH INTELLECTUAL DISABILITY

1. Think of what you want him to learn before you give any command, and directions must be clear and extremely simple. Give short, concise and clear instructions. Use simple words.
2. Attract his attention by calling his name and establishing eye contact.
3. Provide a task within his ability.
4. Break down the task into simple steps.
5. Show rather than tell him how to do a task. Modeling an action/ the steps for your client helps him/her understand what you want your client to do and imitate your action. Your demonstrations help your verbal instructions. For instance, 'do this..' as you stack blocks or as you cut with the knife. This is only effective when done slowly with careful exaggeration.
6. Work on one thing at a time.

7. Try to have each task completed as finishing the job keeps up his interest, gives him pride and creates a good work habit.
8. Keep supervising him while allowing him to do as much as possible for himself.
9. Give positive reinforcement e.g. Praise or a tap on his shoulder when a task is done well or a good attempt has been made.
10. be very patient because the trainee may not readily carry over ideas from one situation to another, thus we usually have to treat each situation individually.
11. Be firm to the trainees. Do not let them manipulate you.
12. Have faith in the trainees regardless of their capabilities.

3.17 Discussion

For students with disabilities as well as those who are at-risk, functional skills are an essential instructional component. Functional skills instruction should include a focus on transition to adulthood, functional literacy, social skills, self-determination, and community involvement. These components of functional skills instruction help facilitate independent living skills that are an integral part of adult living.

There is a recognized need for functional skills instruction in education that focuses on teaching the skills necessary for independent adult living. The teaching and learning of these skills can aid in the development of productive citizens. A functional academic program that integrates the teaching of functional skills provides opportunities for at-risk students, such as students with disabilities, to experience more successful post-school outcomes.

The present study has revealed that children in special schools have achieved higher social skills compared to children in home based program. This may be due to the positive environment in schools which foster social skills in children. In integrated setups or in special schools children interact with each other. Peer interaction is high compared to home environment. In home based programs teachers and professionals should provide an environment where the children get to interact with each other. Emphasis on small group instructions where the children are brought to the center should be promoted. Results have indicated that children learn social skills better

when they are in groups. In India lot of community based rehabilitation programs are there to provide education to children with mental retardation. These programs should make effort to provide group instructions and organize activities where not only the children from the community but the parents also can work together in a social context. This will encourage parent participation and involvement in the overall development of the children. In Home based programs teachers should plan activities that focus on social skill development. Though teachers and parents are aware of the importance of social skills lot needs to be done. Further implications extend the need for teachers to prepare themselves for a diversity of social and behavioral problems and receive training on appropriate intervention and management techniques (Leffert&Siperstein, 2000) [11]. Training is needed in ways to incorporate social skills into the curriculum and in methods and materials by which skills may be taught. Therefore early childhood programs should focus on the social development right from childhood.

3.18 Summary

With the constant changes and shift in the education paradigm, educational programs brought a change in the quality of life of persons with mental retardation. The quality and quantity of educational services for children with mental retardation has been of growing concern since past few years. The field of special education has become increasingly aware of the needs of these children and their potential learning abilities. As such it leads to the evolution of a cascade of service delivery options for children with mental retardation. Research studies have indicated that home, school and environment play an important role in the development of social skills in children with mental retardation. Positive environments foster growth and development in the life of individuals. This enhances their social participation and interaction with fellow beings. Social skills as one of the dimensions of adaptive behavior has a strong and lasting influence on the lives of children with mental retardation, particularly during adolescence. However since every child has the right to education “Education for all”, there is a strong urge that teachers and parents should work collaboratively irrespective of educational service options(special school, home based, CBR, integrated school) in bringing out the potentialities of children with mental retardation to the maximum in the overall development.

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Unit 4

Assistive Devices, Adaptations, Individualized Education Plan, Person Centered Plan, Life Skill Education

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4.1 Introduction

Of the many acronyms used in special education none is as well known as “IEP.” Educators and parents as well as many people in the general public know that if a student has an IEP, an individualized education program, he/she is receiving special education and related services.

Guidelines: Individualized Education Program Planning Process (June 2007) were developed by the North Dakota Department of Public Instruction to assist educators who are developing IEPs for students with disabilities. Changes in the Individuals with Disabilities Education Act (IDEA) resulting from the 2004 reauthorization of the federal special education law, and the publication of accompanying final regulations in 2006 by the U.S. Department of Education, prompted the revision of this document. As defined in federal regulation 300.22 an individualized education program or IEP means “a written statement for a child with a disability that is developed, reviewed, and revised in accordance with 300.320 through 300.324.” It sounds so simple. Yet the IEP process is the focus of intense work by millions of educators, students and parents across America. A frequent criticism of the IEP process is that it is unnecessarily detailed. However, because of the unique needs of each child with a disability, there is a need to thoughtfully consider the special--individualized— needs of that child. The IEP is the process for a thoughtful discussion of those needs and a documentation of the plan for that child to receive a free and appropriate public education.

Guidelines: The Individualized Education Program Process (June 2007) incorporates major themes of the 2004 reauthorization of the IDEA. In this document you will see an increased emphasis on :

- Early intervention;
- Access to the general education curriculum;
- Scientifically based instruction;
- Paperwork reduction;
- Streamlining of the IEP process; and

- Caution related to disproportionality.

Throughout this document, quotes from the IDEA 2004 law, regulations and other supportive documents are included and in some cases, the description of the intent of Congress in making changes to the law. A significant effort has been made at the federal level to make the law and regulations more understandable to parents and school personnel.

Another feature of the IDEA 2004 and the final regulations is an increased relationship between special education and the Elementary and Secondary Education Act (ESEA). Perhaps nowhere is this more evident than in the heightened accountability that is expected. The North Dakota State Performance Plan for Special Education (SPP) is our six-year plan for improving the educational results for all children with disabilities. There are 20 indicators and each indicator detailed in the State Performance Plan contains information such as baseline data, measurable and rigorous annual targets, and improvement activities. collecting these data from each of the state's school districts and from parent surveys. This information is used to monitor and continuously improve both state and school district activities thus improving results for all children with disabilities. Please see Appendix A for the complete list of special education performance indicators.

The North Dakota Department of Public Instruction is committed to assisting educators and parents work together in the best interest of children and youth who have disabilities. We share a common goal of improving the education of young people who have disabilities so that they can be better prepared for postsecondary education, employment, increased independence, and more fulfilling lives. Appendices that are attached present additional guidance for IEP teams. The appendices include the following materials:

A. State Performance Plan 2005-2010

B. Secondary Transition IEP Process and Forms

- Transition IEP Form (Ages 16-21)
- Instructions for Completing IEP Form

- Sample Parental Consent Form for Agency Invitation to IEP Meeting
- Sample Summary of Performance Form
- Sample Transfer of Rights Form

C. Consideration of Specific Student Needs

- Students who are blind or visually impaired
- Students who are deaf or hearing impaired
- Students with limited English proficiency
- Students who demonstrate behaviors which impede learning
- Students who may need assistive technology
- Students Ages 18-21

4.2 Importance of Fitness and Fitness for Children with DisabilitiesCommon Fitness Barriers for Children with Disabilities

- Participants may use the Phone Desk Mat to tap the letters. For those who are not mobile:
- Participants may perform the activities from a seated position on the mat.
- A Desk Mat may be placed on a participant's lap for ease of use.
- A Learnerercise Mat or a Desk Mat may be mounted to the wall so that it is able to be touched with the hands from a seated position.

The Surgeon General's report (1996) on the status of the country's physical activity and health provides two key observations regarding the importance of physical activity for individuals with disabilities:

1. People with disabilities are less likely to engage in regular moderate physical activity than people without disabilities, yet they have similar needs to promote their health and prevent unnecessary disease.

2. Quality physical education, preferably daily and K-12, must be made available to children and youth with disabilities.

Although there has been great emphasis on physical activity and health over the past two decades, information specific to adaptations and strategies for exercise in children with disabilities remains scarce. However, there are some reports of model fitness programs that have demonstrated success in children with disabilities. For example, the Learnercise Mats are designed to engage children of all ability levels while promoting cross-disciplinary collaboration.

The use of activities that promote cardiovascular fitness, are inclusive, and connect with key concepts across the discipline promotes a “win-win” for educators who struggle to include children with disabilities. Activities that can address the full continuum of abilities often found in the general physical education environment are critical for the success of today’s children and teachers (Davis, 2007).

There are many definitions of physical fitness, but almost all of them have been directed at adults. When defining fitness in the context of a child’s life, and in particular a child with a disability, a variation of the definition proposed by Pate, Pratt, and Blair (1996) seems appropriate:

- Provides an emotionally safe, non-violent/non-aggressive learning environment
- Provides opportunities to develop the “whole child” and the use of different learning styles
- Uses the latest brain research to develop curriculum
- Offers options so that each individual can be successful

For physical educators, safety is of utmost concern when teaching all students. Because of the limited number of fitness programs available for children with disabilities, they are often limited in their understanding and knowledge of fitness concepts. When conducting fitness units teachers should be aware that children with disabilities may not wish to participate because it is difficult for them to keep up with their non-disabled peers (Block, 2006). Whether the child with a disability is accompanied by the adapted physical educator, assisted by a peer helper, or is independent, it is important to establish individual safety guidelines for exercise. Most children

with physical and intellectual disabilities should not have difficulty participating in high intensity fitness activities unless there is evidence of a cardiac or pulmonary disorder (Rimmer, 1997). In this case, a physician should screen the child. Specific fitness protocols may have to be followed to ensure safe and successful exercise. Nevertheless, it is always wise to start off slowly and follow procedural guidelines to ensure personal safety. The following general guidelines for conducting safe fitness programs were developed by Dr. Jim Rimmer (1997). General Safety Guidelines for Conducting Fitness Activities for Children with Disabilities

- Check medical history to make sure there are no contraindications to fitness-related activities (e.g., heart condition, exercise-induced asthma, osteopenia, drug interaction).
- Monitor cardiovascular exercise by using rating of perceived exertion (RPE) and an intensity level between 60 and 85% of target heart rate range to assure safety in the activity. If discomfort is exhibited at the 60% threshold, drop to 50%.
- Avoid conditions where there is a likelihood of hypothermia or hyperthermia. Some children with paralysis have difficulty retaining or dissipating heat.
- Teach the physical education instructor how to transfer a child from a wheelchair so that fitness activities can be performed in different positions (e.g., mat, exercise machine).

4.3 Adapted Physical Education

When conducting aerobic activity with students who have limited mobility, balance, or who may use a mobility device to ambulate, an essential question is whether they should sit or stand when conducting the activity. Depending on the ability of the individual, basic adaptations can be made without much difficulty for either. Consider the individual's balance first when assessing their ability. If possible, consider using parallel bars when focusing on lower extremity movement and a seated position when focusing on upper extremity workouts. Generally, wheelchair users or those with mobility impairment need to work on cardio-respiratory endurance and upper body mobility, including muscular strength. Consider the following when conducting aerobic exercise and using Learnercise Mats:

A child with balance issues may use a chair (with rubber tips), a walker, or crutches placed directly on the mat. Movements should be slowed down so the individual has the chance to step

and gain balance before moving to the next step. A child may be taken out of his or her wheelchair and may sit in the middle of the mat. The child may make responses by tapping their hands or feet on the appropriate locations.

- Whenever possible, focus on functional movements of the individual (e.g., leg, trunk, upper extremity mobility). Have the individual move as much as they can independently and without pain.
- Try upper body mobility exercises, including lifting arms straight out in front for a few seconds, then lowering. Increase time with successful performance.
- Try lifting arms to the sides (stopping at shoulder level), holding, and then lowering.
- Additional upper body activities include shoulder shrugs, overhead “air” presses, wrist curls, finger extensions/dexterity, bicep curls, and triceps extensions.
- For students who are able to stand, a proper warm up and a cool down promoting flexibility are also important for reducing the chance of injury. Work on flexibility and range of motion of the shoulders, arms, back, and neck. Rehab Team Site offers examples of stretches for the upper body for individuals with spinal cord injury.

4.4 Students with Mobility Impairments

The specific concerns for those with visual and/or sensory impairments (vision or hearing loss) involve environmental awareness and exercising safely. As with any student, the first concern should be on using proper form. Start slowly with exercise and, whenever possible, use the educational medium that is most familiar to the student. For example, for the individual with visual impairment who uses braille, provide a list of movements that describe the proper fitness form as well as hand over hand modeling.

- Place raised tape on the middle of the mat. This will be the “home” position for the child. Familiarize the individual with the mat and the location of each visual on the mat.
- The Learnercise Mats are very bright in color. Therefore, some children may be able to detect areas on the mat.

- Give directions in reference to the home position. For example, “Directly in front of home is the number ‘2.’ Step forward and balance on one foot.” Activity cards providing the directions can be developed using braille.
- Provide more activities that use music and are performed to a steady beat. Many children may be able to feel the vibration of the music and move to the beat.
- It is important to keep the area safe and clear.

The following are suggestions as outlined by PE Central’s disability fact sheets for physical educators (www.PEcentral.org):

- Ask the student what he or she is able to see.
- Use positive reinforcement to shape the movement performance.
- Use bright colors and/or auditory materials when possible. Consider that some individuals will be able to feel the vibrations of music.
- Use materials that are self-explanatory or post the sequence of movements (Braille card or poster).
- Familiarize the student to the environment (e.g., personal fitness mat). Consider adding raised, taped bumps or sensory targets to the mat so that students with visual impairments can maintain orientation.
- Consider using facilitated communication for students who are blind and nonverbal.
- Keep the environment clear of clutter and equipment. In addition, students with sensory impairments that are included in general classes commonly feel segregated or left out.

4.5 Students with Social/Emotional Delays

The Flixercise curriculum helps provide structure needed for children with intellectual disabilities by working to:

- Increase body awareness by placing different body parts on different locations on the mat and by tapping, bending, and stretching different body parts.

- Improve pre-locomotion skills by clapping, snapping, and tapping on the mat or on different body parts.
- Improve object manipulation skills by adding beanbags, streamers, paper plates, shakers, rhythm sticks, balls, etc.
- Develop posture by using directions such as “stand up tall,” “move on your tiptoes,” “freeze like a statue,” etc.
- Develop locomotor skills by marching, jumping, hopping, running, skipping, galloping, sliding, bouncing, dancing, etc. around the mat or on the mat.
- Stimulate the visual and auditory senses by using visual or auditory directions alone or by using additional tactile manipulation to develop each learning style.
- Focus on functional skills by having children pretend (e.g., “pretend you are cooking and stirring a stew,” “pretend you are a washing machine,” “pretend you are a basketball player and jump up to reach the ball,” etc.).
- Give a personal mat to each participant to provide a “security blanket” (personal space). Remind students that they need to stay in their designated spaces. Keep directions simple.
- Provide one direction or “piece” of information about the activity at one time so the students do not feel overwhelmed.
- Have clear “go” and “stop” signals and practice those signals in the context of a game.
- Use movements to “control” an individual or group that is getting overly excited (e.g., “Let me see how you can balance on three body parts quietly on your mat,” “Show me how you would move slowly around the edge of your mat. Pretend you are in slow motion.”).
- Provide structured routines and maintain consistent rules.

4.6 Students with Intellectual Disabilities

- On the first day, explain class goals and a specific routine to be followed.

- Discuss goals often with each student and develop a contract of what each student must do to achieve a goal.
- Keep rules simple. Set as few as possible while encouraging students to participate in the development and selection of rules.
- Clearly explain the consequences of not following the rules or regulations, as well as the rewards for following them.
- Allow participants to be involved in the consequence process. Post the process in the room or allow them to take part in making it.
- Demonstrate consistency in enforcing rules and providing feedback.
- Target behaviors that need to change and define the components of these behaviors.
- Observe, chart, and analyze behaviors to change.
- Select and apply specific strategies to achieve behavior changes (e.g., start and stop signals, routines for transitions, techniques for forming groups, and strategies for coping with disruptive behaviors).
- Periodically evaluate progress toward changing an individual's behaviors and revise their behavior change plan.
- Allow individuals to feel safe by task analyzing and structuring activities to assure personal mastery.

Asperger's Syndrome is a specific type of pervasive developmental disorder characterized by problems in the development of social skills and behavior.

4.7 CONTRAINDICATED ACTIVITIES

Students and athletes with Down's syndrome should be restricted from participation in gymnastics, diving, the butterfly stroke in swimming, the high jump, "heading" in soccer, and any exercise which places pressure on the muscles of the neck.

4.8 EFFECTIVE TEACHING STRATEGIES

Preschool

- Teach individuals in a highly structured environment.
- Allow for touching and feeling to learn.
- Use lighter weight equipment.
- Use a smaller teaching space.
- Use visual and auditory aids.

Elementary

- Allow students to make choices with some activities to help their decision-making skills.
- Keep the same routine for class structure.
- Keep directions specific and brief.
- Use visual and auditory aids.
- Demonstrate skills.
- Use lighter weight equipment.
- Breakdown the task into simple, small steps.
- Use peer partners.
- Use positive behavior management strategies.

Secondary

- Allow students to make choices with some activities to help their decision-making skills.
- Keep the same routine for class structure.
- Use visual and auditory aids.
- Demonstrate skills.
- Breakdown the task into simple, small steps.

- Use peer partners.
- Use positive behavior management strategies.

4.9 IMPLICATIONS FOR PHYSICAL ACTIVITY

Students with epileptic disorders are eligible for special education and related services under the IDEA. Epilepsy is classified as “other impaired” and an IEP would be developed to specify appropriate services. Some students may have additional conditions such as learning disabilities along with the seizures. Seizures may interfere with the child’s ability to learn, and he or she may be missing parts of what the teacher is saying. Depending on the type of seizure and/or how often they occur, some children may need additional assistance to help them keep up with classmates. Assistance can include adaptations in instruction, first aid on seizure management, and counseling, all of which should be written in the IEP. It is important that teachers and school staff are informed about the child’s condition, possible effects of medication, and what to do in case a seizure occurs at school.

4.10 EFFECTIVE TEACHING STRATEGIES

Before actually setting up a strategy, check the district’s medical history sheet and contact the student’s physician, past teachers, and parents. Activities can improve both mental and physical health and should be encouraged for people with epilepsy. Most individuals with epilepsy can safely exercise in a gymnasium, pursue sports, and use equipment even though seizures aren’t completely under control, but a buddy system may be needed. All activities should be monitored and individually adjusted to each person’s exercise tolerance and medical history. Through the use of the buddy system and consistent use of safety equipment (helmet, knee, and elbow pads), the student can participate in most activities.

These are some activities where it is suggested that one require a physician’s permission or should otherwise be modified:

- Water sports
- Activities that place the student a few feet above ground
- Archery

- Activities that have repeated blows to the head

Children and youth with epilepsy must also deal with the psychological and social aspects of the condition. These include misperceptions and fear of seizures, uncertain occurrence, loss of self-control during the episode, and medications. To help children feel more confident about them and accept their epilepsy, there should be education programs for staff and students, including information on seizure recognition and first aid. A pattern of mental and physical defects that develop in some unborn babies when the mother drinks too much alcohol during pregnancy

- Teachers will need to modify leisure, recreation, fitness, and sport activities.
- Provide short and clear instructions.
- Provide frequent feedback to the individual.
- Repeat directions.
- Have the individual demonstrate the task to show a clear understanding of it.
- Keep the learning environment consistent.
- Eliminate distractions (visual and auditory).
- Demonstrate tasks for visual understanding.
- Use peers as partners for the individual.
- Slow down the speed of the activity, especially if it is concerning a noncompetitive activity, (e.g., stretching, exercises, etc).

EFFECTIVE BEHAVIOR MANAGEMENT STRATEGIES

- Set limits and follow them consistently.
- Change rewards often to keep high interest in reward procurement.
- Have pre-established consequences for misbehavior.
- Review and repeat the consequences of behaviors. Ask the students to tell you the consequences.

- Do not debate or argue over rules already established. "Just do it."
- Notice and comment when students are doing well or behaving appropriately.
- Avoid threats.
- Redirect behavior.
- Intervene before behavior escalates.
- Avoid situations where the students will be overstimulated.
- Have students repeat back directions to show understanding.
- Protect students from being exploited. Individuals with FAS are naive. "An impairment in hearing, whether permanent or fluctuating, that adversely affects a child's educational performance," from IDEA, PL 101-476 (Winnick, 1995)

4.11 IMPLICATIONS FOR PHYSICAL ACTIVITY

Individuals with hearing impairments can do just about everything with the same motor ability as non-disabled individuals. However, a hearing impaired individual may experience static and dynamic balance challenges, some delays in gross and fine motor skills, and some difficulty with applied force and coordination. Making small adaptations for individuals with hearing impairments will allow them to be able to be just as successful as anyone else in their classes. These adaptations include the following:

- Make the environment more conducive to them by eliminating excess noise (if you use music, have it at a reasonable level for all students), use an auditory device, have clear instructions, and keep a routine for the students.
- Educate all the students in the individual's class about his or her disability.
- Provide disability awareness for the entire school to increase acceptance, social skills, and tolerance of individuals that may be a little different.
- Simulating the student's disability can help everyone feel more comfortable and aware of what it is like to be hearing impaired.

CONTRAINDICATED ACTIVITIES

- Swimming is a contraindicated activity due to the nature of the individual's disability. Too much water pressure could further damage the individual's hearing.

EFFECTIVE TEACHING STRATEGIES

- It is necessary to have a safe environment in which individuals can perform activities.

Keep the area free of obstacles and loud noises.

- Assistance may be needed for some activities to get the most out of the student.
- Learn basic sign language and keep a positive attitude.
- Use other teachers as a valuable resource.
- Make sure hearing impaired students can see your lips when you talk.
- Use visual demonstrations when you teach.
- Use an alphanumeric pager. An alphanumeric pager is basically like any other pager that you call and text a message to, except there are more options and a clearer display.
- Teach standing in one place and use visual attention-getters to help get the individual on task.
- Ensure that the individual feels safe and comfortable in his or her environment and with the people around him or her. Juvenile diabetes, also called type 1 diabetes or insulin-dependent diabetes, is a disorder of the body's immune system (the body's system for protecting itself from viruses, bacteria, or any foreign substances).

4.12 What is Assistive Technology?

- The Federal Definition of Assistive Technology
- "any item, piece of equipment, or product system, whether acquired commercially off the shelf, modified, or customized, that is used to increase, maintain, or improve functional capabilities of a individual with a disability" (Technology-related Assistance to Individuals with Disabilities Act, 1988).

4.12.1 Assistive Technology Devices

- AT can fall between Low-tech, Mid-Tech, & High-Tech
- Low-Tech, Mid-Tech, & High-Tech are categories that refer to the ***degree of technology*** used in the device/system and the ***amount of training*** the learner requires to use it.

4.12.2 In most cases, the best solution is the simplest solution.

Considerations for Selecting AT

☐When selecting AT, it is important to consider:

☐Supported Employee

☐Environment

☐Task

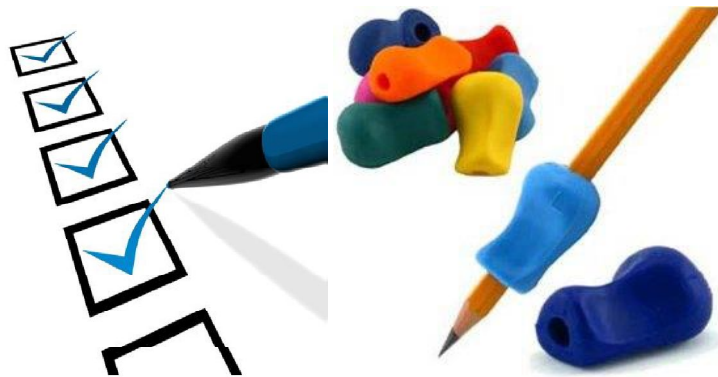
☐Tools

■SETT framework was developed by Joy Zabala www.joyzabala.com

Low Tech Examples

- ✓ Checklists
- ✓ Line Drawings/Pictures/Visual Supports
- ✓ Visual Timers
- ✓ Color Coding
- ✓ Templates
- ✓ Pencil Grips/Enlarged Writing Instruments
- ✓ Highlighters/Highlighting Tape

- ✓ Slant Boards/Book Holders
- ✓ Velcro
- ✓ Non-Slip Materials such as Dycem
- ✓ Handheld Magnifiers



Mid-Tech Examples

- Digital Recorders
- Franklin Dictionary/Thesaurus
- Portable Word-Processors
- Reading Pens and Calculators
- Simple AAC Devices
- Switches/Environmental Control



Other Assistive Devices to Support Employment

- Visual and talking timers
- Picture/text messages on cell phones
- Built in calendar functions on cell phones
- Watchminder Vibrating alarm watch
- Mini recorder to record assigned duties



High-Tech Software/Device Examples

- Mobile Devices/Tablets/iPads - Hundreds of apps
- Word Prediction - Co-Writer, Read and Write Gold,

- Text-to-Speech - Kurzweil, Don Johnston SOLO
- Screen Magnification Software - ZoomText, Magic, Windows, Mac
- Screen Readers - JAWS, NVDA, Voiceover,
- Speech Recognition Software - Dragon Naturally Speaking,
- Expanded Keyboards - Big Keys



4.13 Use of Picture & Video Supports

- Video Modeling:** Review video examples before engaging in tasks at work.
- Video Prompting:** Watch short video segments of steps needed to complete task while performing task (watch step- go do; watch next step- go do). This typically requires the use of portable devices.
- Continuous Video Modeling** Continuous Video Modeling (CVM) is similar to a procedure called Simultaneous Video Modeling where a video is played while the individual is expected to “follow along” to model the behaviors. With SVM the video is only presented once whereby in CVM the video is played on a loop (Mechling, Ayres, Purrazella, & Purrazalla, in press). The continuous presentation of the video model allows individuals to work “hands-free” and to look at the video as needed and refer back to specific steps while completing the task.

•**Video Feedback:** Videotape employee as they engage in tasks and have them evaluate performance when watching video.

4.14 Use of Mobile Devices and Associated Apps that Support Picture & Video

•Several apps can support video-based instruction using either video modeling or video prompting. Some include the following:

•Picture Scheduler

•Keynote

•Video Scheduler

•Storykit

•Go Talk Now



•All of the above also provide picture and auditory prompts. Additional apps that support picture/auditory prompts include:

•Pictello

•iPrompts

•First-Then



4.15 THE INDIVIDUALIZED EDUCATION PROGRAM (IEP) PROCESS

The individualized education program, or IEP, is the result of a process that is essential to ensure that individuals with disabilities have appropriate educational planning to accommodate their unique instructional needs, and that these needs are met in an appropriate learning environment. Frequently the term is applied only to the planning document, with limited emphasis given to the process itself. Yet it is vital that the IEP team members fully understand the process that will result in a plan that will guide the education of a child over the course of a full school year.

Definitions of individualized education program.

(a) *General.* As used in this part, the term individualized education program or IEP means a written statement for each child with a disability that is developed, reviewed, and revised in a meeting in accordance with This section of the *Guidelines: Individualized Education Program Process* (June 2007) includes five components:

- an explanation of the purposes and principles underlying the IEP process;
- a description of children and youth who are eligible for the IEP process;
- a brief description of the sequence or “flow” of the process;
- identification of timelines that apply to the IEP process and planning document; and
- a description of the roles and responsibilities of persons who will participate in development, review, and revision of the IEP document.

Purposes and Principles Underlying the IEP Process

The IEP process serves multiple purposes. First, the IEP documents a child's needs and services and how they affect involvement and progress in the general education curriculum. A simple framework that organizes the IEP process is based on three questions:

- What do we know about the child's skills? This question is answered through the present levels of academic achievement and functional performance.
- What are we going to do to help the student receive an appropriate education? The IEP team addresses this question through the following components: goals, adaptations, least restrictive environment, and special education and related services.
- How will we know if we are succeeding? This is ultimately answered through measurements based on the goals and, when appropriate, short-term objectives of the IEP and other results of the total education experience.

The second purpose underlying the IEP process is its value in providing access to procedural safeguards. It ensures that parents are informed active members in developing, reviewing and revising the annual educational plan for their child, and in monitoring progress. The third purpose of the IEP process provides a means of decision-making that is efficient and instructionally useful. These three purposes are grounded in educational and legal principles. Foremost among the underlying principles is the requirement that the appropriate educational placement and the type, amount, and location of services must be based on the child's unique individual characteristics and not on the category of disability. The documentation of the IEP process – the written plan - should be a child-focused, working document used by all team members throughout the duration of the year. Therefore, the process must reflect pertinent information on the child's present levels of academic achievement and functional performance which are obtained from multiple sources including: (a) parent reports; (b) general education-based interventions; (c) the child's interests and abilities; and

(d) a comprehensive assessment following a special education referral. In addition, an essential consideration is the extent to which the child's instructional needs can be met by general education staff in all environments with or without assistance from special educators. Finally, children with disabilities are to be educated with those who do not have disabilities to the maximum extent appropriate, and should attend general education classes unless it is inappropriate and otherwise indicated on the IEP.

4.16 Children and Youth Who Are Eligible for the IEP Process

The Individuals with Disabilities Education Improvement Act (IDEA 2004) requires that a free appropriate public education (FAPE) be made available to all children and youth with a disability, ages 3 through 21. The IDEA 2004 identifies and defines the specific disability categories that make a child eligible. To be eligible for the IEP process, a child must have been identified through the evaluation process as having one or more of the disabilities. This process is described in *Guidelines: Evaluation Process* (June 2007), North Dakota Department of Public Instruction. In addition to the eligibility categories listed above, the IDEA 2004 includes regulatory language that allows states to identify younger children in need of special education who are determined to have "developmental delays". In North Dakota, the term non-categorical delay (NCD) is used, rather than developmental delay (DD), for the purpose of avoiding confusion with the Developmental Disabilities (DD) system in the Department of Human Services. Detailed information regarding the approved state eligibility criteria can be found in the North Dakota *Guidelines: Identification and Evaluation of Students with Noncategorical Delay for ages 3 through 9* (March 2007).

Children Turning Three. A child with a disability transitions from early intervention services to early childhood special education services when the child turns three. The individualized family services plan (IFSP) that is developed in accordance with the IEP procedures, may continue to be used in lieu of an IEP if the services identified in the IFSP continue to be appropriate. Because of the importance of the IEP as the statutory vehicle for ensuring a free appropriate public education (FAPE) to a child with a disability, it is important that the parents give consent to continued use of an IFSP. An explanation of the differences between an IFSP and an IEP must be provided to parents. The North Dakota state guidelines document, *Understanding Early Childhood Transition: A Guide for Families and Professionals* (August 2006), provides a comprehensive description of these differences.

4.17 Good Practice Guidelines for Person-Centred Planning and Goal Setting for People with Psychosocial Disability

The '**Purpose**' of the project was to identify good practice approaches to assisting people living with psychosocial disability to exercise choice and control in an NDIS environment, and was to particularly focus on identifying –

- person-centred planning and goal setting;
- Management and co-ordination of a package of supports; and,
- models which integrated planning and on-going support co-ordination.

Its '**Objectives**' were to identify and document good practice elements or approaches to –

- person-centred planning and goal setting;
- working with and assisting people living with a psychosocial disability to manage and co-ordinate a package of supports and to monitor their progress in achieving their goals; and,
- to identify the benefits and challenges of integrated planning and on-going support co-ordination and how the benefits can be achieved in an NDIS environment and the challenges overcome.

The '**Outcomes or Deliverables**' involved -

- a set of '**Guidelines**' which outlined how people living with a psychosocial disability could be supported to exercise choice and control in an NDIS environment, including guidance on good practice approaches to the development of person-centred plans and co-ordination of supports and services; and,
- an '**Evaluation framework and an approach to testing**' that would enable the guidelines to be appropriately reviewed and evaluated from 2013.

4.18 The Development of Guidelines for Person-Centred Planning and Goal Setting:

This entailed –

- the knowledge gained from the Literature and Research Review being applied to the development of a set of 'Guidelines' based on a 'person-centred' approach to planning and 'Good Practices';

- the outcome of consultations with a wide range of experts, peak bodies and academia along with focus groups and interviews with people living with a psychosocial disability, their families and careers during the development and drafting of the Guidelines;
- consideration of the NDIS environment that any such Guidelines would need to be adapted to; and
- Understanding of what would be needed in terms of a Plan in such an NDIS environment.

The Guidelines explain the origin of person-centred planning; the concept, philosophy, values and beliefs underpinning it; and what its major or essential elements are. The Guidelines set out a ‘framework’ for person-centred planning and then adapt that ‘framework’ to the planning requirements of the NDIS Act 2013 and the Draft Rules.

Potential barriers and blocks to person-centered planning are identified.

Although known and envisaged ‘good practices’ are incorporated in the Guidelines, a separate description of ‘good practices’ concerning a participant; a service that seeks to provide person-centred services to one or more of its clients or consumers; and for an organisation that fully wants to become a person-centred one by effective consultations about its strategic, service planning and development, are also provided.

4.19 The Evaluation Framework and Approach to Testing:

This entailed –

- building on the Literature and Evaluation Review and Guidelines developed to determine a means by which the Guidelines could be appropriately evaluated and tested at a ‘high level’;
- Incorporating the feedback and comments on the Guidelines from consultations conducted on them to inform such a framework for the evaluation and testing of them.

The approach taken has been to evaluate and test how well the guidelines were implemented; the quality of the plans developed; the achievement of the participant’s goals, objectives and aspirations; and, the contribution of the ‘Plans’ developed to the strategic, service planning and development objectives of the NDIS Launch Transition Agency or DisabilityCare Australia.

The focus was therefore on whether the planning was done well, and if so, then on ‘personal goals’ and ‘policy goals’ facilitated or achieved. It sought to do these multiple tasks through a series of questions (which could be described as checklists) requiring answers.

The proposed ‘Framework’ although in some ways ‘high level’ as intended, has morphed somewhat by drilling down as well into issues about ‘what happened’; ‘how it happened’; ‘what resulted’; ‘what personal gains or benefits were there for the participant’; and ‘what benefit the Plans developed could have for NDIS policy goals’.

4.20 The Literature on Person-Centred Planning and Recovery

2.3.1 What is ‘person-centred planning’?

“...it is at the heart of much recent policy relating to the provision of social care services. It refers to a family of approaches aimed at enabling people who use services to plan their own futures and get the services that they need.” (Dowling et al, 2007 pvi)

“...(it is) a process of continual listening, and learning; focused on what is important to someone now, and for the future; and acting upon this in alliance with their family and friends. It is not simply a set of new techniques to replace Individual Program Planning. It is based on a different way of seeing and working with people with disabilities, which is fundamentally about sharing power and community inclusion.” (Sanderson 2000, p2)

“...(it is) a process that finds out what an individual wants to do with his/her life, helps the person decide on goals and then plans what action needs to be taken to achieve those goals. It is an ongoing process which means that the person can regularly set new goals as his/her skills and experience grows and ambitions change. The core principle in all these methods (of person-centred planning such as PATH, ELP etc) is to look at the whole person, rather than viewing them as a series of ‘needs’- such as for residential, day or leisure services – and to consider all aspects of a person’s life including friendships and relationships. The focus (is) on each person as a unique individual...” (Cole et al, 2000 p30)

“...(the) listening and learning (in person-centred planning) is used to understand a person’s capacities and choices (and) is a basis for problem-solving and negotiation to mobilise the resources to pursue a person’s aspirations. These resources may come from a person’s personal network, from service agencies or from a range of non-specialist and non-service sources.” (Dept of Health ‘Valuing People Guidance’, 2002 p2)

“ it provides a way to understand the experiences of people with disabilities and, with the help of allies, to expand those experiences through reducing social isolation and segregation, facilitating

the establishment of friendships, increasing opportunities to engage in preferred activities, developing competence and promoting respect” (Holburn 2002 as cited in DADHC 2008, p5)

The origin of person-centred planning

The literature generally attributes the development of ‘person-centred planning’ approaches to the independent living movement generated by people with disabilities and their advocates during the 1980’s in North America. (O’Brien and Mount 1989; Mount 1992; Garner and Dietz 1996; O’Brien and O’Brien 2000).

This movement had as its core beliefs that –

- all human life is of value;
- Anyone, whatever their impairment, is capable of exerting choices;
- People who are disabled by society’s reaction to physical, intellectual and sensory impairment and to emotional distress have the right to exert control over their lives;
- People with a disability have the right to participate fully in society (Morris 1993, p21)

The Productivity Commission noted that –

“The concept of so-called ‘person-centred’ and ‘independent living’ approaches...is based on the view that people with a disability should be recognised as active participants in the community, needing support to achieve their lifetime goals based on their strengths.” (Productivity Commission 2011, p344)

The ‘independence’ sought was however, never about limited notions of personal autonomy or self-determination; rather it was about removing barriers (attitudinal, physical and communication) to as full participation in the community as possible with the tailored support necessary to do so.

It was and continues to be a ‘rights-based’ philosophy seeking to ensure that each person with a disability receives the support that they need and have control over that support so as to enable them to live their lives as fully and on terms as equal as possible as those of other members of the community

The 'Human Rights Movement' has also made a marked contribution to policies and procedures embracing 'person-centred' approaches. Of particular importance is the United Nations Convention on the Rights of Persons with Disabilities. The Convention which entered into force on 3 May, 2008 and which has been ratified by Australia provides the following guiding principles -

- respect for inherent dignity, individual autonomy including the freedom to make one's own choices, and independence of persons
- non-discrimination
- Full and effective participation and inclusion in society
- Respect for difference and acceptance of persons with disabilities as part of human diversity and humanity
- Equality of opportunity
- Accessibility
- Equality between men and women
- Respect for the evolving capacities of children with disabilities and respect for the right of children with disabilities to preserve their identities

“The convention states that persons with disabilities enjoy the same human rights as everyone. Without being exhaustive, these rights include equality before the law without discrimination, the right to live in the community, and the right to education and work. The convention entails certain obligations on parties ratifying it, such as adopting legislation and administrative measures to promote the human rights of persons with disabilities, eliminating discrimination in workplaces and society; providing information to people with disabilities, undertaking disability-relevant R&D and appropriate consultation with people with disabilities in developing and implementing legislation and policies and in decision-making processes that concern them.” (United Nations Convention on the Rights of Persons with Disabilities 2008)

The Person-centred planning approach

A range of slightly different approaches to 'Person-centred planning' have developed and are continuing to evolve as knowledge about it and its efficacy and effectiveness is progressively established by evidenced-based practice.

One reason for the advocacy for a more ‘person-centred’ approach was dissatisfaction with the Individual Programme Planning approach which limited ‘choice’ to effectively ‘no choice’ as individuals with a disability were invariably fitted into ‘take-it or leave-it’ services. (Greasley P. 1995 and Mount B. 1987)

The same applied to earlier theories associated with ‘Normalisation’ and ‘Social Role Valorisation’ which heralded a new vision at that time. (Nirje B. 1969; Nirje B. 1980; Wolfensberger 1972; Wolfensberger 1983A; Wolfensberger 1983B)

4.21 Intellectual Disability: promoting daily living skills in adults

Mature adults who have not previously learnt daily living skills can, with appropriate supports, make significant steps towards independence. Age is no barrier to learning.

Adults living at home with parents

It has become common for adults (with and without intellectual disability) to continue living at home with their parents, who may continue to take responsibility for the household chores such as cleaning and preparing meals.

The sharing of household chores can lead to greater independence for your adult son or daughter and also reduces the burden on others in the household. It is important to provide opportunities for your son or daughter to practice and develop daily living skills, even when tasks take longer and require a certain amount of effort.

4.22 Intellectual disability and daily living skills

People with intellectual disability are more likely than their age related peers to need extra support to carry out everyday activities. It is important to have realistic expectations and consider what kind of support is most likely to help develop their independence.

While others may learn household tasks by watching their parents or carers, adults with intellectual disability may require more step by step methods to help them learn tasks.

4.23 Daily activities

The following is a list of activities that adults may be expected to carry out independently:

Helping around the house

Load and start a washing machine, hang out the washing

Fold a basket of washing

Iron clothes

Sweep, mop and/or vacuum the floor

Cook a meal, wash the dishes

Wash the car

Mow the lawn.

In the community

Catch public transport or take a taxi

Handle money, use an Automatic Teller Machine (ATM), pay a bill

Meet a friend for lunch

Shop for clothes and groceries.

4.24 Teaching techniques

Variety

Use a number of different ways to teach tasks, if appropriate. The person may find it easier to learn a particular task if they:

Join an independent living skills group/help out in local community groups

Watch videos or DVDs

Join in helping another family member with the task, or

Have a photo sequence of the task with the steps written as a story or prompt.

Location

Teach the person the task in the place where the task will usually be done.

For example, if person is learning to iron their clothes, teach them how to iron in the room in which they will usually iron. This helps the person become familiar with the layout of the room, the places where items are kept, the way in which to prepare to begin ironing and how to pack everything away.

Step-by-step

Sometimes breaking a task into small steps and teaching one step at a time can make it easier for a person to learn the whole activity.

For example:

Get the ironing board and iron

Set up the ironing board

Plug in the iron

Iron a shirt

Turn the iron off, set it aside to cool

Fold up ironing board

Put the ironing board away.

Backward Chaining

‘Backward chaining’ is a technique you can use to help a person learn a new skill. It means the person is taught the task in the reverse order. Take the person through the first steps of the task and begin teaching at the point where the last step begins.

‘Put the ironing board away’. The person then completes this final step. The process continues over time as you complete one less step and the person you are supporting does one more until they can do the whole task on their own.

Backward chaining means the person receives instant success, which is likely to increase their motivation and confidence.

4.25 Summary

Person-centered planning does not override a professional's/practitioner's ethical and societal obligation to intervene on a person's or the community's behalf should someone pose a serious and imminent threat to self or others. In such cases, just as in the case of an automobile accident or traumatic brain injury, professionals/practitioners are sanctioned to intervene on the person's behalf without getting prior consent.

In reality, as in most other branches of care and support, however, such cases are the extremes and the exceptions, not the norm. The literature suggests that most people with psychosocial disabilities arising from mental illnesses pose few if any risks most of the time. Risk can be exacerbated by substance use and by non-adherence to medication, but even then the risk posed by people with serious issues pales in comparison to the risks they face from others, as it is much more common for a person with a serious mental illness or psychosocial disability to be the victim of a crime than to be a perpetrator.

What this suggests is that heightened concerns about increased risk and liability are misplaced when applied to most people most of the time. In the circumstances in which they are warranted, prudent risk assessment and management are central and crucial aspects of effective care. When not warranted, though, they place undue restrictions on the liberty of persons with serious mental illnesses or psychosocial disability.

Issues of risk and liability put aside for the moment, how do we respond to the concern that people with serious issues will still make bad decisions if left up to their own devices? Initial studies in shared-decision making in fact point to the opposite and indicate that people with schizophrenia, for example, make decisions in similar ways as those with other medical issues. Simply put, some people with mental illnesses or psychosocial disability make good decisions most of the time, some make good decisions some of the time, and some make good decisions only rarely; but the same is true of the general population. At this time, the only main legal or statutorily justified way to interfere with an individual's personal sovereignty (other than based on serious and imminent risk to the person or to others) is when the person has been determined

to be incapable of making (some of) his or her own decisions by a Court or Tribunal, and therefore has been assigned a legal guardian or financial manager. Even in these cases, in many states a Court or Tribunal's decision needs to (or should) outline those specific areas in which the person is unable to make his or her own decisions. Short of this, the vast majority of individuals with serious mental illnesses or psychosocial disabilities have both the right and the responsibility of making their own decisions and of dealing with and learning from the consequences of these decisions.

Where then does all this leave the professional/practitioner who wishes to support someone in his or her personal choice but fears the person is making potentially detrimental decisions that will jeopardize his or her recovery and well-being; e.g., a person is choosing not to take medications that seem to be helpful or is spending time with someone who has physically abused them or provided them drugs? In these situations, the professionals/practitioners should not sit silently on the sidelines in the name of being person-centred or in the hope the individual will ultimately learn from suffering the “natural consequences” of an apparent self-defeating choice. Rather, in keeping with emerging best-practices in recovery-oriented care, the role of the professional/practitioner in such situations is to remain fully engaged with the person to explore what the choice means and why it is important to him or her; to identify potential pros and cons; to brainstorm alternative choices; and to ensure the person has all the information necessary in order to make an informed decision. But in the end, barring any immediate safety concerns, it is the person's decision to make, just as it is for any other person.

Following the type of collaborative dialogue described above, the person, in fact, might arrive at a different decision that both parties are comfortable with. However, there also will be circumstances in which the person and the professional/practitioner may need to “agree to disagree” moving forward. In these circumstances, some professionals/practitioners have expressed concern that they are exposing themselves to liability or failing to meet their professional or clinical obligations, if the issue at hand is not identified in writing as an active problem area or treatment goal on the recovery plan.

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Unit 5

Vocational Training and Independent Living

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5.1 Introduction

The rights of people with disabilities have been given new attention with the entry into force of the United Nations Convention on the Rights of Persons with Disabilities (CRPD) in May 2008. The provisions of the CRPD contribute to other international standards concerning persons with disabilities, signalling a dramatic shift in international policy terms. In relation to training and employment, for example, states are called on to provide opportunities for disabled people alongside non-disabled people. Many countries have already declared their commitment to the goal of inclusion of persons with disabilities through ratification of the CRPD, while others have signed it with a view to ratification. Many have also ratified ILO conventions, committing themselves to the goals of equal opportunity and non-discrimination.

People with intellectual disabilities are entitled to benefit from the provisions of the CRPD and ILO Convention concerning the Vocational Rehabilitation and Employment (Disabled Persons), 1983 (No. 159), as well as people with other kinds of disabilities. Yet, they are frequently not well placed to gain from this changed emphasis on inclusion. In many developing countries, in particular, they are often excluded from school and deprived of opportunities to acquire relevant vocational skills at all, presenting further disadvantages when it comes to seeking jobs. Yet, experience in many countries shows that, with the right training, support in the workplace as required, and the right opportunities, they can make valued contributions in the workplace and to a country's economy.

Measures to open employment opportunities for this group of persons with disabilities in line with the CRPD and ILO Convention No. 159 can build on extensive experience in recent decades in developing new approaches to training and employment. The review of international experience carried out for this working paper highlights good practice in supporting people with intellectual disabilities in integrated employment settings. Evidence clearly points to better outcomes for employees with intellectual disabilities, when they work in integrated settings, with appropriate supports.

The aims of this paper are to:

- examine changes over time in the understanding of intellectual disability and the capacity of persons with disabilities to learn;
- provide an international overview of employment options for people with intellectual disabilities, with special emphasis on Supported Employment (SE) models;
- examine and critically analyse from a research perspective examples of SE across a range of low- and high-income countries; and
- make recommendations for the expansion and future development of inclusive employment options for this population.

Section 1 provides a brief overview of the shift from classifying people with intellectual disabilities on the basis of IQ bands to a support needs framework. It then explores the impact of research initiatives that demonstrated the learning capacity of people with intellectual disabilities in a work environment.

Section 2 discusses the range of employment models currently accessed by people with intellectual disabilities in countries around the world. It highlights that, despite the emergence of a number of more inclusive practices, the predominant model continues to be sheltered employment in segregated settings, and that a high proportion of people with disabilities are unemployed.

Section 3 reviews recent and emerging developments in promoting training and employment opportunities for people with intellectual disabilities, drawing from examples in both high- and low-income countries, with reference to factors facilitating and posing challenges to the spread of supported employment. The findings of research on these developments are summarized.

Section 4 examines the policy implications of research findings and draws some conclusions about the way forward. The Working Paper has been informed by the report of the ILO/Irish Aid sub-regional conference *People with Intellectual Disabilities – Opening Pathways to Training and Employment*, held in Lusaka, Zambia 9-11 March 2010 (ILO, 2010a), and the accompanying Lusaka Declaration *People with Intellectual Disabilities: Achieving Full Participation in Training and Employment* (ILO, 2010b).

The views of several individuals with intellectual disabilities on what work means to them are presented in different parts of the report, so that their voices are reflected, in line with the self-advocacy movement which is gaining momentum.

5.2 People with intellectual disabilities

People with intellectual disabilities have existed throughout the world across human history, making up a part of all cultures. They represent a small part of the extremely wide variety of people in the human population at any one time. They are reported to make up one per cent of the population globally, although this may be an underestimate of the true prevalence, even in terms of the criteria of impairment and activity limitations and there is no clear indication of the level of uniformity of the definition or the comprehensiveness of national data. Different cultural environments may also affect the definition and a decision on whom to include and whom to exclude (Solarsh and Hofman, 2006). Cultural factors may present difficulties in the identification of people with disabilities, as families often persist in hiding their child with a disability because of negative community prejudices (Mung'omba, 2008).

People with intellectual disabilities would like to take part in a range of life experiences, like any non-disabled person or people with other types of disability. Yet they face difficulties in doing so, in one or more of the following areas:

- learning;
- communication (receptive and expressive language; verbal and non-verbal);
- social skills;
- academic skills;
- vocational skills;
- independent living.

Research has shown that people with intellectual disabilities often have multiple secondary impairments, including complex physical and mental health problems (Beange et al., 1999; Einfeld et al., 2006; Hofer et al., 2009). These co-existing disabilities are particularly likely to be

problems of agility, mobility, speech and language; and emotional problems including anger control, anxiety and depression. They are also more likely to experience loneliness and lack of friends (McVilly et al., 2004).

It is acknowledged that people with all categories of disabilities have experienced discrimination and stigmatization throughout the millennia. Yet, an intact intellect has historically been highly valued and often seen as the essential characteristic of being fully human (Parmenter, 2001), and people with intellectual disabilities are often the last within the community of disabled persons to receive attention. In the not-too-distant past a person with an intellectual disability “was viewed not as a second-class citizen, but rather as one who possessed *no citizenship*” (Stevens, 1967).

5.3 Changes in approach

There have been many changes in recent decades in the way people with intellectual disabilities have been named and categorized. Terms formerly used such as *idiot*, *imbecile*, *feeble-minded*, *mentally subnormal*, *moron*, *mentally deficient* and *retard*, are now seen as highly pejorative and stigmatizing, although at the time of their use they were acceptable terms in the scientific literature. For instance, the term ‘mental retardation’ has come to be seen as a derogatory term and has been gradually replaced by the term ‘intellectual disability’ in many parts of the English-speaking world. In the United Kingdom, the terms ‘learning disability’ or ‘learning difficulty’, which have both social and definitional qualities, are used. In Canada, ‘intellectual disability’ and ‘developmental disability’ are used synonymously, whereas in the US the latter term also encompasses conditions such as cerebral palsy and Asperger’s syndrome, which do not necessarily involve an intellectual disability, and ‘learning disability’ is understood to include people with dyslexia. Representative organizations have responded, changing their names to reflect more contemporary acceptable language. Examples are: the American Association on Intellectual and Developmental Disabilities (AAIDD) (formerly the American Association on Mental Retardation, and earlier, the American Association on Mental Deficiency); the International Association for the Scientific Study of Intellectual Disabilities (IASSID) (also, formerly ‘of Mental Deficiency’); and Inclusion International (formerly the International League of Societies for Persons with a Mental Handicap). Eminent researchers from a psychology background, such as Ellis (1963, p. xi) in the US, commented that “mental retardation was a social problem” and Clarke and Clarke (1958, p. xiv) in the UK suggested that, “mental

deficiency is a social-administrative, rather than a scientific, concept, varying between different countries and within a given country at different times.” Self-advocacy groups, representing people with intellectual disabilities, have become very instrumental in changing the way language is used to refer to the condition, arguing that terms used demean them because of the implicit negative connotations. Even the term ‘disability’ implies a condition less than ‘normal’. These groups have also been instrumental in highlighting the aspirations of men and women with intellectual disabilities in recent years.

5.4 Identifying people with intellectual disabilities

5.4.1 IQ testing

In the nineteenth and early twentieth century’s, approaches to the education and training of people with intellectual disabilities were influenced by the rising discipline of psychology. It was Alfred Binet (1857-1911), the director of the psychology laboratory at the Sorbonne in Paris, who is credited with the development of the first scale to measure ‘intelligence’. Developed in 1905, this scale involved ranking individuals on a number of everyday problems of life requiring processes of reasoning. The tasks were arranged in an ascending order of difficulty, with an age-level assigned to each task, giving rise to the concept of mental age and subsequently, the intelligence quotient (IQ). Binet’s motives for the development of a scale were prompted by the need to identify those children whose lack of success in general classrooms may have given rise to the need for a special education programme.

The combined impact of Charles Darwin’s *Origin of the Species*, published in 1859, and the genetic discoveries of Gregor Mendel (1822-1884) reinforced the view that intellectual disability was a single inheritable condition, and not one that can be attributed to a variety of genetic and other causes. Eugenics societies sprang up over much of the Western world, emphasizing the possibility of improving the qualities of the human population, especially by such means as discouraging reproduction by persons having genetic defects or presumed to have inheritable undesirable traits. This led to pressure being placed on parents of people with intellectual disabilities to discourage their marriage and procreation. Arbitrary sterilization laws were enacted in many countries, including Canada, the United States, Sweden and France. Most of these laws were repealed by the late 1970s.

The classification of intellectual disability, or mental deficiency, as it was then described, became a topic of popular debate in the early twentieth century. Three categories emerged: ‘idiots’, who did not develop speech and had mental ages below 3; ‘imbeciles’, who did not become literate and had mental ages between 3 and 7; and ‘high grade defectives’ or ‘morons’, who could be trained to function in society. The net effect of the eugenics movement was a rapid increase in the numbers of people who were institutionalized across the Western world. It was in these institutions that the first recorded vocational training for people with intellectual disabilities was conducted (Sandys, 2007).

From the mid-twentieth century onwards, several significant advances were made in the definition and classification of intellectual disabilities (Parmenter, 2004). These were influenced by at least three factors. First, the results of research demonstrated the learning potential of people with intellectual disabilities. Second, the proclamation by the United Nations in 1971 of the *Declaration of the Rights of Mentally Retarded Persons* provided an impetus for countries to re-examine their laws to ensure that the rights of people with intellectual disabilities were being safeguarded. Third, the growing de-institutionalization movement in several western countries provided a spirit of optimism which challenged commonly held beliefs about the moral status of people with intellectual disabilities.

5.4.2 Support needs framework

A significant development occurred with the publication of the ninth edition of the AAIDD manual where the traditional classification system linked to IQ levels (mild, moderate, severe and profound) was replaced by the concept of levels of support required (Luckasson et al., 1992). This development represented a significant shift in the way intellectual disability was conceptualized. The reformulated definition and the accompanying classification of supports acknowledged that the disability resulted from the interaction of the person with his/her environment, rather than being an absolute trait of an individual.

This development was also consistent with the emerging strength of the social, as opposed to the medical, model of disability (Mercer, 1992). While the medical model focuses on the person’s impairment and steps to reduce or remove this, the social model recognizes that barriers to a person’s participation in general community activities are not solely determined by a person’s

impairment, but arise in combination with elements of the environment. For instance, in the case of a person with an intellectual disability, community attitudes are generally the greatest barrier to participation, involving low expectations of the person's capacity to participate and contribute.

Subsequent revisions of the AAIDD position (Luckasson et al., 2002; Schalock et al., 2010) have further developed the conceptualization and planning of individualized supports and the measurement of adaptive behaviour (Schalock, 1999; 2004a). Throughout these recent revisions there has been a deeper appreciation of the relationship between the AAIDD approach to definition and classification of intellectual disability and that of other relevant international classification systems such as the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV), the WHO International Classification of Diseases (ICD-10) and the WHO International Classification of Functioning, Disability and Health (ICF).

5.4.3 Relevance to training and employment programmes

In the planning and delivery of educational and vocational programmes for people with intellectual disabilities, a support needs approach to classification has significant benefits in contrast to the earlier practice of classification based on IQ levels (mild, moderate, severe and profound). First, it calls for a personal and individualized approach to service delivery that meets the individual support needs of that person. This is important since rigid stereotypes concerning the behaviours of people within a specific IQ band fail to predict the actual support needs of an individual in a work context. Nor does the IQ approach fit comfortably with the social and rights models of disability where environmental factors must be taken into consideration. Rather than addressing a person's particular cognitive impairment, it may be more effective to make an accommodation to the person's environment by the provision of support that lessens the impact of the impairment. Support needs assessment instruments, which assess a person's needs across a range of everyday life domains, are now available (Arnold et al., 2009; Riches et al., 2009a, 2009b; Thompson et al., 2004).

Cultural considerations

The growth of the IQ testing movement throughout the twentieth century within western industrialized countries has been heavily influenced by social, political and economic forces linked to the perceived needs of an industrialized economy. Using a more anthropological

approach, Cianciolo and Sternberg (2004, p. 22), commented that “...people in different cultures may develop somewhat different intellectual abilities, depending on what types of intellectual competence are valued in their particular culture”. Competence and one’s ability to navigate effectively within a culture depends very much on the environmental contexts of that culture. Emerson, Fujiura and Hatton (2007, p.607) suggested that: In summary, there is strong evidence that socio-cultural factors largely determine what is seen as competent behaviour. Within western high-income countries, driven by the values of utilitarian individualism, the construct of intellectual disability has been largely determined to meet the needs of urban, industrialized societies. In contrast, competence in non-industrialized societies may be better reflected in collaborative, interpersonal problem-solving skills, such as those found among Nigerian students labelled as intellectually disabled (Edeh and Hickson, 2002). However, as Emerson, Fujiura and Hatton (2007) pointed out, people who are judged to be incompetent or ‘obtrusive’ in countries deficient in support services are often neglected and consigned to a life in poorly managed segregated institutions.

5.5 Learning and working capacity

The first reported formal employment programmes for people with intellectual disabilities were those conducted within the large institutions established in the Western world in the first half of the twentieth century. The residents, often referred to as ‘inmates’, a custodial term generally applied to prisoners, were employed in both indoor and outdoor occupations, many of which contributed to the economic operation of the facility. For instance, many residents were employed in the facility’s laundry or in small farming activities that contributed to the facility’s food supply.

For people with intellectual disabilities living at home in rural non urbanized communities, daily activities would not be dissimilar to those of other people living in the community. The phenomenon of the ‘6-hour retarded child’, described by Edgerton (1967, 2001) emerged once children with intellectual disabilities were enrolled at school, where the demands of literacy and numeracy skills challenged their cognitive limitations during the six hours they spent in the classroom. Outside of school activities, however, except for those with high-intensity support needs, most people with intellectual disabilities blended into the general community, being able to perform basic everyday living skills. This pattern underscores the problem of using IQ tests as

predictors of the capacity of people to participate in community living activities such as open employment, since many of these tests were designed specifically to predict school performance rather than anything else.

Normalization, deinstitutionalization, community living and working

Developments in the early 1970s, particularly in the US, led to the development of alternatives to sheltered workshops and day activity centres as the major employment facility for people with intellectual disabilities. A number of underlying factors can be identified which are still relevant to the present-day international context. First, the widespread acceptance of the principle of normalization led to the deinstitutionalization movement, which called for people with disabilities to live in normal community settings (Nirje, 1969, 1985; Wolfensberger, 1972). Second, a number of projects across the US during the 1970s demonstrated the employment potential of people with intellectual disabilities (Kiernan and Schalock, 1997; Kiernan and Stark, 1986). A third factor was the increasing dissatisfaction with the major model of adult services prevalent at the time, namely sheltered workshops. A large US study of sheltered workshops found that only 200,000 people were being served, whilst an additional two million people with intellectual disabilities were in need of work (Whitehead, 1979). People with high support needs, in particular, were denied access to sheltered workshops as they were deemed to be insufficiently productive. Studies conducted by Greenleigh Associates (1975) indicated that workshops were seriously limited by: (a) the absence of adequate varieties and quantities of work; (b) prevalence of low-challenge assembly work; (c) the absence of modern tools and machines on which to train workers; (d) a limited number of skilled and competent staff; and (e) funding mechanisms which encouraged continued service to employees in a segregated environment.

5.6 Employment options

Despite a more enlightened approach to the needs of people with disabilities generally fostered by human rights initiatives and the application of research and technological findings which have led to increases in their overall quality of life, their participation in the open labour market falls considerably lower than that of the general population. In industrialized countries, the unemployment rate among people with disabilities of working age is 50 to 70 per cent, which is at least twice the rate of those without a disability (International Disability Rights Monitor, 2004).

In developing countries, the situation is somewhat worse: it is estimated that 80 to 90 per cent of people with disabilities of working age are unemployed.

Evidence indicates that, irrespective of the culture or the economic circumstances of a country, the employment circumstances for persons with intellectual disabilities are equally alarming, if not worse (Dempsey and Ford, 2009; Department of Health, 2009). Just as they have been denied education, they have been denied access to jobs. This dismal state of affairs has come about largely through ignorance and superstition.

In Canada, for example, a survey reported in 2006 found that people with disabilities continue to experience some of the lowest rates of employment in the country, a situation common across the world (Canadian Association for Community Living, 2006). The survey found that: people with disabilities who *are* working work only half as many weeks per year as people who do not have a disability; are unemployed longer; and spend three times as long outside the labour force. People with intellectual disabilities were found to experience even lower rates of employment, with only 27 per cent being employed at the time of the survey, while 40 per cent had never worked. In terms of income, nearly 50 per cent of people with intellectual disabilities had incomes below the Statistics Canada low-income cut-off, a widely used measure of poverty.

In Australia, people with intellectual disabilities made up 41 per cent of those served in government funded employment services in 2010, with more males than females benefitting (64 per cent vs. 36 per cent). They represented 27 per cent of all participants with disabilities in open employment, but 73 per cent of those in segregated business enterprises, formerly called sheltered workshops. There has been a gradual decline in the proportion of people with intellectual disabilities accessing inclusive employment services. And finally, there is evidence of ‘creaming’ - that is, providing services to people who have relatively low support needs and are easier to place in employment – by providers of both employment options, leading to increasing numbers of people with high support needs being referred to state/territory day option programmes.

In European Union countries, the unemployment rate for people with intellectual disabilities was found to approach almost 100 per cent (Greve, 2009). The vast majority who do work are found in sheltered workshops and those with high support needs are generally directed to day activity

centres. A similar picture was reported by Verdonschot et al. (2009), who found from a systematic review of the literature that people with intellectual disabilities were three to four times less likely to be employed than people without intellectual disabilities. In Northern Finland a follow-up of those born with intellectual disability in 1966 revealed a similar position (Taanila et al., 2005). Employment options for people with disabilities, and those with intellectual disabilities in particular, can be broadly distinguished as those in which people with disabilities work in a segregated environment of only workers with disabilities; and those in which they work in an integrated or inclusive environment of workers largely without disabilities (Kregel and Dean, 2002). Employment models under these headings, currently accessed by people with intellectual disabilities in countries around the world, are described below. The section starts with a brief overview of day centres, which are an option for people judged not suited to sheltered or open employment.

- **Day programmes/centres**

Under a medical model of service delivery, people with intellectual disabilities who were assessed or deemed to be unsuitable for employment were catered for in “activity therapy centre” programmes, where one of the goals was to prepare them for employment options. Typically, only a very small number “graduate” to sheltered employment. In practice, these centres in the past became the life-long option for people assessed as not being suitable for employments.

A combination of personal and programme influences are likely to underlie this pattern. Some of the constraints to moving to other employment options may arise from the reluctance of persons with intellectual disabilities themselves to move from a day centre where they have built up friendships over many years. This reluctance acts as a barrier to change, unless opportunities are open to maintain friendships outside of work. This pattern has been highlighted in the UK context by Beyer et al. (2004), who also found that people with intellectual disabilities in day centres were considered to have had restricted lives, leading to a poor attitude to employment, low desire to work, and little self-confidence, in combination with the lack of the experience to make a judgement about what work they might want to do. Traditional day centres were initially based on the principles of occupational therapy and were created as respite for family members caring for their adult children or siblings with disabilities. This is slowly changing, however, as programmes offered become more skills-based and focused on increasing independence. For

instance, in New South Wales, Australia, one of the post school options initiatives for school leavers classified as not being ready for open employment, is the “transition to work” programme. Since the programme’s inception in 2004, Jobsupport, the most successful agency delivering the programme, placed 60 per cent of the 179 school leavers it supported into paid open employment. This agency specifically targets young people with intellectual disabilities who have been classified as having moderate support needs, referred to under the older classification system as “moderately intellectually disabled”⁶.

- **Sheltered workshops**

Sheltered workshops, initially established by parent groups, were essentially segregated facilities established for people considered unable or unlikely to obtain or retain a job in the open labour market. The majority of those employed have tended to have intellectual disabilities. The workshops engage in remunerative work to cover some of their costs. Workers are often paid a training allowance in addition to any welfare payments they might be entitled to, and very minimal or no wages - sometimes in the form of a bonus if production targets are met. Up until recently, contracts of employment were not issued and the provisions of general employment legislation did not apply. In many countries, these facilities are described as rehabilitation and training facilities, but very few workers ever graduate to the general employment market, even when financial incentives are offered. In fact, there was a major disincentive to progress the more skilled workers to real jobs in the open labour market, as their productivity was crucial for the financial viability of the organization (Murphy and Rogan, 1995; Parmenter, 1980). ILO Vocational Rehabilitation (Disabled) Recommendation (No. 99), adopted in 1955, was the first international instrument regarding national legislation and practice concerning vocational guidance, vocational training and placement of disabled persons, guiding practice for almost 30 years. While recommending mainstreaming of vocational training, equality of opportunity and no discrimination in pay for equal work, the Recommendation saw the establishment of sheltered workshops as one method of widening employment opportunities for workers with disabilities. This option was also cited by the UN Standard Rules on the Equalization of Opportunities for Persons with Disabilities; the Council of Europe Social Charter (1961); the Council of Europe Action Plan to promote the rights and full participation of people with disabilities in society in the period 2006 to 2015; as well as the ILO Recommendation concerning Vocational

Rehabilitation and Employment of Disabled Persons, 1983 (No. 168) that provides guidance on the implementation of ILO Convention No. 159. It is not envisaged, however, in the UN CRPD. Given the results of research studies, in combination with the realization that a real job in the community gives people a strong sense of identity, it came to be recognized that segregated settings send a message to people with disabilities, and to the population at large, that they are different from the rest of society, and that alternatives were needed to allow them to have the same conditions of life as everyone else (Nirje, 1985). This gave rise to a debate about sheltered workshops that continues to the present.

Arguments have been put forward in favour of sheltered employment. These include the avoidance of perceived physical risks in the outside world (Dudley and Schatz 1985; Migliore et al. 2008); the high complexity of jobs in outside employment being beyond the skills range and psychological capacity of people with intellectual disabilities (McConkey and Mezza 2001; Visier, 1998); that sheltered workshops provide greater opportunities for the development of friendships than outside jobs (Dudley and Schatz 1985; Weikle 2008); and that sheltered employment tends to provide surety of employment across the person's life span without affecting disability benefits. Sheltered workshops are also seen to provide an option for people who fail to secure or maintain employment in the open labour market.

Yet it is widely recognized that change was and continues to be required. In recent years, for example, high-income countries, such as Australia, have re-badged sheltered workshops as "Australian Disability Enterprises" where a much greater emphasis has been placed on good business practices and productivity. Wages and working conditions for the disabled employees have improved, in some cases as a result of Trade Union support. In New Zealand, legislation has ensured workers receive the minimum wage unless an exemption is granted in individual cases, based on assessed productivity rates. In Australia, wages are linked to assessed productivity of the individual employee, compared to the normal industrial award. Despite these advances, there is evidence that workers in sheltered workshops do not enjoy the same standards of protection available to workers in the open labour market. A survey of 5,000 workers in sheltered workshops in 24 states in the US revealed that people with disabilities in sheltered workshops earned US\$101 per month based on an average of 74 work hours (NCI, 2008). In

contrast, Kregel and Dean (2002) showed that annual earnings for persons in integrated employment settings was at least twice of that earned by people in sheltered workshops.

- **Supported employment**

Alternatives to sheltered employment and day activity centres emerged in the early 1970s (see 1.2.1 above). In the US, in addition to the factors described above, the establishment of the President's Panel on Mental Retardation (now 'The President's Committee for Intellectual Disabilities') in 1961, and the subsequent research agenda under the auspices of the newly created National Institute on Child Health and Human Development (NICHD), saw the development of energetic research teams which targeted the post-school options for this population.

These influences led to the establishment of an *Employment Initiative for Persons with Developmental Disabilities*, which promoted the concept of supported employment (SE) to business leaders, community groups, and journals, and to the enactment of two public laws that provided a mandate for the national development of SE: *The Developmental Disabilities Assistance and Bill of Rights Act* of 1984, and the *Rehabilitation Amendments* of 1986. These developments emerged from experience with 'competitive employment' that generally focused on people being employed in regular jobs after a period of rehabilitation training, where on-going support was not required or provided. Supported Employment was defined in US legislation as: competitive work in integrated settings for individuals:

(a) with severe handicaps for whom competitive employment has not traditionally occurred, or

(b) for individuals for whom competitive employment has been interrupted as a result of severe disability; and who, because of their handicap, need on-going support services to perform such work While this definition has wide acceptance, it may be interpreted differently in other countries. The essential elements of Supported Employment as defined by the Association for Supported Employment New Zealand (ASENZ) reflect a generally accepted understanding of this form of employment. To be classified as SE, jobs should involve the following characteristics.

- **Open employment:** The employment and inclusion of people with disability in the mainstream workforce.
- **Wages and benefits:** The provision of the same wages and related conditions of employment that are the expected norm in any mainstream workplace.
- **Placement first:** Direct access to the labour market through a precise job/person match and without prolonged ‘getting ready’ activities or training.
- **Inclusiveness:** No exclusions or screening from supported employment programmes on the basis of perceived ‘severity’ of disability.
- **Individualized and on-going support:** Support services and strategies that are not time-limited, are tailored to the individual’s needs, and maximize job retention.
- **Choices and career development:** Services and outcomes based on the preferences and aspirations of the individual and a commitment to on-going pursuit of careers. Researchers working with people with very high support needs developed four alternative pathways, all of which were characterized as ‘supported employment’ (Mank et al., 1986). The first was called the ‘supported jobs’ model, which later became the SE model described above. Other alternatives were the ‘enclave’ model, the ‘mobile crew’ model and the ‘benchwork’ model. Each of these models specifically targeted people with intellectual disabilities who had high support needs.

- **Supported jobs model**

The ‘supported jobs’ employment model involves a ‘place and train’ approach, rather than the practice in rehabilitation facilities where the model is ‘train and then place’. Therefore, to provide employment, a job is required at the outset of the service, and not just at its completion. The ‘place, train and support’ model for SE usually involves job coaches working with the potential employee to identify interests and skills. This is followed by job seeking, matching and placement. The employer might also be encouraged to make adaptations (i.e. *accommodations*) to a specific job in order to match it with the abilities and skills of the employee with an intellectual disability. In many cases the job coach might learn to do the job first, in order to subsequently teach the employee with the disability. In other circumstances, the

employer might prefer for a regular staff member to teach the skill. This person may subsequently become a 'mentor' to the person with the disability, thus becoming part of the natural support for the person within the workplace. Irrespective of who does the initial training, research consistently shows that on-going natural supports (through a staff member) are leading to greater sustainability of job placements than external supports (through a job coach employed by public authorities or an NGO). However, while natural supports are proving to be a promising method of increasing the integration and support of people with intellectual disabilities in the workplace, a combination of job coaching and natural supports may be needed, tailored to individual circumstances and needs.

- **Enclave model**

The enclave model consists of a group of people with disabilities who are trained and supervised among workers who do not have a disability, usually in an industrial or commercial environment. In its original design, the workers were to be paid at a level commensurate to the non-disabled workers, often adjusted according to their relative productivity. As the model was taken up by sheltered workshops, more often than not the contract for the enclave was between the company and the sheltered workshop management. This meant that wages for the person with a disability continued to be paid by the sheltered workshop, at the sheltered workshop rates.

On-going support is provided to the enclave workers of much of the same nature as given by the job coach in SE. The literature does indicate that in some cases companies were willing to take on the job coach role along the principle of 'natural' supports described above. There is a danger that there may be little opportunity for work and social interactions. For instance, in some cases the enclave group may not share the common facilities of the company, such as lunchrooms.

- **Bench work Model**

This model was developed in the early 1970s by the Specialized Training Programme at the University of Oregon, US, as an alternative to day-activity programmes to provide long-term employment to people previously denied access to any vocational services. This model operates as a small, single purpose, not-for-profit commercial operation providing a range of goods or services. The model requires a number of highly qualified staff skilled in instructional technology with no more than a 1:5 staff-to-worker ratio. The size of the business was limited to

20 workers and they were located in close proximity to stores, restaurants, and other community venues that can provide opportunities for integration and participation into the regular community activities during lunch breaks as well as before and after work. At first the model was faithfully replicated in five states of the US and in Australia. The costs of operating the facilities were no more than those incurred in the day-activity centres, but employees received wages based on their productivity for the first time in their lives.

Although designed for people with very high support needs, who would not normally be employed in a sheltered workshop because of their supposed low productivity, the bench work model shares many characteristics and constraints with traditional sheltered workshops. It initially also depended heavily on contracts from the electronics industry, many of which have disappeared as this industry has become more automated. However, in its early phases this model provided opportunities for its workers to have opportunities for greater community participation.

- **Customized employment (CE)**

In the US, another model of employment has recently been developed which aims to place people with disabilities in jobs earning competitive wages. It is known as ‘customized employment’, a term first defined by the Office of Disability Employment Policy in 2001. This model embraces a ‘person-centred’ approach, beginning with the person’s needs, aspirations, talents and skills. This profile serves as a basis for contacting potential employers (Inge, 2006; 2008). It also emphasizes the person’s choices, strengths and abilities. It is employed in One-Stop Service Delivery Systems (Blanck et al., 2009; Inge, 2008) where workforce investment, education, and other human service programmes work together to optimize access to services and long-term employment outcomes (see Box 2.5). Similar principles are used by Jobsupport Inc., one of Australia’s most successful SE programmes, in its transition from school to work initiative (Tuckerman, 2008a). While the process of job carving, as illustrated in Box 2.5, is unique for each individual and each employee, the process generally involves these steps:

- determine the contributions, potential and dreams of the individual;
- seek out employment opportunities that utilize, exploit, or highlight these contributions;

- perform a formal (written) job analysis in order to determine task sequencing, natural supports, operations that may require additional instruction, modification, alternative production methods, or those that may need to be performed in partnership with, or entirely by, another worker;
- engage in interest-based negotiation with the employer highlighting the individual's contribution to the workplace and offering a reasonable and understandable re-arrangement of work tasks in order to employ the individual;
- provide quality consultation to the employer and co-workers so that they can teach the individual the job;
- provide on-going support to the employer and the worker. (Griffin and Hammis, 2002, p. 1; Griffin, Hammis and Geary, 2007).

Beyer and Robinson (2009) reported on a 2005 study conducted for the US Department of Labor, which found that CE had been used successfully with people with high support needs. In the first round of the study 345 people had been placed in competitive employment. The majority had found high quality jobs, with 95 per cent earning the minimum wage or above, and over 50 per cent placed in jobs with the potential for career advancement. A second round of data collection on 536 randomly selected individuals yielded similar results. Approximately 42 per cent of the sample disclosed a psychiatric or emotional disability, with 20 per cent reporting an intellectual disability. In essence, the major contribution of the CE model to the SE model has been the recognition that a person's job choices play a crucial role in achieving a successful job match. It is another example of the application of the social model of disability, in that environmental adjustments may be more important than trying to change the person.

5.7 Self-employment, self-directed employment, microenterprises

Neufeldt and Albright (1998, p. 6) defined self-directed employment as: Income-generating work where disabled people, to a significant degree, have a prime decision-making role in the kind of work that is done, how time is allocated, what kinds of investment in time and money should be made, and how to allocate revenue generated.

Duce and Biersdorff (2002) suggested that business considerations for people with intellectual disabilities are little different from those for people who do not have a disability. In the case of

the former, it is usual for someone to perform duties not dissimilar to the role of a job coach. Sometimes this is a family member. They proposed five major steps in the process of establishing a business: i) developing a business plan; ii) obtaining the required skills; iii) securing start-up capital; iv) implementing the business plan; and v) expanding the business. Again, the skills necessary to support self employment will be very similar to those for SE, and may include job analysis, skills training and on-going provision of support. It is also possible for the support worker to initiate contacts with natural supports such as unpaid volunteers and mentors who have experience as entrepreneurs who can act as an on-going resource. This model has also been described as a micro-enterprise: a small business created around one person. Beyer and Robinson (2009, p. 65-66) reported that the main strengths of the model for people with intellectual disabilities are that:

- it respects the capacity and the assets of people with intellectual disabilities, by focusing on people's interests and strengths, and can be more flexible than mainstream employment and working conditions;
- it pursues equality by opening up the self-employment sector of the labour market, where people with intellectual disabilities are not represented;
- it reflects aspects of governmental thinking on increasing employment for disadvantaged groups, and on developing services to meet individual needs;
- it is for some people a way of gaining income from a hobby or an interest, with support and funding through person-centred planning, direct payments and individualized budgets; and
- it provides another way for a person with intellectual disability to move from being a client to being a citizen.

5.8 Community Economic Development (CED)

CED has emerged as an alternative to conventional approaches to economic development. It is founded on the belief that problems facing communities - unemployment, poverty, job loss, environmental degradation, economic instability, and loss of community control - need to be addressed in a holistic and participatory way. There are varied definitions of CED, but the following captures the essential characteristics. CED projects may provide an alternative

approach to providing support to people with disabilities. In the highly urbanized communities of high income countries, policies and services are generally managed in a top-down fashion and controlled by large government bureaucracies, increasingly delivered through large bureaucratized not-for-profit organizations. Stainton et al. (2006) suggested that to overcome barriers to the employment of people with disabilities, connections must be made with existing groups of people in order to overcome prejudice and discrimination. Decision-making must take into account the perspectives of people with disabilities as far as possible in a 'bottom-up' fashion. For instance, Wilson (1996, p. 617) commented that "...community economic development, if it is truly to empower people, must build community from the inside out – i.e. from the individual's realization of self-efficacy and interconnectedness with the larger community".

CED has proven successful in addressing the needs of various marginalized groups in low-income countries. Each community is encouraged to develop viable initiatives based on an appreciation of its strengths and capabilities through a process that Kretzman and McKnight (1997) have described as 'asset mapping'. McCall (2003) suggested that, unlike a needs based approach, which often looks to external professional input for solutions, asset mapping looks to the community's own assets that may be utilized to solve the local problem.

- **Employment options - summary**

Recent decades have seen growing recognition of the working capacity of persons with intellectual disabilities in many countries around the world and a flourishing of initiatives to provide them with opportunities to perform meaningful, productive work in ordinary workplaces. Supported Employment has been the main approach adopted, in a variety of forms reflecting the different levels of support required by different persons with intellectual disabilities. The experience in implementing these approaches in different countries around the world is summarized in the following section, along with the main findings of research carried out. It must be recognized, though, that the SE model was developed in high income countries with well-established formal services. In low-income countries, the informal economy, especially in rural and less developed areas, is the major source of employment. People with intellectual disabilities are often employed in village-based industries, including farm work, similar to the pattern in western countries in the pre-industrial era.

5.9 Development of Supported Employment

Initiatives

The implementation of supported employment in several high-income countries and on a smaller scale in some low-income countries is described in this section. It has not been possible to detail developments in all countries, but those with the most significant research activity in this field have been chosen, together with those that have demonstrated some innovative initiatives. The research findings are summarized in Section 3.3 below.

What makes for successful inclusive employment?

The focus of the research studies from Australia, Canada, the United States and the European Union summarized here has been on factors which have facilitated the growth and sustainability of the Supported Employment model of inclusive employment. As the ‘home’ of the SE model, it is not surprising that the bulk of research studies has also emanated from research centres in the US. These research studies have identified a number of factors that influence the impact of SE and characteristics of SE programmes which have led to successful outcomes. These are outlined below.

Barriers to participation

In respect to those people with high support needs, their prospects for employment are hampered by a range of factors that are not directly attributable to their impairments. Stainton et al. (2006) pointed out that factors such as disincentives to employment arising from income and disability support programmes, lack of information about job availability, inadequate training and comparatively low education levels, lack of accessible transportation, and employer discrimination were limitations to being recruited into SE programmes. In countries which have implemented a quota system for the employment of employees with disabilities, employers will often declare that they have not reached the quota and elect to pay a fine for non-compliance. This money is then used to support sheltered or “special” employment.

Preparedness of individuals with disabilities

Riches and Green (2003) found that people with disabilities were generally well accepted by supervisors and co-workers in integrated employment settings. However, such acceptance was

contingent on people with disabilities ‘blending in’ or ‘fitting in’ and not drawing attention to themselves – that is, having acceptable social skills. As people with intellectual disabilities experience additional physical and mental health problems to those of the general population, research has shown that regular health checks prevent more serious problems occurring (Beange et al., 1999). As the prevalence of mental health problems for people with intellectual disabilities can be as high as 40 per cent, it is imperative that mental health screening be conducted so that those requiring specific support can be identified (Einfeld et al., 2006; Hofer et al., 2009). As indicated above, inappropriate social and emotional behaviours can be a significant barrier to successful employment outcomes.

O’Brien and Dempsey (2004), in their comparative analysis of employment services for people with intellectual disabilities in Australia, Finland and Sweden, emphasized the necessity for education and training as a key factor in improving the participation of people with intellectual disabilities in the general labour force.

Individualized financial support

Allocating financial support to the individual in conjunction with person centred planning (see below) provides the person with a disability greater opportunity to exercise self-determination and freedom of choice (O’Brien et al., 2005; Stainton et al., 2006). Canada has been at the forefront in adopting an individualized funding approach through which government financial support is directed to the individual rather than to an agency.

School to work transition initiatives

The acquisition of job-related skills is a process which should begin long before a student with intellectual disabilities leaves school. Research in the area of transition from school to work and adult living has a long history, reaching back to the early 1970s (Parmenter, 1986; 1990). For instance, Brolin (1972) studied the effects of a school curriculum which facilitated the subsequent employment of people with intellectual disabilities, including programmes such as work study or work experience. A decade later, the emphasis swung to people with much higher cognitive support needs, leading to the first SE initiatives for this population. Rutkowski et al. (2006) provide a description of Project SEARCH, of which a large component is work experience. The *Americans with Disabilities Act* of 1990 (ADA) supported these initiatives in the

US by targeting discriminative practices in the workforce. One of the keys to a successful transition process is for the secondary school to implement a transition policy, an ingredient of which should be the development of individual transition plans (ITPs) for each student early in the secondary school years. Gradually, inter-agency involvement of agencies such as employment providers and post-school education providers need to be involved in the planning. Towards the end of schooling, part-time work experience has been shown to be a factor predicting later employment success (Parmenter, 1986, 1990; Parmenter and Fraser, 1980; Parmenter and Knox, 1991).

5.10 Characteristics of successful programmes

Values base

Programmes in which all the stakeholders, including government, employers, service providers and families, were committed to the principle of equity for people with intellectual disabilities led to successful and sustainable integrated employment outcomes.

Programme size

Research in the US has paid particular attention to factors which facilitate the growth of SE programmes, in an effort to correct the imbalance with facility-based models of employment. Programme size appears to play a role in the type and range of day and/or employment services offered. In a national survey of rehabilitation agencies, Kiernan, Gilmore and Butterworth (1997) reported that smaller agencies, providing support to 50 people or fewer, were found to be more likely to provide integrated employment exclusively than larger organizations. In the current globalized climate of business amalgamations, based on the principle of economies of scale, the loss of a personalized service militates against the delivery of effective outcomes for individual beneficiaries.

Support needs assessment

A comprehensive support needs assessment of the person with intellectual disabilities has been shown to be important, not only for the purposes of employment, but also for other life activities, including community living and leisure activities (Riches et al., 2009a, 2009b; Thompson, et al., 2004). It has been shown that work cannot be dealt with in isolation from other life activities (Halpern, et al., 1986). In terms of the sustainability of integrated employment, there is

overwhelming evidence that people with intellectual disabilities tend to lose jobs, not because of their inability to perform the job tasks, but because of negative social behaviours (Chadsey et al., 1999; Knox and Parmenter, 1993).

A field-based study which examined the relationships between general community living skills and the degree of community independence in working and living domains of adults with intellectual disabilities found that people who displayed higher levels of basic living skills generally worked and lived more independently. These findings support the inclusion of specific training in basic living skills in school curricula during the transitional school years of a student with intellectual disabilities (Woolf et al., 2010). On the other hand, Chadsey et al. (1997) warned against too much emphasis being placed on a ‘deficit-remedial model’, rather than emphasizing the congruence or ‘ecological fit’ between the person and his/her environment. It may be more efficacious to find the right environment, rather than changing a particular social behaviour. This principle is especially relevant for people with intellectual disabilities and high support needs.

5.11 Person-centred planning

Person-centred planning puts the person with the disability at the centre of the decision-making processes, taking into account the person’s wishes and desires (Coyle and Moloney, 1999; Wiese and Parmenter, 2009). In the employment area, strong motivation on the part of the person seeking a job has been shown to be a significant predictor of a satisfactory outcome. Whilst more work remains to be done on assessing the impact of person-centred planning on client outcomes and satisfaction, research to date has shown promising results (Robertson, et al., 2006). In the pilot SE project in Zambia, Koistinen (2008) found, that good job matches were made when job-seekers’ preferences and needs were recognized, leading to better long-term outcomes.

On-job support

The ‘place, train and support’ model for SE usually involves job coaches working with the potential employee to identify interests and skills. This is followed by job seeking, matching and placement. The employer might also be encouraged to make adaptations (i.e. accommodations) to a specific job in order to match it with the abilities and skills of the employee with an intellectual disability. In many cases the job coach might learn to do the job first, in order to

subsequently teach the employee with the disability. In other circumstances, the employer might prefer for a regular staff member to teach the skill. This person may subsequently become a 'mentor' to the person with the disability. In Zambia, availability of support programmes for job placements was a crucial element for success not only in obtaining but also keeping jobs. It was not sufficient to rely solely on vocational training programmes to place people in employment.

Irrespective of who does the initial training, research is consistently showing that on-going natural or informal, supports are leading to greater sustainability of the job placements than external/formal supports. In practical terms this means that job coaches need to gradually withdraw their support, but at the same time ensuring there are networks within the workplace that will provide support to the employee when required. For example, they could identify someone in the workplace who is prepared to be a "mentor" to the employee with a disability. However, while natural supports are proving to be a promising method of increasing the integration and support of people with intellectual disabilities in the workplace, Storey (2003) concluded that a combination of job coaching and natural supports may be needed, tailored to individual circumstances and needs.

Links to parents

Griffin et al. (2008) and Lindstrom et al. (2007) stressed the importance of families' attitudes toward their adult children's capabilities as being a critical determinant in their support for community-based employment options. Since the first point of contact in the diagnostic process is the medical and allied health professions, limited expectations are often entrenched in the child's early years.

Lack of career choice was highlighted in a study of attitudes of parents and primary caregivers of people with high support needs towards SE (Ford et al., 1995). This study also emphasized the need for greater communication between the service providers and families. Donnelly et al. (2010) have shown the importance of family networks to the person with intellectual disabilities achieving satisfactory employment outcomes. Koistinen (2008) concluded that parental support was important, though often lacking in Zambia, especially at the vocational training, job-seeking and placement levels.

Benchmarking quality

In order to abide by the original SE model, commentators are increasingly calling for effective benchmark indicators to be established against which the quality of SE programmes can be measured (Jenaro et al., 2002; Wehman et al., 2003) (See Box 3.7). Timmons et al. (2002), who investigated the characteristics of effective state service systems in the US, by examining the experiences of people with disabilities who have used state agencies to find employment, indicated five key components to effective service delivery: i) agency culture; ii) consumer directedness; iii) access to resources; iv) quality personnel; and v) coordinated services. In a particularly rigorous economic analysis of data in the state of Southern Carolina in the period 1999–2005, geared to determine whether job coaching leads to stable employment in community settings, McInnes et al. (2010) found that:

- 56 per cent of individuals with job coaches were working one year later, compared to nine per cent of those without job coaches;
- those who received job coaching were likely to have favourable job characteristics such as higher IQs and an absence of emotional or behavioural problems; and
- when these endogenous factors were corrected for in subsequent analyses, a sizeable and statistically significant effect remained. It is possibly not surprising that agencies are tempted to recruit ‘easier to place’ individuals into SE programmes, such as those with higher IQs and good social skills, despite the fact that these programmes were initially meant to target the more disabled population.

Sustainability

Based on the findings and recommendations of a survey of the situation in 25 US states, together with a detailed investigation of over a dozen states which reported integrated employment outcomes well above the national average, Mills (2006, p. 3) suggested that seven factors explained the impressive integrated employment outcomes achieved by these states:

- the existence of strong, clear and unambiguous state developmental disabilities agency policies, rules and programmatic requirements intended to support a clearly articulated agency

preference for, and commitment to, integrated employment for people with developmental disabilities;

- the use of funding incentives to encourage the expansion of integrated employment opportunities, and/or funding disincentives to discourage the use of facility-based employment and non-work services;
- a liberal definition of the kinds of employment arrangements which qualify for SE funding;
- adequate state agency staffing dedicated to employment;
- investment in on-going training and technical assistance for agency staff;
- a commitment to supporting organizational change among facility-based (sheltered) providers moving to integrated employment settings; and
- the use of a comprehensive data-tracking system focused on integrated employment outcomes.

5.12 Workplace characteristics *Employer attitudes*

A US study on employer attitudes (Olson et al., 2001) showed that employers from firms of all sizes have favourable experiences in employing people with intellectual disabilities and value the training done by SE service providers. Accommodations frequently made include extra supervision time, the provision of flexible hours and the use of a job coach. These accommodations were perceived to be of minimal cost to the company. In terms of human resource management, employees with intellectual disabilities were viewed as costing companies the same or less than employees without a disability. However, negative employer attitudes can be one of the most serious threats to the success of a SE programme. Nevertheless, one variable appears the most reliable in predicting employers' favourable attitudes to people with intellectual disabilities; that is, previous positive job-related experience with a person with a disability (Rimmerman, 1998).

Negative employer attitudes can be one of the most serious threats to the success of a SE programme. Disability may be perceived as taboo. Employing persons with disabilities may be expected to affect the prosperity of businesses. Negative stereotypical views on the concept of intellectual disability are deep-seated and prevail in most countries (Yazbeck et al., 2004). This

suggests the importance of working closely with employers to ensure their satisfaction with the employment outcome. Hence, advertising 'success stories' involving the employment of people with disabilities has been found to be an important strategy. Research has shown that, contrary to the negative stereotypes portrayed throughout history; people with intellectual disabilities can be loyal, trustworthy and diligent workers who seldom fail to turn up for work (Ward et al., 1978). Tuckerman et al. (1999) found very positive employer attitudes toward workers with intellectual disabilities with high support needs. Smith et al. (2004) reported that the existence of policies on employing people with disabilities, previous contacts with a person with a disability, and the level of the disability were associated with more favourable attitudes. A survey of 642 Australian employers who had employed a person with a disability, including those with intellectual disabilities, concluded that:

Workplace culture

A congenial workplace culture has been found to be critical for the successful placement and maintenance of people with intellectual disabilities in integrated work environments. Butterworth et al. (2000) outlined four salient characteristics of a supportive and interactive work setting: a) multiple-context relationships, for instance opportunities to mix with co-workers after work has finished for the day; b) specific social opportunities, including designated lunch/break rooms where everyone is welcome; c) personal and team-building management style, where managers build a sense of teamwork, take a personal interest in employees, and schedule work around workers' personal strengths and needs; and d) interdependent job designs, which may include 'cross training' employees on several different tasks. This workplace culture also has the opportunity to create career pathways for supported workers, who in many cases, do not progress beyond entry-level jobs. Such a culture also helps the development of strong relationships between workers with and without disabilities. These relationships are often characterized by the presence of humour and informality, which ultimately assist the processes of job retention (Parmenter, 2002). To assist potential employers to be more accepting of people with intellectual disabilities as genuine workers, and to counter the negative views often rooted in the stereotypical attitudes of the general community, Fabian, Edelman and Leedy (1993, p. 32) have suggested the need for intervention strategies at three levels:

Storey (2002) and Parmenter et al. (2006) have provided best practice suggestions for increasing the social interaction of supported employees through strategies targeting social skills instruction, communication instruction, problem-solving and co-worker assistance. A Dutch study found that supported employees were perceived on an equal basis within the workplace by their non-disabled colleagues (Cramm et al., 2008). Most co-workers considered supported employees to be at least 'work acquaintances'. Interestingly, some of them identified supported employees as 'social friends', indicating a higher level of inclusion.

Outcomes

Most of the studies which have addressed outcomes for supported employees have made comparisons between SE and segregated programmes, including sheltered workshops and day centres, on a range of indices.

Performance

An Institute of Employment Studies Survey in the UK in 2005 (Dewson et al., 2005) reported that:

- disabled employees show a high level of commitment and loyalty to an organization, and as a result,
 - disabled employees are hardworking, have low sickness absence rates and high retention rates.
- Whilst these findings refer to disabled workers generally, they are especially true of workers with intellectual disabilities (Ward et al., 1978).

Consumer satisfaction

Surveys of satisfaction from the perspective of people receiving SE services have found that the majority of consumers like their jobs. Parent, Kregel and Johnson (1996) found, from face-to-face interviews with people with high-support needs receiving SE services in Virginia, that many would like to change some aspect of their job to make it better. More than half indicated that their current job is not the career they would like to have permanently. The study concluded that there was a need to increase consumer involvement in all phases of the SE service delivery. Lack of career choice was also highlighted in a study of attitudes towards SE of parents and primary

caregivers of people with high support needs (Ford et al., 1995). This study also emphasized the need for greater communication between the service providers and families.

Earnings

SE programmes by their very definition give people with intellectual disabilities access to real wages. These are, especially in programmes for people with high support needs, adjusted according to the person's productivity and pro-rata amounts are paid. There is also a high rate of part time jobs reported in SE programmes. Overall, however, people in SE programmes earn significantly more than those employed in segregated programmes.

Quality of life

In terms of quality of life and life satisfaction, studies generally report that those in SE programmes had a higher rating than those unemployed or in alternative disability programmes. In an Australian study, Eggleton et al. (1999) found that those in open employment had a significantly higher quality of life than those in sheltered workshops or who were unemployed. In a Spanish study, Verdugo et al. (2006) found no differences in quality of life between those in SE and those in sheltered workshops. However, there was a direct relationship between the typicality of the employment setting and quality of life for those in SE programmes. There was also a negative relationship between the intensity of external support and quality of life, suggesting that such support should be used when only absolutely necessary. Kober (2005), in a later Australian study, compared the quality of life and level of empowerment/independence of people in integrated and segregated employment settings. For people with low functional work ability there was no difference in quality of life, while for those with high functional work ability those in integrated work settings were higher on both quality of life and empowerment/independence scales. Interestingly, living in the family home had a relationship to quality of life. For those living at home, irrespective of the type of employment, there were no differences on quality of life. However, in responses of participants living independently or in group homes, quality of life was found to be significantly higher for those in competitive employment settings. This confirms an earlier observation concerning the interdependence of life activities: work outcomes cannot be seen in isolation from other life events.

Economic benefits

While cost-benefit/cost efficiency analyses of SE have produced conflicting results (Schneider, 2003; Beyer and Kilsby, 1998; Beyer and Robinson, 2009), other studies have reported clear gains to both the individual and the taxpayer (Kregel, 1999; Kregel and Dean, 2002; Kregel et al., 2000; McCaughrin et al., 1991). In a comprehensive review of the cost efficiency of SE programmes, Cimera and Rusch (1999) reported that:

- over time, SE is a good investment for taxpayers and society;
- generally, workers earn more in SE than in sheltered employment; and
- cost efficiency of SE programmes varies among regions of countries.

Cimera's comprehensive research of all supported employees funded by vocational rehabilitation throughout the entire US from 2002 to 2007 (2010) returned an average net benefit to taxpayers of US\$251 per month per supported employee. Cimera noted that this figure, if extrapolated to the total number of Americans with disabilities served in segregated day services, would amount to over US\$1.5 billion in cost benefits for SE. Disappointingly, SE wages still lagged behind those of the general population, and often did not represent a living wage. High rates of part-time employment may have contributed to this finding

Tuckerman, Smith and Borland (1999) also found that the placement of people with high support needs in an Australian context was a cost-effective option for government. As the programme grew in size, the costs per client stabilized, together with a reduction in disability pension costs. The study revealed that the SE programme was significantly less costly than a day activity programme catering for a similar population. Of special significance to the principle of inclusive communities and the acceptance of diversity was the finding by Graffam et al. (2002b) that employers experience both material and non-material benefits to their organizations from employing a person with a disability, with those benefits being financially cost-neutral or cost-beneficial in a large proportion of cases. In terms of non-material benefits, the presence of people with disabilities in the workforce highlights the need for overall attention to training and supervisory practices, basic work practices, health and safety issues and an improvement of organizational performance. Graffam et al. (2002b) also suggested that their findings provided

some evidence of employers acknowledging and accepting diversity in the workplace. Overall, they concluded that, "...an employee with a disability can be seen as a catalyst for positive change, a catalyst for improved organization performance" (p260). Beyer and colleagues at the Welsh Centre for Learning Disabilities have carried out extensive cost-benefit analyses of SE over the past two decades (Beyer et al., 1996; Beyer and Kilsby, 1998; Beyer et al., 2002; Beyer and Seeböhm, 2003; Beyer, 2008; Shearn et al., 2000). Major conclusions were as follows:

- Initial costs of SE can be higher in the start-up period than in the traditional sheltered employment options. Taxpayer costs⁵⁷ improved over time. It may take up to five years before financial benefits outstrip the costs of SE.
- Welfare benefits regulations and the absence of mainstream funding for SE were seen as significant restrictions on the growth of SE in the UK.
- Higher rates of part-time work, with supported workers working for a few hours and also being on welfare benefits, led to relatively unfavourable levels of cost-benefits, compared to the US.
- Comparison of the costs of SE with those of day activity centres was unfavourable to SE, but SE was found to generate greater outcomes for money spent in terms of social integration.
- Economies of scale in terms of the numbers of workers employed and the hours worked per person were critical factors in determining cost-benefit.
- When factors such as the taxes paid by supported workers and their higher spending capacity are taken into account, community placements represent a net gain for the taxpayer. In a report commissioned by the UK Department for Work and Pensions, Rangarajan et al. (2008) questioned the findings of cost-benefit studies of SE on the grounds of methodological flaws, including the lack of randomized control trials..

5.13 Moving forward

Since its inception in the 1970s, SE has provided inclusive employment options for people with intellectual disabilities across several continents. The personal stories of what having a job means to people with intellectual disabilities is testimony to how important this issue is in terms of social justice and human dignity. A key factor in the growth of initiatives has been a clear

policy commitment by governments to support such initiatives. However, experience from both high- and low-income countries indicates that on-going support is required for these initiatives to be sustainable. Low-income countries, given their rural nature and severe fiscal constraints, may be in a better position to test a number of the alternative models described above. For instance, self-employment, small business enterprises, and, at a regional level, CED, may be viable options, but these have not as yet been critically analysed. The evidence from a study conducted in Zambia questioned the viability of the small business enterprise model for people with intellectual disabilities, as it was suggested they lack the necessary business skills (see Koistinen 2008, p. 32). An alternative would be to employ a person with business experience and a strong commitment to assisting people with intellectual disabilities, as the manager. The SE principles, including the important 'place and train' and on-going support features, can be used in each of these approaches.

A positive feature is the growth of national SE organizations which provide advocacy and training support to their constituents. As these organizations gain strength, they may be in a position to mentor initiatives in low-income countries.

Pointers for Policy Development

Various countries have clear policies concerning employment for people with disabilities in general, which affirm the goal of community integration and support for individual choice. Despite this, however, this review has highlighted the reality that the vast majority of people with intellectual disabilities still do not have access to integrated employment; and in many cases, no access to any meaningful day activities. The following pointers relevant to policy development have been informed by the research outcomes and field initiatives reviewed for this working paper and can act as a guide for the further expansion of SE to facilitate the employment of people with intellectual disabilities in inclusive workplaces. These are relevant to countries presently attempting to revise old policies and/or develop new policies which will go some way in addressing the inequalities experienced by people with intellectual disabilities.

- Programmes need to abide by the principles of Supported Employment and Customized Employment, to achieve good outcomes for people with intellectual disabilities, especially the need for on-going formal and informal supports.

- Long-term success is influenced by sound initial planning, which must start in the transition years at the secondary school level. A well-planned transition process from school is one of the better indicators of successful employment outcomes, especially if work experience is a feature.
- There is a need for a balance between structure and over structure in the planning and delivery of employment options. The intensive, clinically-driven rehabilitation model is generally not recommended for people with intellectual disabilities, especially for those with moderate to high support needs.
- Good policies are based on good information. There is a dearth of data on the actual incidence and prevalence of intellectual disability, especially in low-income countries. The use of typical epidemiological methodologies may not be entirely relevant, given that disability is a culturally specific phenomenon. What counts as a disability in an urban context may not be seen as such in a rural community.
- Effective data management systems assist in informing programme needs, programme monitoring and programme development for people with intellectual disabilities, especially those that are embedded in data-gathering systems for the general population.
- Planning which is person-focused and which involves all stakeholders, including families, is also a key factor in its success. Emphasis needs to be placed on the person's strengths, interests and the support system that needs to be provided to achieve personal goals. The importance of matching the person's preferences, wherever possible, to the job placement cannot be stressed too strongly.
- Career planning is recommended, as many people with intellectual disabilities do not progress beyond entry-level jobs.
- Adequate staff training of job coaches, with an emphasis on marketing and on-site training skills, is critical to the success of SE.
- Assessing outcomes on the dimension of quality of life reflect the inter-relationship between work and other life activities, including where the person lives. The reciprocal relationship

between levels of independence and feelings of self determination and empowerment is well established.

- Success stories of positive employment outcomes are one of the best ways to convince potential employers that people with intellectual disabilities can be loyal and productive workers.
- As employment has been shown to impact on the quality of life of people with intellectual disabilities; it may be an important factor in the possible reduction of emotional disorders in this population.

Action is required at both the national and regional levels to give effect to the lessons learned in a way which will lead to improved opportunities for this group of people with disabilities

National level

Governments in consultation and collaboration with civil society and social partners bear responsibility for promoting opportunities for people with intellectual disabilities to take part in their communities and the broader society. This section outlines the main steps towards meeting this responsibility.

Conduct a systems analysis of current situations

- In collaboration with all relevant stakeholders, including schools, conduct surveys, consultation meetings and individual interviews to determine the current status of employment for people with intellectual disabilities.
- Analyse barriers and facilitators to the achievement of integrated employment for this population across the various geographical areas of the country in order to tap the experiences in urban and rural communities.
- Inform this process through reference to documentation on current models of employment for other disability groups.

Develop goals and objectives for the way forward

- Using the same consultative processes, with the input of all relevant stakeholders, develop a set of goals and objectives to guide the implementation of SE initiatives for people with intellectual disabilities.
- Explore a variety of models to cater to the particular circumstances of local communities. Elements of the Community Economic Development (CED) model are worthy of investigation, as are aspects of the social enterprise approach.
- Agree a statement of the underlying value system which has informed this process.

Develop a national policy on integrated employment for people with intellectual disabilities

- Develop a clear written policy articulating its underlying values, and the corresponding goals and objectives of integrated employment services for people with intellectual disabilities which flow from these values, in consultation with relevant stakeholders, including different government departments, social partners and civil society agencies.

Development of an effective data collection system designed to track progress

- Develop a data collection system to provide information on people supported (inputs); details of the various programme models (processes); details of wages earned, days worked, level of community integration, and skills achieved (outputs); and measures of satisfaction and quality of life of the employees (outcomes).
- Design this data collection system to allow for programme modifications following feedback, and the identification of environmental barriers and facilitators.
- Develop local research capacity to help maintain the on-going sustainability of programmes by developing coalitions with local university research centres, encouraging national universities to collaborate on issues such as programme development and evaluation and staff training, as well as exploring alliances with universities in other countries which have a track record in research into employment of people with intellectual disabilities.

5.14 Summary

Despite enormous challenges, including extreme poverty and difficulties faced from attitudinal and policy perspectives in the provision of support to people with disabilities in general, and those with intellectual disabilities in particular, there are some positive trends. Of special significance is that the majority of countries around the world have subscribed to the various contemporary human rights principles relating to people with disabilities. What remains is the need for a genuine commitment to ensuring these principles are not only incorporated into national disability policies, but are put into practice and monitored to the best level possible in terms of the socio-economic conditions prevailing.

The opportunity to engage in real work in the general community has been denied to the vast majority of people with intellectual disabilities, for reasons explored above. The first step in bringing about change in this situation is a belief that change is desirable and a genuine commitment to effect change. Most countries have signed and ratified the UN *Convention on the Rights of Persons with Disabilities*. Article 27 of the Convention states:

States parties recognize the right of persons with disabilities to work on an equal basis with others; this includes the right to the opportunity to gain a living by work freely chosen or accepted in a labour market and work environment that is open, inclusive and accessible to persons with disabilities. States parties shall safeguard and promote the realization of the right to work. This statement provides a compelling reason to explore how a country, even with limited resources, might embark on a programme to assist one of the most disadvantaged groups in its society to become engaged in work ‘that is open, inclusive and accessible’. It is crucial that, where ratified, the implementation of the CRPD is closely monitored. The fact that countries around the world have enacted specific disability legislation is a positive feature, but as noted throughout this paper, people with intellectual disabilities are often left until last in the provision of support.

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Block 3: Autism Spectrum Disorder: Nature, Needs and Intervention

Unit 1: Definition, Types and Characteristics

Unit 2: Tools and Areas of Assessment

Unit 3: Instructional Approaches

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Unit 1

Definition, Types and Characteristics

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1.1 Introduction

Autism spectrum disorders are life-long neuron developmental disabilities with onset before 36 months and characterized by: Impairments in reciprocal social interactions. Impairments in verbal and non-verbal communication skills. Stereotyped behavior, interests and activities. Autism spectrum disorders represent an abnormality of brain development and function, appearing within the first three years of life. Although the detailed causal mechanism(s) are not known, autism is likely to have multiple aetiologies including genetic factors. A range of studies has found in 10 - 37% of cases there may be an associated medical condition (eg tuberous sclerosis).

1.2 Presentation

The clinical picture may vary depending on factors such as age, sex, IQ and personality. There may be a wide range of overlapping presentations. Discussed below are:

- 1) Autistic Disorder.
- 2) Asperger's Disorder.
- 3) Atypical Autism.

1.2.1 Autistic Disorder

Sometimes designated as classical or childhood autism. It involves the following:

- **Impairments of Social Interaction**

Aloofness and failure to develop friendships. Seeking people's company but lacking ability to engage in two-way social interactions. Stilted, one-sided or repetitive social interactions. Being socially passive while tolerating social approaches. Awkward, avoidant or indifferent eye contact. Inability to understand social rules eg make socially embarrassing comments unintentionally. Impaired understanding of other people's motivations, perspectives or feelings. Markedly impaired use of non-verbal behaviors to regulate social interaction eg gestures. Lack of spontaneously seeking to share enjoyment with others.

- **Impairments of Communication and Play**

A delay in or lack of development of speech without any compensatory forms of communication eg gesture or mime. Significant difficulty in initiating and sustaining a conversation (in those with speech). Stereotyped or idiosyncratic use of language. Lack of imitation of others. In children, play lacks the varied or make-believe qualities appropriate to the developmental level.

- **Restricted or Repetitive Interests and Activities**

- o Interests that is excessively narrow, intense or unusual.
- o Adherence to rigid routines.
- o Intolerance of change.
- o Stereotyped and repetitive motor mannerisms eg hand flapping.
- o Persistent preoccupation with parts of objects.

- **Other Commonly Associated Features**

Intellectual disability occurs in 70% of people with autistic disorder. Autistic disorder is more common in males than females (3:1). Epilepsy is common and can develop at any age.

Unusual sensory responses eg sensitivity/aversion to particular sounds or tactile sensations, intolerance to foods, or fascination with spinning objects or lights. Can form affectionate attachments to those they know well.

May develop behaviour disorders eg angry outbursts, self-injurious behaviour, feeding difficulties/fads, sleep problems, and over activity (particularly as children). Anxiety due to poor communication skills, over-stimulation etc.

1.2.2 Asperger's Disorder

Asperger's disorder is diagnosed by the presence of social interaction impairments and repetitive and restricted interests as described in Autistic Disorder above. There is usually no significant language delay, however, subtle impairments in the social use of language are present and often disabling, eg by leading to teasing and social isolation. This disorder is more common in males (13:1) but may be underdiagnosed in females.

- **Typical characteristics of Asperger's disorder are:**

- o Normal or borderline intellectual ability.
- o Clumsiness.

- o Concrete, pedantic speech.
- o Lack of common sense.
- o Normal or even precocious speech development.
- o Better verbal than non-verbal skills on psychological assessments.
- o Intolerance of change.
- o Anxiety.

1.2.3 Atypical Autism

This diagnostic grouping is used when core autistic behaviours are present but the criteria for autistic disorder are not fully met. Management follows the same principles as for Autistic Disorder.

- **Diagnosis**

The accurate diagnosis of autism spectrum disorder is important for a number of reasons:

- o Parents often suspect that their child is "different", through assessment; they can understand their child's needs and strengths better.
- o Early intervention and appropriate educational programs can be planned.
- o Access to specialist and other support services can be facilitated.
- o Undiagnosed adults without significant intellectual disability often find a diagnostic assessment is helpful in explaining their experience and enabling access to supports.
- o All support persons are able to better understand the management process

- **Influences on Behavior**

- o Common difficulties encountered by parents include the management of:
- o Difficult behaviour eg obsessions, tantrums and aggression.
- o Communication problems eg repetitive speech.
- o Disturbed sleep and fussy eating.
- o Socially inappropriate behaviour and social isolation or teasing and bullying.

Referral to appropriate services may be necessary. Experience has shown that early, intensive intervention produces the better outcomes for children with autism.

Be Aware: Deterioration in behaviour may be due to causes other than autism such as a physical problem (eg middle ear infection), epilepsy, family or school stress or changes in routine.

- **Psychiatric Disorders**

Emotional and behavioural problems are common in people with autism spectrum disorders, but the exact prevalence is not known. Depression, mood disorders, anxiety disorders and psychosis may emerge during adolescence and continue into adulthood, requiring psychotropic medication and psychological treatment.

- **Medication**

No medications are available to treat the core symptoms of autism spectrum disorders. Medication can occasionally be used for symptomatic control if behavioural treatment and environmental modification have failed, and if the benefits of medicating outweigh the costs.

1.3 Types of Autism Spectrum Disorders

1.3.1 Autistic Disorder (Classic Autism)

Autistic Disorder (sometimes called “classical autism”) is the most common condition in a group of developmental disorders known as the autism spectrum disorders (ASD). Classic autism is characterized by impaired social interaction, problems with verbal and nonverbal communication, and unusual, repetitive, or severely limited activities and interests (National Institute of Neurological Disorders and Stroke, 2005a).

There are three distinctive behaviors that characterize children with classic autism. These children have difficulties with social interaction, problems with verbal and nonverbal communication, and repetitive behaviors or narrow, obsessive interests. The behaviors can range in impact from mild to disabling. Other common symptoms of classic autism include (Westling & Fox, 2004; Deutsch-Smith, 2004; Heward, 2006; Gargiulo, 2004):

- o Difficulty in expressing needs; uses gestures or pointing instead of words
- o Difficulty in relating to and mixing in with others
- o Does not respond to normal teaching methods
- o Does not respond verbal cues

- o Exhibits minimal or no eye contact
- o Exhibits over-sensitivity or under-sensitivity to pain
- o Exhibits physical over-activity or extreme under-activity
- o Has difficulty expressing and receiving physical affection
- o Has no real fears of danger
- o Maintains inappropriate attachments to objects
- o Motor skills are uneven
- o Often exhibits tantrums
- o Prefers to be alone; aloof manner
- o Repetition of words or phrases in place of normal, responsive language
- o Resistance to change
- o Spins objects
- o Sustained odd play

According to the Autism Society of America (2005), there is no known single cause for autistic disorder, but it is generally accepted that it is caused by abnormalities in brain structure or function. Brain scans show differences in the shape and structure of the brain in autistic versus non-autistic children.

Researchers are investigating a number of theories, including the link between heredity, genetics and medical problems. In many families, there appears to be a pattern of autism or related disabilities, further supporting a genetic basis to the disorder (National Institute of Neurological Disorders and Stroke, 2005a).

While no one gene has been identified as causing autism, researchers are searching for irregular segments of genetic code that autistic children may have inherited. It also appears that some children are born with a susceptibility to autism, but researchers have not yet identified a single "trigger" that causes autism to develop.

1.4 Childhood Disintegrative Disorder

Childhood disintegrative disorder is a condition occurring in 3- and 4-year-olds who have developed normally to age 2. Over several months, a child with this disorder will deteriorate in

intellectual, social, and language functioning from previously normal behavior (U.S. National Library of Medicine 2004b).

Childhood disintegrative disorder develops in children who have previously seemed perfectly normal. Typically language, interest in the social environment, and often toileting and self-care abilities are lost, and there may be a general loss of interest in the environment. The child usually comes to look very 'autistic', i.e., the clinical presentation (but not the history) is then typical of a child with ASD (Yale Developmental Disabilities Clinic, 2006). An affected child shows a loss of communication skills, has regression in nonverbal behaviors, and significant loss of previously-acquired skills. The condition is very similar to classic autism.

Symptoms of Childhood Disintegrative Disorder may include (U.S. National Library of Medicine 2004b):

- o Loss of social skills
- o Loss of bowel and bladder control
- o Loss of expressive or receptive language
- o Loss of motor skills
- o Lack of play
- o Failure to develop peer relationships
- o Impairment in nonverbal behaviors
- o Delay or lack of spoken language
- o Inability to start or sustain a conversation

Childhood Disintegrative Disorder must be differentiated from both childhood schizophrenia and pervasive developmental disorder (PDD). The most important signs of childhood disintegrative disorder are loss of developmental milestones (U.S. National Library of Medicine, 2004b). The child tends to have normal development through age 3 to 4 and then over a few months undergoes a gradual loss of previously established abilities listed above in Diagnostic Symptoms (e.g., language, motor, or social skills). The cause is unknown, but it has been linked to neurological problems (Yale Developmental Disabilities Clinic, 2006).

1.5 Rett Syndrome

Rett Syndrome (RS) is a neurological disorder seen almost exclusively in females, and found in a variety of racial and ethnic groups worldwide (International Rett Syndrome Association, 2005).

An inability to perform motor functions is perhaps the most severely disabling feature of Rett syndrome, interfering with every body movement, including eye gaze and speech. Other diagnostic symptoms of Rett Syndrome include (International Rett Syndrome Association, 2005; National Institute on Neurological Disorders and Stroke, 2005c):

- o An early period of apparently normal or near normal development until 6–18 months of life.
- o A period of temporary stagnation or regression follows during which the child loses communication skills and purposeful use of the hands.
- o stereotyped hand movements
- o gait disturbances
- o slowing of the rate of head growth become apparent
- o Seizures
- o Disorganized breathing patterns which occur when awake
- o Compulsive hand movements such as wringing and washing follow the loss of functional use of the hands

1.6 Pervasive Developmental Disorder, Not Otherwise Specified (PDD-NOS)

Pervasive Developmental Disorder, Not Otherwise Specified (PDD-NOS) is a developmental-neurologic spectrum disorder, occurs in 10 to 12 of every 10,000 children. Children with PDD-NOS either (a) do not fully meet the criteria of symptoms clinicians use to diagnose any of the four specific types of PDD above, and/or (b) do not have the degree of impairment described in any of the above four PDD specific types. PDD-NOS will present similarly to the kids who have ASD (some people argue that these conditions should be combined as one), but will have a lesser degree of a severe impairment. These kids are more likely to be verbal and have some degree of verbal or non-verbal effective communication, yet they must have the autistic features and a severe impairment in social interaction, communication, or repetitive stereotype behavior. This term is reserved for children with a severe impairment who do not fully qualify for any other autistic diagnosis, due to age of onset or combination of autistic features.

1.7 Asperger Syndrome

In Asperger syndrome, a young child experiences impaired social interactions and develops limited repetitive patterns of behavior (Nemours Foundation, 2005). Motor milestones may be delayed and clumsiness is often observed (U.S. National Library of Medicine, 2004a). However, people with Asperger syndrome usually have fewer problems with language than those with ASD, often speaking fluently, though their words can sometimes sound formal or stilted. People with Asperger syndrome do not usually have the accompanying learning disabilities associated with ASD; in fact, people with Asperger syndrome are often of average or above average intelligence (National Autistic Society, 2005).

School-age children with Asperger syndrome exhibit a range of characteristics, with varying degrees of severity. Diagnostic Symptoms of Asperger Disorder includes (Hallahan & Kauffman, 2006; Turnbull, Turnbull, & Wehmeyer, 2006; Friend, 2005; Westling & Fox, 2004;

National Institute of Neurological Disorders and Stroke, 2005a; U.S. National Library of Medicine, 2004a; Mayo Clinic, 2004):

1.8 Social skills

- Has difficulty making friends
- Engages in one-sided, long-winded conversations, without noticing if the listener is listening or trying to change the subject
- Displays unusual nonverbal communication, such as lack of eye contact, few facial expressions, or awkward body postures and gestures
- Doesn't empathize with or seem sensitive to others' feelings and has a hard time "reading" other people or understanding humor
- Doesn't understand the give-and-take of conversation or engage in "small talk"
- Seems egocentric or self-absorbed
- May speak in a voice that is monotone, rigid, jerky or unusually fast
- Can be extremely literal or have difficulty understanding the nuances of language, despite having a good vocabulary

1.9 Behavior

- Shows an intense obsession with one or two specific, narrow subjects, such as baseball statistics, train schedules, weather or snakes
- Likes repetitive routines or rituals
- May memorize information and facts easily, especially information related to a topic of interest
- May have clumsy, uncoordinated movements, an odd posture or a rigid gait
- May perform repetitive movements, such as hand or finger flapping
- May engage in violent outbursts, self-injurious behaviors, tantrums or meltdowns
- May be hypersensitive to sensory stimulation, such as light, sound and texture

Asperger Syndrome is a neurobiological disorder named for a Viennese physician, Hans Asperger, who in 1944 published a paper which described a pattern of behaviors in several young boys who had normal intelligence and language development, but who also exhibited autistic-like behaviors and marked deficiencies in social and communication skills (National Institute of Neurological Disorders and Stroke, 2005a).

Overall, people with AS are quite capable of functioning in everyday life, but they tend to be somewhat socially immature and may be seen by others as odd or eccentric (Nemours Foundation, 2005). They have higher intelligence and communication skills than those with classic or more severe forms of ASD, but they display most, if not all of the other characteristics of ASD, with their primary difficulties manifested in poor social interactions (Hallahan & Kauffman, 2006).

1.10 Autism spectrum disorders in adolescence and early adulthood: Characteristics and issues

Jason McElwain shot his way into history at Greece- Athena High School one cold February day in 2006 by scoring six three-point shots and one two-point shot in the waning minutes of the last basketball game of his senior year [15]. What made his 20-point feat particularly stunning was that Jason was the team's manager, not a regular roster player. His coach put him on the roster and let him play at the end of the game as a thank you for his service to the team. While his shooting prowess was notable, Jason made the national news and met the President of the United States because he has autism. The juxtaposition of his accomplishment in the face of his

disability made his story newsworthy. In fact, some saw his interview on CNN or saw pictures of him meeting LeBron James and other celebrities and thought that perhaps Jason was misdiagnosed, or that Jason was unique among individuals with autism. Jason did not appear to flap, he seemed to make occasional eye contact, and he was able to answer questions from the reporter who interviewed him [3]. In short, Jason did not display those classic behaviors that have become emblematic of a young child with autism. Yet those who understand autism in adolescence and adulthood saw the signs. While Jason answered the reporter's questions, his vocal 'prosody' was different. When he described the game, he was not as enthusiastic as another senior might have been. He seemed more polite, maybe even pedantic, for a young man his age. Today, Jason works 14 hours a week at a local grocery store stocking the bakery shelves with bread and lives at home with his parents. He does not receive benefits at his job and may not be working to his full potential. He is also a volunteer coach for the junior varsity basketball team at his alma mater [15]. The promise that he showed in those final minutes of that basketball game seem lost in the underemployment he is experiencing now. In order to help Jason and others like him achieve their full potential in adulthood, we must understand the characteristics of autism spectrum disorders (ASD) and the issues in providing rehabilitation services to adolescents and young adults with ASD that lead to improvement in employment and community living outcomes.

Jason represents the beginning of a wave of students With ASD working their way through school. During his time in school, the recorded prevalence of ASD's has increased significantly. The most recent figure places the prevalence of ASD in school aged children at 1% of all children [4]. Early intervention providers and elementary school teachers have been most affected by the increased prevalence [12]. Thus, most of the research data describes characteristics and interventions for very young and elementary aged children [7]. Despite this fact, ASD is a collection of lifelong disabilities that continues to impact affected individuals in all areas of living, yet there has been a paucity of literature describing these disorders in late adolescence and adulthood [9]. The purpose of this article is to describe how the symptoms of autism, Aspergers disorder, and Pervasive Developmental Disorder-Not Otherwise Specified (PDD-NOS) change in late adolescence and early adulthood and to present three case studies illuminating those characteristics. Finally, the authors will discuss the impact of the disorder on transition aged youth and young adults and identify issues for future research.

1.11 The characteristics of autism in adolescence and early adulthood

Autism Spectrum Disorders is the diagnostic category that includes autism, Aspergers disorder, and PDD-NOS (APA, 2001). These three represent a spectrum from autism where most individuals are affected by many more communication challenges and are more likely to have an intellectual disability to Aspergers disorder where individuals acquire language in toddlerhood, and tend to have higher measured IQ's. PDD-NOS represents those individuals on either end of the spectrum who display many but not all required behavioral symptoms of either Aspergers disorder or autism. Table 1 presents a comparison of the three disorders across symptom patterns. These three disorders have behavioral characteristics in three areas; communication deficits, social skill deficits, and restricted, repetitive, and stereotyped patterns of activities, interests, behavior [9]. In addition to these primary characteristics, many individuals with ASD experience secondary characteristics including displaying sensory sensitivities, behavior challenges, and additional mental health diagnoses such as anxiety disorders and depression [2, 9, 16].

The primary characteristics of autism are most salient in young children [12]. That is, those characteristics most associated with autism including difficulty acquiring language, unusual motor movements, and avoidance of interacting with others, are most problematic between the ages of 3 to 6. With education these characteristics are mutable and the individual's communication and social skills improve, while their stereotypic patterns of responding subside [13]. Therefore, by adolescence and early adulthood, the symptoms that lead to diagnosis usually have changed [14]. Nevertheless, there are usually enough symptoms present to validate the diagnosis of ASD in adolescence and early adulthood [5, 6].

Gilchrist et al. found that, by adolescence, individuals with Asperger syndrome show similar behavioral characteristics to individuals with high functioning autism in adolescence [5]. They found that delays in language development for individuals with high functioning autism were ameliorated by adolescence; leading to the finding that individuals with Asperger disorder were similar to individuals with High Functioning Autism. Seltzer et al. suggest that, while there is an abatement of symptoms, the developmental trajectory for individuals with ASD is splintered with improvement in some behaviors that define autism. They also noted that some individuals with ASD experience periods of regression in the areas of behavioral challenges and insistence

of sameness. Finally, they note that some individuals experience a worsening of symptoms [13]. Thus, the developmental trajectory for individuals with ASD through adolescence is neither uniform nor linearly ascending. They report findings indicating improvement in communication, but continued impairment, particularly in social communication, persists into adulthood. Seltzer et al. also note that, as they age, individuals with ASD adjust to change and decrease the presence of stereotyped behavior, but impairments in social interaction remain into adolescence and adulthood [13]. On the point of social interaction, McGovern and Sigman note that individuals with ASD who were more socially engaged with their peers without disabilities gained more adaptive skills than their peers with ASD who were not socially engaged with non-disabled peers. They conclude that “engagement with peers improves the social skills of children with autism as is true for typically developing children” ([5], p. 407.) Further, they note that as children with ASD enter adolescence, they show increased social interest.

Thus, the literature indicates that through adolescence and early adulthood, most individuals with ASD continue to display the behavioral characteristics of ASD. Additionally, many individuals with ASD show improvement in communication and decreases in challenges related to stereotyped movements and patterns of behavior. Finally, while many individuals with ASD continue to struggle with social interaction, those who have had engagement with peers without disabilities show more improvement in social skills. Table 2 shows a listing of symptom patterns and how they change through adolescence into early adulthood. Despite these general findings, individuals may respond differently. There are some individuals who demonstrate noteworthy improvement to the point where they no longer display the symptoms of the disorder. Likewise there are some individuals who continue to display significant debilitation by the disorder. Nevertheless, the majority of individuals on the autism spectrum display a varied pattern of improvement into adolescence and early adulthood. The next section of this article will present three case studies to further illuminate the disorder in adolescence and early adulthood.

1.12 Three case studies of adolescents and young adults with ASD

Mary Ann

Mary Ann is 20 years old. She had a diagnosis of PDD-NOS. While she met all of her motor milestones on time, she did not speak her first word until she was 5 years old. She did not

compensate for her lack of spoken language with gestures or facial expressions. Her parents described her as “in a world of her own”. Once she began to talk, though, she also became more interested in interaction with others, although her attempts to play were uncoordinated and awkward. She is currently performing slightly below grade level in school. Her greatest challenge in academic work is comprehension of written material. On her individualized education program (IEP), she is receiving services under the category “other health impaired”.

As a young adult, Mary Ann is unique in her classroom. She is the only girl out of 6 students in a senior year transition program for youth with ASD modeled after Project Search at Cincinnati Children’s Hospital in Ohio [8]. Upon first meeting, Mary Ann seems overly polite and engaging. After a few moments, however, a few issues stand out. First, Mary Ann sounds different when she speaks. Her vocal prosody is at times pedantic and monotone, and other times sing song in nature. When she laughs, her laugh is very giggly and louder than expected. While she makes eye contact, her eye contact is fleeting and uncoordinated with her conversation. She frequently looks through rather than at persons with whom she speaks. Finally, she is dependent on her conversation partner to maintain conversations. She responds to questions but does not initiate topics or ask questions that would extend a conversation. Thus, she displays communication differences that indicate the possible presence of an ASD, but they are not currently excessive or severe.

Her social skills are a little different as well. Her parents report that she has only one friend that is her age. Most of her friends are either younger or older than she is. She has difficulty relating to friends her own age. Otherwise, she is able to share enjoyment with others appropriate to her age. Because of her social differences, she displays behaviors that are further evidence of a possible ASD. In the final category of symptoms for autism, Mary Ann displays few restricted or stereotypic behaviors. While she does not like change, she is able to adjust to changes in her routines with relative ease. She does not engage in any motor stereo types. She also does not display any excessive interests that frequently characterize Aspergers disorder. In short, she displays symptoms consistent with an ASD in the areas of communication and social skill deficits but not in the area of stereotypic and repetitive patterns of behavior, interests, or activities. Thus, the best diagnostic match is PDDNOS. Regardless of the specific diagnostic

category, it is perhaps more important to discuss Mary Ann's future goals and how to improve the likelihood that she will achieve those goals.

When Mary Ann graduates, she wants a job. She has not defined her particular career interest, but is clear that she doesn't want to go to college. While attending high school, she took classes at the technical center in printing and print shop. She had difficulty learning all of the computer programs and struggled with the social aspects of the job. In short, she has limited knowledge of her strengths and skills, and an even more limited knowledge of work and adulthood. Her parents worry about her future and are anxious to find services and supports that will help her become successfully employed.

While Mary Ann is not quite ready for her transition out of school and into work, the clock is ticking and time is running out. Thus, her participation in the specially designed "Project SEARCH" modeled classroom seems particularly important. In this classroom, she will rotate through three internships that, together, will result in her having marketable skills toward a job in patient care, materials management, or specialized room and cart stocking. The program is located in the hospital and will result in repeated 'trials' where she will practice employment skills and social behaviors as well as experience working. She will develop a portfolio where she will list her skills, strengths and further identify desired future careers. She will interact regularly with job coaches and adult services case managers. Finally, she will network with potential employers through her internships at the hospital. These types of experiences will increase the chances that Mary Ann will achieve her graduation goal. Mary Ann's case raises a number of questions. She has not developed an understanding of career development and therefore will struggle with making good decisions regarding her job and career paths. Thus she will likely have difficulty displaying the behavioral components of self determination. She is typical of many individuals with ASD. While she does not display many of the most debilitating challenges associated with the disorders, she also does not initiate requests or see herself as an actor in her own life. Thus, the question raised by Mary Ann's case is what instruction and interventions would result in increasing her career awareness and self determination? These interventions would likely be implemented across her entire school career and would change the priorities from becoming compliant to becoming self determined [9].

Patrick

Patrick was diagnosed with autism at the age of four. He developed some language by 18 months, but lost most of his words by two years old. He was severely withdrawn and avoided interaction with peers and adults alike. He displayed temper tantrums when ever his routine was changed unexpectedly or when he was rushed through a favored task. He lined all of his toys up in his bedroom and never played with them. He spent most of his childhood leisure time pacing and flapping his hands. Thus, he displayed all of the classic signs of autism before he turned three. Although his diagnosis at the age of four was delayed by today's standards, at the time his diagnosis in the early 1990's was as early as most.

Patrick has made progress in communication and behavior since that time. Specifically, Patrick uses words to communicate, although he continues to display communication deficits. He can make requests and respond to direct questions. He has difficulty engaging in conversations and frequently uses phrases in a rote fashion. For example, he asks to use the restroom in the same way, even when he is not required to ask. He also uses echolalia phrases to express his emotions. Sometimes it is difficult for the unfamiliar person to understand those phrases. For example, when he repeatedly says "Your Fired!" that generally means that he is frustrated or unhappy with a person or situation. That phrase usually precedes darting.

He will occasionally dart when his schedule changes, when he is rushed to stop a desired task, or when he encounters an unexpected problem. There were two episodes of this behavior across four months. Nevertheless, his team is working on developing a positive behavior support plan to teach him how to accept changes in his routine and to teach him a "calm down" strategy when he becomes frustrated or upset. Patrick prefers solitary activities such as playing computer games, looking at preferred books about NASCAR racing, or playing on his Nintendo DS. Additionally, Patrick still struggles with minor changes in his schedule or routine. For example, he became upset when he missed completing a page in his work book because he was absent from school. Despite these challenges and deficits, Patrick has many strengths. He is an excellent typist and is able to transcribe notes and text with a high degree of accuracy. His reading ability is measured at the 4th grade level, while his comprehension of written material is somewhat lower. He is also able to complete basic mathematical operations and is skilled at using a calculator and an adding machine.

Patrick is also in Mary Ann's classroom at the local hospital. His internship rotations have been in materials management and nutrition. In materials management, he counted inventory in floor supply closets and uploaded that data to a central computer system. He was skilled at the rote aspects of the job, but struggled with activities that required decision making. At this particular internship, there were tags that required rote counting and tags that required a decision based on an estimation of the number of items used in a day. Patrick was fast and accurate when inputting rote counts into the computer system, but was not skilled at completing the estimations required in each closet.

To address this aspect of the task the team worked to make this decision algorithmic. They took pictures of the bins at full, three quarters full, half full, one quarter full and empty. Then they affixed each picture to a page in a 'picture dictionary'. Next to each picture of the bin, they put the number that Patrick should enter on the inventory counter he used. For example, if, when a bin was empty Patrick should input the number 500 to order 500 units of the supply in the bin, then the dictionary would have a picture of an empty bin next to an equal sign "=" and the number 500 next to it. That way, instead of making decisions, Patrick just had to match the bin to the picture in his dictionary and input the number listed. Patrick performed best with visual supports like this including a visual schedule and written instructions. His team anticipates using more visual supports in all aspects of Patrick's internships.

Patrick's case review also raises questions related to supporting individuals with ASD in transition and at work. Specifically, there are few studies that describe positive behavior supports in work place settings [10]. What are the types of interventions that are most effective in the context of work? What types of supports are necessary to implement these interventions? Does implementation of positive behavior supports in the work place change because of the unique nature of the environment? How does a person like Patrick access those supports when there are few funding options for behavioral services for adults? Access to positive behavior support is critical to the successful employment of individuals like Patrick who display many work strengths, but also have intensive support needs. There is a critical need for research in this area to assure the successful transition of youth and young adults with ASD.

Jackson

Jackson is 19 years old. Throughout his life, he has had numerous medical complications that resulted in him receiving an IEP to address his learning needs related to these complications. For example, he has a heart defect and scoliosis that resulted in multiple surgeries. In addition to a hearing loss, he also had learning, behavioral, and social skill problems that were attributed to anoxia due to heart complications in infancy. Over time though, it became clear that the medical complications and subsequent learning problems did not explain all of Jackson's needs. In fact, at the age of 17, Jackson's mother requested a re-evaluation due to on-going social and behavioral challenges. That re-evaluation resulted in a diagnosis of Aspergers Disorder. This diagnosis seemed to provide closer match and helped account for the behavioral and social difficulties that Jackson displays. Jackson's story is common among individuals his age with Aspergers disorder. That he received this diagnosis in adolescence is not unusual. After all, the disorder was not well understood in his youth and teams failed to consider other explanations for his learning problems. It was only when his mother requested further evaluation that this diagnosis was considered. This is an issue of which transition and adult services should be aware. That is some individuals with Aspergers disorder were not diagnosed as youngsters because their early diagnostic evaluations predated full understanding of the disorder [11]. Thus, transition and adult services teams should be alert to the possibility of under diagnosis or misdiagnosis of Asperger disorder in youth who's learning, social, and behavioral characteristics seem to match this profile. Like Mary Ann, Jackson does not know what he wants to do upon graduation from high school. While he performs adequately in school, he does not want to go to college. He would like to work, but does not have a clear idea of a career that he likes or a job that would match his skills and strengths. He has a penchant for data entry and but has never tried work in a data entry position. Unlike Mary Ann and Patrick, Jackson has worked menial jobs in fast food, and a large retail store. These experiences did not result in advanced insight into his future career.

While working at these jobs, Jackson also appeared vulnerable to inappropriate suggestions from coworkers and acquaintances who attempted to take advantage of his position to acquire free food and goods. In all of his jobs, he had to be taught to say no to peers who sought to take advantage of him. Other social behaviors included misunderstanding verbal directions and failing to ask for help or clarification. His mother reports, "he doesn't seem to know what he doesn't know!" Finally, Jackson can sometimes exasperate his supervisors. While he doesn't argue often,

if he perceives he is right and his supervisor is incorrect, he will insist that the supervisor correct himself, even on small points of disagreement.

Jackson also attends school in the hospital based transition program. He is currently in his second internship rotation. His first internship capitalized on his visual organization skills and his data entry abilities. He was assigned to a cardiac care unit where he stocked carts, entered patient and inventory data into an electronic record, and organized patient charts at the nurses' station. His success in this environment was likely due to the nature of the job. That is, he was able to capitalize on his strengths and avoid public interaction where peers might take advantage of him. The next step in his career development is to have Jackson evaluate his performance and identify why he was successful in this environment. This would be the beginning of developing a portfolio of his skills.

Like Mary Ann and Patrick, Jackson presents unique challenges for his transition team. He also requires intensive supports. He especially requires in depth social skill instruction. His vulnerability is of particular concern. Because he is vulnerable to suggestion and may not be able to discriminate between friends and acquaintances who would do him harm, he requires instruction combined with monitoring. Thus, his support needs are different, but equally intensive. Again, the issues for Jackson require further research to assist the service community in understanding the types of supports that result in successful independent or supported employment and independent living.

1.13 Summary

The purpose of this article was to explore the characteristics of ASD in adolescence and early adulthood and to present three cases to illuminate the spectrum. All three case studies pose important issues related to transition to adulthood and employment. Specifically, all three individuals require higher intensity services than are typically offered to transition aged youth. Also, they require intensive services and instruction in communication and social skills. Research in the area of transition services is becoming critical. As the first wave approaches graduation, there are at least four times as many students identified with ASD who are in elementary and middle school. In order to change the outcomes for this group of individuals, we have to develop scientifically-based practices now.

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Unit 2

Tools and Areas of Assessment

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2.1 INTRODUCTION

Assessment is a necessary step in the overall program planning for students with Autism Spectrum Disorders (ASD). However, students with ASD are likely to present unique challenges and issues during formal evaluation, which may impact the success of the evaluation process and the utility of the assessment data (Klin, Saulnier, Tsatsanis & Volkmar, 2005). Since instructional programming is based on information regarding a student's present levels of performance and identification of priority educational needs (PENS), accurate and useful assessment data are critical to programming success (see Table 2.1)

This chapter will provide a summary of the symptoms and characteristics of ASD that will impact the assessment process and describe the components of a core "best practices" assessment of individuals with ASD. Formal and informal assessment procedures and instruments for the evaluation of ASD will be described. Finally, this chapter will close with a description of recommendations for necessary areas of intervention and curricula.

2.2 SYMPTOM PRESENTATION IN ASD

Autism Spectrum Disorders are included under a class of disorders referred to as Pervasive Developmental Disorders (PDDs) within the Diagnostic and Statistical Manual of Mental Disorders–Fourth Edition–TR (DSM–IV–TR; APA, 2000). The *DSM–IV–TR* lists five specific

PDDs: Autistic Disorder, Asperger's Disorder, Pervasive Developmental Disorder–Not Otherwise Specified (PDD-NOS), Rett Disorder, and Childhood Disintegrative Disorder. These disorders are diagnosed based on a set of criteria provided in the *DSM–IV* and include impairments in three general domains of functioning:

- Delayed and atypical development in social reciprocity and relatedness
- Delayed and atypical development of language, communication, and play skills
- The presence of repetitive and restricted patterns of behaviors, interests, and activities

In addition, individuals with ASD may also exhibit associated disturbances in areas such as cognition and information processing, attention, executive functioning, and sensory processing.

These core ASD deficits and associated impairments will ultimately have a direct effect on the assessment process.

2.3 Social Impairments

Due to deficits in the social domain, clinicians will often find that it is more difficult to establish rapport, or a comfortable cooperative relationship, with the student during the evaluation process. Students with ASD may not respond to the usual methods of establishing rapport, including the use of informal conversation and the use of social praise to motivate the child to cooperate with the testing procedures. In addition, children with ASD (particularly younger children) may be unfamiliar with basic social behaviors that will impact the evaluation process and the child's performance, such as sitting quietly at a table, taking turns, or returning test materials after an item is completed. Finally, impairments in the use and understanding of nonverbal communication (such as eye contact and gesture use) may lead the child to miss important components of task instructions and/or respond in line with test specifications (such as pointing to the correct answer).

2.4 Language/Communication Impairments

Language and communication deficits also present major challenges to conducting an effective evaluation. The majority of children with ASD have receptive language difficulties. This is true even for "higher-functioning" individuals with ASD who have cognitive and expressive language abilities that are within the expected range for their age (Filipek et al., 1999). These difficulties may impact the ability of the child to adequately understand orally presented instructions, which will certainly impact task performance. For some children, there is also a great deal of scatter in their receptive language skills, with understanding of more difficult words and concepts in the face of unfamiliarity with easier vocabulary. This discrepancy often results in inconsistent responding patterns that may be interpreted as noncompliance by the examiner. Such receptive language problems, and attempts by the examiner to "gain compliance," increase the likelihood of behavioral problems during the evaluation, which stem from language expectations which are too high for the child's abilities.

Further, even for those individuals with adequate receptive language skills, deficits in pragmatics (use of language in a social context) may still be apparent. Such difficulties may include a tendency to be overly literal and concrete in interpreting instructions and problems with interpreting nonverbal behaviors and turn-taking skills. The effect of these pragmatic language difficulties on the evaluation process, however, may be more difficult to initially notice. As an example, a student who was asked to produce a written essay on an academic achievement test repeatedly refused to begin the task, stating that he didn't "know how to." At this point, it would be common for the examiner to move on to the next item and/or task with the assumption that the student lacked the necessary skills to correctly respond to the test item. Upon follow-up questioning by the examiner, it became apparent that the student did, in fact, have experience writing essays. However, he knew how to write an essay only in French. Apparently, the student had written essays only in French class and was unable to spontaneously apply that experience to the current testing situation. The student readily began the task when the examiner indicated to "write the essay just like you would in French, but use English instead."

Finally, expressive language difficulties also affect task performance in the testing environment. Many cognitive assessment measures assess verbal as well as nonverbal problem-solving skills. For students who are nonverbal or minimally verbal, requirements of verbal subtests will likely be beyond their language abilities. Students with better language abilities are likely to be able to respond to such tasks, but may still encounter problems. Difficulties may include word-finding problems and difficulty formulating and generating answers and/or the use of unusual or idiosyncratic language which may impact the examiner's understanding of their responses. Presence of Restricted, Repetitive, and Stereotyped Interests, Behaviors, and Activities The behavioral challenges inherent to ASD also create unique problems during the testing session.

Students with ASD often exhibit strong interest in idiosyncratic objects and events that will affect manipulation of the testing materials (such as flapping materials, rubbing surfaces, and/or visual inspection of items). Thus, it may be quite difficult to obtain cooperation from the child in using materials as intended, retrieving materials from the child to demonstrate the task, and/or removing the materials in order to present the next task. In addition, students may be overly reliant on routines and often have extreme reactions to changes. Interrupting or redirecting the

child in such instances may trigger a variety of behavioral responses that can interrupt assessment procedures.

2.5 THE SPECTRUM NATURE OF ASD

Another important consideration for assessment is the spectrum nature of ASD. Autism is referred to as a “spectrum” condition, meaning that the symptoms of ASD fall along a spectrum, or continuum. As the word “spectrum” implies, ASD affects each individual differently and to varying degrees. Figure 2.1 presents the characteristics of ASD that may impact the assessment process. Symptoms can occur in any combination and can range from the very mild to the severe. In addition, individuals may differ significantly in their overall language and cognitive abilities. In the area of language development, children may range from completely nonverbal to highly verbose. Further, individuals also vary along a continuum with respect to cognitive functioning, from the mental retardation/intellectual disability range to the gifted range. However, there is some evidence that there may be changes in the number of individuals who fall within the intellectually disabled range.

While the prevalence of comorbid mental retardation/intellectual disability in ASD was estimated to be approximately 70% to 75% of individuals during the previous decade (Filipek, et al., 1999; National Research Council, NRC, 2001), more recent prevalence studies now report rates of comorbid intellectual disability of just under 50% (Yeargin-Allsopp, Rice, Karapurkar, Doernberg, Boyle, & Murphy, 2003).

Thus, the spectrum nature of ASD results in a great degree of heterogeneity across children which is likely to affect the process of conducting the evaluation; there will be no “one size fits all” approach to testing strategies and instrument selection, requiring a great deal of knowledge and flexibility on the part of the examiner.

BOX 2.1 Diversity Notes

A commonly accepted premise today is that autism “knows no racial, ethnic, or social boundaries” according to the Autism Society of America. Unfortunately, the accuracy of this premise is called into question as researchers (and clinicians alike) have tended to pay little attention to racial and/or cultural differences in autism. Dyches and colleagues present a

provocative paper raising a number of questions for further examination in the field. Specific questions for consideration included:

- Are there differences in the prevalence of autism across race/ethnicity?
- Are there factors that may influence children from different races/cultures who are identified as having an ASD (including cultural differences in how symptoms are viewed and defined, stigmatization of disability status, potential bias from clinicians in diagnosis, etc.)?
- Are there differences in family adaptation to the autism diagnosis?

Dyches and colleagues suggest that the relative paucity of research into the specific challenges faced by families from diverse backgrounds should be a concern to the professionals who work with these families. Because of the paucity of research in this area, what we know about the characteristics of best practice in assessment and intervention for ASDs may not apply to individuals and families of different cultural, racial, and/or ethnic backgrounds.

2.6 Learning Challenges Associated with ASD

Students with ASD often exhibit uneven learning profiles marked by “scatter” in skill development. Such students may function at or above age level in some areas, but well below age level in others. This pattern also results in a scattered profile of results on formal testing measures, as well as inconsistent performance within individual subtests that will affect not only test results but instructional programming as well.

Coupled with inconsistent skill acquisition, students with ASD also exhibit inconsistencies in responding, even for skills that have been mastered. Such difficulties are frequently referred to as problems with generalization. Despite having acquired given skills, individuals with ASD often have difficulties “showing what they know” in the evaluation setting and fail to demonstrate skills that they can successfully perform in the home and/or at school. Generalization difficulties also can manifest as an inability to demonstrate a skill under different conditions than those in which the skill was learned (e.g., with different materials and/or verbal instructions). Deficits in skill maintenance are also characteristic of children with ASD; children may “lose” skills if they are not consistently practiced and/or used in the child’s daily life. Again, these difficulties have implications for the testing setting, as parents often proclaim that the child possesses skills that

were not exhibited during the evaluation session. This is often an accurate statement; however, difficulties with generalization and maintenance of skills have direct relation to instructional programming. Skills that have been “mastered” but cannot be demonstrated across people, settings, and materials should take first priority as instructional goals and objectives.

- Difficulty establishing rapport with examiner
- Lack of motivation to please examiner
- Limited flexibility and/or overly reliant on nonfunctional routines or rituals
- Difficulty understanding and following instructions and generating verbal responses due to language deficits
- Unique learning profiles
- Stimulus over selectivity (such as attending to irrelevant stimuli, difficulty switching between tasks, and a desire to use materials in unique or unusual ways)
- Attending to materials, and persisting in completing tasks
- Inconsistent responding across items on a task
- Inability to demonstrate skills that child can do at home (generalization problems)
- Interfering and challenging behaviors Challenges may also be displayed in areas associated with the core deficits of ASD.

For example, there may be significant problems with orienting to the examiner and task materials and distractibility. Further, attention and persistence may vary significantly across tasks with reduced motivation for non preferred activities. Individuals may also be hypersensitive to sounds and visual stimuli in the environment, display self-stimulatory behaviors, and become preoccupied by oral exploration of items.

Difficulties with tasks requiring sequential steps or that have multiple stimuli and stimulus over selectivity may also be observed. Over selectivity refers to the tendency of individuals with ASD to focus on a restricted range of available environmental cues, such as focusing on one feature of

an object while ignoring other equally important features. For example, a student may respond to extraneous and/or irrelevant details (such as the model of the car in a picture) and fail to pay attention to the more salient and important aspects of stimuli on which the task depends (such as object identification, color identification, and/or object function). Over selectivity will have a significant impact on behavior during both assessment and instructional tasks.

2.7 THE ASSESSMENT PROCESS

Assessment is more than the administration and interpretation of test results. In contrast to “testing,” assessment is much broader, being conceptualized as a systematic process for gathering information (or data) for use in making diagnostic, legal, and/or educationally relevant decisions. Data collected during the assessment process generally falls within two main types of procedures: formal and informal.

Formal Assessment Approaches

Simply put, formal assessments are the conventional methods of testing that the majority of people are familiar with, such as tests like the SAT and their other aptitude measures. Formal assessments are generally used to compare a student’s performance in some domain with that of his or her peers. Norm-referenced tests are the most commonly used formal assessment procedure; such tests have specific standards that outline the basis of comparison and administration guidelines, resulting in them also being referred to as **standardized tests**.

The standard of comparison for **norm-referenced tests** is a representative “student body” comprising individuals of the same age, grade-level, gender and/or disability category; comparing one student’s performance to others in a similar category or categories. The development of norm based standardized tests requires that a representative population of students (the standardization sample) is assessed in order to derive the norms to be used for comparison. While this is a process that is more complex than described herein, it involves ensuring that a large enough number of students are assessed across age and grade levels, racial and ethnic categories, socioeconomic status, and geographical region, among others. Based on this information, data is mathematically computed and summarized through statistical procedures in order to generate a range of standardized scores upon which comparisons can be made.

Standardized scores are converted from raw scores and are then used to interpret the student's performance; scores are then reported as scaled scores, standard scores, stanines, percentiles, and/or age-equivalents. Standard scores are the most commonly reported type of score, in which the mean (or average) and the standard deviation (a measure of variation) have been assigned preset values, typically with a mean of 100 and the standard deviation at 15. Percentiles allow performance to be compared to other students by ranking performance within a range of 1 to 99, with the 50th percentile indicating the median (or middle) score. A percentile rank reflects the percentage of students who scored as well as or lower than the student being tested. As an example, a percentile score of indicates that the student's performance exceeded that of 82 percent of the comparison group.

In addition, the standardization procedures for norm-referenced tests also apply to administration of test items, with strict guidelines for implementation. Since these tests are used as comparisons between students, they must be administered under similar circumstances in each instance of test taking. This is a major advantage of standardized tests; because they are so specific with respect to their implementation, they are quite easy to administer. All the materials are provided and are held constant across administrations, both over time and across child, allowing for every administration to be conducted in same manner, thereby increasing reliability of results.

However, this approach is not without its disadvantages. Such measures often do not provide an adequate amount of information in order to fully assess the student's level of achievement with respect to what has and has not been learned, or to reach conclusions regarding whether the student meets the standards set for specific grade levels and/or content areas. Thus, this information is only moderately helpful in describing **present levels of performance** during the IEP process (described in more detail later in the chapter).

Also included under formal assessment procedures are **criterion-referenced tests** (which are discussed under informal measures as well). In contrast to norm-based assessments, criterion-referenced tests measure a child's performance and compare it to a specific, typically curricular, standard instead of to other students' performance. Criterion-referenced tests occur when individuals are measured against defined (and objective) criteria and are often, but not always, used to establish a person's competence (whether he or she can do something). Essentially, criterion-referenced tests are able to track or measure a student's mastery of specific skills. The

main basis of comparison is between students' performance and a specified level of mastery or achievement. Placement tests and state "accountability tests" (such as the Florida Comprehensive Assessment Test or FCAT) are examples of criterion-referenced competency tests that are commercially available and have standardized administration procedures.

2.8 Informal Assessment Approaches

Informal assessments are centered on content and individual performance, rather than on comparisons to other students. As such, these approaches do not necessarily require a defined reference group, but rather compare the child's performance to expected skills and abilities as set forth by developmental standards and/or the curriculum. Most informal assessment tools and measures are not standardized and therefore do not provide information about psychometric properties of the assessment (such as reliability and validity). Nonetheless, this approach does have some advantages over standardized tests. Standardized measures generally assess a larger breadth of skills than informal measures; however the skills included in the assessment may be selective rather than comprehensive and there may be relatively few items for each individual skill area. Informal measures tend to focus on sub skills within a curriculum or developmental area in much greater depth (representing a smaller range of skills) and much more thoroughly. Therefore, informal measures are often preferred for goal setting and in determining appropriate instructional strategies.

2.9 Types and Goals of Informal Assessments

Informal assessments are sometimes referred to as performance-based measures whose main goal is to inform instruction. Since informal assessments make up such a broad range of procedures, there are a number of different types of assessments that fall under this category.

Criterion-referenced assessments are often considered to be an informal assessment procedure as well. In criterion-referenced assessments, specific criterion serve as the standard for what every student is expected to know, and scores are set to allow comparisons to these benchmarks. Thus, an individual's skill mastery is measured and performance is compared to curricular standards, rather than to a normative group. Criterion-referenced assessments tell us how well students are performing with respect to specific goals or standards, rather than just indicating

how their performance compares to a norm group of local or national students. Therefore, in criterion-referenced assessments, it is possible that none (or all) of the examinees will reach a particular goal or performance standard. This approach allows for determination of individual needs and abilities which aid in the selection of areas to target in intervention, allowing instructional goals and approaches to be specifically tailored to a child.

Many informal assessments are **curriculum-based**. As discussed earlier, the standard of comparison for such measures is the curriculum and students are evaluated as to whether or not they have mastered the skills specified in the curriculum. Thus, these measures directly assess “school skills,” which easily translate to instructionally relevant goals and objectives. **Curriculum-based measurement (CBM)** is a type of curriculum-based assessment strategy that differs primarily with respect to the types of skills selected for assessment and the frequency with which assessment takes place. Brief probes (behavior/data sampling) are collected to sample critical target behaviors and are administered frequently; data are then graphed and the results are analyzed. This process of graphing student performance allows for the identification of whether or not adequate progress is being made and/or when changes need to be made to improve the rate of skill acquisition.

Some other informal assessment procedures include inventories and quizzes, work-sample analysis, task analysis, and portfolio assessment. In **work sample analysis**, samples of the student’s work are observed to study responses (both correct and incorrect) in order to shed light on areas of successful performance and those areas in which the student still needs assistance. Errors are often examined in more detail in order to identify error patterns that may suggest where the learning process has broken down for the student. **Task analysis** approaches involve the study of task demands, in which complex tasks are broken into teachable subcomponents or steps. Because of this feature, task analysis is considered to be an instructional technique as much as it is an assessment strategy. The major goal of this strategy is to allow for a direct relation to specifying the instructional sequence for a specific task. A further informal strategy is **portfolio assessment**. A portfolio assessment involves collecting work products that demonstrate improvement over time in the objectives and goals that have been set for the student. This approach is the most common type of alternative assessment used in classrooms

today. It is important to keep in mind that the products selected for inclusion in the portfolio should be consistent with the instructional goals for the student.

Final informal assessment approaches consist of dynamic assessment and diagnostic teaching procedures. Dynamic assessment can be conceptualized as a “learning potential” assessment that differs from traditional testing in terms of the nature of the examiner–student relationship and the type and content of feedback offered. In addition, while formal assessment (and other informal assessment approaches) tend to emphasize the *product* of student learning (i.e., the student’s level of performance), in a dynamic assessment, the examiner is interested in both the product and the *process* of student learning. During **dynamic assessment**, the examiner not only gives performance-contingent feedback on the student’s correct and incorrect responses, but also offers instruction in response to student failure aimed at enhancing the student’s achievement. Diagnostic teaching involves very similar concepts and approaches. In this approach, the focus is on interpreting the student’s interactions with the learning environment and involves taking observational notes regarding the way in which the student approaches a task, handles frustration, self-corrects errors, and engages in problem-solving strategies (Carlson et al., 1998). Another way in which diagnostic teaching can be conducted is to present the same task to the student, while systematically altering task requirements, such as presentation or response modes and observing their effects on student performance. In essence, both approaches involve the systematic manipulation of the testing and/or instructional conditions to determine the most appropriate strategy for teaching a particular skill to a given student; performance is assessed under standard instructional conditions, aspects of the task or teaching approach are modified, and performance is assessed again under these modified conditions and compared with previous performance. These strategies offer a clear advantage for instructional programming in that they provide necessary insight that will allow educators to alter how they teach according to the type of conditions under which the student is most likely to be successful.

2.10 SPECIAL ASSESSMENT GUIDELINES FOR STUDENTS WITH ASD

A thorough and comprehensive assessment, including both formal and informal assessment procedures, is crucial to understanding and appropriately assisting individuals with ASD. Specifically, assessment of individuals with ASD should provide information as to how the individual’s development compares to other children his or her age and the specific symptoms,

strengths, and challenges exhibited by the child. Further, changes in the individual's development over time should be documented, including skills mastered, rate of skill acquisition, and the maintenance and generalization of mastered skills. Finally, there should be a clearly articulated plan at the outset for linking assessment results to intervention planning and program monitoring. The above-mentioned assessment goals are best accomplished by utilizing a process-oriented approach.

A **process-oriented assessment** is based on idea that the assessment process itself represents a feedback mechanism or a circular process, whereby some portion of the system's output is returned (or fed back) to the input system. An overview of a process-oriented assessment for students with ASD is presented in Figure 2.2.

The aim of a process-oriented assessment is to measure the student's learning and performance over time, allowing data to continually guide goal setting and programming. As such, this approach is particularly well suited for data collection and evaluation that will be ongoing in nature. A major assumption of this approach is that instruction is most effective when it is based on (a) identification of objectives that match student strengths, weakness, and learning styles; (b) instruction that matches clearly defined and measurable objectives; (c) continual assessment of the student's performance relative to objectives; and (d) adjustments to goal selection, instructional programming, and teaching strategies based on feedback on student performance.

Further, best practice assessment guidelines for individuals with ASD have been published, emphasizing four general themes (Klin, Saulnier, Tsatsanis, & Volkmar, 2005; Ozonoff, Goodlin- Jones, & Solomon, 2005). First, assessment must utilize a developmental perspective. Using a developmental framework provides a benchmark for understanding the severity or quality of delays or deviance. Delays in one developmental area can significantly impact the acquisition of later developmental milestones. In addition, whenever possible, assessments of ASD should be multidisciplinary, including a range of professionals with specific perspectives and areas of expertise (such as psychologists, speech and language pathologists, psychiatrists, neurologists and other medical specialists, educational specialists, and/or social workers).

Further, information should be collected from multiple sources, across multiple contexts/settings, and utilizing multiple strategies. Sources include interactions with the individuals with ASD;

their parents, teachers, therapists, and service providers; and their siblings and/or peers. Settings for assessment include not only the evaluation room, but home, school, and/or community settings as well. There are many important reasons to collect assessment data across multiple settings. First, symptoms of ASD are often dependent on characteristics of the environment (e.g., situation-specific).

Therefore, the child may demonstrate more adaptive and functional skills within a one-to-one testing situation, but have much more difficulty in less structured settings, such as with peers on a playground or in a distracting classroom situation. In contrast, due to difficulties with adapting to changes in routines and environments, along with generalization difficulties, structured one-to-one evaluation sessions may fail to capture the full range of skills that the individual possesses. Therefore, the child may look much more competent in a known environment (such as the classroom) than in an evaluation room without familiar, well-practiced routines and materials. Assessment must also utilize a wide range of strategies and assessment procedures, including observations, administration of formal standardized and informal curriculum-based assessments, and completion of ratings scales and inventories by individuals familiar with the child's day-to-day functioning, such as parents and teachers.

Finally, and perhaps most importantly, decisions regarding assessment methods and approaches to be used need to be individualized based on the child's age, developmental level, specific diagnosis, and areas of need. Necessary components include a review of available records, including anecdotal reports, work samples, behavioral charting systems, report cards, previous diagnostic evaluations, and **Individualized Education Programs (IEPs)**. Additional information can be obtained through parental interviews regarding the child's developmental, educational, and medical histories, as well as by conducting observations across multiple settings.

2.11 Types and Goals of Formal Assessment of Individuals with ASD

Formal assessments generally involve both **diagnostic** as well as **psycho educational** components. Guidelines for goals within each type of assessment, specific to individuals with ASD, are offered.

BOX 2.2 Research Notes # 1

With the increased interest in early identification and the belief that early intervention for ASD will lead to more optimal outcomes, researchers have begun to focus on studying the long-term outcomes of young children with ASD who have participated in specialized early intervention services. Turner and colleagues followed 26 children (22 male, 4 female) who received clinical diagnoses of either autism ($n = 18$) or PDD-NOS ($n = 8$) under the age of 3 and evaluated them 5_7 years after their initial evaluations.

2.12 DIAGNOSTIC ASSESSMENT

Diagnostic assessment for ASD is typically the first formal assessment process that a family will undergo. The overall goal of such an assessment is to obtain an initial diagnosis, through consideration of other disorders with similar presenting problems and symptoms. Through the diagnostic evaluation, clinicians seek to determine whether the child's behavioral difficulties are consistent with a diagnosis with the autism spectrum, or rather, can be better explained by another disorder such as an intellectual disability/mental retardation or a language disorder. The process of receiving a diagnosis or label is important for a number of reasons. Most importantly, accurate diagnosis facilitates the selection of additional assessment components, as well as identification of appropriate intervention approaches. Further, symptom presentation is often related to other domains of interest, including cognitive abilities, joint attention, language, and social development. Early diagnosis also has far-reaching implications for prognosis and family adjustment. Early identification allows for early entry into intervention as well as assisting families in understanding their child's difficulties.

Diagnosis is the first step in providing families with the tools they will need to advocate for services for their child and become active participants in their child's intervention. Documentation of diagnostic status is often necessary for access to services or funding mechanisms. Accurate classification is important from an educational standpoint in that it is used to determine eligibility for school-based special educational placement and related services. State and federally funding social service and health care agencies also rely on diagnostic reports in order to document eligibility for services.

Finally, diagnostic evaluations offer a basis from which to prescribe or plan interventions, by specifying child characteristics across core and associated impairment areas in ASD, such as

specific deficits and skills in the areas of social, communication, and play skills as well as restricted, repetitive, and stereotyped behaviors and sensory processing disturbances that may interfere with instruction. Such information, collected during the initial evaluations stage, can also be important in documenting progress and evaluating intervention efforts. In this regard, it is equally important that intervention consider goals for outcomes that reflect areas that are relevant to symptom presentation.

2.13 PSYCHOEDUCATIONAL EVALUATION

Psycho educational evaluation should also be incorporated into the initial assessment stages, as the emphasis of this approach is to gather information. Results indicated considerable improvement in cognitive scores for a large number of children in the sample, compared with initial cognitive scores at age 3. At age 3, 84% of the children had cognitive scores below the threshold for intellectual disability (less than 70), which decreased to only 28% of children by follow-up (mean age = 7.9 years). In addition, at follow-up, 72% of children obtained scores in the borderline to average ranges in comparison to only 16% at age 3. Further, and most encouragingly, 56% of the children were scoring in the average range or above at follow-up.

Results from this study are important because they suggest that cognitive scores may change dramatically between the ages of 2 and 9 for students with ASD. Cognitive scores increased for most children, with 68% of scores rising at least 15 points. Findings indicated that, on average, cognitive scores at outcome increased 23 points, and nine children increased more than 30 points. Higher initial cognitive scores appeared to be more stable than lower scores, as all four children who attained cognitive scores above 70 at age 2 obtained average scores at age 9. Results further revealed that there were some initial characteristics of children that were associated with better outcomes at follow-up. Children with better outcomes tended to have a younger age of diagnosis, higher initial development quotient and language scores, and received more hours of speech and language therapy services.

Across multiple domains of functioning that can be used to guide educational programming. The major goal of a psycho educational evaluation is to obtain a better understanding of the student's strengths and weaknesses in order to provide a baseline of skills from which to plan intervention (Klin, et al., 2005). Ultimately, this information will be used to create an individualized

education program (IEP) that takes into account developmental level, strengths and needs, likes and dislikes behavioral difficulties, effective teaching strategies, and motivational systems. Information relevant for program evaluation can be provided through the psycho educational process and can be complemented with consistent application of informal assessment strategies during instruction. Such informal procedures allow for documentation of progress toward goals over time through continued evaluation of a child's characteristics (learning readiness, academic strengths and weaknesses) with frequent comparisons between current and previous functioning.

A "Best Practices" Assessment Battery for ASD Best practices in the assessment of ASD specify that an adequate assessment must lead to an appropriate intervention plan (Perry, Condillac, & Freeman 2002). Therefore, it will be important to use guidelines for priority areas for intervention in ASD as a basis for the evaluation, in order to ensure that all important areas are assessed. According to recommendations put for by the National Research Council (NRC, 2001), areas that should be prioritized in intervention programs for students with ASD include language and communication development, social and play skills, fine and gross motor development, development of cognitive and functional academic skills, and skills needed for success in a regular classroom, including flexibility, organization, and engagement in the learning process. An overview of Best Practices in the assessment of individuals with ASD is presented in Figure 2.3.

COMPONENTS OF A CORE ASSESSMENT BATTERY

Klin et al. (2005) offer that "developmentally based assessment of cognitive, social communicative and adaptive skills provides the essential bases on which decisions on diagnosis, eligibility for services, and program planning have to be made" (p. 772). Elements of a core assessment battery include those domains that are both necessary and sufficient for an evaluation of a suspected ASD, and are typically included as part of the diagnostic evaluation. These core domains include developmental and/or intellectual assessment of cognitive abilities, speech and language skills, and adaptive behavior functioning. Figure 2.4 provides a recommended evidence-based assessment battery for students with ASD.

Cognitive/Intellectual Assessment

Cognitive/intellectual assessment is necessary for the evaluation of individuals suspected of meeting criteria for ASD and for differentiating between individuals with ASD and those with a primary diagnosis of mental retardation/intellectual disability. Thus, intellectual assessment allows for the differential diagnosis of conditions that may affect language and cognitive development, including ASD, general developmental delay, intellectual disability, and language disorders. Cognitive assessment is also necessary because many diagnostic assessment measures for ASD require that behaviors

Core Assessment Battery—Necessary assessment domains for diagnosis and eligibility determination, as well as preliminary planning of priority instructional needs.

- Assessment of specific symptoms of autism using a variety of autism diagnostic measures, combined with parent interview
- Assessment of **intellectual functioning** to allow clinicians to frame the interpretation of other evaluation results
- Assessment of **language**, including receptive, expressive, and pragmatic language abilities
- **Adaptive behavior** assessment in order to rule out co-occurring intellectual disability/mental retardation as well as setting appropriate goals in treatment planning. This domain should include self-help, recreational/leisure and community-based skills

Additional Assessment Domains—Additional assessment domains that will provide information about additional priority educational needs to guide IEP development

- **Neuropsychological assessments** including attention and executive functioning, memory, and processing
- **Developmental, pre-academic and academic functioning** in order to identify patterns of strengths and weaknesses in the student's individual learning profile
- **Prevocational and vocational skills** for students approaching transition-age to plan for post school activities and outcomes
- **Emotional functioning** to assess co morbid anxiety and depression that may need to be targeted

- **Motor skills and sensory processing** to set appropriate instructional objectives, supports, and accommodations
- **Reinforcement assessment** in order to determine effective learning and behavioral supports
- **School, family, and community contexts** because the goal of assessment should be to understand how ASD affects individuals in the course of daily life be interpreted against the child's developmental level; without a measure of intellectual ability, it is often very difficult to ascertain whether a given behavior is appropriate for the child's level of development and skills. Further, intellectual ability is one of the best predictors of long-term prognosis for individuals with ASD, and therefore can be used to provide family members with preliminary information regarding the child's expected developmental trajectory.

The inclusion of standardized intelligence tests is especially important given prevailing misconceptions (or “myths”) regarding the measurement of cognitive abilities in individuals with ASD. Such “myths” include the idea that measurement of IQ is irrelevant for individuals with autism; many erroneously believe that because of motivational difficulties, individuals with ASD are often “untestable” on standardized tests and therefore, it is not possible to accurately measure their cognitive abilities. This assumption results in the viewpoints that it is not possible to know the true intellectual capabilities of someone with ASD and/or that all individuals with ASD are actually quite intelligent, but do not demonstrate the skills they have due to problems with motivation and attention.

BOX 2.3 Research Notes # 2

“Despite the general agreement that cognitive ability is a critical component of treatment research in addition to the clinical need for IQ information, there is no consensus about the most appropriate test instrument(s) to measure cognitive functioning in this population” (Delmolino, 2006, p. 959). So, a study was conducted to assess the usefulness of the PEP-R as a measure of general cognitive development in young children with autism. Participants consisted of 27 children with autism or PDD-NOS (23 boys and 4 girls) with a mean age of 44 months (3.8 years) at the time of testing.

In order to answer this question, Delmolino administered the PEP-R along with a standardized intelligence test (the Stanford-Binet Intelligence Scales—Fourth Edition) and compared the scores. Findings of the current study indicate that the PEP-R is an instrument that is correlated significantly with estimates of cognitive functioning obtained with the SB-FE. Correlations between the DQ scores obtained on the PEP-R and the Composite IQ score on the SB-FE were .73, indicating a high degree of consistency in scores across the 2 measures.

Further, Delmonico offered that the PEP-R has “some potential advantages for testing this population including flexibility range of appealing materials, lack of timed items, and separation of language items from general assessment items that are not dependent on language and its utility for educational programming” (p. 962). These findings suggest that developmental assessment instruments, such as the PEP-R, can be useful tools to use when estimating cognitive functioning of children with autism, in addition to serving a role in educational programming.

Speech and Language Assessment

Speech and language assessment is the second critical area to include in a core assessment battery for the diagnosis of ASD. As with intellectual ability, understanding of the child’s language abilities, particularly when compared against their cognitive abilities, is important for the differential diagnosis process (specifying the most appropriate diagnosis). Further, language abilities are also important indicators of prognosis. Research suggests that individuals with the best prognosis for outcome are those who, by the age 5, have developed the spontaneous and communicative use of language and the ability to speak in phrases.

Assessment of Adaptive Behavior

Assessment of adaptive behavior serves as the final core assessment battery component, as it is also important for the differential diagnosis of ASD from intellectual disability/mental retardation (ID/MR). Adaptive behavior refers to “real life” skills that are necessary for people to live independently and to function safely and appropriately in daily life, including skills such as self-care, socialization, communication, safety, food preparation, ability to follow community rules, money management, shopping, and cleaning. By definition, individuals with ID/MR must exhibit deficits in adaptive behavior in addition to impaired intellectual functioning (APA; *DSM-IV-TR*). In addition to differential diagnosis, ASD can co-occur (or be comorbid) with ID/MR.

Adaptive behavior assessment is critical to diagnosing the presence of both conditions, if appropriate. Further, adaptive behavior is important to include in diagnostic evaluations because such measures give a more comprehensive picture of the students' various capabilities and weaknesses than measures of IQ alone. Functional skills that should be the focus in instructional programs can also be revealed through this approach.

2.14 COGNITIVE ASSESSMENT

Developmental and Intelligence Tests

According to the practice parameters for the screening and diagnosis of ASD, "knowing the child's cognitive status is important in determining his overall level of functioning. This is, in turn, important when trying to establish a discrepancy between the child's level of social function and the overall cognitive and adaptive function, a key criterion in the diagnosis of autism" (Filipek et al, 1999, p. 462). As mentioned previously, intellectual functioning is generally considered as a predictor of outcome; however, it should be recognized that the predictive validity of such scores is not necessarily high. Research has indicated that there may be relatively low stability in IQ scores for children with ASD, with a significant proportion of children exhibiting changes in cognitive functioning over time (Sigman & Ruskin, 1999; Turner et al., 2006). Scores are relatively less stable, and reliable, the younger the child is at the time of administration of the intelligence test. For a list of commonly used measures of intelligence for students with ASD, refer to Table 2.2.

Recommendations for test selection include choosing measures that (a) are appropriate for both the child's chronological age and estimated mental age; (b) provide a full range (in the lower direction) of raw to standard scores conversions for the child's chronological age; (c) sample both verbal and nonverbal intellectual skills; and (d) measure and score verbal and nonverbal skills separately from one another.

In addition to the above specifications, it is often useful for tests to also have visually interesting materials and allow for subtests to be administered in flexible, rather than fixed, sequence. Some assessment instruments may be more useful than others in maintaining a child's interest, tapping into strengths as well as weaknesses, gathering information about learning style, and generating meaningful (e.g., valid and reliable) test scores. With respect to the latter, it is critical to consider

both floor and ceiling effects when selecting measures. The majority of assessment tests organize test items in a sequential manner, with easier/earlier occurring skills preceding more difficult ones. However, due to the uneven developmental profile observed in many children with ASD, this approach may not accurately reflect the full range of skills the child possesses. For example, due to language difficulties, a child may reach the ceiling (the item in which administration of a subtest ceases due to a series of incorrect responses) on a certain test item, but be able to pass visual-spatial and memory tasks at a higher level.

Therefore, it is recommended that the following aspects of instruments be evaluated when attempting to select an appropriate measure for a particular child: the range of easy to difficult items, the number of items at each difficulty level, the normative age range and raw to standard score conversion charts, whether the test provides “out of level” norms, the manner in which test items are presented to the child (verbally, visually, with demonstration or modeling etc.) and/or the presence of “practice” or “teaching” items. Further, to obtain the most useful results, it is also recommended that clinicians choose measures that minimize language demands and/or assess nonverbal problem solving independently from verbal skills, have a “backup” instrument readily accessible to allow for switching tests depending on the child’s ability level and/or interests, and consider administering subtests from different instruments to get a more thorough picture of ability level.

With these suggestions in mind, appropriate and widely used developmental tests, including those for younger, and/or less verbal individuals with ASD, are listed in Table 2.4 and described in detail below. Because educators are less likely to administer and interpret the results of intelligence (IQ) tests, only brief details regarding the most commonly used tests are provided.

Developmental Assessments

Developmental assessments are similar to intelligence tests, but are typically intended for infants through early school-age students. Like IQ tests, developmental assessment measures are usually norm referenced standardized assessments. However, because they are intended for a younger age group, the types of skills measured are often different from those included on IQ tests. The types of skills assessed on developmental measures relate easily to the domains frequently targeted in preschool and early elementary school, such as fine motor skills,

vocabulary, pre academic concepts (color and shape identification, matching, sorting, classification/categorization) and academic skills (counting, one-to-one correspondence, number and letter identification, sight word identification, etc.). Thus, these batteries are especially well-suited for use by educators and school psychologists alike.

On developmental measures, the child's current abilities are compared to the expected scope and sequence of skill acquisition in typically developing students, and a developmental quotient (DQ) norm-based score is derived. This DQ provides an estimate of cognitive functioning; however, "the concept of IQ is avoided in young children because of the close interdependence of cognitive functioning with other domains of development" (Klin et al., 2005, p. 790). Nonetheless, developmental quotient scores and IQ scores are **correlated** with one another (Delmolino, 2006).

2.15 ADDITIONAL COMPONENTS OF AN ASSESSMENT BATTERY FOR ASD

In addition to the core assessment components, evaluation of students with ASD may benefit from inclusion of other measures and domains that can serve to enhance understanding of the student's strengths and difficulties, as well as approach to problem-solving and learning style. Inclusion of these measures is consistent with a process-oriented assessment approach that is focused on understanding how the child learns, rather than what the child has learned. A major assessment area in this approach includes **information processing**, which can include some or all of the following areas:

- **Auditory/Phonological processing**, including auditory memory, letter-sound correspondence, phonemic discrimination and segmentation, and the synthesis of sounds into words. These abilities underlie fundamental reading processes and can indicate when the student may have difficulty with the attainment of literacy skills.
- **Visual Processing and Visual-Motor Integration**, including visual discrimination, spatial awareness, eye-hand coordination, and ability to translate visual perception into motor functioning (which involves motor control, accuracy, and coordination as well as psychomotor speed). These abilities typically underlie successful performance in academic areas such as mathematics and writing.

- **Memory and Learning** processes are critical for adequate functioning in school and daily life. Memory refers to the encoding, storage, and retrieval of information. The types of memory systems that appear to be most important, and which may be the focus of assessment, include short term memory, working memory, and long-term memory and retrieval. Such assessment tasks generally repeat the same information and/or stimuli over a number of times to assess the efficiency and accuracy of storage and retrieval of new information. These tests can provide insight into the manner in which the individual is likely to learn best and the ease with which they are likely to learn new tasks. A process-oriented approach to learning and memory tasks is critical as it will reveal important information about the strategies that the individual is using to complete the task; dynamic assessment approaches may be especially useful in highlighting additional strategies that may improve performance and thus enhance efficiency of learning.

- **Attention and Executive Functions** include the areas of sustained and selective attention, initiation and inhibition, planning and organization, problem solving, self-monitoring, and cognitive flexibility. Research has consistently indicated that individuals with ASD often have a pattern of executive functioning impairments including selective focus on irrelevant details, initiation and planning difficulties, and the ability to be flexible in problem-solving strategies and behavior. Since attention and executive functioning are critical to school (and workplace) success, they should be a focus of both assessment and intervention efforts.

Research with individuals with ASD has consistently demonstrated specific information processing profiles, including relatively good performance on tasks that rely on rote, mechanical, or perceptual processes, and impaired performance on tasks requiring higher-order conceptual processes, reasoning, interpretation, integration, or abstraction (Minshew & Goldstein, 1998 as cited in Ozonoff et al., 2005).

In addition, other areas that may impact behavior and instructional planning include sensory processing (including under- and oversensitivities to specific stimuli including sounds, lighting, touch, temperature, etc.), emotional and behavioral functioning (including interfering or challenging behaviors and emotional issues such as anxiety and/or depression), preference assessments (including information about what the child will “work” to obtain). These should be included in the assessment process when relevant.

Assessment Strategies for Individuals with ASD

Due to learning challenges outlined earlier in the chapter, children with ASD may present particular challenges during assessment that may make generating meaningful estimates of performance on standardized tests difficult. Such children are often colloquially referred to as “untestable.” However, Ozonoff and colleagues believe that “un test ability” primarily reflects a lack of availability of appropriate tests and/or clinician experience; “when experienced clinicians evaluate children with autism, few should be un test able” (Ozonoff et al., 2005, p. 529).

While the focus of the chapter thus far has been primarily on standardized assessment procedures and instruments, it is important to keep in mind the overarching purpose of assessment—namely, to provide information that will be useful for program planning. Therefore, while standardized, norm based scores are important for documenting eligibility for special education services and revealing areas of educational priority, scores do not “tell the whole picture.” In fact, for some children, the obtained “scores will hardly convey the most important information to the special educators whose mission is to address the child’s needs while capitalizing on their strengths” (Klin et al., 2005, p. 791). Therefore, no child is truly “untestable.” Even if scores generated are not considered to be a valid reflection of the child’s overall skills, useful information can still be obtained. The likelihood of the latter is significantly increased when the assessment process includes effective use of behavioral observations and the provision of adaptations to evaluation procedures.

Clinicians have an ethical obligation to make such adaptations when working with special populations, both in instruction as well as in assessment. Adaptations will inform instructional planning by providing the most accurate possible estimate of the child’s abilities in the constructs of interest, by minimizing the impact of test-taker attributes. In addition, the inclusion of adaptations to test procedures during the assessment process provides some ecological (or “face”) validity to parents that the child was given every opportunity to demonstrate the skill in question. Further, all of the strategies provided below are applicable for use during instructional tasks. Adaptations provided during assessment (whether formal or informal) and their subsequent effect on the student’s performance directly relate to the process of developing an IEP. Adaptations allow for “testing the limits” and provide more valid information regarding what the student knows, providing a more accurate picture of the child’s present level of performance. In

addition, by trying out different strategies, educators are armed with more complete knowledge regarding potential accommodations to include in the IEP.

2.16 ADAPTATIONS FOR TESTING DIFFICULT TO TEST CHILDREN

Suggestions for testing difficult to test children include (a) selecting appropriate measures; (b) presenting detailed descriptions of behavioral observations during the assessment procedures; (c) utilizing environmental supports and reinforcement strategies; and (d) altering or making modifications to standardization procedures.

Behavioral observations during testing often provide critical information regarding preferred types of tasks, task persistence and frustration level, attention to details, instructions and materials, attention span across preferred and non-preferred tasks, length of time the child will attend and work before becoming disengaged, and approach to problem solving (i.e., trial and error, deliberation, etc.). In addition, useful information can be obtained regarding strategies that increase cooperation and performance.

Environmental supports and reinforcement strategies improve cooperation and performance during the testing situation by making adjustments to the testing environment, in order to increase the child's attention and success on tasks. Such approaches inform instruction by systematically assessing the level and type of environmental structure needed for task completion and social engagement. The first step in this process involves creating a comfortable testing environment by conducting the evaluation in a child-friendly environment with familiar adults present and/or in familiar locations (e.g., the classroom rather than the clinician's office). Testing can be completed with the child standing near the table, seated on a familiar person's lap (for younger children), and/or on the floor if needed. It is also helpful, if space allows, to create clearly identified "work" and "play/break" areas. Visual strategies are also an effective addition to the testing session and may include a "firstthen" or "work-play" visual schedule, the use of work systems (e.g., "finished basket") to demonstrate task completion, and visual, token, or word schedules to communicate the number and type of tasks to be completed. Visual cues for requests (e.g., a "break" card, reinforcer choice boards, and/or token or penny boards) can also help less verbal students to signal the need for a break and to increase motivation for task

completion. With respect to reinforcement, there are three main goals for the use of reinforcers during testing:

(a) to improve task compliance and effort by increasing motivation, (b) to assist in the identification of functional reinforcers for the child that can be incorporated into his or her instructional program, and (c) to assess for preferred stimuli, any self-stimulatory behaviors, preoccupations, and circumscribed interests. Some strategies and guidelines for using reinforcers during testing include asking the child's parent and/or teacher about preferred and non-preferred stimuli and having them bring preferred snacks and favorite toys or materials to the testing session, trying out a full range of different reinforcers to assess their value, varying reinforcers throughout the session to prevent satiation, assessing the schedule of reinforcement that results in optimal performance, evaluating the child's response to social praise as a reinforcer, using planned ignoring for minor disruptive behavior, and considering using escape from tasks or "work" as a reinforcer. In contrast to the above mentioned adaptations, breaking standardization during testing is typically done through making modifications to the test procedures.

BOX 2.4 Trends and Issues # 2

When, Why, and How to Break Standardization Procedures

Perry and colleagues (2002) provide some thoughts and guidelines related to the issue of "breaking standardization," along with a number of useful testing strategies. According to the authors, making adaptations when required for persons from special populations is ethically responsible practice in order to reduce the effect of characteristics of the test-taker that are not relevant to the construct being tested and/or the primary focus of the assessment. Such adaptations may include presentation or response formatting changes, modifications to timing or the testing environment, using portions of tests, or using substitution tests.

2.17 Linking Assessment to Intervention

Once the data is collected and the evaluation is completed, the assessment results must be linked to instructional programming. The NRC (2001) guidelines for necessary components of interventions for ASD provide a good starting place for creating an instructional plan for a student with ASD (see Box 2.5). The NRC (2001) recommendations for preschool children with

ASD specified that children be enrolled in early intervention programs as soon as an ASD is seriously considered. In addition, active engagement in intensive instructional programming (a minimum of a full school day, at least 5 days a week – 25 hours a week) for the full year was considered essential. Further, the

BOX 2.5 Trends and Issues # 1

Guidelines for Selecting Appropriate Educational Objectives for Children with ASD

Appropriate objectives should be observable, measurable behaviors that can be reasonably accomplished within a 1-year time frame. In addition, these objectives should have a direct impact on enhancing the child's ability to participate fully in education, the community, and family life. The following areas should be targeted for development:

- ***Social skills*** that will enhance participation in family, school, and community activities. Suggestions for target areas include imitation skills, responding to and initiating interactions with peers and adults, and parallel and interactive play skills with peers and siblings.
- ***Language and nonverbal communication skills*** including expressive (verbal) language, receptive language, and use of eye contact and gestures to communicate with others.
- ***Development of a functional and symbolic communication system*** that should emphasize identifying a system that allows children to communicate their wants and needs, and to make choices that affect them, to the greatest extent possible.
- ***Fine and gross motor skills*** that will assist in achieving age-appropriate functional activities.
- ***Cognitive skills*** including the development of basic concepts, life skills, symbolic play, and academics. Goals for cognitive development should be carried out in the context in which the skills are expected to be used and functional academic skills should be taught when appropriate.
- ***Behavioral goals*** that are focused on skill development (communication skills, self-regulation skills, etc.) to replace more challenging and/or problematic behaviors. Behavioral strategies should be positive and proactive, incorporate information about the contexts in which the behaviors occur, and include a range of behavioral techniques that have empirical support.

- ***Independent organizational and self-help skills*** that underlie successful participation in the home, school, and broader community (independent task completion, asking for assistance, following directions and instructions, etc.).

2.18 DEVELOPMENT OF THE INDIVIDUALIZED EDUCATION PROGRAM (IEP)

The assessment process leads directly to the development of an IEP for the student. The IEP is a written document that describes the special education and related services to be provided in order to meet the specific needs of a child with a disability, and is comprised of six required parts:

1. *Description of the child's present level of performance, or functioning:* The first step in developing an IEP involves a description of the child as he or she is at the present time. This section of the IEP highlights the child's current academic and behavioral skills, interests, and learning style, and discusses the implications of the child's disability on academic and non-academic (e.g., social, communication) achievement. From this description, the IEP team develops a list of areas of instructional priority, often referred to as priority educational needs (PENS). Since instructional programming is based on information regarding a student's present levels of performance, learning style, and preferences, and identification of PENS, the assessment process and its resulting data are critical to successful instructional planning.

2. *Annual goals and objectives:* Goals and objectives are based upon the information described above, namely descriptions of the child's present level of performance and identification of priority educational needs. Goals and objectives are descriptions of the skills that the child will attain within a specified period of time. Goals are typically written to reflect annual expectations for progress, whereas objectives typically reflect short-term steps (often quarterly) toward the attainment of the annual goal. Goals and objectives should take into account the above referenced NRC (2001) recommendations for programming for students with ASD, and target each identified area of priority educational need.

3. *Related services:* The IEP also describes services that will be provided in order to supplement the educational services provided in the classroom. Related services are those services that are necessary in order to effectively implement the IEP and designed to ensure that the child is able

to benefit from special education in the least restrictive environment. Examples of possible related services include counseling; occupational, physical, and/or speech and language therapies; parent training; and assistive technology.

4. Educational placement: Placement refers to the educational setting in which the IEP will be implemented and is chosen based on the setting in which the goals and objectives will be appropriately met. If the classroom setting is not a general education classroom, the IEP must specify the amount of time (if any) that the child will participate in the general education classroom and include a statement ensuring that the least restrictive environment was considered.

5. Time and duration of services: This step in the IEP process specifies starting and ending dates for goals, objectives, and related services. In addition, the frequency of any related service is also specified. For example, the IEP might specify that speech and language therapy will be delivered twice a week for 30 minutes each session for the duration of the IEP. Because special education law (IDEA) requires annual review of services provided in the IEP, long-term duration for services should be projected no further than 1 year.

6. Evaluation of the IEP: The final step in IEP development is specifying how student progress toward short-term objectives and annual goals will be measured or evaluated. Following a process-oriented assessment approach will ensure that this step is not overlooked. In addition, well-written goals and objectives (described below) automatically build in an evaluation process by allowing for growth and development to be tracked.

Creating Annual Goals and Objectives

Annual goals and objectives will be based upon comprehensive assessment data (both formal and informal) and should incorporate the NRC (2001) recommendations referenced in this chapter. Given the challenges associated with ASD, annual goals should address, at minimum, the following areas: social functioning; nonverbal communication skills; receptive, expressive and pragmatic language; fine and/or gross motor skills; academic skills; and behavioral, organizational, and/or self-help skills.

Annual goals should (a) clearly state what the student is expected to accomplish in a 1-year time frame, (b) be worded as a positive statement (what the student *will* accomplish, versus what the

student will no longer do), and (c) be worded to be clearly observable and measurable. Well-written IEP goals and objectives tend to follow a specific formula consisting of five questions, which when used, will result in a statement that is objective, observable, and measurable.

1. Who will demonstrate what behavior or skill?
2. How will this skill be demonstrated? What will the skill look like as it is demonstrated and/or at what level will the skill be demonstrated?
3. Where or under what condition will the skill be demonstrated?
4. How frequently will the skill be demonstrated (What are the criteria for mastery)?
5. By when will the skill be demonstrated?

Curriculum/Assessment Instrument Description Skills Addressed

- Information obtained from the completed assessment can be used to develop an effective Individualized Education

Program (IEP)

- Social Interaction
- Group Instruction
- Classroom Routines
- Generalized Responding
- Reading, Math, Writing, Spelling
- Eating
- Dressing, Grooming, Toileting
- Gross Motor Skills
- Fine Motor Skills

A Work in Progress: Behavior Management Strategies and a Curriculum for Intensive Behavioral Treatment of Autism

Leaf & McEachin (1999)

- A curriculum based on principles of Applied behavior analysis
- Instructional programs fit roughly into four categories
- Also provides suggestions for toilet training, eating, play and social skills, sleep problems, and dealing with disruptive and self stimulatory behaviors
- Imitation: verbal and motor
- Language and Communicative Concepts: labeling, pronouns, yes/no, following instructions, prepositions, plurals, attributes, asking questions, conversation
- Cognitive: same and different, first and last, before and after, quantities
- Academic and Pre-academics: reading, writing, stories, sequencing, recall

Behavioral Intervention for Young Children with Autism

Maurice, Green, & Luce (1996)

- A curriculum based on principles of applied behavior analysis
- Attending
- Imitation
- Receptive Language
- Expressive Language
- Abstract Language
- Pre-academics
- Academics

- School Readiness
- Self-help
- Socialization

2.19 Summary

Instructional planning for students with ASD requires knowledge and experience and a range of approaches and strategies. Programming is only as useful as the foundation on which it is based; assessment provides that solid foundation. The ultimate goal of assessment is to obtain data that can be used to devise an appropriate intervention plan. There are many approaches for assessment of individuals with ASD from diagnostic to psycho educational evaluations, and including formal and informal measures. The range of available assessment tools and strategies is wide, allowing clinicians the flexibility to select a battery of measures that is individualized to the assessment questions and student's characteristics. In addition to assessment scores, the use of modifications during testing greatly increases the chance of success in formal testing situations and provides insight in to the student's learning process and preferences. These strategies can provide a comprehensive picture of the student that will ensure the greatest success in developing the most appropriate instructional plan possible.

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Unit 3

Instructional Approaches

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3.1 Introduction

Autism Spectrum Disorders (ASD) are complex neurological disorders that have a lifelong effect on the development of various abilities and skills. ASD is characterized by impairments in communication and social interaction, as well as unusual patterns of behaviours, interests, and activities. The term “spectrum” is used to recognize a range of disorders that include a continuum of developmental severity. The symptoms of ASD can range from mild to severe impairments in several areas of development. Many professionals in the medical, educational, and vocational fields are still discovering how ASD affects people and how to work effectively with individuals with ASD.

3.2 Understanding Autism Spectrum Disorders

Terminology

The terms “autism”, “Pervasive Developmental Disorders (PDD)”, and “ASD” have been used interchangeably within the literature and by professionals and parents, and may cause confusion. Autism was first identified in 1943 by Leo Kanner, an American psychologist. Kanner noticed distinctive, common characteristics within a subgroup of children in whom other mental disorders had been diagnosed originally. Kanner recognized the inability of this subgroup to relate in the ordinary way to other people and situations, and he described this behaviour as “extreme autistic aloneness” (Kanner, 1943). As a result, for several decades the disorder was referred to as autism.

Pervasive Developmental Disorders (PDD) was first used in the *Diagnostic and Statistical Manual of Mental Disorders–III (DSM-III)* by the American Psychiatric Association (APA) in 1980 as a general term to describe a class of disorders that shared the following characteristics: impairments in social interaction, imaginative activity, and verbal and non-verbal communication skills, and a limited number of interests and activities that tend to be repetitive. In 1994, five disorders were identified in the updated *DSM-IV* (APA, 1994) under the category of PDD: Autistic Disorder, Rett’s Disorder, Childhood Disintegrative Disorder, Asperger’s Disorder, and Pervasive Developmental Disorder Not Otherwise Specified (PDD NOS). Autism is recognized by the Ontario Ministry of Education as communication exceptionality for the purposes of student identification and placement. This

term continues to be used frequently as a shorthand term to include various conditions that are now recognized as a range of disorders.

Autism Spectrum Disorders (ASD) was first used in 1988 (Wing, Allen) and is now commonly used to describe a subset of PDD that includes only Autistic Disorder, Asperger's Disorder, and PDD NOS. In some cases, the term ASD is used to acknowledge the complete range or spectrum of associated characteristics and disorders that are included in PDD. The term "ASD" will be used throughout this document in reference to students who have Autistic Disorder, Asperger's Disorder, or PDD NOS.

3.3 ASD

The following is a brief summary of each of the three disorders considered ASD. More specific information regarding the diagnostic criteria, such as the characteristics that must be present in both quantity and type for diagnosis, can be found in the *Diagnostic and Statistical Manual of Mental Disorders-IV Text Revision (DSM-IV-TR)* (APA, 2000).

- **Autistic Disorder**

Students with Autistic Disorder have noticeably abnormal or impaired development in social interaction and communication and a restricted repertoire of activity and interests. Such students often show a preoccupation with one narrow interest and an insistence on following routines. Abnormalities in the development of cognitive skills and in posture and body movements may be present. These impairments are accompanied by a delay or abnormal functioning in social interaction, language used in social communication, or symbolic or imaginative play that was recognized prior to three years of age.

- **Asperger's Disorder**

Like students with Autistic Disorder, students with Asperger's Disorder show impaired social interaction and display a limited field of interests and activities prior to three years of age. This impairment causes difficulties in social and/or occupational functioning. The difference between students with Autistic Disorder is four to five times more common in boys than girls. Asperger's Disorder is also known as Asperger's syndrome and is diagnosed at least five times more frequently in boys than girls. Disorder and Asperger's Disorder is that students with Asperger's

Disorder show no significant delay in language acquisition, although the more subtle aspects of social communication may be affected. There are no significant delays in cognitive development or in the acquisition of age-appropriate learning skills or adaptive behaviours. Restricted, repetitive patterns of behaviour, interests, and activities are common. Students with Asperger's Disorder may experience feelings of social isolation, which may contribute to depression or anxiety in adolescence.

- **Pervasive Developmental Disorder Not Otherwise Specified (PDD NOS)**

Students with PDD NOS have severe impairments in the development of reciprocal social interaction, including impairments in either verbal or non-verbal communication skills, or have stereotyped behaviours, interests, and activities. PDD NOS, also referred to as “atypical autism”, includes individuals who do not fit into the other categories because they do not meet all of the diagnostic criteria for a specific disorder; for example, diagnostic symptoms may occur at a later age.

Causes of ASD

There are several theories about the cause or causes of ASD. Researchers are exploring various explanations but, to date, no definitive answers or specific causes have been linked scientifically to the onset of ASD. Research suggests that individuals with ASD experience biological or neurological differences in the brain. In many families, there appears to be a pattern of ASD-related disabilities, which suggests that ASD is an inherited genetic disorder. Current research studies show that certain classes of genes may be involved or work in combination to cause ASD. There appear to be many different forms of genetic susceptibility but, to date, no single gene has been directly related to ASD (Autism Genome Project Consortium, 2007). Ongoing research is being done to further investigate the cause of ASD.

It is important to note that ASD is not a mental illness, and there are no known psychological factors in the development of the child that have been shown to cause ASD.

Prevalence of ASD

ASD was once viewed as a rare disorder, but recent studies suggest that the prevalence rate for all forms of ASD is much higher than previously thought. The most current estimates from studies in Canada and the United Kingdom are that ASD is diagnosed in 60 of every 10,000 children, or one in 165 (Fombonne et al., 2006). ASD is now recognized as the most common neurological disorder (Geneva Centre for Autism, 2006) and has been found throughout the world in families of all racial, ethnic, and social backgrounds. It is diagnosed more frequently in males than females; worldwide, males are affected four times as often as females (Chakrabarti & Fombonne, 2005). An increase in the numbers of individuals in whom ASD is diagnosed can be linked to a combination of factors. With the broader definition of PDD that was provided in *DSM-IV* (APA, 1994) and an improved recognition of the symptoms, changes in diagnostic practices have occurred (Chakrabarti & Fombonne, 2005). As a result, the diagnosis of PDD is being made more frequently and at an earlier age (Fombonne, 2003). Also, in recent years, more rigorous methods are being used in surveys to find and gather data on cases of ASD. While other possible causes for the increasing rates of diagnosis and resulting identification continue to be investigated, there is no direct evidence to support an increased incidence, or epidemic, of ASD. The research cited above suggests that ASD appears to be more common now because the tools used for diagnosing ASD in children are better now than before.

Diagnosis of ASD

The complex nature of ASD makes it difficult to diagnose, and there is no single medical test to determine if a student has ASD. In general, the perspectives of various professionals are required as part of the diagnostic process, which usually includes an assessment conducted by a qualified professional

ASD in Ontario

In the 2005-06 school year, school boards in Ontario reported that 7,888 students in publicly funded elementary and secondary schools were identified by an Identification, Placement and Review Committee (IPRC) as exceptional under the Communication–Autism category. This is more than double the 3,348 students who were reported to be identified in the same category in the 1998-99 school year (Ontario Ministry of Education, School October Reports). who specializes in developmental disorders, such as a child psychiatrist, developmental pediatrician, child

psychologist, or neuropsychologist. Accurately diagnosing ASD in a student typically includes the following:

- Assessments of multiple areas of functioning, such as intellectual and communication skills
- A review of developmental history
- Parental input

Diagnosis is not a quick process and is much like putting together pieces of a puzzle. ASD is diagnosed through the presence or absence of certain behaviours, characteristic symptoms, or developmental delays. Often several tests may be performed to rule out other medical conditions, such as a loss of hearing that may be causing the social and communication impairments, before considering a diagnosis of ASD. In many cases, ASD will be diagnosed in children before they begin attending school, often when they are between two and three years of age, although in some cases the child will start school prior to receiving a medical diagnosis of ASD. Parents often share with health professionals the information provided by educators about concerns related to the development of a child's social, communication, and behaviour skills in the school setting. This information may assist in the determination of an appropriate medical diagnosis.

3.4 Characteristics of Students with ASD

Several disorders are included in the diagnosis of ASD, and the symptoms and characteristics of each disorder can be present in a variety of combinations and develop within a continuum of severity. The degree of impairments can range from mild to profound and will affect individuals very differently. Characteristics vary widely. For example, some students may be overly sensitive and display extreme reactions to sensory stimuli, There is no single impairment, behaviour, or ability level that identifies ASD. Although ASD is typically characterized by impairments in communication and social interaction, as well as unusual patterns of behaviours, interests, and activities, the extent of the difficulties will vary considerably across individuals and within an individual over time. (National Research Council, 2001)

3.5 Area of Impairment The student:

Social skills

- demonstrates difficulties interacting with peers and adults
- has difficulty reading and understanding social cues or situations
- withdraws from or provides unusual responses in social situations
- engages in play that is lacking in the imaginative qualities of social play

Communication skills

- has difficulty communicating thoughts and needs verbally and non-verbally
- has difficulty with non-verbal communication, such as use of gestures, pictures, eye contact, and facial expressions
- uses speech that includes repetitive, echolalic, or unusual language

Behaviour

- displays obsessions or preoccupations with specific themes or objects
- likes order and may line up toys repeatedly
- engages in unusual behaviors, such as rocking, spinning, or hand flapping
- gets extremely upset with changes in routine or schedules
- has an unusual response to loud noises or other sensory stimuli while others do not respond at all to the same stimuli.

Also, although students with ASD generally have impairments in both verbal and non-verbal communication skills, those with Asperger's Disorder usually have relatively good verbal skills. The following chart describes some characteristics, related to the above-noted impairments, that students with ASD may exhibit in the classroom. The general level of intelligence can vary significantly across the population of students with ASD. Some individuals will demonstrate normal levels of cognitive development, while others will have profoundly impaired or abnormal development of cognitive skills. For example, about 75 per cent to 80 per cent of individuals with Autistic Disorder will have significant cognitive impairments. Many individuals with

Asperger's Disorder, however, will have average to above-average intellectual functioning (Perry & Condillac, 2003).

The profile of various cognitive abilities in an individual may also be uneven, regardless of the individual's general level of intelligence. In some cases, special or "splinter" skills are present: an individual has highly developed skills in a very specific area of ability, such as long-term memory, but the same individual also has severe impairments in the development of other skills. For example, some individuals with Autistic Disorder may have excellent recall of technical data or mathematical formulas, but they repeat the information over and over again in a context that is socially inappropriate. In many cases, comorbid (or additional) disorders, such as an anxiety or mental disorder, are diagnosed in individuals with ASD. The effects of the symptoms of other disorders can increase the severity of impairments for individuals with ASD.

3.6 Parent Involvement

Parents play a vital role in the education of their children. As partners in the process, parents can provide perspectives and information that will broaden educators' understanding of the student. Parent participation will enhance program planning and assist in the determination of educational goals, methods, and motivational strategies that are most appropriate and effective for a student. A consultative partnership between home and school is developed through regular, frequent opportunities for discussions about the student's unique learning needs, evidence of progress, and any adjustments to the educational program that may need to be considered.

Students with ASD generally have difficulty transferring or generalizing skills from one situation or environment to another. The learning process for a student with ASD is supported when the same skills and concepts are reinforced in both the home and school environments. Involving families in planning their child's educational program may assist in developing the student's ability to learn. Families are often able to provide additional opportunities for the student to practise skills. This may also enhance the student's ability to generalize skills to various settings. Students will be more successful in learning and generalizing skills when families and schools share a common focus, approach, and goals. Having a child with ASD is a challenge for any family and may have an emotional, physical, and financial impact on the family (Higgins,

Bailey, & Pearce, 2005). The impact on families can vary considerably across situations. Some families are able to cope with these additional stressors; however, for others it can be

3.7 Program Planning

- Developmental history
- Health issues
- The range of professionals who are or have been involved with the student and the services provided
- Their child's likes dislikes, special interests, and sensory sensitivities
- Effective positive reinforcers and motivators
- How the student has learned skills at home
- Behavior and communication strategies that have been successful at home and in other environments
- Student performance across settings and over different periods of time
- Perspectives on the student's personality. more difficult. It is important to keep in mind that many parents are on their own journey of acceptance of the diagnosis and its significant implications for both their own lives and the life of their child. The needs of families who have children with ASD can depend on:
 - The particular characteristics of their child (such as age, level of functioning);
 - The parents' own interpersonal resources;
 - The availability and effectiveness of supports and services.

When interacting with families, it is important to be sensitive to these issues (Perry & Condillac, 2003). Parents are advocates for their child's best interests. It is essential that they have opportunities to be involved as partners in the ongoing planning and review of educational programs for students with ASD. A strong partnership between schools and families is the foundation for a student's ongoing, successful experience at school.

- **Home/School Communication**

Positive relationships between home and school have been shown to improve parental involvement and increase parents' comfort level in working with the school.

- Activities in which the student participated
- Any new or particular skills that were demonstrated
- Nature of play with friends and classmates
- Songs and stories of the day
- New themes or areas of learning
- Upcoming, special events, trips, or snack days.

The nature of the communication between home and school can have a significant impact on the quality of this relationship, as well as on a student's program at school and the ongoing development and generalization of skills. Many parents report that they wait expectantly to read the communication book at the end of each day and that their emotional state can be considerably influenced by its content. Thus, it is important to be cognizant of the messages that are sent and how the messages are stated. It is important that the teacher, TA, principal, and parent meet to discuss and establish methods for home/school communication. Consideration should be given to the format, information to be included, information from home that will be shared, and the constraints of time for all parties. As well, guidelines for reporting any significant behavioural issues need to be clearly established. The classroom teacher is responsible for the content of the home/school communication. In some situations, while a TA may guide the student through the process of reporting daily events, this should be reviewed by the teacher. As a guideline to positive practice, the teacher should use methods such as a home/school communication book or phone call to maintain contact with the parent on a regular basis to provide updates on the student's activities and progress.

- **Health Issues**

Some children have serious health issues, such as seizures or allergies, about which parents need more frequent information. Any serious occurrence should be reported to parents immediately. Collaboration between parent and school is essential to mutually decide on a way to report on health-related issues that is efficient for both home and school. In any situation where health issues are a serious concern, safety protocols must be created and shared with all staff.

- **Reporting Behavioral Difficulties**

Most parents are aware of the behavioral difficulties of their child, and it can be a disheartening process to regularly receive a listing of their child's misbehaviors throughout the day. In most situations, it is unnecessary to report the daily incidences of non-compliance, off-task behaviour, and other occurrences if these are an ongoing or typical component of the child's behavioural profile.

There will be times when reporting and discussion of behaviour are necessary and unavoidable. When significant behavioral incidents occur, they must be reported to parents. However, the communication book is *not* the forum for sharing this kind of information. The principal decides how this information is to be shared with parents, and often assumes this important role. In composing the content of daily communication, it can be helpful to ask yourself these questions: What if the communication book became lost and was found by another student or a stranger? Would the child's or the family's dignity or privacy be compromised?

While it will take a few extra minutes each week to manage the home/school communication process, the responsive and responsible reporting of the events of a student's days at school makes a positive contribution to both the home and the school experience. Parents feel informed, supported, and more able to support the learning needs of their child.

3.8 Individual Learning Profile

Decisions about what to teach or how to teach an individual student should not be based solely on the diagnosis of ASD. No one method or intervention will meet the needs of all students with ASD, as individual students differ in their abilities and their needs vary considerably (National Research Council, 2001). Some students with ASD may have developmental delays or an intellectual disability and experience challenges with a number of basic, pre-learning,

developmental skills, including imitation, joint attention, and generalization, while others may have much

- Involve the student in creating the system. Students can be partners in preparing pictures or picture symbols, writing, stamping, or applying stickers in the home/school communication book. This gives the student a sense of ownership and responsibility for the process.
- Often, the job of writing in the communication book is left until the end of the day. In some cases, however, it is possible to connect the student's visual schedule to the daily home reporting process. This method is helpful to reinforce the schedule and involve the student directly in communication with parents. Make the reporting process as positive as possible. The preparation and maintenance of the communication book can be a valuable component of the literacy program for some students.
- Respect the privacy of families.
- Comments should be objective. Always imagine yourself in the position of the parent receiving the information.
- When possible, pictures of the student involved in school activities can provide an excellent stimulus for information retrieval and communication. Reviewing pictures of school activities while at home will help some students to generalize thinking between the home and school environments.

Establishing a Successful Home/School Communication Process more highly developed skills.

Program planning for students with ASD, as for all students with special education needs, should be individualized and focused on developing skills that will be of use in the student's current and future life in school, home, and community. This requires careful planning, preparation, and teamwork so that the programs provided are appropriate and effective for the individual student. Determining the educational interventions that may be required to meet the learning needs of students with ASD begins with an understanding of the unique learning profile of each student. This requires consideration of information from a range of sources, such as the following:

- Data and reports from assessments

- Observations of skills and behaviors through the student's participation in assignments, activities, tasks, and projects
 - Ontario Student Record (OSR)
 - information and reports provided by parents, former teachers, and other involved professionals, such as service providers from outside the school system
- In developing the learning profile for a student with ASD, it is important to consider information gained from observations of the student. Observations should be as objective and descriptive as possible and avoid value judgments. In particular, educators should note the student's:
- Progress towards learning outcomes;
 - Behaviors and interactions that occur in the school and classroom environment, including recess and lunchtime;
 - Age and social appropriateness of interactions with peers and adults;
 - Social language skills;
 - Reactions to the learning environment and the setup of the classroom;
 - Responses to transitions in the school day – between activities in the classroom, for example.

3.9 Assessment

Assessment information from various sources will need to be considered to develop individual learning profiles and establish appropriate educational programs for students with ASD. Comprehensive assessments are necessary to recognize and understand the various strengths and needs of individual students. It is important that generalizations or unrealistic expectations about overall abilities are not made on the basis of the diagnosis, individual skill impairments, or splinter skills. Assessment data and results that have been gathered across a diverse range of skills by a variety of professionals will provide comprehensive information that will be useful in making accurate decisions about programs for students. A diagnosis of ASD is associated with impairments in communication, social, and behavioural skills, and multidisciplinary assessments

in these skill areas will provide information that will be used to determine the extent of the impairments and how the difficulties interfere with the educational process.

When a student in whom ASD is diagnosed is first enrolled in school, parents should be invited to participate in a transition planning process that includes the sharing of any relevant assessment information with the school. This information will be considered in:

- determining the student's strengths and needs;
- deciding learning goals for the student;
- making decisions regarding programs, services, and supports that may be required to meet the student's needs;
- establishing records against which to measure future achievement.

Parents should be encouraged to participate in an ongoing process of sharing any relevant, updated assessment information with the school. requires school boards to have procedures to identify each child's level of development, learning abilities, and needs. A continuous assessment and program planning process should be initiated when a child is first enrolled in school and continue throughout the child's school life. Assessment of student progress by a classroom teacher is a continuous, complex process that is an integral part of the learning-teaching process every day. Teachers use a variety of methods to gather information about a student's achievement, the level of the student understands, and the effectiveness of a particular teaching technique. Examples of classroom-based assessment methods that are used by teachers on a regular basis to assess student learning include the following:

- Observations
- Teacher-designed tasks
- Interviews with the student
- Criterion-referenced academic tests
- Functional assessments

During the school day, there are ongoing opportunities for teachers to assess students across a variety of settings and situations. The information and data collected by teachers are primarily used for planning programs that will be appropriate to each student's strengths, interests, needs, and level of functioning. The choice of assessment instruments is a complex one and depends on the student's:

- Verbal skills;
- Ability to respond to complex instructions and social expectations;
- Ability to work rapidly;
- Ability to cope with transitions in test activities. (National Research Council, 2001:28)

The responsiveness of a student with ASD to an assessment task may be affected by the novelty and structure of the assessment situation. Consideration should be given to the possibility that the results of an assessment may indicate the student's response to the assessment task or situation and may not be an accurate reflection of the student's abilities. Assessment accommodations may be required to allow the student with ASD the opportunity to demonstrate achievement of specific skills or expectations. Examples of assessment accommodations that may be required for students with ASD include providing:

- Visual supports to clarify verbal instructions;
- Additional time for student responses;
- Alternative methods for the student to demonstrate achievement of skills;
- Alternative environments for assessment tasks. An effective assessment process is continuous and includes ongoing, systematic data collection that is necessary to:
 - monitor student progress;
 - evaluate instructional effectiveness;
 - update goals as a student learns and masters a skill. Information about skill development in multiple areas of functioning should be considered when decisions are made about students'

programs. This would include collecting and analyzing data and other information related to progress within various skill areas such as communication, social, and behavioural skills,

- Recognize that verbal responses may be the most difficult and least accurate.
- If a verbal response is required, do not insist on eye contact.
- Provide the student with methods to answer questions non-verbally, if possible. For example, provide Yes/No or True/False cards that the student can touch or point out to indicate the answer.
- Allow the student several readings of a passage before asking comprehension questions. This may include reading the passage to the student.
- Begin an assessment process by starting with material two or more grades below the ability level of the student. Gradually work up to the student's level.
- Provide the student with sample questions to practice with the assessment format.
- Allow the student to answer questions using the computer.

3.10 The Individual Education Plan (IEP)

Many students with ASD will be identified by an Identification, Placement and Review Committee (IPRC) as exceptional students under the Communication–Autism category. As described in *Special Education: A Guide for Educators* (Ontario Ministry of Education, 2001), the definition that the IPRC considers in determining the exceptionality of Communication–Autism is: A severe learning disorder that is characterized by:

a) disturbances in:

- Rate of educational development
- Ability to relate to the environment
- Mobility
- Perception, speech, and language

b) Lack of representational symbolic behavior that precedes language.

The IPRC will determine the most appropriate placement to meet the individual needs of students with ASD who are identified as exceptional pupils. Parents and students (if 16 years of age or older) are invited to attend the IPRC meetings and participate in the committee discussions. A range of placement options and services should be available for students with ASD to address the broad range of needs of these students. The needs of many students with ASD are met in regular class placements with appropriate supports. Other students require placement in special education classes for all or part of the school day. The IEP is a written program plan that describes the special education program and/or services required by a student on the basis of a thorough assessment and understanding of the student's strengths and needs. The information gathered to understand a student's individual learning profile will be an important resource in the development of an appropriate program plan for the student. The IEP should be reviewed and updated regularly, at least once in every reporting period, and based on ongoing and continuous evaluation of the student's progress. The IEP is a working document that identifies the accommodations that are required to help the individual student achieve learning expectations and demonstrate learning. The IEP also identifies the modified or alternative learning expectations, where appropriate, that are planned for a student's educational program and the specific knowledge and skills to be assessed and evaluated for the purpose of reporting student achievement. It is helpful, for purposes of planning and IEP development, to classify the subjects or courses and alternative programs in which the student will receive instruction according to the following categories, as appropriate to the student's individual requirements:

- No accommodations or modifications
- Accommodated only
- Modified and/or
- Alternative.

For some students with ASD, the most appropriate program is based on expectations from the Ontario curriculum with minimal or moderate accommodations or modifications. Other students may require a program that includes significantly modified expectations or is mainly composed of expectations that are an alternative to the Ontario curriculum.

Accommodations are the special teaching and assessment strategies, human supports and/or individualized equipment required to enable a student to learn and demonstrate learning. Accommodations do not alter the provincial curriculum expectations for the grade.

Modified expectations differ in some way from the regular grade expectations as outlined in the Ministry of Education's curriculum policy documents.

Alternative expectations are developed to help students acquire knowledge and skills that are not represented in the Ontario curriculum.

(The Individual Education Plan (IEP): A Resource Guide, Ontario Ministry of Education, 2004) goals for students with ASD often need to address social, communication, and adaptive skills that are not part of standard curricula (National Research Council, 2001). For many students with ASD, the most effective education program includes a combination of instructional goals based on the Ontario curriculum with accommodations or modifications, as required, and includes alternative programs with specific goals and activities to support the development of functional skills that are useful and meaningful for the student.

Parents and relevant school board and community personnel who have previously worked with and/or are currently working with the student should be invited to provide input and participate in the IEP process. This may include, but is not limited to, consultations with current and previous teachers, the principal, the student, a psychologist, special education staff, an OT, service providers from community agencies, and autism program providers, as appropriate. All members of the IEP team have important roles and responsibilities in the IEP process. It is important that the teacher(s) responsible for the direct instruction and assessment of the student's progress work collaboratively with parents and other involved professionals to determine the student's programming needs and appropriate learning expectations. Effective educational programming for students with ASD should be based on a student's abilities and gradually increase in complexity as skills develop. In order to achieve this, programs should be carefully planned and constantly evaluated using a variety of formal and informal assessments. The program should then be modified on the basis of assessment results and student progress.

3.11 Collaborative Planning

A collaborative planning approach to support students with ASD is most effective and promotes the best outcomes for students. Keeping this in mind, it is important for school staff to invite input from and the participation of the parent(s) and, with parental consent, other professionals who have previously worked with and those who are currently working with the student. Students with ASD have a broad range of needs and abilities. The perspectives, information, and resources from parents and various in-school, school board, and community professionals will enhance the effectiveness of the program planning process.

Many school boards in Ontario have established school-based support teams that play a significant role in helping teachers to plan and implement programs for students with special learning needs. The school team is made up of school staff members who work together with the family to collaborate, consult, and share information and knowledge to identify strategies that may increase the student's learning success. As circumstances require, the team may also seek assistance from other resources, such as the following:

- Community associations and agencies (such as Autism Ontario)
- Service providers from the Ministry of Health and Long-Term Care (such as CCACs, which coordinate service providers such as OTs)
- Service providers from the Ministry of Children and Youth Services (such as staff of the regional autism service provider agencies)
- Children's Treatment Centres (CTCs)
- Children's Mental Health Centres

Multidisciplinary teams composed of professionals with expertise in a variety of areas have also been developed by some school boards to provide additional support to schools in the planning of effective programs for students with ASD. The membership of the multidisciplinary team could include a consultant, an SLP, a psychologist, an OT, and others who have the experience and knowledge to provide information, resources, and recommendations related to meeting the needs of students with ASD. Through working collaboratively with other individuals, staff are able to better understand a student's learning profile and become aware of interventions that have

successfully supported other students or the student in other environments. With this knowledge, staff can plan appropriate program goals and interventions for the student.

Some school boards have found it helpful to develop protocols with local community agencies to identify responsibilities processes for working together. These local agreements are used to support collaborative partnerships by identifying and resolving issues that may arise. Collaborative efforts will be enhanced and most effective when those involved are committed to working together for the best interests of the student.

3.12 Universal Design for Learning

Education for All(Ontario Ministry of Education, 2005a), provides educators with recommendations on a broad range of techniques to enhance the instruction of students with special education needs, including those with ASD. The report provides information on the use of the principles of Universal Design for Learning (UDL) and differentiated instruction to plan for and respond to students with various needs. Learning is a continuum and every student is a unique learner. Flexible, supportive, and adjustable classrooms and programs must be planned and developed to meet the learning needs of individual students. These are important considerations in the planning of programs for students with ASD.

UDL is recommended for consideration by teachers to guide the planning of the various components of teaching, such as defining the expectations of learning situations and determining the teaching strategies and assessment methods that will be required. Programs designed according to the core concepts of UDL are based on considerations of the following:

- Universality and equity
- Flexibility and inclusion
- An appropriately designed space
- Simplicity
- Safety

Universality and Equity

Teachers are encouraged to develop a class profile to identify the strengths, challenges, and needs of all students and to determine the stage that each student has reached in his or her learning. The instructional methods and classroom accommodations should be planned to ensure that the needs of all students are met. Students with ASD vary in their cognitive level, communication ability, social skills, and behavioural characteristics. They have a wide range of skills and diverse needs. It is important for teachers to gather information to understand the individual strengths, needs, and interests of students to identify appropriate curriculum expectations, required accommodations, and effective instructional approaches.

Flexibility and Inclusion

To ensure that all students are provided opportunities for real learning experiences, flexibility and options should be built into the planning of teaching strategies, materials, and student activities.

The ability of students with ASD to participate in or respond to a learning experience may be affected by limitations in communication and social skills. Consideration of the likes, dislikes, strengths, needs, and interests of a student with ASD can help to make materials and tasks more engaging and provide motivation for the student to participate in and complete the desired task or activity. Concrete examples and hands-on activities provide students with opportunities to learn by seeing and doing and have been found to increase the motivation and engagement levels of students with ASD. For example, a student who has difficulty with concepts in mathematics may be motivated to learn graphing techniques by gathering data on items of personal or special interest.

Many students with ASD have difficulty processing information and are unable to respond immediately and “on demand” to expected tasks. They often require flexibility regarding the timing and method used to demonstrate their knowledge and skills. Teachers will need to consider the various alternatives, such as extended timelines and additional activities that may need to be planned to ensure that appropriate learning experiences are provided for all students.

An Appropriately Designed Space

Consideration should be given to the size, space, and arrangement of the physical and visual elements in the learning environment to ensure that they are conducive to student learning. Students with ASD may be unusually sensitive to sensory stimulation, which can be reflected in an increased sensitivity to the physical environment of the learning situation. The classroom is filled with many sensory demands that can be overwhelming for some students. Although some of these demands, such as noise in the hallway and fire alarms, are unpredictable, teachers should monitor the physical environment to ensure that the sensory distractions (such as auditory and visual stimuli) for a student with ASD are minimized as much as possible. For example, tennis balls can be used to cover the bottoms of chair legs to reduce classroom noise.

Some students with ASD are very aware of and need to know where things belong. The organization of materials, furniture, and resources should be carefully considered in relation to their effect on the learning environment of the student with ASD.

Simplicity

Teachers should ensure that the information provided in learning situations is presented clearly and is easily understood by the students. Unnecessarily complex and distracting information should be reduced as much as possible. The communication impairments that are often present in students with ASD may affect their ability to process verbal information. They will often have difficulty understanding complex, abstract language and may misinterpret metaphors, slang terms, and colloquialisms. Effective methods to simplify information and make it easier for the student to understand include using clear and concise language, breaking instructions and tasks into smaller steps, and using visual supports, such as written or picture schedules. Information and materials should be organized in such a way that important or key components are highlighted and easily identified by students.

Safety

Teachers need to consider possible safety hazards and elements with the potential to cause accidents in the classroom. Staff should be aware of and able to act on any safety assessments, safety plans, or safety protocols that may apply to specific students in the classroom.

If a student with ASD is considered to be a safety risk in the school setting, adults involved with the student need to have access to the information and supports that are required to ensure a safe learning environment. A safety assessment can help to identify the factors that may lead to or cause situations and provide an assessment of the potential risks in given situations. A safety plan outlines the appropriate responses and supports that are required during specific situations with students.

3.13 Planning for Transitions

Common characteristics associated with a diagnosis of ASD are restricted, repetitive patterns of behaviour, interests, and activities. Many individuals with ASD have difficulty coping with novel and unexpected events. Change, including transitions between activities and environments, is often difficult for students with ASD and can lead to increased anxiety and unusual or inappropriate behaviours.

In school, transitions happen at various stages and levels for students. Some transitions occur on a regular basis between activities and settings within the routines of the school day. Other transitions, such as class excursions, occur less frequently. Significant transitions such as entry to school, between grades and divisions, from elementary to secondary school, and from secondary school to the postsecondary destination happen periodically, are more complex, and include significant changes to many aspects of a student's routines. Planning for transitions provides the foundations for successful transition experiences that help a student learn to cope with change and adapt to a variety of settings. Transitions cannot be avoided, The National Research Council (2001) found that no one specific method or intervention is effective for all individuals with ASD and that integrating a variety of approaches leads to the development of programs that promote the best outcomes for students.

3.14 Differentiated Instruction

As recommended in *Education for All* (Ontario Ministry of Education, 2005a), teachers can effectively respond to a learner's needs and strengths through the use of differentiated instruction. Through this approach, the specific skills or difficulties of students with ASD can be addressed by employing a variety of methods to differentiate (or vary) the following:

The content:The depth or breadth of the information or skills to be taught.

The processes:The instructional approaches used with the student, as well as the materials used to deliver or illustrate the content.

The products of the learning situation:What the end product will be or look like. This product may be tangible (a worksheet, project, composition), a skill that has been acquired, or knowledge that has been gained. To determine the most effective strategies for students in a learning situation, it is necessary to consider the learning goals for the student in the context of the following questions:

- **What** do we want the student to learn? What essential skills or understandings do we want the student to acquire?
- **Why** is the student learning this? How does the learning goal fit into the goals outlined in the IEP for the student?
- **How** will the student best learn this? What type of activities, materials, and supports are appropriate and effective for the student? How will the student demonstrate learning?

3.15 INSTRUCTIONAL STRATEGIES

To differentiate instruction, teachers should consider adaptations to the curriculum, instruction, or expectations that may be required according to a student's readiness, interests, and learning profile. Elements that can be varied to differentiate learning activities Examples of differentiation for students with ASD

Student groupings

- Establish multiple or similar ability grouping.
- Divide groups according to similar interests.
- Clearly define the role for each student in a group (e.g., note-taker, reporter, artist).

Instruction activities

- Support verbal information with visuals.

- Include explicit (or direct) teaching of information.
- Provide opportunities for the student to practice and rehearse.

Level and type of support provided to the student

- Include tasks that the student can participate in or complete independently.
- Provide opportunities for peer support through small-group activities.
- Have more intensive levels of assistance available as required.

Task expectations (time, type of task)

- Build in flexible or extended timelines for task participation and completion.
- Chunk longer activity into smaller segments (e.g., divide a 45-minute activity into several shorter sections).
- Encourage use of technology to complete a task (e.g., keyboarding for written tasks).

Materials and resources

- Match materials and resources to the readiness level, interests, and learning profiles of students (e.g., a student with keen interest in vehicles collects data on cars in a parking lot and graphs results).

Assessment activities

- Provide a variety of opportunities and ways for the student to demonstrate learning.
- Allow alternatives to written tasks, such as drawings, fill-in-the-blank activities, questions with multiple-choice answers, pointing to the correct answers, scribing.
- Include visual supports such as graphic organizers or highlighting of key words.

Differentiation involves an ongoing process of monitoring student response to the differentiated strategies and evaluating student progress on a regular basis. Strategies that are found to be effective for a student during one activity may be less effective over time or during another

activity. The level and type of differentiation will need to be varied according to the student's response and progress. Data from assessments and observations should be used to inform decisions about the effectiveness of methods being used and further differentiation that may be required.

3.16 Visual Supports

The use of visual supports is one of the most widely recommended strategies for teaching students with ASD, as they usually process visual information more efficiently and effectively than information that is presented verbally. Some students may require extra time to process verbal language and understand the message. Speech is transient: once information or instructions have been spoken, the message is no longer available and students must recall the information from memory.

Visual images help students to understand information as they provide a source that can be referred to as often as necessary and for the length of time that is required in order to process the content of the information. Visuals are mainly used to:

- **Improve communication:** provide a reminder or cue of what to say or do in a situation
 - **Provide information:** supplement or provide an alternative to verbal information
- Visual supports can include visual schedules, checklists, task exemplar sheets, choice boards, illustrated task sequences, printed instructions, rules or topical/content materials, tip sheets, safety signs and messages, videos, and story illustrations, as well as illustrations based on theme content (e.g., the Pioneer Unit). These are two effective methods of visual support:

1. **Passive modelling** (Biederman et al., 1998): Students are able to see what is expected in a task by being provided with visual examples and demonstrations of how tasks are performed. Instructional language is reduced as much as possible.

2. **Video modelling** (Bellini & Akullian, 2007): Further to and building on the concept of passive modelling, video modeling provides the students with a video example of how tasks are performed and task sequences. This format allows the student to watch the instructional sequence over and over. In terms of practice, both passive modelling and video modeling are viable tools. Through the use of a digital camera, short instructional sequences on CD-ROMs can be prepared

quickly and easily. These sequences can be a powerful addition to other instructional methods and materials. Visual supports are usually simple and inexpensive. They will be most effective for a student with ASD if they are used consistently and across various settings. For example, a visual schedule used at home to organize the after-school activities for a student will be easier for the student to understand and use if it is referred to daily and follows a format similar to the one used at school to structure the student's school day.

3.17 Structured Learning Environment

All children function better in a predictable environment. Students with ASD require a structured learning environment to know what is expected of them in specific situations, to assist them in anticipating what comes next, and to learn and generalize a variety of skills (Iovannone et al., 2003). Rules and expectations should be clear and consistent and include specific information regarding the expectations for appropriate behaviour. It is also important to structure the physical environment so that it is organized with "a place for everything and everything in its place". The student's seating arrangement needs to be consistent and in a location that affords as few distractions or exposure to sensory irritants as possible. It may be necessary to do environmental scans in all areas of the school that the student will access (e.g., the gym, library, music room) to determine what elements may have an impact on the student's ability to participate effectively. Appropriate accommodations can then be made to the environment. Developing as much consistency as possible in the environment, schedule, and instructional approaches provides structure and routines that may increase the comfort level and reduce anxiety for students with ASD. However, there are situations in which changes to the environment and routines during the school day are inevitable. Providing students with what they need to be prepared for these changes, such as advance warnings and concrete visual information, can help them to become more flexible and adaptable to change. Once students know the expectations of their visual schedule and transition system, flexibility training can be built into the process.

- Posting the student's individual rules or guidelines, as well as classroom rules, in a visually accessible location (e.g., beside the student's desk or workspace)
- Providing clear information about task expectations
- Introducing new or unfamiliar tasks in a familiar environment

- Establishing routines and regularly scheduled activities
- Developing a visual schedule of activities
- Providing warnings before transitions or changes to routines
- Having specific places where materials are stored
- Following a consistent work system.

Strategies for structuring the learning environment for a student with ASD include:

Recording changes in the student's visual schedule and providing opportunities for the student to rehearse and review upcoming changes have been found to be effective in preparing students with ASD for changes to the learning environment and routines. For example, if the student's regularly scheduled gym class is cancelled because the gym is being used for another activity, the student will need preparation to understand this change. The change could be explained to the student and visually recorded in the student's daily schedule with an alternative activity, ideally one that is highly preferred by the student. Involving the student in making the change to the visual schedule and visiting the gym to see the other activity being set up may help the student to understand the change.

Students with ASD are most successful in an organized environment that is structured to be as predictable and understandable as possible.

3.18 Assistive Technology

In *Education for All* (Ontario Ministry of Education, 2005a), assistive technology is defined as any technology that allows one to increase, maintain, or improve the functional capabilities of an individual with special learning needs (Edyburn, 2000). Its applications and adaptations can help open doors to previously inaccessible learning opportunities for many children with special needs (Judge, 2001).

Assistive technology includes highly technical (commonly referred to as "high-tech") computerized devices such as speech generating software, as well as less technical ("low-tech") resources such as visual supports. Technology can be used by students to provide alternative

methods to access information, demonstrate and reinforce learning, and interact with others. It can also be used by adults as a tool to support the teaching and learning process. A research study investigating the use of assistive technology with students with ASD in British Columbia (Randle, 2005) used a survey to identify a wide variety of uses of technology that include supporting the following:

- written output
- Academic concept development
- Motivation
- Communication
- The development of social skills

As the learning needs of students with ASD are diverse, it is important that a student's specific technological needs be evaluated and that the use of assistive technology be carefully planned. The potential benefits of assistive technology for individual students should be considered from a multidisciplinary and cross-curricular perspective. For example, technology that is used to support a student's communication skills may also be found helpful to accommodate a student's fine motor difficulties and increase written output.

The support that is provided through the use of assistive technology will change over time for a student and may vary across activities. For example, a student may require the use of text-to-speech software to support the understanding of information in one subject area, but may not require it to read and understand materials on a topic of interest.

- Match between the technology and the student's need
- Positive attitude towards the student's use of technology to access learning
- Interest and comfort level of the teacher and TA in using technology
- Reliable and timely access to technical support, when required
- Student comfort level in using technology with gradual introduction of new hardware and/or software, when required

- Perseverance through problems and challenges.

The survey responses from school-based teams in a study on the use of assistive technology with students with ASD in British Columbia indicated that successful implementation of technology appears to depend on several factors such as:

Ongoing monitoring of students' needs and use of assistive technology is important to determine if the technology is being used effectively and is providing the student with the intended support. Collaboration between parents and professionals will help to ensure that the technology that is being used is appropriate to meet the needs of the student and is as multi-functional as possible.

3.19 Sensory Considerations

Students with ASD vary in their sensitivity and tolerance to sensory stimulation in the environment. It is important to be aware of the sensory preferences or sensitivities of a student and to determine possible elements in the environment that might have an impact on a student's learning and level of anxiety. Some students are very ("hyper-") sensitive in one or more sensory areas and may be more comfortable in environments with reduced levels of sensory stimulation. Other individuals are under ("hypo-") sensitive and seek enhanced sensory experience. For example, some students become anxious or upset because of an extreme sensitivity to certain sounds or have a difficulty processing more than one sense at a time. Other students seek additional sensory experiences in order to become or maintain calm. It is usually possible to determine strategies or accommodations that are effective to manage sensory issues.

Special Equipment Amount Funding

Special Equipment Amount (SEA) funding assists school boards with the costs of equipment that is not available through the board's normal textbook, supply, and/or computer purchasing arrangements. Equipment funded through SEA must be recommended by a qualified professional to be essential to support students with special needs and provide students with accommodations to access the Ontario curriculum and/or a board-determined alternative program and/or course and/or to attend school. Evidence of the use of equipment must be documented in the student's IEP. For additional information on Special Equipment Amount funding, see *Special Education Funding Guidelines: Special Equipment Amount (SEA) and Special Incidence Portion (SIP)*,

2007–08 (Ontario Ministry of Education, 2007b). environmental accommodations can be made quite easily. In others, students can be taught socially appropriate ways to access sensory materials and experiences. Proactive strategies to manage sensory issues include:

- providing predictable, scheduled breaks for sensory input;
- providing a variety of sensory materials and/or equipment that will mitigate a student's particular sensory needs;
- using engagement in sensory activities as reinforcement for task completion and other classroom requirements or expectations;
- performing environmental scans across all environments in a school that the student may access in order to determine possible sensory irritants and make adjustments accommodations as appropriate.

Parents are a valuable resource in providing detailed information about their child's sensory issues. OTs are usually involved in determining the sensory-based activities and materials that will enable a student to regulate his or her arousal level and associated behaviour.

3.20 Applied Behaviour Analysis (ABA)

As outlined in Policy/Program Memorandum No. 140, "Incorporating Methods of Applied Behaviour Analysis (ABA) into Programs for Students with Autism Spectrum Disorders (ASD)", 2007, ABA is an effective approach to understanding and changing behaviour, and teaching new skills. ABA uses methods based on scientific principles of learning and behaviour to build useful repertoires and reduce problematic ones. In this approach, the behaviour(s) to be changed are clearly defined and recorded, and the antecedents and reinforcers that might be maintaining an undesirable behaviour, or that could be used to help develop alternative or new behaviours, are analysed. Interventions based on principles of learning and behaviour are then designed and implemented to develop appropriate behaviours. Progress is assessed and the program is altered if necessary (adapted from Perry, A. & Condillac, R., 2003). ABA can be used as an instructional approach with students of every age. It can be applied in a variety of situations, and it can be employed for very limited and specific purposes such as the development or reduction of single behaviours or sets of behaviours. ABA methods can be used with varying degrees of intensity along a student learning continuum. ABA is used according to

the individual needs of each student, and may be applied to developing academic skills or behaviours related to social skills, communication, or self-care. The focus in an instructional approach that uses ABA is on measuring and tracking behaviours over time, determining the function of the target behaviour (for the student), and altering the behaviour (either increasing or decreasing its occurrence) by providing intervention. Student progress is based on ongoing data collection and assessment, as measured against the identified objectives. The ultimate goal of using ABA methods is the generalization of the learned skills and behaviours to other settings and situations.

ABA methods can be used to:

- increase positive behaviors;
- teach new skills;
- maintain behaviors;
- generalize or transfer behaviour from one situation to another;
- restrict or narrow conditions under which interfering behavior occurs.

Examples of teaching strategies using ABA methods include prompts, modelling, reinforcement, task analysis, forward chaining, backward chaining, Discrete Trial Training, and shaping. These are discussed below.

Prompts: Prompts are cues or assistance to encourage the student to provide the desired response. There are many natural prompts in the environment, for example, seeing the school bus arrive, hearing a bell ring, or noticing that students are lining up at the door.

Prompts can be:

- direct (“Put your books on the shelf in the locker”) or indirect (“Where do books need to go?”);
- Verbal, physical, or visual (gestures or pictures);
- Brief (to get the student started on a task) or intensive (step by- step through the task).

Some students have difficulty recognizing, understanding, or responding to some prompts. It is important to determine the type and intensity of prompting that the individual student requires. Students can become dependent on prompts from others, so it is important to plan for the fading of prompts when appropriate. Visual supports can provide a method of prompting that many students can learn to use independently. Here, the goal is to help the student become as independent as possible in participating and completing tasks.

Modeling: A form of prompting is modeling. It provides a visual example of what is expected in a task by having students see the task being performed. It can also help the student see the sequence of steps in the task. For example, the student learns the actions to a song by first watching a demonstration by an adult.

Reinforcement: The target behaviour is encouraged through the use of reinforcement. Reinforcement is provided after the target behaviour to increase the likelihood that the behavior will reoccur. Reinforces can be:

- Tangible (such as stickers);
- Activity-based (the student is able to participate in preferred activity);
- Social (praise or thumbs up sign).

Reinforcements must be motivating to the individual student. What motivates one student may not motivate another. Similarly, what is motivating to an individual student may change over time. It is important that monitoring be used to ensure that the reinforcer remains effective. In some cases, developing a “menu” of reinforcers may be helpful so that students may select from several options. To encourage students to be as independent as possible, it is important to gradually change reinforcers from others to more natural reinforcers, such as the satisfaction of completing a task.

3.21 Summary

Asperger’s Disorder, commonly known as Asperger’s syndrome or Asperger’s, is one of the disorders found within the range of ASDs. Some unique features related to a diagnosis of this

disorder are outlined in *DSM-IV-TR* (APA, 2000), and specific strategies should be considered for students in whom this disorder has been diagnosed.

The most significant differences between students with Asperger's Disorder and those in whom another ASD is diagnosed are that these students generally are considered higher functioning in many areas and do not have clinically significant delays in the development of early language or cognitive skills. Early language skills, such as using single words and spontaneous communication phrases, are usually within normal limits and in some cases may be seen as advanced or precocious during the preschool years. Many students with Asperger's Disorder have average intellectual abilities, and they may perform as well as or better than typical students in some academic areas, especially those that involve rote memory skills such as the recall of factual information.

However, students in whom Asperger's Disorder is diagnosed have a severe social deficit, which makes understanding and using age-appropriate social behaviours and social communication skills difficult. They often have difficulty learning social skills incidentally by watching and interacting with others, and often misunderstand social situations. These difficulties affect the development of friendships and relationships with peers and adults. Students with Asperger's Disorder usually speak fluently and are able to use routine language for social interactions.

They are intensely motivated to gather and share a great deal of detailed information on this topic. However, the social communication difficulties of the student usually affect the student's ability to recognize or respond to the interest level of the audience with whom the information is being shared. Usually, students in whom Asperger's Disorder is diagnosed do not have clinically significant delays in the development of age-appropriate self-help skills. However, the effects of unusual behaviours and impairments in social interaction can cause significant impairments in social, occupational, or other important areas of functioning. Motor coordination difficulties that may affect fine motor and gross motor skills, as well as organizational skills, are areas of concern for some students.

The difficulties that students with Asperger's Disorder have in academic or social situations often increase or become more obvious during adolescence. As students at this stage are expected to learn, analyse, and use more abstract and complex information, the difficulties of

students with Asperger's Disorder have an increased effect on their participation in learning activities. They often become aware of their social or academic difficulties or differences, and as a result become self-conscious or anxious in social and academic situations. Some students develop a sense of social isolation in adolescence, with increased anxieties, depression, or withdrawn behaviors.

It is not unusual for the behaviours of students with Asperger's Disorder to be misinterpreted by adults and peers. For example, a student may be able to read advanced books on a topic of interest but unable to read another book at the same reading level on a topic of limited interest to the student. This may be seen as a behaviour that demonstrates deliberate task avoidance or refusal. Because of the strengths and "normally" developing skills that students with Asperger's Disorder may have, the impairments and difficulties in social and behavioural functioning can be overlooked or misunderstood. These students often want to establish friendships or relate to others but may not have developed the appropriate skills, and their interactions or responses are perceived as being unusual. For example, in an attempt to be accurate or honest or to converse with others, a student may make an observational comment that is considered offensive or inappropriate to others. Students with Asperger's Disorder may be unaware of the impact on others of what they are saying or doing, and are at an increased risk of becoming victims of teasing and bullying by peers.

3.22 References

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Unit 4

Teaching Methods

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4.1 Introduction

The Report of the Task Group on Autism states that: 'An Autistic Spectrum Disorder is a complex developmental disability that essentially affects the way a person communicates and relates to people. The term 'autistic spectrum' is often used because the condition varies from person to person. Asperger's Syndrome is a condition at the more able end of the spectrum. At the 'less able' end of the spectrum is Kanner's Syndrome, sometimes referred to as 'classic autism'.

4.2 The Autistic Spectrum

ASD is the term used to describe the variables of autism. While there is a wide range of differences among children with an ASD, they share common impairments in the ability to:

- interact socially and appropriately;
- understand and use language and to communicate;
- think imaginatively, enjoy and participate in a range of activities; often shown in repetitive and restricted behavior patterns.

Children with an ASD also differ in their intellectual abilities. Many have severe learning difficulties and are educated in special schools, while others have average or above abilities and are taught in mainstream schools. Those with a higher level of ability are described as having Asperger's Syndrome or high functioning autism. There is much debate about the differences between both categories and readers interested in further information may find the Recommended Reading List at the end of this section of use. The following section clarifies the areas of impairment although it is important to remember that they are interdependent and can vary widely from person to person.

1. Impairments in social interaction can present as:

- apparent aloofness and indifference to other people. Pupils can sometimes relate on a simple level to an adult they know well, but not to their peers;
- Acceptance and enjoyment of social contact, but the inability to initiate and maintain it;

- inappropriate or odd approaches to other people, paying little or no attention to the response given;

- stilted and overly formal interaction, even in the most able individuals. In other words, children with an ASD think literally; they find it tiring to communicate and are often confused by the rules that govern social behaviour.

Key Teaching Points

One of the key characteristics of best teaching is remembering that children with an ASD have significant difficulty recognizing what others know and think; they do not readily grasp that others have viewpoints. It is without doubt the most problematic area for these children and will most likely present him/her with difficulties throughout their lives especially in their teenage years. Teachers need to understand that a child with an ASD will not learn social skills casually and it will be necessary to include specific strategies in the teaching lessons and to provide opportunities for the child to use skills in different situations.

Impairments in language and communication can present as:

- a wide variation in communicative ability from individuals who have not developed speech to those who have but find difficulty using their language to communicate effectively. Even those pupils with an extensive vocabulary will have great difficulties in using it in an appropriate manner;

- Difficulty with the social aspects of language, for example, turn-taking and timing;

- Impairment of social communication. Pupils often interact in socially inappropriate ways such as shouting out in class, interrupting adult conversations or constantly asking questions;

- An inability to “read” or understand the significance of gestures, facial expression, vocal intonation or body language;

- an inability to empathise with others;

- a literal understanding of language.

Key Teaching Points

With these impairments, it is essential that the teacher supports the child in understanding the purpose and value of communication. Teachers need to be aware of the problems which they may inadvertently cause by talking at length when giving instructions and by not ensuring that the individual with an ASD understands the purpose of the conversation. The best practice will include a report from a speech and language therapist detailing the child's communication difficulties and suggested interventions. In general terms, teaching instructions should focus on paying attention, imitating, understanding and using language in play and social settings. Using visual material and prompts will also promote success.

Impairments in imagination with a restricted range of behaviours, activities and interests can present as:

- An adherence to repetitive play activities, often including lining up or spinning toys. Play materials may be used in an unusual manner or the child may become preoccupied with irrelevant detail;
- engagement in stereotypical behaviour, for example, spinning, rocking, toe walking and making noises;
- encompassing narrow interests to the exclusion of other activities. These obsessive interests often include dinosaurs, computers, trains and factual information on any subject;
- Difficulty with planning and organization;
- Difficulty with change in routine. Children may show excessive anxiety or repeatedly ask questions;
- a lack of appreciation on the part of children of the consequences of their own actions and problems with the concept of cause and effect;
- a blurred distinction between fantasy and reality. Individuals with an ASD may be anxious if asked to play with others and prefer to play alone for long periods.

Key Teaching Points

The children with an ASD will have developed few coping skills to deal with situations or factors which cause them stress. Often these children will employ a number of unusual behaviours which will interfere with their ability to learn. Some behaviour may be extreme and challenging. In planning lessons, teachers need to observe the child over a period of time to ensure that they understand the conditions which give rise to the child's stress and behaviour. Best plans include strategies to ensure classroom conditions favour the child, are conducive to learning and are complemented by helping the child to cope with new and varying activities.

4.3 Additional difficulties might indicate:

- sensory and perceptual abnormalities often including over- or under-sensitivity to noise, smell, taste, light, touch or movement;
- Fine motor control problems such as handwriting difficulties;
- Gross motor control problems such as clumsiness;
- Eating, drinking and sleeping irregularities;
- limited attention span and inability to block out distractions;
- Lack of appropriate eye contact;
- speaking out of turn or interrupting;
- Over-compliance; the individual is too easily led;
- Anger or aggression when things do not go their way;
- a lack of organizational skills, for example, many children with an ASD forget important resources needed for a lesson such as books, pens, physical education kit, or glasses;
- Anxiety: unstructured or unfamiliar situations can be difficult for the pupil. The playground, corridors, dining hall or educational outings are common sources of anxiety;
- A difficulty with managing time, finishing work, handing in or completing homework, or getting to lessons on time;

- A tendency to compartmentalize their thinking: they may complete the homework but not have regard to the presentation of the work.

Key Teaching Points

The additional difficulties need to be understood by the teacher and adjustments made to the classroom, the noise level, the materials in use during lessons. An awareness of the possible problems of the environment for learning is crucial to ensure the individual with an ASD is at ease and interested.

4.4 Key Underpinning Principles

Children with an ASD differ from their peers in a variety of ways, though they may outshine them in particular areas. Many have stronger rote abilities and can do spatial, matching and perceptual tasks with greater ease than their peers. Many, for example, excel at mathematics and science. The task for the teacher is to recognize that the child with an ASD will have a mixed profile of strengths and weaknesses. In the best practice, the teacher will focus on the child's strengths while attending to the learning weaknesses and ideally aim to assist the child to manage or generalise the strategies for himself. In summary, the “**autism-aware**” teacher will:

develop an interest in and an understanding of autism
collate information profiling the child's autism and its implications for teaching
plan an individualized programme focusing on the child's communication, social and behavioral needs and providing access to the curriculum of the classroom
involve the parents
ensure all staff share information and are supportive
evaluate the outcomes of the work in the classroom

4.4.1 Top Teaching Tips

- Get to know the child as an individual. Understand his autism and assess how it interrupts his learning. Use his strengths to teach new skills.
- Give the child an opportunity to explain a situation from his point of view. A good idea is to share viewpoints using visual information such as ‘comic strip conversations’.

- Cut down on the amount of language you use when giving instructions. Keep the instructions clear, simple and direct. Avoid idioms. Tell the child what should be done rather than what should not be done.
- Emphasize routine and structure. Teaching strategies which incorporate visual structure and clarity will be of great help.
- Set targets that are realistic, attainable and short term.
- Teach social skills as they do not come naturally.
- Keep all staff informed of the strategies you are using. Encourage consistency in the implementation of strategies.
- Establish regular communication between home and school: this is of paramount importance.
- Do not be afraid to make mistakes, this is a learning process.
- A calm disposition and a sense of humour are helpful!

4.4.2 Classroom Assistants: Helpful Guidance

Good communication between the teacher and classroom assistant is paramount. The classroom assistant's contribution is very important and he/she should be valued as part of the teaching team:

- The classroom assistant should be fully informed of the classroom planning. It is helpful if the classroom assistant is involved with the classroom planning.
- The classroom assistant should work with the teacher to create visual strategies to develop independence; children with an ASD learn visually, verbal instruction may not be enough to ensure learning.
- A classroom assistant should avoid being 'velcroed' to the child; create opportunities for independence and social interaction.
- It will help the child's self esteem if his/her classroom assistant is seen working with other pupils.

- Cut down on the amount of language you use, especially when you are giving instructions.
- Think about your use of language and how the pupil will interpret it. The child may think literally and in concrete terms.
- Give the child enough time to process information rather than expecting an immediate response.
- Usual strategies for calming a child may not work. The child may need time alone rather than being comforted by another person. If the child has had a bad day it is not your fault; tomorrow is another day!

4.5 Strategies for Teaching Social Skills Aims of this Section

This section aims to highlight the social difficulties which you may observe in your classroom and provides practical examples of effective strategies.

Key Messages of this Section

At the end of this section, you should be more aware of the types of social problems a child with an ASD faces in a class of 20+ pupils and be able to employ some strategies to help the child interact and experience a sense of belonging.

4.6 The Friendly Social Skills Classroom

Social skills are an area of complexity for children with an ASD. Social interaction with understanding is an innate ability for most pupils, however, this does not come naturally to those diagnosed as having an ASD. This causes great difficulties for them in social situations such as:

- Interaction with others;
- codes of conduct;
- Friendships;
- Eye contact;
- Emotion.

It is a pervasive difficulty, affecting all areas of their lives. Consequently social understanding must be taught specifically.

Difficulties with ***social interaction*** will be most obvious in less structured situations, such as:

- Break and lunchtime;
- In the corridor;
- Transition between lessons.

Lack of social judgement is likely to be most apparent in the following situations:

- Whole class lessons when the pupils with an ASD may interrupt or shout out inappropriately and repeatedly.
- Speaking to teachers and peers in the same manner. Having no recognition of social status.
- Commenting inappropriately on someone's appearance or actions.
- Adhering to set rules in a rigid way.
- Interpreting language in a literal manner.
- Initiating and maintaining friendships may prove difficult.
- Combining or integrating ideas. Understanding that two opposing concepts can be acceptable.

Example

One young man, a member of a conservation club, was dismayed when a lady picked a blossom as a surprise for him. He could not appreciate that a 'nice' lady would damage the environment by picking flowers, and told her so in an abrupt manner. He felt that if she were a 'nice' person she would want to know it was wrong. It is important, when dealing with situations that arise from lack of social judgment, that the teacher understands the perception of the child with an ASD and does not expect his/her social understanding to match his/her level of academic ability. Therefore each situation needs to be dealt with individually. Ask, or help the child explore what

happened from their perspective. In knowing this we can help them understand the effect it had on others and how they could change their reaction in the future. This does not happen immediately, it requires much practice, as their generalisation skills are limited. Concepts often have to be relearned in varying social situations. The following are examples of misunderstandings arising from lack of social judgement. Brendan is a pupil aged twelve in a mainstream post-primary school. He has a classroom assistant. On leaving class one morning he made a loud comment about the substitute teacher being 'an old bat'. Unfortunately, the teacher heard the comment and reported it to the head of discipline in school. This resulted in Brendan being suspended for two days. When asked about the incident Brendan said that he was trying to make the other boys laugh. He made two errors. He did not judge the teacher's whereabouts before making the comment, nor did he consider the end result of his actions. There was no intent to offend the teacher. Brendan is a very pleasant pupil. This incident reflects his lack of social judgment due to him having Asperger's Syndrome. This led him into trouble on a regular basis. Another pupil, during lunch break watched his peers participating in rough and tumble activities, which got more and more boisterous. Without a thought John suddenly pushed his classroom assistant to the ground. This was his way of joining in. He failed to see that, from the school's perspective, this is regarded as abuse towards a staff member and a serious breach of discipline.

A strategy that often helps is known as a buddy system. Asking sympathetic peers to guide the pupils with an ASD in social situations that are proving difficult for them, for example,

- moving to different locations;
- Break and dinner times;
- Organizing belongings;
- Group work;
- Time keeping;
- PE.

4.7 Social Skills for Pupils with a Learning Difficulty

Pupils with an ASD and a diagnosed learning difficulty are often misunderstood as having purely challenging behaviour. However, it is much more likely that they have underlying difficulties linked to the fact that they have a diagnosis of autism.

4.7.1 Areas of Difficulty

- Transitioning between activities and locations.
- Change of routine (timetable or staff).
- Sharing.
- Waiting.
- Sensory difficulties.
- Turn taking.
- Unstructured time (free play, assembly, break and lunchtime).
- Group work.

Understanding the uniqueness of the child with an ASD and implementing individualised visual strategies, is the key to help the child make sense of the world and fit in more easily. This is social skill teaching with strong links to behaviour management. For useful strategies in dealing with these problems please refer to the section on behaviour in this pack. The “**social skills**” aware teacher is likely to have:

- a good understanding of the child’s social interaction problems: know the things which may upset the child;
- established a relationship with the child based on consistency and routine: be a friend;
- Careful planning, using real social situations, to encourage the child to work with a peer partner, in small groups and in whole class situations: help the child to communicate with others - exaggerate social gestures and expressions;

- considered how to encourage the child to develop his/her ability to understand social interaction, accept change and to ask for help when confused: stay near the child when activities are new or likely to cause confusion;

- plans to involve others as appropriate, eg peers, other staff and parents: plan personal social lessons to emphasise social rules.

In order to help children with an ASD, it is necessary to observe and assess their social competencies, to decide what skills need to be taught explicitly. Good use can be made of a variety of methods and materials to focus on the child's social skills needs. These approaches include:

- The use of social skills stories which can be easily adapted from the strategies developed by Carol Gray and taught by teachers, assistants and parents. Non-readers can access the stories on tape;

- the teaching of key social skills in all lessons will help the child with an ASD to understand and feel at ease with the social context, eg:

- Turn taking: use visual rules/cues such as objects, pictures or written sequences and practise as a game;

- accepting change: let the child know if you are about to change an activity, using a visual cue the child recognises and encourage him/her to know that a task can be completed later;

- accepting a time limit to a task: use an egg timer for younger children;

- approaching others: use social stories to help the child initiate communication, make a request and finish a conversation; encourage the child to observe how others behave and to follow the example;

- behaving well: visual rules/social stories can help the child to be quiet. To know when to speak and what can be said by, eg taking him/her repeatedly through a sequence of appropriate behaviors in pictorial form.

- using peer support can help the child with an ASD to develop his social skills. This requires the permission of the child and his parents and needs careful explanation to the participating peers. This is a most useful strategy during free time and transitions from one activity to another or during school outings;

- developing a social skills counselling programme: provide the child with 1:1 or small group training for different situations; this is suitable during teenage years when relationships are developing significantly;

- using circle time and buddying approaches: this approach can be used to assist the child to establish friendships and to maintain these over time; this also encourages the child to join clubs, leisure centres and is a good preparation for life after school; older children can be encouraged to develop their vocational and functional skills, including recognizing dangers, for the world of employment or further study. For further ideas on assessment procedures and strategies to address social skills issues refer to the guidance materials suggested in the recommended reading list.

4.8 Strategies for Teaching Communication Aims of this Section

This section provides a brief introduction to the language difficulties associated with an ASD and indicates the key factors that should be considered when planning to meet the communication needs of a child with an ASD in the classroom.

Key Messages

By the end of this section, you should be more aware of the important aspects of language development in children with an ASD and be able to consider a range of teaching strategies to facilitate the development of communication in children.

4.9 The “Communication-Friendly” Classroom The problems.....

Children with an ASD show huge variations in their language abilities. Teachers working with these children recognise the difficulties the children have in making progress in stark contrast to their peers; this poses the greatest challenge to the teacher and to parents. In all instances, the child with an ASD experiences subtle difficulties with the understanding and use of language.

Many have not developed the necessary skills to initiate communication; they must be taught if the child is to be able to express his needs and to interact effectively. Some children with an ASD repeat the words and conversations of others; this echolalia can be used to shape the child's more appropriate language. Many other children with an ASD rarely have difficulty acquiring speech and rarely have articulation difficulties. They often have excellent memories and can acquire extensive vocabularies. Superficially their language appears to be well developed and this can lead people to over-estimate their understanding of language. They tend to be very concrete in their understanding and use of language. They are often described as having what speech and language therapists refer to as "semantic-pragmatic language difficulties". However, it is not possible to have this disorder of communication without an underlying social communication problem.

An effective communicator has few problems in language use, understands the rules of social interaction and knows how and when to apply them, eg understanding that one speaks differently depending on the relative status of the person being spoken to. Such individuals are aware of the impact of their utterances on the person being spoken to and will modify speech accordingly. Children with an ASD find this task particularly difficult. A child with an ASD may experience any of the following difficulties:

- ▶ Attending to instructions or explanations - this is a particular problem in group settings. The child may not be "tuning-in" to the relevant aspects of the "message" or may be distracted because of the tendency to focus on non-essential details. When the instruction or explanation is longer or more complex the problem is made worse.

- ▶ dealing with abstract ideas or concepts - children with an ASD find ideas and concepts difficult to understand because of their concrete and literal understanding of language. Idiomatic speech and metaphors are likely to be interpreted literally, eg:

- She hit the roof;
- get your skates on.

They will not appreciate sarcasm or humour for the same reason, eg:

- You're driving me around the bend.

▸ **Social understanding** - as noted above, the social difficulties experienced by children with an ASD may affect their understanding of language. Even if they understand the actual words they may not be able to grasp the speaker's underlying meaning or intentions. This is a particular problem with sarcastic language where the speaker may say one thing and yet will be communicating something entirely different by their non-verbal signals. This makes pupils with an ASD particularly vulnerable to teasing and ridicule, eg teenagers using "wicked" when they mean something is actually really good, or "Oh Brilliant" when in fact it is far from brilliant.

▸ **respecting others' viewpoints**- children with an ASD tend to launch into conversation with others without assessing the listener's previous knowledge of the topic.

▸ **Progressing a conversation from a single aspect**— a feature of the language of children with an ASD is their tendency to talk around the subject, which often has the effect of making their accounts of events boring, repetitive and at times incoherent.

▸ **Recognising and repairing communication breakdown**- children with an ASD often do not realise when a communication situation has broken down because they do not recognise the listener's non-verbal messages. They do not identify signs of boredom, confusion, etc. When alerted to a breakdown in communication children with an ASD are unable to alter their conversation to repair the breakdown, eg they might simply repeat exactly the same sentence.

▸ **Casual and particularly temporal aspect of the communication**- they tend to slip from past to present tense during speaking or writing activities.

▸ **Apparent inattention when spoken to in a group**- it is thought that this may result from difficulty in understanding fully the question or what has been said. They may try to analyze what has been said or be "buying time" to formulate a reply. They may not be aware that they are part of the group being addressed.

▸ **Dwelling on obsession interests** - it is often noted that children with an ASD introduce a favorite topic into virtually every conversation, even when it is inappropriate to do so.

- Inability to ask for help successfully - such children may react inappropriately to a situation that is causing a problem; his/her attempts at seeking help are inadequate and ineffective, eg they may mumble to themselves "I have got a problem."

- Literal interpretation - children with an ASD may need explicit instructions, they are confused by statements such as:

Watch the puddles which should read - be careful not to stand in the puddles.

- Facial expression and tone of voice - difficulty using or interpreting facial expression or tone of voice during conversation.

- Insensitivity to social rules - the child does not understand the conventions of social and linguistic interaction.

- Humour - many children with an ASD do not develop an understanding of humour beyond the obvious slapstick type.

The best practice.....

In the best practice, the speech and language therapist will provide supportive information and assessment indicating the child's communicative level and suggesting ways to structure the classroom environment to promote language in a variety of situations. The teacher should:

- use simple language that is concise and clearly understood; provide visual supports where practical; avoid sarcasm;

- Talk to the child directly; praise all attempts to communicate; speak slowly to give time for understanding and response;

- use familiar words and teach them to ask when they are confused;

- teach new words in a variety of contexts;

- encourage the child to listen;

- accept the child's best verbal efforts;

- create a home-school communication book for the child who is reluctant to speak or who does not use speech;
- use simple drawings to encourage conversation;
- provide opportunities for the child to practise his language skills;
- with some children who are non-verbal or are reluctant to use speech, provide other means of developing their ability to communicate, eg using physical prompts, pictures or computer programmes to teach communication skills, encourage those communicating with the child with an ASD to adjust their skills, helping the pupil to become an effective participant in talking and listening activities. For further ideas on promoting communication in the classroom see the recommended reading list.

4.10 Strategies to Promote Positive Behaviour and Deal with Challenging Behaviours

Aims of this Section

This section provides an overview of the behaviours associated with ASD and describes some strategies to promote positive interest in learning and appropriate behaviour.

Key Messages

By the end of this section you should be more aware of the particular behavior problems associated with ASD and be able to use your teaching skills to develop effective management strategies for the classroom. You will realise that many children with an ASD do not exhibit any behaviour problems of significance in the classroom.

4.11 The “Autism-Friendly” Classroom: Promoting Good Behaviour

Typically, the child with an ASD will have fewer skills to cope with difficult situations and may have learned a number of unusual behaviours to deal with them. It is important to remember that the child is not the problem; it is the conditions that give rise to them. The challenge to the teacher is to understand what might give rise to problem behaviours and respond appropriately. In some instances, the child may be oversensitive to certain stimuli or may experience anxiety, fears, eating and sleeping problems and may engage in rituals and obsessions or aggressive

behaviour to themselves or others. The evidence of research suggests that, where appropriate intervention is established, there is likely to be a reduction in such behaviours. The teenager with an ASD can also expect the usual adolescent difficulties, but may not be able to understand what is happening. What is helpful to recognise is that there are sometimes other reasons for unwanted behaviours. However, in the classroom children with an ASD tend to react to difficult situations in a number of ways such as:

- running from their seats or out of the room;
- feeling anxious of change, shouting or withdrawing;
- showing destructiveness and aggression.

The child with an ASD does not always respond to the usual forms of discipline. It may be necessary to develop a systematic plan to improve behaviour. A behaviour plan must be based on an understanding of the child's ASD and recognise that the behaviour, in most instances, is an attempt to communicate. To begin, the teacher needs to think of the behaviour as a 'clue' to understanding its occurrence. Behaviours may indicate that something is difficult for the child, he/she perhaps does not understand or his/her autism is interfering with more appropriate behaviours. Traditional approaches to behaviour modification may not work with the child with an ASD, for example:

1. Communication difficulties make it hard to understand 'reasoning'.
2. The child with autism may not be motivated to please others or may be motivated by unusual things.
3. The child may not understand the cause and effect relationship between behaviour and rewards. Similarly, the child is unlikely to understand that the punishment is a consequence of their inappropriate behaviour.
4. The child may not generalise positive behaviour changes from one situation to another.
5. The child may have established routines that may include negative behaviours.

The first step in solving a behaviour problem is searching for the clues. The following questions may help:

1. What are the underlying difficulties that might cause this problem?
2. In what situations, or at what times, does this behaviour occur?
3. How often does this behaviour occur?
4. What does this behaviour mean from the child's perspective?
5. How does the pupil's behaviour differ from that of other children of the same age?
6. Is this a behaviour that is important to change?
7. What might be confusing to the pupil about this situation?
8. What new skill could the child learn to better handle this situation?

WHAT DOES WORK?

1. Identify the problem behaviour and the possible communication intent:

- What usually happens before the behaviour occurs?
- How does it inhibit the child with his learning?
- What are the consequences?
- When and where does the behaviour occur?
- What else is happening?
- Who else is there?

Describe the frequency, intensity and duration of the behavior incident: this will help you analyze the characteristics of the behavior and indicate possible solutions.

2. Develop teaching strategies to address the behavior and ease communication:

- Talk to the child and his/her parents about his/her behavior.

- Analyze the information you have gathered (if serious consult a psychologist).
- Observe the child and identify an alternative and more appropriate behaviour.
- Give reward.
- Provide interesting activities to engage the child.

3. Adapt the classroom to support the child's understanding and sense of security:

- Change distracting stimuli.
- Provide a place for the child to take a break.
- Offer choices.
- Plan predictable routines.
- Introduce calming moments into the class routine.
- Help the child to manage his/her behaviour by using social stories and shape behaviour toward appropriate outcome.
- Plan a crisis strategy when things get out of hand and know the triggers which can upset. Never confront the child; offer "time out" space.
- in more serious instances, develop, with the support of other professionals, a behaviour support plan based on a formal functional assessment which can detail:
 - Classroom adaptations;
 - programme interventions;
 - inform further planning.

4. Assessing the behaviour situation

What are the cues that tell you the underlying reasons for the behaviour?

4.12 TEACHING STRUCTURE

What skills can you teach How can the environment be that might eliminate the altered so that the child need for this behaviour? better understands the situation?

Use strategies and structures to make a situation more visually clear:

1. Use schedules, timetables or calendars that help to show when something will happen.
2. Use accessible pictorial or written rules to remind a child what he/she should do.
3. Physically arrange the classroom to show where something should happen.
4. Provide a checklist using objects of reference, photographs or a written list to help the pupil follow directions independently.

The above strategies are essential to planning and will benefit those children who do not have an obvious behavioural difficulty. The inclusion of the strategies in classroom practice will assist the development of social and communication skills.

5. Replace inappropriate with appropriate behaviours:

1. Teach the child to communicate; in difficult situations, for example, the child needs to be able to ask for help, to take a break, to express emotions such as fear or anger.
2. Teach social skills, which will help the child initiate, sustain or escape anxious social situations, such as going to a quiet place, completing a calming activity or using relaxation techniques.
3. Teach specific play and leisure skills which might occupy his interest.
4. Develop behaviour targets which shape the child's behavior toward more appropriate behaviour.

4.13 Anecdotes and Addresses Nursery

Peter was in his second year of the nursery. At the finish of an outside play session, the teacher would blow a whistle and the children were told to freeze and then to walk to the line. Peter would freeze and walk carefully to the line. The other children would run or walk quickly to the line getting there before Peter. Peter would then throw himself down on the ground screaming

and kicking because he was last. The teacher found that she had to be flexible in changing her ideas and rules: after all, she can be flexible, the child with an ASD cannot!

Key Stage

John looked forward to going swimming with his class every week. At the end of the session, the class was given some free time to play but was not allowed to jump in the pool. John got out of the pool and was running to jump in when the teacher stopped him and told him off for jumping in. As a consequence, the teacher told John he would not be allowed to go swimming the following week. John stomped off, refused to get dressed and delayed the school bus for an hour. When he calmed down he was able to say, “but I didn’t jump in” True, he hadn’t actually jumped in and from his perspective he was being punished for something he hadn’t done. The ASD child can be so literal in his interpretations. He also may not realise or understand that sometimes you can foresee what the child’s actions are going to be.

4.14 The Autism Working Group:

Belfast Education and Library Board Ros Millar North-Eastern Education and Library Board Janet McCann Southern Education and Library Board Lorraine Scott South-Eastern Education and Library Board Kate Doherty Glynnis McSorley Western Education and Library Board Caroline O’Hara Department of Education John Hunter Education and Training Inspectorate

4.15 Role of teachers

The role of teachers in helping students prepare for transitions out of the school system is to provide opportunities for students to develop skills for work and independent living. The range of expectations depends on students’ abilities and needs. Some students with autism spectrum disorders plan to go on to further education or training following high school. Consequently, there will be a greater emphasis on academic preparation, in addition to work experience and development of job-related skills, and skills for leisure and recreation. For others, the program may focus on work experience, community-based training and self care. To facilitate transition planning, it is important to be familiar with regional systems and services for adults with developmental disabilities. In general, school programs can prepare students for transition by: providing a variety of work experiences to help individuals determine preferences encouraging

participation in extracurricular activities and social events encouraging volunteer work helping with developing résumés training in social skills for the job place providing on-the-job preparation, once preferences have been established training in the use of public transportation training in self-management teaching functional academics appropriate to the ability levels of students.

4.16 Students with Asperger's Syndrome

Students with Asperger's syndrome often display many of the characteristics associated with autism spectrum disorders. The Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (American Psychiatric Association, 1994) criteria stipulate that students must display a qualitative impairment in social interaction, and restricted, repetitive and stereotyped patterns of behaviour in order to be diagnosed with Asperger's syndrome. In addition, the student must possess at least average cognitive abilities and adaptive functioning skills. The main differences between people with autism spectrum disorders and those with Asperger's syndrome is that individuals with Asperger's syndrome do not have clinically significant delays in early language development or significant delays in cognitive development. They usually do not have the same degree of difficulty in the development of age-appropriate self-help skills, adaptive behaviour and curiosity about the environment in childhood. However, even though students with Asperger's syndrome may obtain scores in the average or above average range, they continue to demonstrate subtle but important differences in cognitive and social processing. Individuals with Asperger's syndrome constitute a diverse group, so treatment and educational approaches must be individualized. It is important to know that individuals with Asperger's syndrome are at risk for developing other psychiatric conditions. Historically approximately 30–35 percent of students with Asperger's syndrome develop secondary psychiatric conditions, such as depression, anxiety disorders or obsessive-compulsive disorders. Pushing students too hard to “be normal” can exacerbate existing anxieties and social difficulties.

4.17 Learning and Behavioural Characteristics

Asperger's syndrome is characterized by a qualitative impairment in social interaction. People with Asperger's syndrome may be keen to relate to others but do not have the necessary skills and may approach others in peculiar ways.⁶⁹ They frequently lack understanding of social

customs and may appear socially awkward, have difficulty with empathy and misinterpret social cues. They often have the same difficulties as individuals with Students with Asperger's Syndrome /133 © Alberta Learning, Alberta, Canada 2003 autism spectrum disorders in understanding that other people have their own perceptions, thoughts and feelings. People with Asperger's syndrome do not acquire social skills efficiently through incidental learning and frequently need explicit instruction in the area of socialization. Although children with Asperger's syndrome usually speak fluently by the time they enter kindergarten, they often have problems with the complexities of language, including:

pragmatics—the use of language in social contexts semantics—multiple meanings prosody—the pitch, stress and rhythm of speech.⁷⁰ One common characteristic of people with Asperger's syndrome is that they have difficulty carrying on social conversations. They may have an advanced vocabulary and talk incessantly about a favourite subject, however the topic may be somewhat narrowly defined and they may have difficulty switching to other topics. People with Asperger's syndrome may have problems communicating with others because they do not naturally learn the rules of conversation. They may: interrupt or talk over the speech of others make irrelevant comments have difficulty initiating and terminating conversations use speech characterized by a lack of variation in pitch, stress and rhythm use overly pedantic or formal speech, particularly as they reach adolescence stand too close when talking to someone stare, use abnormal body posture or other strange body language fail to understand gestures and facial expressions of others. Students with Asperger's syndrome are of average to aboveaverage intelligence and may appear quite capable. Many students with Asperger's syndrome are relatively proficient in their knowledge of facts and may have extensive factual information about a subject that absorbs them. However, they demonstrate relative weaknesses in comprehension and abstract thought, as well as in social cognition. Consequently, they experience academic problems, particularly with: reading comprehension problem solving organizational skills concept development making inferences and judgements.

4.18 Summary

Students with autism spectrum disorders often have difficulty with transitions. It is hard for them to take in all the information presented by a new situation, determine expectations and generate appropriate responses. As a result, even minor transitions are often difficult and result in

increased anxiety and inappropriate or resistant behaviours. Transitions should be carefully and thoughtfully planned. This includes transitions between activities and settings throughout the day, from grade to grade, from school to school and beyond school. All programs and environments involve transition and change. The goal is to help students cope with these changes and adapt to a variety of settings. Anxiety can often be decreased and inappropriate behaviours prevented or reduced if students are prepared for change and transition. The strategies for communication development and suggestions for instructional approaches can be used to help students understand and cope with change.

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Unit 5

Vocational Training and Career Opportunities

Content

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5.1 Introduction

Autism spectrum disorders (ASD) represent a distinct category of developmental disabilities that share many of the same characteristics. Individuals with an ASD have difficulties with reciprocal social interactions and interpersonal communication, and may demonstrate unusual or repetitive patterns of behaviors. We use the term “autism spectrum disorders” because the symptoms can occur in any combination and can range from very mild to quite severe [1, 58]. Autism spectrum disorders include autism and two related disorders; Asperger Syndrome (AS) and pervasive developmental disorder-not otherwise specified (PDD-NOS). It is often regarded that autism is the most severe of the three; however, any individual diagnosed with a spectrum disorder may be profoundly affected. Autism spectrum disorders are believed to be permanent developmental disorders that will continue into adulthood, creating lifelong challenges for the individual [87].

The number of individuals in the United States diagnosed with an ASD has increased dramatically in recent years [18]. Current data from the United States Department of Education indicate this disorder is increasing at a rate of 10–17% annually [63]. This annual growth increase certainly impacts service delivery within the public education system; however, also extends to postsecondary services as these individuals leave high school. Despite a great deal of data and ubiquitous media reports publicizing an increase in numbers, there is currently still very little known about how best to provide support that will result in postsecondary success. The majority of research and programmatic efforts have focused on young children, with relatively little focus placed on adolescents and adults with ASD [55]. Successful employment is perhaps the primary aspiration for most individuals as they enter their postsecondary years, whether such employment occurs directly after high school or following a training or educational program. According to the Individuals with Disabilities Education Act [34], the primary purpose of a public education is to prepare an individual for employment. This law clearly outlines post-school employment as the measure of accountability for individuals with disabilities. Postsecondary employment opportunities, however, have traditionally been very limited for individuals with ASD [62, 70]. Obtaining and maintaining employment can be difficult for any person, but is particularly complicated for an adolescent or adult with ASD due to unique communication and social impairments [56, 60]. While employment outcomes have received

growing attention for many students with disabilities in the educational research (e.g., [74]), with the expanded numbers of individuals with ASD, there is a stark demand to assess what we know currently about their employment needs and understand better what services and supports may help facilitate employment success in the years after high school.

5.2 Purpose

In this paper we will review evidence based research related to employment for individuals with ASD. A spectrum of employment options is now available which include segregated workshops, supported employment, and competitive employment [80]. The outcomes of supported and competitive employment are far superior to sheltered workshop or other day service options in terms of financial gains, wider social integration, and worker satisfaction [12, 13, 67, 68]. Therefore, for the purpose of this paper, employment will refer to work within the natural community. Specific areas addressed in this review include benefits of employment, state of employment, obstacles to employment, current service options, and in depth review of supports for success.

Information on employment and ASD was obtained by conducting electronic and ancestral searches using ERIC and Psyc Info. Searches concentrated on research published from 2000 through 2009. Hand searches were conducted on six journals which frequently publish research involving persons with ASD (*Autism, Disorders*). Because of the continuity across autistic disorders, this review addresses both the narrowly defined disorder of autism and the broader range of autism spectrum disorders, including AS and PDD-NOS.

5.2.1 Benefits of employment

There are many reasons to examine factors related to employment success. Individuals with ASD should have the same rights and entitlements enjoyed by the rest of society. Employment enables adults with and without disabilities to earn wages they can use to support themselves and pursue their interests. Employment provides a forum that promotes personal dignity and has been demonstrated to improve quality of life in individuals with ASD [20, 21, 65] as well as improve cognitive performance [19].

Despite the humanitarian reasons, increased employment also provides a number of economic advantages. With employment, there is less reliance on government funds and more contribution

to taxes [28, 35]. Jarbrink et al. [36] conducted a study looking at the societal cost-impact of adults with high-functioning autism and found that lack of employment results in lost work time and productivity that supports a society and helps keep it moving. Further, employment has a significant cost impact on the economy due to resource utilization. Jarbrink et al. reported that with employment, the cost of community supports would decrease creating less reliance on day programs and activities, such as adult day care. Wages would provide the means to pay bills decreasing the need for living support. Health benefits would cover the costs of medical concerns, a critical issue for this population, as increased health care costs are common due to a need for medication and psychiatric services that address comorbid disorders [3, 6, 15, 40].

There is also ample evidence to support the potential benefits to employers and companies when they hire individuals with ASD [25, 26, 28]. Employers often value the trustworthiness, reliability, and low absenteeism of this group [30]. Once employed, individuals with ASD often demonstrate particular strengths, such as attention to detail and intense focus that result in increased work output and may enjoy performing jobs often shunned by others due to social isolation or the repetitive nature of the task [80].

5.2.2 State of employment

Follow-up studies suggest that long-term employment outcomes are poor for the majority of this group. It is estimated that 50–75% of adults with ASD are unemployed [29, 32, 52, 57]. Even for individuals who are considered to be higher functioning, employment results are appalling. Adults experience underemployment, switch jobs frequently, have difficulty adjusting to new job settings, make less money than their counterparts, and are much less likely to be employed than typically developing peers, individuals with less severe language disorders, or individuals with learning disabilities [9]. Even for those individuals who have postsecondary educational experience, employment difficulties are common [27].

5.2.3 Obstacles to successful employment

Individuals with ASD have markedly different vocational needs than individuals with other developmental disabilities [5, 56]. Because each person demonstrates a variety of characteristics

across the diagnostic criteria, each individual is unique. The spectrum of need and ability makes the provision of successful employment a great challenge indeed [38]. As noted by Simpson: The countless permutations and combinations of social interactions, language, learning, sensory, and behavior deficits and excesses found in these individuals, in combination with their wide range of abilities, developmental levels, isolated skills, and unique personalities make autism an especially baffling disability ([73], p. 69).

Interactional difficulties associated with ASD account for the biggest vocational impact. In interviews with adults with ASD, communication and social difficulties with supervisors and coworkers consistently emerge as a primary hindrance to job performance. Bolman [6] found that such difficulties often lead to termination. Communication obstacles include difficulty understanding directions; inability to “read between the lines”, read facial expressions, and tone of voice; asking too many questions; and communicating in an inappropriate manner [32, 33]. Social impairment involves a wide array of deficits intrinsic to successful interactions and may include inappropriate hygiene and grooming skills, difficulty following social rules, inability to understand affect, working alone, and acting inappropriately with individuals of the opposite sex. Navigating social interactions are often problematic even before employment is procured and begins with mastering the job application and interview process [56]. There are areas of cognitive functioning known to be impacted in individuals with ASD which can affect job performance. Impairments in executive functioning are well documented. Many exhibit difficulties in task execution due to problems with attention, motor planning, response shifting, and working memory. Acclimating to new job routines and managing changes in the work setting often pose a challenge [38]. Individuals with AS demonstrate difficulties with both problem-solving and organization, despite having average or above-average intelligence [2]. According to Howlin [27], many may experience difficulty in fulfilling employment roles as they get older due to deterioration in skills that may occur in early adulthood for those who have low IQs or develop epilepsy.

Behavioral difficulties which may include tantrums, aggression, self-injury, property destruction, ritualistic behaviors, or pica can create an employment barrier. Interfering behaviors create a complicated issue as many are misinterpreted, have multiple functions, and require multi-faceted behavior management strategies [39]. Such behaviors are not well tolerated in the workplace and

may prevent employment all together and lead to segregation [77]. Individuals with ASD report encountering high levels of stress and anxiety in the workplace which may interfere with performance. Hurlbutt and Chalmers [33] interviewed six adults with AS who reported high levels of anxiety due to the task of trying to fit in socially with the neurotypical world. Burt et al. [7] found that individuals with autism had increased anxiety due to sensitivity to workplace noise and other sensory stimuli. Camarena and Sarigiani [8] interviewed the parents of 21 high functioning adolescents with ASD regarding postsecondary aspirations. Parents reported anxiety as a major obstacle for postsecondary success due to their child's fear of the unknown and navigating social interactions. The comorbidity with a variety of psychiatric symptoms including depression, anxiety, and bipolar disorder in individuals with ASD has been widely reported [23, 40, 72, 84]. Further, the onset of epilepsy occurs in approximately one-third of the population [5, 18]. Such mental and physical health problems may significantly interfere with the capacity for positive employment outcomes. Schaller and Yang [71] used the 2001 Rehabilitation Services Administration database to describe the demographic characteristics related to employment by the 815 individuals with ASD who achieved competitive and supported employment outcomes. Absence of a secondary disability was significantly related to successful competitive employment.

5.2.4 Current service options

Given the heterogeneity found in this group, for employment to be procured and maintained in the community, there must be a variety of services and supports provided. Vocational Rehabilitation (VR) programs provide supports and services designed to help individuals with disabilities meet their employment goals by assisting individuals prepare for, get, keep, or regain employment. The number of individuals with ASD utilizing VR services increased by more than 121 percent from 2002 to 2006 [14]. The small body of research evaluating employment outcomes of the VR system suggests that services provided through these programs are less than optimal for individuals with ASD and do not provide sufficient support [56]. Lawer et al. [47] examined the employment outcomes of 1,707 adults with ASD in the United States VR system whose cases were closed in 2005. Individuals with ASD were much more likely than adults with other impairments to be denied services as a result of being considered to have a disability too severe to benefit from services. Cimera and Cowan examined employment outcomes obtained by

adults with ASD whose cases were closed between 2002 and 2006. The number of cases closed reached 41%. While this percentage was similar or higher to other disability groups investigated, those with ASD worked far fewer hours and earned lower wages. In addition to having limited efficacy, generic employment services and supports are not cost effective. Research indicates employment services offered through our current VR system for individuals with ASD are expensive and may cost more than services for other disability groups. Cimera and Cowan found service costs for those with ASD were among the most costly of nine disability groups served. Only participants with sensory impairments obtained services that cost more. Lawer et al. found services obtained by adults with ASD were the most costly of all groups served.

5.3 Strategies for success

Assistance provided through the provision of supported employment is often necessary for job success. With a scheme providing on-the-job support services, employment outcomes for this population improve dramatically. Howlin et al. [28] tracked employment of adults with HFA and AS over an eight year period and found that approximately 68% found permanent employment, jobs were commensurate with the individual's intellectual abilities and academic attainments, and salaries were increased. Hillier et al. [26] followed nine participants through two years of a vocational support program. Employment levels increased by 78% and income levels increased by 443%.

There is some evidence indicating programs specifically targeting the needs of this population may indeed be a more appropriate and cost effective solution [28]. There is limited research outlining the specific services and supports those with ASD require; however, examples of successful supported employment programs for individuals with ASD are described in the literature (e.g. [26, 38, 59]). Additionally, case studies conducted with adults with ASD provide tremendous insight regarding vocational needs [32, 33, 56]. From this research, recommendations for vocational supports can be ascertained and grouped into four major themes: (1) job placement, (2) supervisors and co-workers, (3) on-the-job provisions, (4) work place modifications, and (5) long-term support. The section to follow provides a summary of research for each area of work place support and identified strategies.

5.3.1 Job placement

Mounting research indicates supports related to job placement are the most impactful when it comes to successful employment [71]. Particularly crucial is ensuring an appropriate job match. In their description of the Treatment and Education of Autistic and related Communication-handicapped Children (TEACCH) supported employment program, Keel et al. [38] emphasized the need for job placement to be individualized and based on the person's strengths and interests. Research describing Prospects, a supported employment program for adults with AS, stresses the need to identify jobs that are appropriate to the individual's intellectual and educational background and that match his or her social skills and abilities [28, 51]. Placement also requires considerations regarding the job tasks and work environment [38]. An appropriate job is predictable and can be adapted to a schedule for the person with ASD. Work tasks are clearly defined and in areas with minimal distractions. According to personal opinion, jobs should require minimal social skills, allow adequate time for learning, and be void of excessive sensory stimulation [56].

To assist in ensuring an appropriate job placement, a variety of strategies have been highlighted in the literature. Job preference has been determined through exposure to a range of tasks in a variety of businesses [61]. Also effective are preference assessments. Lattimore et al. [44] found paired-item preference assessments determined more or less preferred tasks for workers with autism, while Lattimore et al. [45] found efficacy in multi-stimulus assessments. Nuehring and Sitlington described a formal process for using assessments to job match. These assessments took the form of behavior observations, situational assessments, and analysis of background information. The assessment data were used to determine individual interests, jobs the individual could do well, jobs requiring assistance and training, and work habits needing further development. In a vocational support program described by Hillier et al. [26] job placement skills, including job matching, were taught to adults with ASD. These skills covered job searching, identifying appropriate jobs, preparing resumes, and interviewing. Instruction was provided one on one for a minimum of one hour per week until employment was secured.

5.3.2 Supervisors and co-workers

Employment success can be contingent on employers and co-workers who provide a supportive environment. Employment retention may require flexibility and tolerance from individuals who supervise, interact, or collaborate with the person with ASD [7, 51]. Modifications to job tasks or

the work environment, adjustments to communication exchanges, and assistance with social interaction will likely be required [25].

Both knowledge of the disorder as well as job supports available are important prerequisites for employing individuals on the spectrum. Nesbitt et al. [59] surveyed employers to determine differences between those who did or did not employ an individual with high functioning autism or asperger syndrome. Differences were related to awareness or understanding of the disorder. Those not employing an individual on the spectrum focused on potential negative behaviors rather than the person's capabilities. These same organizations also failed to acknowledge the importance of employment supports that may be provided. Adults with ASD interviewed about employment described significant difficulty in the workplace due to insufficient knowledge. They recommended providing autism awareness training to employers and co-workers to help foster understanding of who they are and why they behave in a certain way [6, 56].

A primary component of existing supported employment programs is receptivity and knowledge of the employers and co-workers. TEACCH staff provides training to everyone at the job site who is willing to learn about ASD, resulting in an environment where the individual is more likely to be successful [38]. Informal training to supervisors on ASD characteristics and support strategies is a component of the Prospects program [28, 51]. In the program described by Hillier et al. [26], information was provided to supervisors and co-workers emphasizing how to interact socially and to maximize social integration for the individual with ASD.

5.3.3 On-the-job training

Experience has proven that with proper intervention and training, individuals with ASD can work in a variety of businesses and industries. Workers may need extensive training and/or instruction using special methods to learn job tasks as well as other skills related to employment [25, 61]. For the individual with ASD, such skills may include communication, interpersonal skills, and management of stereotyped patterns of behavior that will foster a positive work experience [7, 16]. While it should be expected that the person with ASD enters the work place with a variety of skills intact, it is highly likely, the individual, regardless of ability level, will require on the job support [6, 16, 61]. Due to the unique learner profile associated with ASD, training may be most beneficial when provided on the actual job site. This allows for learning to

take place in the natural setting where real jobs are performed increasing the chance for success [11, 89]. Lattimore et al. [46] demonstrated the importance of onsite training, when they increased skill acquisition in adults with autism by pairing simulation tasks with on the job training. An excellent example of providing onsite training is Project Search at the Cincinnati Children's Hospital [69]. Project Search provides a demand side employment model, where vocational training and placement occurs with a wide array of jobs within the hospital setting.

Supported employment programs serving this population emphasize providing individualized on the job training that targets job tasks, acclimation to the job site, and social integration [26, 28, 38]. Typically, intensive training is provided upon commencement by a job coach who slowly and systematically lessens support. Hillier et al. provide a description of common training objectives targeted by the employment specialist. These objectives include ensuring the worker with ASD: understands his or her job tasks; can complete all job tasks to the satisfaction of his or her supervisor; understands workplace rules; knows the start time, end time, and break times; understands sick leave and vacation policies; is aware of emergency procedures; knows his or her way to pivotal areas in the building; and knows how to get to and from work.

The uneven cognitive and social abilities in those with ASD require a diverse set of specialized interventions and treatments [22]. Research on instructional strategies with adolescents or adults with ASD is scarce. Even more uncommon is research on teaching skills to this group in the workplace. Available research supports the use of strategies derived from the principles of applied behavior analysis and includes modeling, structured reward systems, video modeling, errorless learning, graduated guidance, and systems of prompts [7, 80]. These strategies are not new, but are essential to teach individuals with a variety of disabilities in the workplace.

More recent research has utilized visual supports implemented via low-tech and high-tech modalities. Riffel et al. [66] used a palmtop computer to teach an adult with autism to roll silverware. The program enabled the worker to view step-by-step pictures paired with auditory instructions on the screen at his own pace. Foley and Staples [17] increased communication skills through the use of augmentative and alternative communication interventions. Adults were taught to use a voice output system to greet, name coworkers, request, comment, give directions, and make choices. In this same study, social stories [24] were used to increase appropriate

conversational interactions. Targeted literacy instruction using written text was used to increase language comprehension and reduce preservative commenting and questioning.

The literature promotes the use of behavior management strategies to reduce inappropriate behavior, including aggression, self-injury, property destruction, and pica in the workplace . Kemp and Carr reduced inappropriate behaviors in three adults with autism when they conducted a functional behavior assessment (FBA) in the workplace. A multi component intervention package was implemented that included functional communication, building rapport, making choices, imbedding demands, and increasing tolerance for reinforcement. Other successful strategies have included teaching the worker to take a short break when upset [7] and implementation of a picture schedule [17].

5.3.4 Work place modifications

In addition to training and instruction to learn job tasks and other skills related to employment, the worker with ASD may also require workplace modifications and adaptations [25, 61]. A variety of strategies may be employed to make the job a better match for the employee by adapting work areas and work tasks. Due to difficulties individuals with ASD experience in modulating sensory input, the actual work site can prove to be an obstacle for success. Environmental stimuli can be distracting and disorienting. Supported employment programs for individuals with ASD incorporate the use of environmental assessments to identify distractions and determine ways to lessen their impact in each relevant work area [38]. Hillier et al. [26] recommend that at a minimum, assessments should evaluate noise level, interruptions, crowding, lighting, and space navigation and should be conducted with the individual's unique needs in mind.

The worker with ASD will likely benefit from clearly defined work tasks and may require restructuring actual job duties or the way a job is performed [16, 56]. There are many types of modifications highlighted in the literature. Providing a consistent schedule for completion of work tasks has increased job predictability [7, 17]. Organizers have been utilized to help structure and keep track of work. Simple organizers have included the use of instruction sheets, notebooks, labels, and checklists [7, 17, 25]. Mc Clannahan et al. [53] utilized activity schedules based on a task analysis of work assignments while Hume and Odom [31] used a structured work

system to organize the job task. Idle time in the workplace may prove to be problematic and also require restructuring. Down time alternatives including taking a walk or playing a computer game successfully reduced the amount of unstructured time for adults with autism [25].

5.3.5 Long-term support

The intensive individualized training provided by a job coach should slowly and systematically lessen over time. However, workers with ASD may need extended support to avoid later deterioration in work performance which can lead to employment failure [38]. Long-term support services are especially important to this group since unexpected changes can be disruptive. Personal accounts by adults with ASD recommend ongoing vocational services that would help prepare for and negotiate the types of challenges that arise from day to day [56].

Supported employment programs serving this population implement both natural and external long-term supports [38]. Natural supports include identified supervisors and company employees [80, 82]. As the job coach fades, active strategies are employed to transfer relationships and support responsibilities to these individuals resulting in sustained and continual assistance [90]. The importance of onsite support was stressed by supervisors of individuals with ASD who were interviewed regarding their experiences [25]. Specifically, they reported natural supports being essential for explaining and helping the employee deal with changes in the workplace.

Some individuals may need intensive external supports for continued success. External supports often include follow-up and close monitoring by supported employment staff [89]. The TEACCH program provides extensive long-term support indefinitely through the provision of site visits, phone calls to the employer, contact with families and/or residential staff, and retraining when needed [38]. Long-term supports may extend beyond the work day since factors in the whole life of the individual with ASD could potentially influence job performance. Certainly, residential, social, and medical matters may have an impact. Therefore, support staff may also act as a liaison for relevant non-work issues. Other long-term supports may include the provision of targeted instruction in areas of deficit outside of the workplace [26, 28, 38]. Areas of instruction have included social groups to address difficulties in social interaction and/or individual counseling services which focus on developing appropriate coping strategies for the work situation.

5.4 Vocational support approaches in autism spectrum disorder: A synthesis review of the literature

Mounting evidence suggests that employment outcomes among adults with autism spectrum disorder (ASD) are poor with employment estimates in ASD ranging from 10%–50% (Autism Speaks, 2009; Hendricks, 2010; Levy and Perry, 2011; Shattuck et al., 2012). Even if employment is achieved, positions held by persons with ASD tend to be part-time (Taylor and Seltzer, 2011), menial (Taylor and Seltzer, 2011), and low skilled (Levy and Perry, 2011). These findings are particularly concerning given literature suggesting that work potentially offers heightened self-esteem and quality of life (Shogren and Plotner, 2012). Moreover, failing to effectively support this largely untapped labor market risks productivity loss, with potential negative impacts for persons with ASD, families, and communities.

In their review article, VanBergeijk et al. (2008) argue that the transition from high school to employment requires careful planning to ensure that adolescents are well prepared prior to leaving school. They recommend a conscientious effort on the part of the school system and community. Yet in a large population-based sample, Shattuck et al. (2012) report that more than half of the sample with ASD were not employed or in post-secondary education by 2 years after high school graduation, yet that number decreased to 11% by 4 years after high school exit. Youth with ASD had significantly lower rates of employment and higher rates of “no participation” relative to peers with speech language impairment, learning disability, and mental retardation (Shattuck et al., 2012). Of note, cognitively able adults with autism are more likely to achieve employment relative to peers with both ASD and intellectual disabilities (Graetz, 2010; Howlin and Moss, 2012; Seltzer et al., 2004), and cognitively able persons with autism have fewer symptoms and problematic behaviors, greater functional independence (Levy and Perry, 2011), and are more likely to be in post-secondary education or employment (Shattuck et al., 2012).

The literature identifies a range of challenges for adults with ASD in achieving stable employment (Hendricks, 2010; Howlin, 2000; Howlin and Moss, 2012; Hurlbutt, 2004; Nicholas and Roberts, 2012; Shattuck et al., 2012). Challenges include difficulty with the hiring process, and a lack of responsiveness and accommodation in the workplace (Richards, 2012). Job

interviewing requires social skills that can be difficult for many individuals with ASD (Richards, 2012). Beyond the challenges of acquiring employment, work settings offer multiple complex difficulties such as challenging physical, social, and sensory environments, and a general lack of effective supports (Kreiger et al., 2012; Richards, 2012; Van Wieren et al., 2008, 2012).

Overall, the literature suggests dissatisfaction with existing ASD vocational supports (Nicholas and Roberts, 2012; Richards, 2012; Van Wieren et al., 2008, 2012). The pervasiveness and multiplicity of barriers invite greater understanding about key issues and ameliorative ways to navigate vocational possibility for persons with ASD. Although there has been improved recognition of the need for vocational services, this need is not strongly informed by epidemiological research or needs assessment, and hence this field continues to be limited by a generalized lack of a research-based approach. Taylor et al. (2012) recently published a systematic review comprising a synthesis of five vocational interventions for youth and adults with ASD, with a focus on studies addressing the transition to adulthood, published since 1980, and with a sample size of ≥ 20 participants. Unfortunately, their review found little evidence for vocational treatment approaches due to poor methodological quality within existing studies. The authors conclude that “no study used random assignment, making it difficult to draw conclusions about the effectiveness of the programs” (p. 536), and they raise concern that supported employment interventions are particularly understudied.

More broadly, some literature examines interventions and models for intervention for the larger developmental and/or intellectual disability population. A study by Migliore and Butterworth (2008) examined outcomes of a vocational rehabilitation program for individuals with developmental disabilities (DD). The authors found that while employee wages did increase and accommodated inflation; wages were still very close to the poverty threshold. Additionally, between the study periods of 1995–2005, a similar amount of time was needed for service users to obtain employment. Maybe and Swain (2009) present a desired intervention model in which supported employment extends beyond job coaching and training, and lasts for the long term, including job progress monitoring, financial assistance, and planning such as when the employee draws a wage, job change, or advancement assistance, support, and training for co-workers, integration into work culture, and monitoring the satisfaction of the employee and employer.

In a longitudinal study by Stephens et al. (2005) involving 2760 individuals with DD, those who became employed increased their adaptive skills, which subsequently decreased if they left the employment. Furthermore, the more competitive the employment, the more that individuals' adaptive skills were bolstered, suggesting that competitive employment may improve an individual's capacity to navigate the community. Jordan de Urries et al. (2005) reported that individuals who were involved in supported employment models, as opposed to sheltered employment, had longer terms of employment and higher salaries.

A meta-analysis by Wehmeyer et al. (2006) examined how technology use impacted the employment outcomes for people with intellectual and DD. A wide range of technologies were used, with a generally positive effect reported. Moreover, interventions that utilized technology with universal design features, such as different input and output options (e.g. visual and audio) or had more accessible features (e.g. larger buttons), were reportedly more effective. Canella-Malone et al. (2006) compared video prompting to video modeling in teaching living skills to adults with DD and found that video prompting (shorter video clips displaying one task each) was effective in all but one case compared to video modeling (one long video showing all skills). In a later study, Goodson et al. (2007) attempted to teach domestic skills to individuals with DD using video clips. However, this was determined to be largely ineffective until an error-correction procedure was initiated involving reviewing the clip and an in-person trainer completing the task, at which point the intervention became consistently effective. Chang et al. (2011) utilized a smart phone for prompting two employees with cognitive and DD at a food service workplace. Both individuals showed a significant increase in job performance when using the prompting tool.

Overall, the studies suggest promise in supporting individuals with disability in the workplace. However, little is known about the application of vocational supports specifically with individuals with ASD. While the broader literature appears to indicate program effect, there is little to ascertain whether persons with ASD have specific barriers that uniquely reflect their profile. This merits a review of evaluation studies that focus specifically on ASD.

5.5 Purpose of the study

To address these gaps and conceptually ascertain the overall state of the vocational support intervention literature in ASD, review of this literature was conducted. The aim of this synthesis was to identify studies addressing employment supports in ASD including studies that examine employment acquisition and retention. The guiding premise of “vocation” entailed an inclusive definition that moved beyond only employment as an outcome, as vocation is deemed to more broadly comprise meaningful, routine, sustained activity that is growth provoking, personally rewarding, and associated with the provision of a living wage.

5.6 Supported employment

Supported employment is defined as the “process of enabling a person with a disability to secure and maintain paid work that is in a regular work environment” (Pozner and Hammond, as cited in Mawhood and Howlin, 1999: 231). It tends to involve some degree of formal training for employment preparedness, job matching according to abilities and propensities of the individual, and ongoing support in the workplace. The supported employment approach has largely been used for facilitating employment success for individuals who are more intellectually able, although success has been demonstrated with individuals with lower IQ, education, and language ability. Supported employment generally focuses on the vocational tasks required of a job, including micro- and mezzo-level skill requirements for employment stability. Job retention and stability are generally increased although detail about these processes is relatively sparse.

Earlier and protracted intervention appears to offer possibilities. With job supports, individuals generally were more likely to get a job, had higher salaries, and more diverse types of jobs, and were employed longer than controls. For instance, Howlin et al. (2005) report that of the 19 individuals working at the time of an original study in 1999 who were contacted at follow-up (2005), 13 were still employed in permanent jobs; 4 were lost to follow-up, 1 had just reregistered with the program, and 1 was receiving further support from the program. Programming notably included three core elements: employment preparation, job finding tailored to the person with ASD, and workplace support. Workplace support included support to the employee as well as colleagues and supervisors.

McClannahan et al. (2002) reported on a longitudinal study that began implementing intervention programs in childhood and followed participants for 15–25 years. Where individuals

were supported long term from childhood to adulthood, training for vocational placements included standard behavioral analysis such as reinforcement procedures, behavior shaping including prompting and fading, and skill generalization. Starting in adolescence, programming focused on vocational skills such as community participation, keyboarding skills, self-care, social interaction, money management, and language development. As young adults, participants became part of the supported employment program, typically serving individuals with mild to moderate cognitive impairment. Findings from this study illuminate benefits from long-term, tailored supported employment programs for persons with ASD, including stable employment and wages, individual satisfaction with work, and employer commitment to employing persons with disabilities.

Garcia-Villamizar and Hughes (2007) studied the effects of a supported employment program for adults with ASD. Adults in the program scored significantly better on more than half of the measures tested, while the unemployed group showed no change in cognitive performance over the duration of the study. These results offer support for the contention that vocational rehabilitation offers promise in potentially increasing cognitive performance for adults with ASD.

The employment support approach emphasizes structured, supported work placements in community settings. In this vocational support modality, the young person is placed in a local organization to work alongside others at the worksite. Individual placement comprises a supported job in the community, and the employee is hired and supervised by the employer, earning similar wages and benefits as fellow employees. Individuals can receive support from staff or from a disability employment support agency. Over time, the goal is for the employee to develop increased independence and for vocational support to taper.

Sheltered work programs differ from this approach in that sheltered workshops tend to place individuals in segregated settings consisting largely only of employees with disabilities. In that model, employees are paid based on their productivity rather than receiving market wages. Better outcomes from supported work models relative to sheltered workshop models have been demonstrated (Garcia-Villamizar et al., 2000; Rusch and Hughes, 1989); hence, there has been a shift away from the sheltered work model in favor of a community placement model.

Supported employment occurs in a variety of service sectors such as food services, hospitality, recycling and delivery, retail, gardening, industrial laundry, and agriculture. Simulation-based training and job coaching appear to enhance learning in the workplace (Lattimore et al., 2006). Of particular importance, supportive employment or job coaching is a specific mechanism that is reported to fulfill a range of roles: to facilitate finding a job, to teach the individual with ASD how to apply and interview for a position, to liaise between an individual with ASD and his or her co-workers or supervisor, to deal with crisis situations that may arise, and in so doing, to support overall vocational success.

Overall, job coaching emerges as a common and consistently important component of employment support. As an example, a determinant of success in the evaluation of a supported employment program was a close liaison between the employers and the job coaches (Mawhood and Howlin, 1999). The coaches were considered integral to finding employment for the individual with ASD and for providing needed support in the work environment. This close contact allowed for exploring appropriate job opportunities and matching the skills and abilities of a specific employee with a suitable job. The greatest difficulty of job coaching was reported to be the large amount of one-on-one time required of the coach. Yet, this investment of time and effort was deemed necessary, as many individuals with ASD experienced difficulties in both employment seeking and their understanding of the social aspects inherent in work settings.

5.7 Technology and media-based support tools

A growing body of literature addresses the application of media and technology to support vocational success for adults with ASD. As an example, the study by Hillier et al. (2007) (also cited above) included a media component using videotaped sessions to support job interview practice. This study entailed comprehensive employment support processes, and hence is listed above, but also utilized media to augment learning and application.

Media and technology are delivered in a variety of forms which are reported to yield distinctive outcomes; some approaches are used for teaching away from the jobsite, while others are implemented onsite. Our review found two studies that specifically cited the use of technology, in the first instance, media use (Allen et al., 2010), and in the second, online application (Burke et al., 2010).

Using media, Allen et al. (2010) explored the use of videotaped modeling of work behaviors as a method to support young adults with ASD. The study focused on three employees (one adolescent and two young adults) who received vocational instruction that featured videotaped demonstrations of skills necessary to conduct work, such as how to promote products in a retail setting while wearing an air-inflated mascot costume. Over a 4-month period, the participants were observed before and after watching a video in which a model performed the various skills of the position. After watching the video, participants were able to correctly use the skills modeled in the video in a work setting. The participants reported that they enjoyed the work, and comments from supervisors were positive.

Another study reported on adolescents with ASD being trained to effectively use personal digital assistant (PDA) tools to prompt them in completing work tasks. Burke et al. (2010) created a performance cue system using PDAs (an adapted iPhone application) and assessed its effectiveness in teaching work-related behaviors. Within this article, two studies were conducted with six young adults with ASD, whereby participants were employed to assist in the delivery of a fire safety training session. They were asked to complete 63 scripted behaviors. In the first study, three participants completed the company's training program and were exposed to the PDA performance cue system only if it became necessary to meet criteria. The second study involved different participants who accessed only the PDA performance cue system to learn the same work behaviors. A multiple baseline design was used to measure the efficacy of the program for each participant over time. Results showed that five of the six participants achieved the desired work behaviors only after the introduction of the performance cue system. Accordingly, such technology-based cuing may support complex task learning by breaking down tasks into discrete steps seen on the iPhone or other PDAs.

5.8 Summary

While the ASD vocational literature offers helpful knowledge at this early phase of its development, further study is needed to ensure stable, welcoming vocational communities in which individuals with ASD can enjoy meaningful work and optimal quality of life. The reviewed studies provide a backdrop for vocational scholarship in ASD, and cumulatively offer an emerging "test of concept" of current vocational approaches for persons with ASD.

This paper provided a review of evidence based research related to employment for individuals with ASD. There are many reasons to examine factors which lead to improving employment outcomes. First and foremost, meaningful integrated employment should be an option for all individuals with ASD who wish to work [49]. Employment enables adults with and without disabilities to earn wages they can use to become self-sufficient. Further, employment provides a forum that leads to improvement in one's quality of life. While personal factors are a sufficient motive, employment also provides a number of economic advantages as it leads to less reliance on government funds and more contribution to taxes [36].

Research overwhelmingly demonstrates disappointing employment outcomes for this group. The vast majority is unemployed and for those who do have gainful employment, underemployment is common. The increased prevalence of ASD coupled with unique social, communication, and behavioral characteristics translate into the need for services to help them achieve employment success. Services provided through the traditional VR system are less than optimal and do not provide sufficient support [14, 47]. Furthermore, most individuals with ASD, because they do not have an intellectual disability, are ineligible to participate in state and/or federally funded programs designed to assist individuals with other types of developmental disabilities [56]. Such inadequacies in services, both because of quantity as well as quality, result in a tremendous shortcoming and a waste of human potential. It is clear that vocational outcome is highly dependent upon the adequacy of the facilities that are available to adults with autism [71]. Fortunately, there is a burgeoning body of research outlining efficacious supports and strategies designed to increase vocational success. This literature has major implications for this population, since many cannot obtain or keep a job without the appropriate assistance and oversight [16]. While this body of research gives shape to this critical topic there are many limitations worth noting. The number of peer reviewed articles on this topic is sparse and is hampered by limited sample sizes and restricted population ranges. Research that does exist covers a wide array of topics, yielding few solid conclusions in any given area. Much can be learned from vocational support programs describing successful supports implemented with individuals with ASD; however, there are only a handful of such descriptions.

Despite these limitations, the research outlined in this paper provides direction as we move to improve employment outcomes for a significant and growing number of people. Further, the

research provided here sets a course for future study on this topic. There is a substantial need for rigorous evaluation of methods of intervention and the means for implementation within the workplace. Such studies should focus not only on job tasks, but also the interpersonal skills that will foster a positive work experience. Longitudinal studies documenting strategies that lead to favorable outcomes and create lasting effects are essential for furthering knowledge and delivering services that are both beneficial and cost effective.

Individuals with all levels of ASD, and all levels of cognitive, social, and behavioral functioning, have the ability and desire to work. Consideration of individual characteristics including strengths, weaknesses, as well as specific interests can lead to an appropriate job placement. This careful match between the individual and the environment, when coupled with implementation of proper supports, can result in successful and ongoing employment.

5.9 References

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